Chaos and bifurcations in continuous language-model feedback

Timothy D. Jones
Praxagent

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Status of this draft. Unless marked otherwise, results on the soft-token map are numerically validated in float64 (rung 1 or 2 of the evidence ladder in Appendix L). Float32 maps are labeled as such. Huginn's spectra are float32 and its accuracy runs bfloat16. **[Planned:** Red**]** marks planned work and **[In progress:** orange**]** running work. The only computer-assisted proof is that of the period-8 orbit of Table 4 (rung 3, Appendix L).

Abstract

Feeding a language model the expected embedding of its own next token, as Soft Thinking does, turns inference into a dynamical system. In Qwen2.5, OLMo-2 and Llama-3 models of 0.5B to 72B parameters, temperature and steering move this loop between repeating text and chaos, with positive finite-time Lyapunov exponents that exact tangents reproduce. The chaos has practical consequences. Rerun with only the batch size changed, chaotic float64 loops separate until their most probable tokens differ, on the schedule the Lyapunov exponent sets, while periodic loops mostly stay together, in four models up to 70B. In arithmetic tests designed before the runs, merely running a problem in a batch with others, which changes only the rounding, changed 56 % of the answers of bfloat16 soft-thinking inference in a 0.5B model and 55 % in a 7B model, far more often than with greedy decoding. An exploratory reset of the fed-back state at every step removed most such changes. In tests fixed before their data, Soft Thinking with its published settings changed 21 % of the answers to fresh GSM8K problems with the batch layout alone, and reasoning models that answer 86 to 90 % of them correctly (DeepSeek-R1-Distill-Qwen-7B, QwQ-32B) changed 14.5 and 19.5 %, deciding correctness for 12 to 15 % of the problems; ordinary sampled reasoning with the same random numbers changed about as many. The two sources differ by model: in float64 the loops of Qwen2.5-7B and DeepSeek-R1-Distill-Qwen-7B mostly contract, so their answer changes are bfloat16 rounding carried to answers that sit near ties, while the loops of QwQ-32B and DeepSeek-R1-Distill-Llama-8B grow slowly in about half of the copies, and a relative displacement of 10^{-10} alone changes about 11 and 18 % of their answers, respectively. Exactly repeating text can hide a chaotic state. In Qwen2.5-0.5B, along recorded orbits, steering creates the chaos mainly by making next tokens compete, which turns the softmax readout from a contraction into an expansion; a logit bias alone suffices. Earlier injection carries chaos to lower temperature, steering at two depths creates chaos neither depth causes alone, and two chaotic loops writing one text lock each other into periodic orbits. Whether a prompt's loop is chaotic does not predict the accuracy of ordinary sampled answers beyond what the model's uncertainty about its first answer token predicts, and we find no edge-of-chaos advantage. The loop obeys quantitative laws of low-dimensional maps: doubling ratios approach Feigenbaum's constant (4.6688 against 4.6692), periodic windows accumulate on folds with a $1/N^2$ law, passage times predicted in advance from fold normal forms came within one iterate in 42 of 48 cases at four new folds (the six misses, at a period-6 fold, 3 to 5 % short), and interval arithmetic proves an attracting period-8 orbit. In the recurrent-depth model Huginn, a frozen test confirmed a predicted accuracy loss. The loop is not ordinary sampled generation.

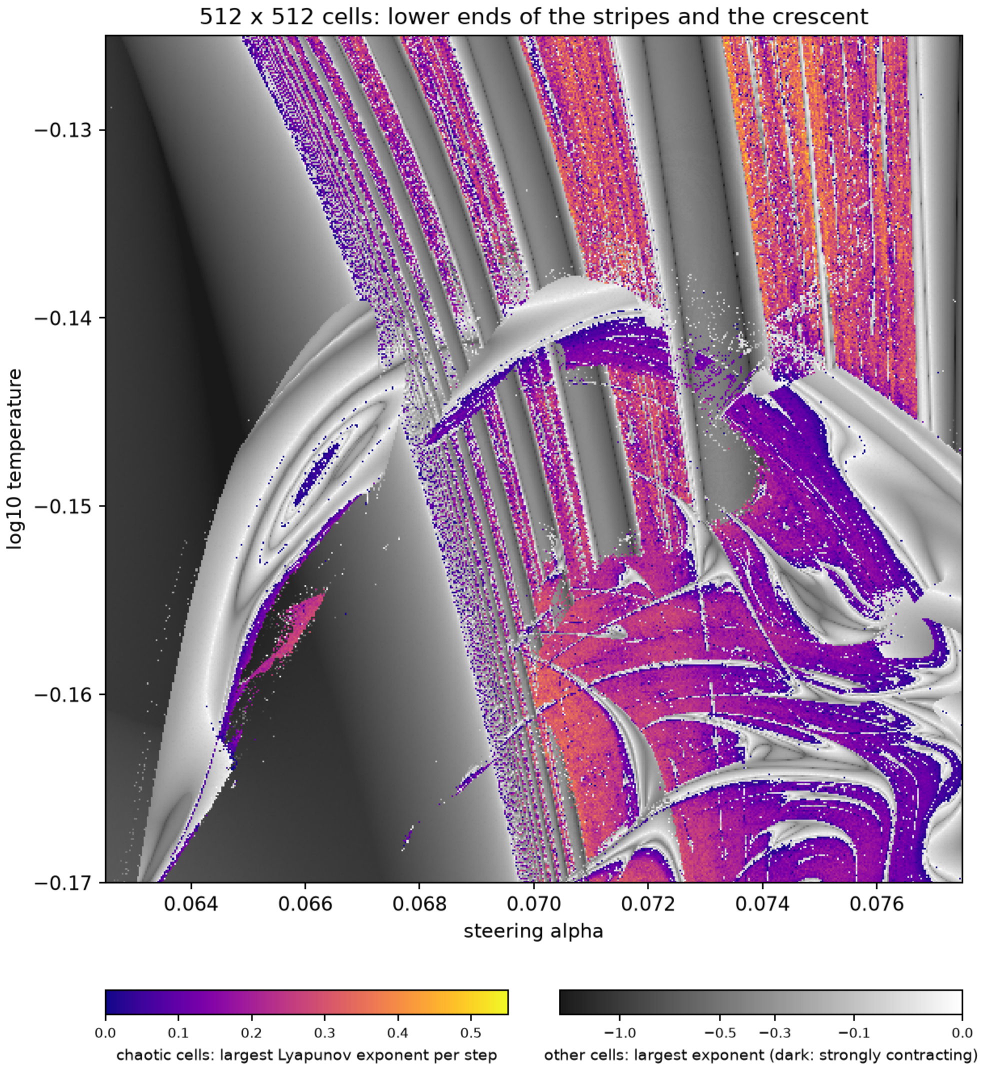


Figure 1: Chaos and repeating text in a language model fed its own output: the soft-token map of Qwen2.5-0.5B-Instruct, steered along the philosophy direction at layer 12, on 512 × 512 float64 cells over the steering strength $\alpha \in [0.0625, 0.0775]$ and the temperature, $\log_{10} T \in [-0.17, -0.125]$. Colored cells are chaotic by the finite-horizon criterion of Appendix B, and the color gives their largest Lyapunov exponent (left color bar). In gray cells the model settles into a repeating loop of text, shaded by its largest exponent, which measures how strongly the loop contracts (right color bar; light gray: close to losing stability); 4.4 % of the cells, also gray, have neither a detected period nor a positive exponent. Figure 9 names the loops.

1 Introduction

Several recent inference methods feed a language model its own continuous output. Soft Thinking replaces each sampled token by the expected embedding under the next-token distribution [1], latent-reasoning methods feed hidden states back [2], and recurrent-depth models iterate a block toward a fixed point [3]. Each turns inference into an iterated map, and an iterated map can be chaotic. We show that the expected-embedding loop of released language models is chaotic over wide ranges of temperature and steering strength, explain why steering makes it so, and measure what the chaos does: in a chaotic regime an irrelevant change of batch layout decides what the loop writes. In bfloat16, where nearly every loop diverges, the batch layout changed the answers of a short soft-thinking pipeline on 56 % and 55 % of 96 arithmetic problems in two models. The same loops obey the quantitative laws of low-dimensional dynamics, down to Feigenbaum's constant.

Three consequences follow for the use of such loops. The first concerns reproducibility: in a chaotic loop, differences far below anything a user controls, such as the batch layout or the numerical precision, decide the output, and in an exploratory test resetting the fed-back state before every soft step removed nearly all such changes (Section 6). Continuous-feedback inference can therefore be irreproducible because of its dynamics, not because of an implementation error. The second concerns steering: practitioners choose steering strengths, and here the strength is a bifurcation parameter, so near a boundary a small change switches the attractor or brings chaos, and a smooth dose-response cannot be assumed (Section 17). The third concerns prediction: where the loop is periodic, normal forms computed from the model's derivatives predicted passage times fixed before they were measured: at four new folds the normal form's passage integral (the secondary predictor of the first test, the primary of the second) came within one iterate of 42 of 48 passages, and the other six were 3.1 to 5.2 % short (Section 8), so the quantitative tools of dynamical systems apply to the inference loops of released models, and larger models do not simply leave the chaos behind (Section 14). We also report a negative result: beyond the model's first-token uncertainty, the loop's regime carries no detectable information about the accuracy of ordinary sampled answers.

Figure 1 shows a small piece of the parameter plane of one such map, built from Qwen2.5-0.5B-Instruct. Cells with a positive finite-horizon Lyapunov exponent lie side by side with windows in which the model repeats a fixed loop of text. Parts of this picture have been seen before. Recent work has measured chaos-like sensitivity in language-model generation [4, 5], temperature-driven changes of output statistics [6–10], periodic attractors of paraphrasing loops [11], attractor states of model-to-model conversations [12], and, through their Jacobians, the stability of looped and recurrent-depth models [13–16], and finite-depth readout dynamics have been used to predict the answers of looped models [17]. Besides Soft Thinking, probability-weighted vocabulary embeddings are fed back in training [18]. In a small task-trained looped transformer, fold coefficients computed from derivatives predict passage times and spectral gaps without fitting [19].

We study the expected-embedding ("soft-token") map of released pretrained language models, defined exactly by their stored weights and run with a fixed chat prefix at a fixed position. We follow it in fifteen models, from Qwen2.5-0.5B to Llama-3.3-70B and Qwen2.5-72B (three dense families and a mixture of experts, with three further models as controls for the latter), in most detail in Qwen2.5-0.5B-Instruct, and in two variations: two models writing one text, and a recurrent-depth model that iterates its own block. We chose these families because each releases open weights at several sizes, so that size can be varied within a family, and because they come from three developers with different training data and recipes; OLMo-2 also releases its training data, and Llama-3 has public sparse autoencoders, which we use for feature controls. The map is smooth. Its Jacobians come from automatic differentiation, and its periodic orbits can be computed and continued like those of any smooth map. Chaos and periodic windows occur in many nonlinear feedback systems. Random-network theory, in which a specified ensemble of recurrent networks becomes chaotic above a critical gain, in continuous time [20] and in discrete time by a quasi-periodic route [21], and the period-doubling cascades and shrimp-shaped domains of low-dimensional maps [22, 23] motivate looking for such structure here, but they do not show that a trained network realizes it. We test this directly. We locate chaos and periodic windows in the models' parameter space, read the texts the loop writes there, and check whether bifurcation theory, applied to derivatives computed from the model, predicts its behavior before the behavior is measured. The models realize this structure, and the predictions largely hold. We also show where the chaos comes from (Section 3; Section 12 assembles the pieces into one picture), and that larger models do not simply leave it behind (Section 14). Ordinary sampled generation with a growing context is a different system; Section 13 also measures how much of the mechanism survives sampling.

Contributions. The results fall into four groups.

1. *Chaos in released models, and what it does to inference* (Sections 5 and 6; Appendix H). Temperature and steering move the loop between repeating text and chaos with embedded windows in all three model families, Qwen2.5, OLMo-2 and Llama-3. The regime decides whether rounding differences between runs grow. Run twice with only the batch layout changed, 190 of 192 chaotic float64 loops of Qwen2.5-0.5B grow past a relative distance of 10^{-2} after a median of 102 steps, in line with the Lyapunov exponent (median ratio 0.91), and at 173 of them the most probable readout token comes to differ; 182 of 192 periodic loops stay within 10^{-2}. Qwen2.5-7B and Llama-3.1-8B without steering (24 of 24 and 20 of 20 chaotic loops diverge, 0 of 24 and 1 of 20 periodic ones) and Llama-3.3-70B with steering (24 of 24, none of 24) behave alike. In bfloat16 nearly every loop diverges, and in a frozen test on 96 additions the batch layout changed the answer of a soft-thinking pipeline in 56 % of problems in Qwen2.5-0.5B and 55 % in Qwen2.5-7B, against never and 10 % for a forced greedy baseline. An exploratory causal test shows that the loop carries this: resetting the fed-back state before every soft step removes nearly all of it (on an H100, 66 against 1.6 % of answers changed in Qwen2.5-0.5B, and 64 against 1.6 % in Qwen2.5-7B), and accuracy changes by at most 5 points. In a test frozen before its data, Soft Thinking with its published settings changed the answers to 21 % of 256 fresh GSM8K problems in Qwen2.5-7B with the batch layout alone, on an H100, mostly between wrong answers; replaying the fed-back tokens cut this to 6 %; with reasoning models that answer 86 to 90 % of the problems correctly through soft thinking (DeepSeek-R1-Distill-Qwen-7B and QwQ-32B), 14.5 and 19.5 % of the soft-thinking answers changed, and a similar share with discrete sampled thinking. A chaotic loop's text reuses the phrases of neighboring loops in no fixed order, and a copy started 10^{-10} away writes the same text for 9 to 150 steps, then a different one (Table 2). Exactly repeating text can hide a chaotic state (Section 10). Beyond the model's first-token uncertainty, the loop's regime carries no detectable information about the accuracy of ordinary sampled answers, and we find no advantage at the edge of chaos.

2. *Why the loop is chaotic, and how steering makes it so* (Section 3). The loop's derivative factorizes into the inverse temperature, the covariance of token embeddings under the model's own next-token distribution, and the network's Jacobian; with tied embeddings and without the network the map would be a modern Hopfield update, which cannot cycle. The factorization proves contraction at sufficiently high temperature, the zero-temperature limit is eventually periodic, and chaos in our sweeps occupies intermediate temperatures. Stability belongs to the loop, not to its steps: along a period-8 orbit every step expands some direction while the cycle contracts. Along the sampled orbits, steering makes next tokens compete. Steering into chaos raises the readout factor of each step's growth by 2.58 nats per step relative to nearly unsteered cells, while the network factor changes by only -0.07 (interval $[-0.23, 0.12]$); a logit bias alone, with the network untouched, makes 3.5 % of the plane chaotic; and an orbit-preserving crossover attributes the expansion at fixed states to steering's change of the readout (+0.81 nats per step in the 0.5B model), not of the network, whose change damps it (-0.49). Random directions also steer into chaos. Earlier steering carries chaos to lower temperature, in Qwen2.5-0.5B and Llama-3.3-70B, and two vectors at two depths create chaos where neither alone does (Section 6).

3. *Low-dimensional organization, predicted from derivatives* (Sections 7 to 11). The windows carry concrete texts: a period-adding family gains one "is X" pair per period step. A fold's normal form, computed from derivatives, predicts without fitting how long trajectories linger near the vanished orbit: in two frozen tests at four new folds, 42 of 48 forecasts fell within one iterate, and both all-row criteria failed (the second because passages from the entrance of a period-6 fold's channel came in 3 to 5 % short), and windows accumulate on the fold with a $1/N^2$ law. On the nine fold curves we continued, all thirteen tangent reversals are located cusps, and near them local unfoldings, frozen before solving, matched the orbits that Newton solves recovered at twelve parameter offsets in number and at eleven in stability. A doubling cascade has superstable points through period 768 and finite-order Feigenbaum ratios that approach δ (the last, 4.6688, against 4.6692). A period-5 window of a plane steered at two depths has the anatomy of a shrimp in a two-dimensional map: fold and flip curves bound a main cell with legs and doubling shells, a trace-zero loop replaces the head of a one-dimensional shrimp, and twice the fold curve has two cusps of opposite kinds and a self-crossing. Around one of these junctions an orbit census fixed in advance matched the predicted counts at 9 of 13 points and found two coexisting attracting orbits at all five points inside it. An interval-arithmetic proof shows that a primitive period-8 orbit exists and is locally attracting, to our knowledge the first proof of a periodic orbit of a full transformer language-model map (Appendix L).

4. *Beyond the one-token loop* (Sections 13 to 16). The chaos is not an artifact of the one-token window: with a memory of 8 or 16 soft tokens Qwen2.5-7B is chaotic at more temperatures than with one, and in Soft Thinking's growing context a tiny early perturbation grows exponentially inside its one-token chaotic band, though not in every model. The organization recurs across families, and size does not set the chaos: across fifteen models from 0.5B to 72B, a larger model's chaos can move to higher temperature instead of vanishing. Resolved through the stack, the network damps a change of the fed-back token more strongly in the larger Qwen2.5 and Llama-3 models, because the change falls behind the growing residual stream; a mixture of experts with 3B of its 35B parameters active per token damps more than a dense 9B model of its family. Two loops writing one text lock each other into periodic orbits: 60 % of pairs of differently steered 0.5B agents that are each chaotic alone are periodic together (twelve such orbits verified by Newton's method). In a second system, the recurrent-depth model Huginn, a spectral estimate ranks held-out questions by the size of their residual under relaxation, and a frozen generation test confirms the predicted accuracy loss at relaxation 1.20 and noninferiority at 1.05.

Related work. Chaos-like sensitivity of generation [4, 5, 24], temperature-driven changes of output statistics [6–9], repetition loops [25, 26] and behavioral attractors of paraphrasing and conversation [11, 12] have been measured in generated text, and one study fits a noisy effective map to a hidden-state coordinate [10], without computing orbits, multipliers or bifurcations of the model's own map. Equilibrium, looped and recurrent-depth models have been analyzed through fixed points and Jacobians [3, 13–16], and the closest methodological precedent, a derivative-based fold analysis of a small task-trained looped transformer, predicts passage times without fitting [19]. Appendix A discusses this work, expected-embedding feedback, and bifurcations, Feigenbaum ratios and shrimps in neural models in detail.

To our knowledge, this is the first work to compute periodic orbits with their multipliers, continue fold and flip bifurcation curves and locate cusps in the feedback map of a released pretrained language model. It is also, to our knowledge, the first to make quantitative predictions from such a model's derivatives, for the objects named here, and confirm them prospectively: passage times at four new folds, where 42 of 48 forecasts fell within one iterate and both all-row criteria failed (by one row, and at the entrance passages of a period-6 fold), and the orbits recovered and their stabilities at frozen offsets near cusps. Derivative-based normal-form predictions in a task-trained looped transformer precede ours [19]. We know of no earlier frozen held-out test of a spectral prediction of a pretrained language model's response to relaxation, which we give for Huginn, and of no earlier computer-assisted proof of a periodic orbit of a full transformer language-model map. These priority statements rest on targeted searches (12 and 15 October 2026) that found no prior work combining these elements for the feedback map of a pretrained language model. That is an absence of evidence within the search, not a proof of priority.

2 The inference map

Definition. We describe the map for Qwen2.5-0.5B-Instruct, the model studied in most detail; the other models enter the same construction with their own embeddings and steering layers (Appendix H). Let $E \in \mathbb{R}^{V \times d}$ be the input embedding matrix ($V = 151{,}936$, $d = 896$). The model reads a fixed chat prefix, with the system prompt "You are an AI talking with another AI. You have complete freedom. Feel free to pursue whatever you want." and the user turn "Hello!", followed by a single input embedding x, and produces next-token logits $z(x) \in \mathbb{R}^V$. The soft-token map feeds the expected next-token embedding back in place of x:

$$F_{T,\alpha}(x) = E^\top \operatorname{softmax}(z_\alpha(x)/T), \tag{1}$$

where z_α adds αv to the residual stream at layer 12 at every position, prefix included. The steering direction v is contrastive: it compares last-token states of "philosophy" prompts with those of neutral prompts and is normalized to the mean residual norm. Positions are fixed and the window does not grow, so F is an autonomous map of $\mathbb{R}^{896}$. As $T \to 0$ it reduces, where the argmax is unique, to greedy decoding of the same fixed-prefix, one-token-window system. It is not ordinary generation with a growing context.

State space and regularity. Attention is causal, so the keys and values of the prefix do not depend on x, and F is an autonomous map of $\mathbb{R}^{896}$. For the dense models studied here it is real-analytic, because RMS normalization (with its positive regularizing constant), the SiLU activation, attention, rotary embeddings at fixed positions and the softmax readout are analytic; the top-k routing of a mixture of experts makes the map only piecewise smooth. Every soft token is a convex combination of embeddings, so F maps $\mathbb{R}^{896}$ into the compact convex polytope $K = \operatorname{conv}\{e_t\}$, and $F(K) \subset K$. Every orbit therefore enters K after one step, F has a fixed point in K by Brouwer's theorem, and, being continuous on a compact set, it has invariant probability measures there. Because $\log^+\|DF\|$ is bounded on K, Oseledets' multiplicative ergodic theorem applies: for every invariant measure the Lyapunov exponents exist at almost every point [27, 28]. The exponents we report estimate them over finite horizons (Appendix B).

Why feed a model its own output? Feedback is not an addition to language-model inference; it is how text is generated. Autoregressive decoding appends each chosen token to the input and conditions the next step on it, and as $T \to 0$ the map studied here becomes greedy decoding, whose familiar repetition loops are its periodic attractors. What the loop changes is what is fed back: the expected embedding instead of one chosen token. That is the step of Soft Thinking [1], and related methods feed back hidden states [2] or iterate a block to a fixed point [3]; continuous feedback of this kind is studied because it can carry more than one token's worth of information. We also hold the prefix and the window fixed. That simplification is deliberate: it makes the loop an autonomous map whose orbits, exponents and bifurcations can be computed, and in one case proved, much as low-dimensional maps made the routes to turbulence tractable. We then test which effects survive in the fuller systems: Soft Thinking's growing context, sampled decoding, longer fed-back memories and two models writing to each other (Appendices G and 16). The answer-level consequences reported below were measured in a soft-thinking pipeline with a growing context, not in the reduced loop. Deployed systems feed outputs back as well, when agents read their own earlier steps and when pipelines chain model calls.

Steering as a probe. Steering is an established way to control a language model without retraining: a vector added to the residual stream, typically the mean difference between activations on contrasting prompts, strengthens or weakens a behavior, such as sycophancy or refusal [29], sentiment or topic [30], or a personal style of writing [31], and representation engineering uses such directions to read and control concepts [32]. Our recipe is the standard contrastive one [29], added at every position, and our pushes are moderate: at most 12 % of the mean residual norm per vector in every map (the continued curves reach 13.1 %), and 6.25 to 7.75 % in Figure 1. Pushes of this size do change what a model writes, which is what steering is for: in Qwen2.5-7B, 1 % of the residual norm is the largest push that kept greedy accuracy on a multiplication task within 5 points (Appendix D). They are far smaller than activation additions reported as effective in steering work, one of which has nearly ten times the norm of the residual stream it is added to [30]. We do not steer to make the model behave in a particular way. We use steering as a probe: α is a control parameter, like the amplitude of a periodic forcing, that moves the model through a one-parameter family of nearby maps without retraining it, so that bifurcation curves can be traced. The chaos does not depend on it: at $\alpha = 0$, temperature alone gives chaos with periodic windows in Llama-3.1-8B, Qwen2.5-7B and OLMo-2-7B, and in a narrow band in Qwen2.5-0.5B (Appendix J). Nor, in the 0.5B and 8B models, does the direction's meaning: random, sparse-autoencoder and single-neuron directions of the same norm make the map chaotic about as often as the concept directions (at 70B the directions differ; Appendix I).

Precision. Hugging Face RMSNorm layers and rotary-embedding tables compute in float32 even when the model is cast to float64, so we patch both. A float32 map is accurate to 3×10^{-6}–2×10^{-5} per step. That is enough for phase maps (245 of 256 test cells agree with float64) but not for derivatives of composed maps: one-step float32 Jacobian-vector products are accurate to 5×10^{-5}, yet the 8-step monodromy of a period-8 orbit is off by 3%. All orbits, multipliers and test functions below are therefore computed in float64.

3 Why the loop can be chaotic

The soft-token map has a simple derivative. With $p = \mathrm{softmax}(z/T)$,

$$DF(x) = \frac{1}{T} E^\top (\operatorname{diag} p - pp^\top) J_z(x). \tag{2}$$

Qwen2.5-0.5B ties its output matrix to the input embedding, so $z = E\tilde{h}(x)$, where $\tilde{h}$ is the final normalized hidden state, and the derivative factorizes:

$$DF(x) = \frac{1}{T} \Sigma_p J_{\tilde{h}}(x), \qquad \Sigma_p = E^\top (\operatorname{diag} p - pp^\top) E = \mathrm{Cov}_{t\sim p}(e_t). \tag{3}$$

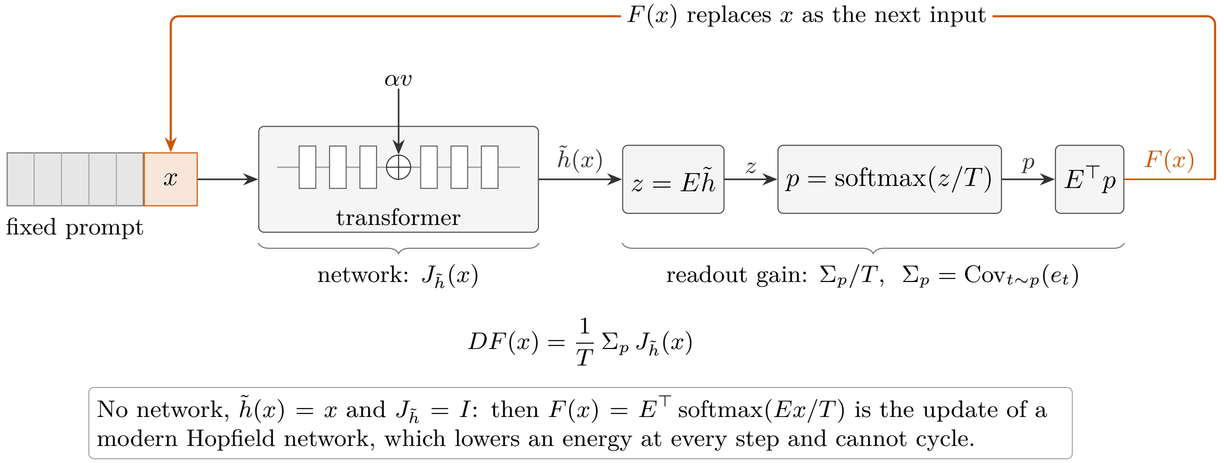

$F(x)$ replaces x as the next input
αv
x
fixed prompt
transformer
$\tilde{h}(x)$
$z = E\tilde{h}$
z
$p = \mathrm{softmax}(z/T)$
p
$E^\top p$
$F(x)$
network: $J_{\tilde{h}}(x)$
readout gain: $\Sigma_p/T,\ \ \Sigma_p = \mathrm{Cov}_{t\sim p}(e_t)$
$DF(x) = \frac{1}{T}\,\Sigma_p\,J_{\tilde{h}}(x)$
No network, $\tilde{h}(x) = x$ and $J_{\tilde{h}} = I$: then $F(x) = E^\top\,\mathrm{softmax}(Ex/T)$ is the update of a modern Hopfield network, which lowers an energy at every step and cannot cycle.

Figure 2: The soft-token loop. A fixed prompt is followed by one input embedding x. The transformer, with αv added to the residual stream at a middle layer (at every position), gives the final normalized hidden state $\tilde{h}(x)$ at the position of x; with tied embeddings the logits are $z = E\tilde{h}$ (untied: $z = W\tilde{h}$), and $p = \mathrm{softmax}(z/T)$ is the next-token distribution. The expected embedding $E^\top p = F(x)$ is fed back as the next input, at the same position after the same prompt. Below: the two factors of the derivative (3), the network's Jacobian $J_{\tilde{h}}$ and the readout gain Σ_p/T. With $J_{\tilde{h}} = I$ the loop is a modern Hopfield update [33], which cannot cycle (Section 3).

The derivative of the loop (Figure 2) is the covariance of the token embeddings under the model's own next-token distribution, amplified by the inverse temperature and composed with the derivative of the network. At a state of the period-8 orbit of Table 4, (3) agrees with forward-mode derivatives of the full map to a relative error of at most 1.35×10^{-13} over eight random directions (Figure 3; `tools/softmax_gain.py`). Without tied embeddings, as in the 7B and 8B models, the logits are $z = W\tilde{h}(x)$ with an output matrix $W \neq E$, and $DF = T^{-1}E^\top(\operatorname{diag} p - pp^\top)WJ_{\tilde{h}}(x)$: Σ_p becomes the cross-covariance $\mathrm{Cov}_{t\sim p}(e_t, w_t)$ of input and output embeddings, which is not symmetric in general. The temperature limits below do not change.

A Hopfield network with a transformer inside. If the network passed its input straight through, $\tilde{h}(x) = x$, the map would be $F(x) = E^\top\,\mathrm{softmax}(Ex/T)$. This is exactly the update rule of a modern Hopfield network whose stored patterns are the token embeddings, at inverse temperature $\beta = 1/T$ [33]. That update lowers an energy function at every step and converges to fixed points. In (3) it is the case $J_{\tilde{h}} = I$, where $DF = \Sigma_p/T$ is symmetric and positive semidefinite. The energy argument covers every map with that property. If DF is symmetric and positive semidefinite everywhere, then $F = \nabla\phi$ for a convex ϕ, and convexity, $\phi(F(x)) \geq \phi(x) + F(x)^\top(F(x) - x)$, gives for $V(x) = \frac{1}{2}\|x\|^2 - \phi(x)$

$$V(F(x)) \leq V(x) - \tfrac{1}{2}\|F(x) - x\|^2. \tag{4}$$

Along a cycle these decreases sum to zero, so every point of the cycle is fixed. F maps every state into the convex hull of the embeddings, where V is bounded below, so the steps of every orbit tend to zero and its limit points are fixed points: such a map has neither cycles nor chaos. In models with tied embeddings, such as Qwen2.5-0.5B, every cycle and every chaotic cell therefore depends on the network factor $J_{\tilde{h}}$ breaking that structure. With untied embeddings the network-free map $x \mapsto E^\top\,\mathrm{softmax}(Wx/T)$ already has the non-symmetric derivative $\mathrm{Cov}_{t\sim p}(e_t, w_t)/T$ and can cycle (two tokens whose output vectors point at each other's input embeddings exchange at low temperature), so the argument does not apply to those models. Attention has itself been read as a Hopfield retrieval [33]; in the soft-token map the retrieval closes a loop around the whole network.

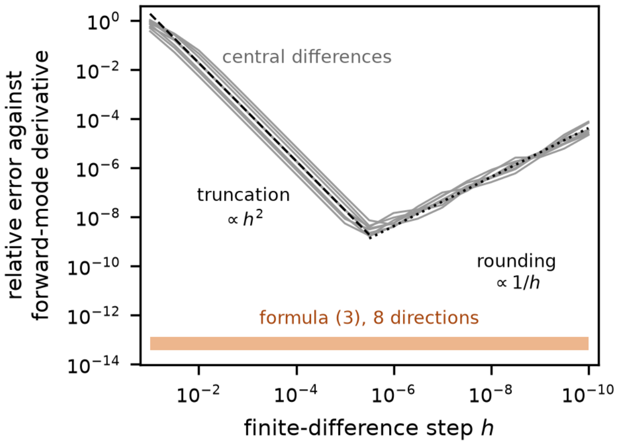


Figure 3: Formula (3) checked against the network's own derivatives at a state of the period-8 orbit of Table 4 (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\alpha = 0.071674$, $\log_{10} T = -0.115$, float64). Gray: relative error of central differences $(F(x+hu)-F(x-hu))/(2h)$ against the forward-mode derivative $DF(x)\,u$ along eight random unit directions u. They converge like h^2 (fitted slopes 1.98 to 2.00 for $10^{-4} \le h \le 10^{-2}$) until rounding takes over near $h = 3 \times 10^{-6}$, where they reach 1.7×10^{-9} to 4.8×10^{-9}. Orange: relative error of $T^{-1}\Sigma_p J_{\tilde{h}}(x)\,u$ against the same derivatives, 3.8×10^{-14} to 1.35×10^{-13}, more than four orders of magnitude below the best finite difference.

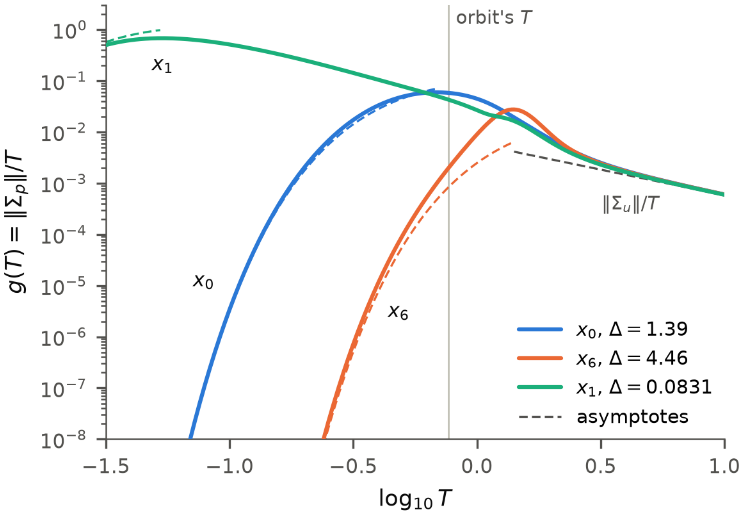


Figure 4: Readout gain $g(T) = \|\Sigma_p\|/T$ (spectral norm) at the fixed logits $z_\alpha(x)$ of three states $x_k = F^k(x_0)$ of the period-8 orbit of Table 4 (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\alpha = 0.071674$), with only the temperature in $p = \mathrm{softmax}(z/T)$ varied. x_0 is the state at which (3) was checked; x_6 and x_1 have the largest and the smallest gap Δ between the two largest logits among the eight states. Dashed: the low-temperature asymptote $\|e_{t_1} - e_{t_2}\|^2 e^{-\Delta/T}/T$ of each state (t_1, t_2 the two leading tokens), drawn below its peak, and the high-temperature asymptote $\|\Sigma_u\|/T$, with Σ_u the covariance of all token embeddings under equal weights. The peaks lie at $\log_{10} T = -1.27$ (x_1, $g = 0.68$), -0.16 (x_0, 0.059) and 0.15 (x_6, 0.028); the vertical line marks the orbit's temperature, $\log_{10} T = -0.115$. Where the two leading logits nearly tie (x_1), g grows roughly like $1/T$ as T falls toward Δ. This is the readout factor alone: the network factor $J_{\tilde{h}}$ is not included, so g does not decide whether DF expands. At the orbit's temperature $g \le 0.064$ at all eight states, while the largest singular value of DF lies between 1.2 and 21 (Section 3).

High-temperature contraction and the greedy limit. The largest Lyapunov exponent of an orbit cannot exceed the orbit average of $\log \|DF\|$, so chaos needs $\|DF\| > 1$ on a substantial part of the attractor. Both ends of the temperature axis work against this, in different ways (Figure 4). As $T \to 0$, p becomes one-hot except near ties between the two leading tokens; away from ties the gain $\|\Sigma_p\|/T$ decays like $Ce^{-\Delta/T}/T$, where Δ is the gap between the two largest logits and the prefactor C is set by the squared embedding distances between the leading token and its nearest competitors, and F tends to the piecewise-constant greedy map, every orbit of which is eventually periodic (Appendix G). This limit is pointwise, not uniform: near a tie the gain is of order $1/T$, so at small positive temperature chaos is excluded only on invariant sets that keep a uniform logit gap, and a chaotic set would have to return to near-ties again and again. At high temperature the bound is explicit. For every distribution p and unit vector u, the projections $u^\top e_t$ range over an interval no longer than the diameter $\operatorname{diam}(E)$ of the set of embeddings, and Popoviciu's inequality bounds a variance by a quarter of the squared range, so $\|\Sigma_p\| \le \operatorname{diam}(E)^2/4$. With $L = \sup_K \|J_{\tilde{h}}\|$, finite because K is compact, $\|DF\| \le \operatorname{diam}(E)^2 L/(4T)$ on K, and F is a contraction of K for $T > T_* = \frac{1}{4}\operatorname{diam}(E)^2 L$, with a single attracting fixed point that tends to the mean embedding as $T \to \infty$; without tied embeddings, $\operatorname{diam}(E)^2$ becomes $\operatorname{diam}(E)\operatorname{diam}(W)$. Here the exclusion is a bound, sufficient but not sharp. It does not cover hard routing or a growing context. Away from near-ties, chaos thus needs temperatures at which the next-token distribution is spread enough to make Σ_p large while $1/T$ still amplifies it. The unsteered sweeps agree (Figure 58). In the four long sweeps (Appendix J) the chaotic temperatures lie in one band, interleaved with periodic windows, and every band lies between $\log_{10} T = -0.20$ and 0.28. Below the bands, all 1547 temperatures swept have negative exponents, and all but five have a detected period; above them, up to the end of the sweeps at $\log_{10} T = 0.3$, 127 of 128 temperatures have a detected period (fixed points at 101) and one has a negative exponent without one. These are finite sweep findings; the wider sweeps of Section 14 find chaos at higher temperatures in some models, and an OLMo-2-32B sweep found three candidate cells below one nat of entropy (at $\log_{10} T = -0.42$, -0.41 and -0.07; one twin offset). A further sweep of the unsteered 0.5B map from $\log_{10} T = 0.3$ to 1.5 finds an attracting fixed point at all 25 temperatures. Its entropy approaches $\log V$ (11.928 against 11.931 nats at $\log_{10} T = 1.5$), and its exponent falls steadily, by 2.14 per decade of temperature at the top of the range, toward the slope $-\ln 10 \approx -2.30$ expected once p is uniform.

Stability belongs to the loop, not to the step. The gain condition is necessary, not sufficient. Along the period-8 orbit of Table 4, the largest singular value of DF lies between 1.2 and 21 at every step, so every step expands some direction, and no step is close to rank one: the second singular value is at least 0.15 of the first, and the second eigenvalue of Σ_p is at least 0.23 of the first (minimum 0.236). Yet the Jacobian of the whole cycle, the product of the eight step Jacobians, has norm between 0.059 and 0.644, depending on the phase at which the cycle is started, against 1.1×10^7 for the product of the step norms. Its eigenvalues, which do not depend on that phase, reproduce the multipliers of Table 4 (0.01104 and -2.96×10^{-7}; the table lists moduli), and its second singular value is 1.7×10^{-6} to 2×10^{-4} of its first. An independent implementation of the model reproduced the factorization (to 8.3×10^{-14}), the step singular values, the signed multipliers and the cycle norms at all eight starting phases. The expansion of each step is undone by the steps that follow, and the strongly anisotropic linear return of the stripe orbits is a property of the composed cycle, not of any single step or of the rank of Σ_p. Nor does the contraction need the actual alignment of the steps: with independent random (Haar) rotations inserted between the eight step Jacobians, all 16 tested products contract, and more strongly than the actual cycle (`tools/alignment_intervention.py`). Stepwise norm or spectral summaries can be inconclusive;

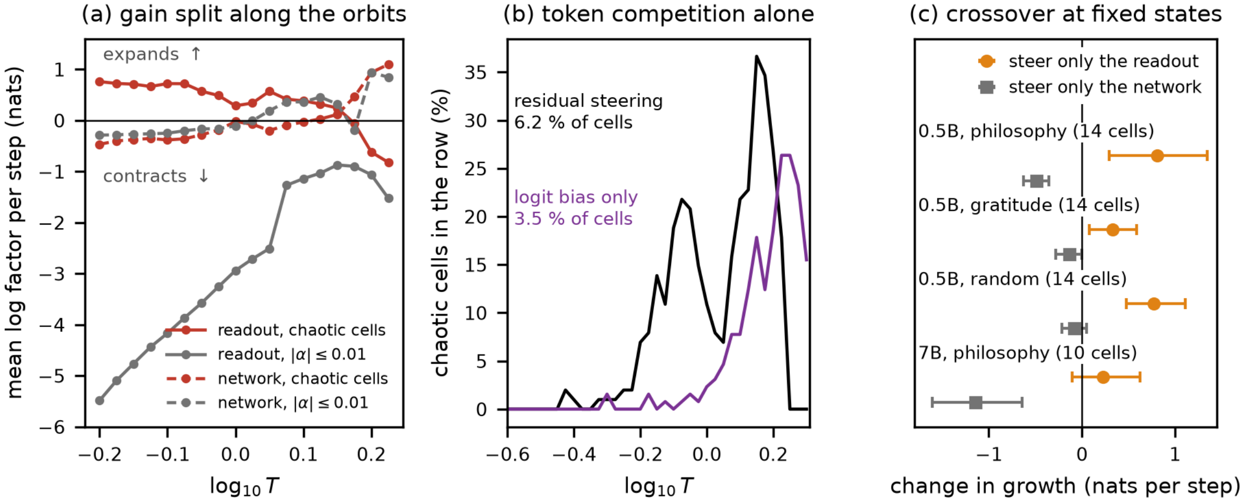


Figure 5: Why steering makes the loop chaotic: the three measurements of this section (Qwen2.5-0.5B-Instruct unless marked 7B, float64). (a) Gain split along the attractors of the block-12 map: mean log readout factor (solid) and network factor (dashed) per step, for chaotic cells (red) and nearly unsteered cells, $|\alpha| \leq 0.01$ (gray), in the 18 temperature rows with at least three cells of each kind. Without steering the readout contracts by 0.87 to 5.49 nats per step; in the chaotic cells it expands, except in the three highest rows, while the network factor changes little. (b) Share of chaotic cells per temperature row under residual steering at block 12 (331 of 5353 cells) and under a logit bias along the direct logit effect of the steering vector, with no residual steering (241 of 6837 cells). (c) Orbit-preserving crossover: change in the finite-time growth along recorded steered orbits when only the readout (orange) or only the network Jacobian (gray) is taken from the steered model; mean over held-out chaotic cells with 95 % bootstrap intervals. `tools/plot_mechanism_summary.py`.

the ordered product of the step Jacobians captures the cycle's linear stability.

Steering makes tokens compete. Steering enters both factors of (3): it moves the logits, and with them Σ_p, and it moves the hidden states, and with them $J_{\tilde{h}}$. Which change makes the loop chaotic? Three measurements on Qwen2.5-0.5B, steered along the philosophy direction, give one answer (Figure 5; details in Figure 6 and Table 1). First, each step's growth splits exactly into a network factor $g_{\text{net}} = (|\delta\tilde{h}|/|\tilde{h}|)/(|\delta x|/|x|)$ and a readout factor $g_{\text{read}} = (|\delta F|/|F|)/(|\delta\tilde{h}|/|\tilde{h}|)$. Averaged along an orbit, their logarithms add up to the largest Lyapunov exponent, up to a boundary term. The split depends on how the intermediate state is measured, so only the sum is intrinsic; we compare each factor between cells within this one convention. Over a 101×53 map in steering strength and temperature (block 12; 5353 cells, 331 chaotic), the chaotic cells' readout factor exceeds that of nearly unsteered cells ($|\alpha| \leq 0.01$) at the same temperature by 2.58 nats per step (95 % interval $[1.78, 3.59]$ over 18 temperature rows; 321 chaotic and 162 nearly unsteered cells). Their network factor differs by only -0.07 $[-0.23, 0.12]$. With nearly no steering the readout contracts strongly, by 0.87 to 5.49 nats per step. In the chaotic cells it expands, except in the three highest temperature rows ($\log_{10} T = 0.175$–0.225), where it still contracts but less. Compared with periodic cells at the same temperature, chaotic cells have a higher next-token entropy (+1.62 nats, $[1.26, 1.91]$), a smaller gap between the two largest logits (-1.37 in units of T, $[-1.98, -0.87]$) and a larger readout factor (+1.36, $[0.91, 1.93]$), each with $p \leq 10^{-4}$ in a permutation test within temperature rows.

Second, a readout-only intervention gives chaos. We add to the logits a bias $\beta\, b$ along the direct logit effect of the steering vector ($b = Wv$, centered and scaled to unit standard deviation over the vocabulary), with no residual steering. This leaves the network untouched and changes only which tokens compete. It makes 3.5 % of the same temperature range chaotic (241 of 6837 cells), against 6.2 % for residual steering, mostly at $\log_{10} T > 0.1$. Creating token competition therefore suffices for chaos, given the unmodified network.

Third, a gain split compares different orbits, so steering could act simply by moving the orbit to other states. A crossover at fixed states separates the two contributions along the recorded orbits. Along recorded steered orbits we evaluated the steered and the unsteered model at the same states and carried four tangent dynamics. Each pairs the steered or unsteered readout $R(p)$ with the steered or unsteered network Jacobian: arms 11, 10, 01 and 00. The frozen validation (arm 11 within 0.02 of the twin exponent at every cell) holds at 23 of 24 cells, all within 0.012; at the remaining periodic cell, which contracts at about -12 per step, the twin offset falls to the rounding floor. The frozen sufficiency criterion (arm 10 within 0.1 of arm 11 on average) fails by overshoot: steering only the readout gives more growth than the full steered map, by 0.23, because steering's change of the network damps it. Over 14 held-out chaotic cells, steering only the readout raises the growth by 0.81 nats per step ($[0.29, 1.35]$). Steering only the network response lowers it by 0.49 ($[-0.63, -0.35]$). The interaction is $+0.26$ ($[0.09, 0.45]$). Below $\log_{10} T = 0$ the steered readout turns a contracting unsteered derivative (-1.03 nats per step) into an expanding one ($+0.72$), more strongly than the full steered map ($+0.34$). Above it, the unsteered derivative already expands at the steered states ($+0.41$), and steering may act by bringing the orbit there. In Qwen2.5-7B, whose embeddings are untied, the same crossover (18 cells of its steered map at block 14; arm 11 equals the twin exponent within 10^{-4} at every cell) reproduces the damping. Over its 10 held-out chaotic cells, steering only the network response lowers the growth by 1.14 nats per step ($[-1.61, -0.64]$), and steering only the readout adds $+0.23$ ($[-0.11, 0.63]$). Eight of the ten lie at $\log_{10} T \geq 0$, where the unsteered derivative already expands at the steered states (mean $+1.33$). The crossover along two more directions of the 0.5B model gives the same signs. For gratitude the readout effect is $+0.33$ ($[0.07, 0.59]$) and the network effect -0.14 ($[-0.29, 0.01]$). For a random direction matched in norm to the philosophy vector, they are $+0.77$ ($[0.47, 1.12]$) and -0.09 ($[-0.22, 0.06]$); 14 held-out chaotic cells each. The readout effect is positive along all three directions, and the network effect is negative but compatible with zero along these two. In these three directions the readout effect does not depend on the direction's meaning.

Along the recorded orbits of Qwen2.5-0.5B, then, steering makes the loop chaotic mainly by making next tokens compete, which turns the softmax readout from a contraction into an expansion. At low temperature it creates that competition at fixed states. At higher temperature the orbit lies in states that are already competitive, consistent with steering having carried it there. Along the philosophy direction, in both models, its change of the network's response damps the growth. These are conditional comparisons on recorded trajectories; they do not establish that the readout change alone would sustain the same attractor. The readout rule holds below $\log_{10} T \approx 0.1$; at higher temperature the network factor itself can expand. Steered at block 4, the chaotic cells at $\log_{10} T = 0.15$ have a network factor of $+0.72$ and a contracting readout (-0.35). Along a line of the hostility-steered map (Section 10) at $\log_{10} T = 0.12$–0.13 the network factor of the chaotic cells is $+0.60$ against $+0.24$ to $+0.38$ at the neighboring periodic cells. The mechanism is not specific to concept-labeled directions. This is consistent with the controls of Appendix J, in which norm-matched random directions also produce chaos in the 0.5B and 8B models, and rotating the steering direction at constant norm moves the chaotic regions but keeps the chaotic share between 13 and 24 % at every one of 103 angles (Figure 7); at 70B the directions differ (Appendix I). What a direction does select is which tokens compete, and therefore what the attractors write.

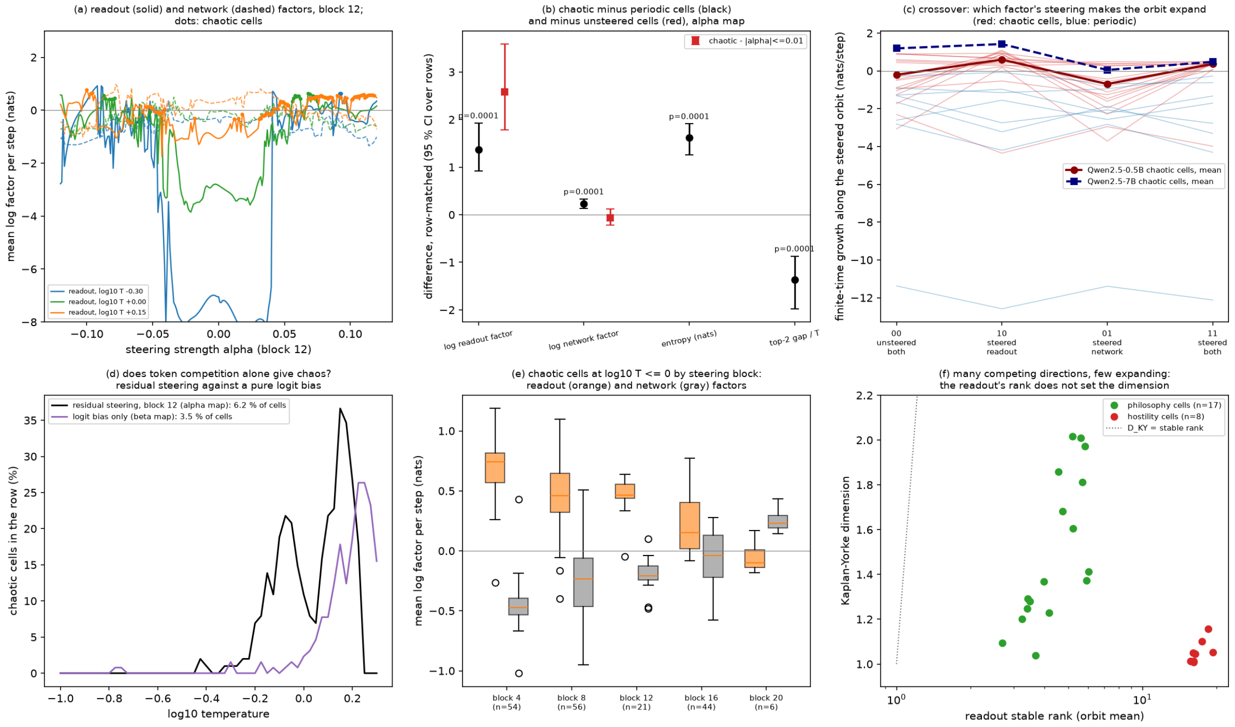


Figure 6: Why steering makes the loop chaotic (Qwen2.5-0.5B-Instruct, philosophy direction, float64). (a) Readout factor (solid) and network factor (dotted) along steering lines at block 12, at five temperatures; dots mark chaotic cells. (b) Row-matched differences between chaotic and periodic cells (black) and between chaotic and nearly unsteered cells (red) on the 101×53 map, with 95 % bootstrap intervals over temperature rows and permutation p-values. (c) Orbit-preserving crossover: finite-time growth along each recorded steered orbit when the readout and the network Jacobian are taken from the unsteered (0) or steered (1) model; red, chaotic cells; blue, periodic cells; dark red, the mean over held-out chaotic cells; navy, the same mean in Qwen2.5-7B. (d) Share of chaotic cells per temperature under residual steering at block 12 and under a pure logit bias along Wv. (e) The two factors of the chaotic cells by steering block, at $\log_{10} T \leq 0$. (f) Kaplan–Yorke dimension against the readout's trace-to-leading-eigenvalue ratio at the chaotic cells of the dimension measurement (Appendix C); dotted, equality.

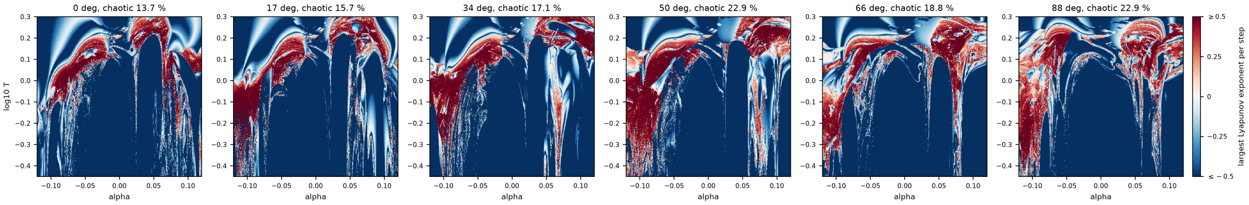


Figure 7: The soft-token map of Qwen2.5-0.5B-Instruct (steering at layer 12) while the steering direction rotates at constant norm from one Gaussian direction (0°) to another (90°): largest Lyapunov exponent (color bar; clipped at ±0.5, red positive) at six of 103 angles, with the chaotic share. Float32 maps of 296×512 cells over $\alpha \in [-0.12, 0.12]$, $\log_{10} T \in [-0.45, 0.3]$; exponents from 80 measured steps at twin offset 10^{-4}; each map continues from the final states of the previous angle after 40 steps, and every tenth starts from the fixed start after 300 steps. Where attractors coexist, continued and fresh starts can reach different ones, so the maps, and the rotation video, can jump at every tenth angle and across the 17 of 120 planned angles that did not finish within the run limit. These short-horizon labels are coarser than those of the float64 maps.

measurement	contrast	estimate (95 % interval)
gain split, 101×53 map	readout factor, chaotic $-$ $\lvert\alpha\rvert \leq 0.01$	$+2.58$ $[1.78, 3.59]$ nats/step
	network factor, chaotic $-$ $\lvert\alpha\rvert \leq 0.01$	-0.07 $[-0.23, 0.12]$ nats/step
	entropy, chaotic $-$ periodic	$+1.62$ $[1.26, 1.91]$ nats
	top-2 logit gap$/T$, chaotic $-$ periodic	-1.37 $[-1.98, -0.87]$
logit bias only, $\alpha = 0$	chaotic share	3.5 % (241 of 6837 cells)
residual steering, block 12	chaotic share	6.2 % (331 of 5353 cells)
crossover, 0.5B, 14 cells	readout effect, arm 10 $-$ arm 00	$+0.81$ $[0.29, 1.35]$ nats/step
	network effect, arm 01 $-$ arm 00	-0.49 $[-0.63, -0.35]$ nats/step
	interaction	$+0.26$ $[0.09, 0.45]$ nats/step
crossover, gratitude	readout effect	$+0.33$ $[0.07, 0.59]$ nats/step
	network effect	-0.14 $[-0.29, 0.01]$ nats/step
crossover, random dir.	readout effect	$+0.77$ $[0.47, 1.12]$ nats/step
	network effect	-0.09 $[-0.22, 0.06]$ nats/step
crossover, 7B, 10 cells	readout effect	$+0.23$ $[-0.11, 0.63]$ nats/step
	network effect	-1.14 $[-1.61, -0.64]$ nats/step

Table 1: Where steering's instability comes from (Qwen2.5-0.5B unless marked 7B, Qwen2.5-7B). Intervals: bootstrap over temperature rows (gain split) or over held-out cells (crossover). Runs MECH and XO; `tools/steer_gain.py`, `tools/crossover.py`, `tools/plot_mechanism.py`.

Why low-dimensional structure can appear. Near a bifurcation of a periodic orbit the center manifold theorem reduces the local dynamics to the critical directions, so the normal forms of folds, flips and cusps, the leading terms of that reduction, apply in any dimension. Near most orbits we computed the non-critical multipliers are small, from about 0.1 at short periods to below 10^{-13} along the cascade. This transverse contraction motivates scalar descriptions of whole attractors, which need additional, global evidence; where the next multiplier is not small (up to 0.4 along the period-5 curves of Section 11, and 0.28 and 0.38 in a high-temperature bubble of Section 9), such a description is only a rough guide. Coexisting attracting orbits constrain such descriptions: in a smooth unimodal interval map with negative Schwarzian derivative, the immediate basin of every attracting periodic orbit contains the turning point or an endpoint, so at most one attracting periodic orbit avoids the endpoints [34], and coexistence calls for a second turning point, a stretch of positive Schwarzian derivative or a second dimension (Appendix C).

4 Methods in brief

Results on the soft-token map labeled float64 are computed in float64, including the normalizations and rotary tables that the library would otherwise compute in float32; two exceptions, float32 storage of some diagnostic step records and a float32 final normalization in the two-GPU runs of 70B-class models before 14 October, are marked where they occur (Appendix B and Section 14). A cell of a phase map is chaotic when the largest finite-horizon Lyapunov exponent of a twin trajectory (offset 10^{-6}, renormalized every step; 300 measured steps after 300 transient ones unless stated) exceeds 10^{-3} and no period up to 32 (48 in later zooms) is found; at 448 checked cells exact tangents reproduce the twin exponents within 0.0041, and the label is a finite-horizon diagnostic, not a proof. Noise in a measured difference can fake chaos but does not hide it: below the noise level a twin exponent rises as the offset shrinks, while a real exponent does not depend on the offset (Appendix B, Figure 36). Periodic orbits are solved by Newton's method with forward-mode derivatives; branches are continued by pseudo-arclength continuation and fold and flip curves by minimally augmented systems [35, 36]; cusps and other codimension-two points are located by bracketing sign changes along the curves; and the normal forms of folds are computed from the network's derivatives without fitting. Appendix B gives the details and the checks on maps with known answers.

5 Chaos with periodic windows

Figure 8 shows a 256 × 256 float64 map over $\alpha \in [0.060, 0.082]$ and $\log_{10} T \in [-0.17, -0.04]$. More than half of its cells are chaotic (35,650 of 65,536), and periodic windows of periods 2–32 are embedded in the chaos. Inside each window the gray darkens toward curves on which the periodic orbit is most strongly attracting (Section 11).

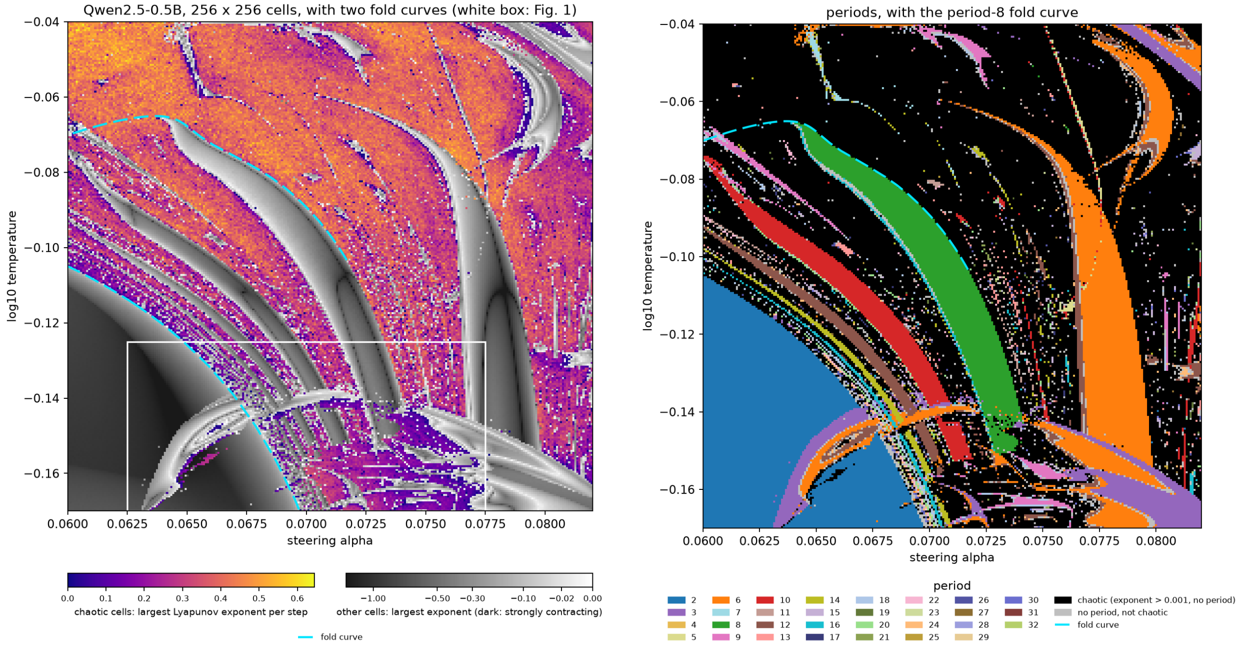


Figure 8: Soft-token map of Qwen2.5-0.5B-Instruct steered along the philosophy direction at layer 12, 256^2 float64 cells. Left: chaotic cells colored by the largest Lyapunov exponent, the other cells in gray by how strongly they contract (light gray: close to losing stability; color bars as in Figure 1). Cyan dashed curves are float64 fold curves of the 2-cycle "is it" (lower left, the boundary on which the stripes accumulate) and of the period-8 orbit (top). Right: periods, with the period-8 fold curve (Appendix E).

Figure 1 resolves the stripes' lower ends at 512 × 512 cells (78,868 chaotic), and Figure 9 labels the same cells by their attractor texts. Three structures stand out. The 2-cycle "is it" fills the left, bounded by its fold curve, on which the stripes accumulate. A closed crescent of the period-3 loop "\n\nThe existence" contains its period doublings and a chaotic core (Section 9). Its continuation, the period-6 band "\n\nThe question\n\nThe existence", crosses the stripes near $\log_{10} T = -0.14$, and where the crescent lies on the 2-cycle's side of that fold curve, the two attractors coexist. The lower right is a chaotic region filled with small hooked windows of "\n\nThe question" and "\n\nThe existence" loops.

These names are what the loops write. A period-3 orbit returns to the same soft state every three steps, and at its three states the most probable next tokens are "\n\n" (a blank line, one token), "The" and " existence", with probabilities 0.74, 0.36 and 0.21 at $\alpha = 0.0642$, $\log_{10} T = -0.156$. Read greedily, the loop writes "The existence" paragraph after paragraph; what it feeds back at each step, however, is the probability-weighted average of all token embeddings, not that token. Coexistence is just as concrete. At that same point the loop started from a state of the “is it” 2-cycle (taken from its attractor at $\alpha = 0.0655$) stays on the 2-cycle and writes “ is it is it…” (most probable tokens with probabilities 0.66 and 0.29; 300-step exponent -1.19), while from the fixed start it settles on the period-3 loop (exponent -0.13). At $\alpha = 0.0652$ the fixed start reaches the crescent’s chaotic attractor, whose text alternates irregularly between “The question” and “The existence” paragraphs (exponent $+0.095$), and the 2-cycle start again writes “ is it” (-1.29): one model, one steering vector and one temperature, and the text written after the transient depends on where the loop starts (`tools/coexist_texts.py`, float64).

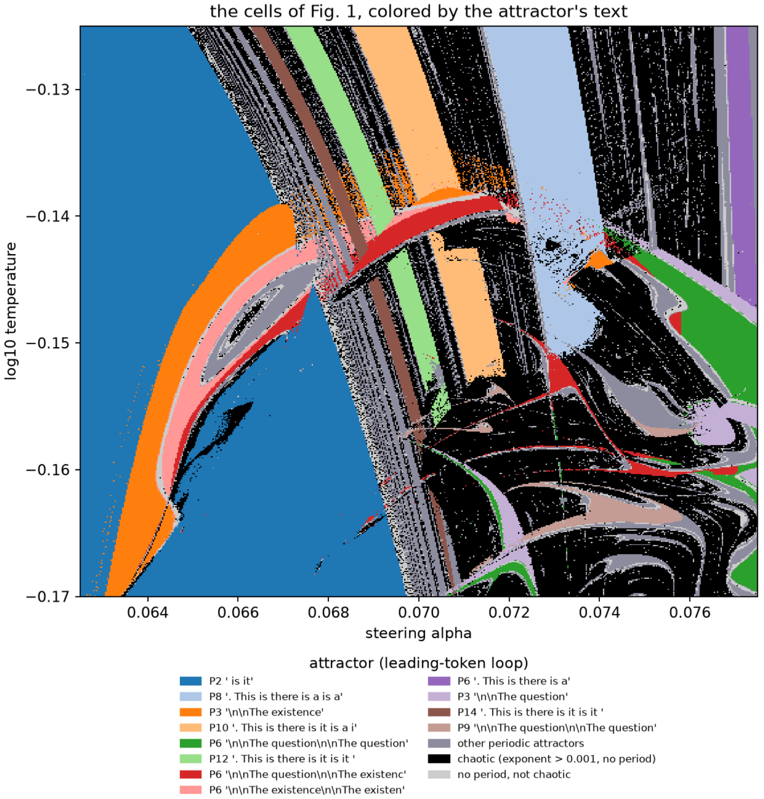


Figure 9: The cells of Figure 1 (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12; 512^2 float64 cells over $\alpha \in [0.0625, 0.0775]$, $\log_{10} T \in [-0.17, -0.125]$), colored by the attractor’s leading-token loop (black: chaotic by the criterion of Appendix B; gray: no period up to 32 and not chaotic; bluish gray: other periodic attractors). Figure 1 shows the largest Lyapunov exponent of the same cells.

Zooms of 256^2 cells (Figure 10) show what the model writes in each window, and every window is a repeating piece of text. The period-8 band repeats “. This is there is a is a”. The neighboring bands add laps of “is it” (periods 10 and 12), and along the band’s flip edge the period-16 loop repeats the sentence twice. The small windows inside the chaos break the “is X” chain with a paragraph break and “The” (periods 11, 12, 13, 14). Inside the period-8 and period-6 bands, two curves of maximal contraction meet in a sharp junction.

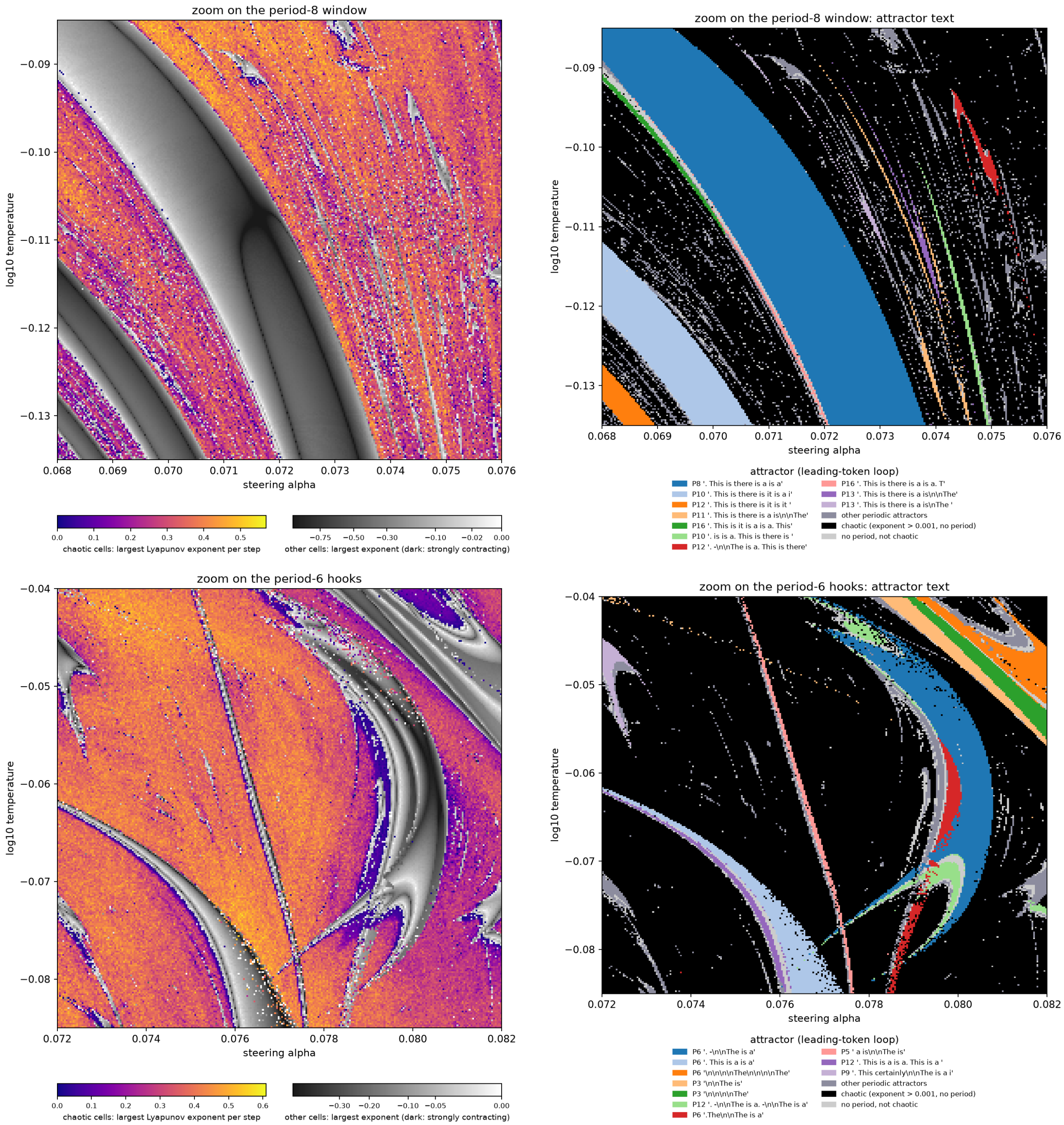


Figure 10: Zooms of 256^2 float64 cells of the Qwen2.5-0.5B-Instruct map (philosophy direction, layer 12), with the largest Lyapunov exponent on the left, colored as in Figure 1, and attractor texts on the right. Top: the period-8 window ($\alpha \in [0.068, 0.076]$, $\log_{10} T \in [-0.135, -0.085]$). Bottom: the period-6 hooks ($\alpha \in [0.072, 0.082]$, $\log_{10} T \in [-0.085, -0.04]$), whose curved period-6 body, period-12 foot and thin leg are analyzed in Appendix E.

What a chaotic loop writes. The chaotic cells write text too (Table 2). Their text reuses the words and phrases of the neighboring loops without settling into an order, and it depends on the state to precisions far below rounding. To show the second property in words, we ran the loop at cells labeled chaotic for 1000 steps in float64. We then recorded it together with a copy started at a relative distance of 10^{-10}, about 600 times smaller than the relative rounding error of float32.

The copy is never reset. Its distance from the loop grows at about the rate the cell's Lyapunov exponent sets, faster or slower over finite stretches (Figure 11). The two write the same text until the distance has grown to between 1.27 % and 176 % of the soft token's norm, and different texts afterwards. In Qwen2.5-0.5B, two cells steered along philosophy, with exponents 0.44 and 0.36, split at steps 44 and 51, and a cell steered along hostility, with exponent 0.35, at step 45. By then the distance had grown by 8.5 to 10.2 decades. The splits come where no token dominates. At each of the seven splits we recorded, in this model and in Qwen2.5-1.5B below, the loop's most probable token has a probability of at most 0.31 and leads the next by at most 0.08, and at five of them the copy writes the loop's second choice. At step 44 of the cell with exponent 0.44, a blank line and " is" have probabilities 0.139 and 0.135; at step 45 of the hostility cell, "Yes" and "No" have 0.303 and 0.298, and the copy writes "No".

The texts show where the chaos lives. Next to the period-3 crescent, where the exponent is only 0.10, the loop keeps the skeleton of the period-3 text, a blank line, "The" and a noun, and only the noun varies, between " existence" and " question", whose probabilities come within 0.0025 of each other; there the texts split at step 151. On the hostility plane (Section 10), the chaotic cell lies between windows whose loops write "!Yes!Yes, that's crap!Yeah" and "!No, that's crap!Yeah!Yes!Yes". It repeats "!Yes" and inserts ", nonsense", ", that" and "!No, that" at irregular intervals. In Qwen2.5-1.5B steered along philosophy, two chaotic cells (exponents 0.83 and 0.80) write short questions separated by line breaks, "What?", "What? Or" and "In". Their copies split at steps 22 and 10; at step 10, "What" and "In" have probabilities 0.091 and 0.090. Finite stretches can expand much faster than the average: that copy's distance grew a hundredfold in its first step. Without steering, at $\log_{10} T = 0.657$, the next-token distribution is nearly uniform, and in the recorded steps the most probable token never exceeds a probability of 0.001. There the chaotic loop (exponent 0.59) writes runs of exclamation marks, in ASCII and in full width, commas, the end-of-turn marker and the Chinese 一下 ("a bit"), and its copy splits at step 43.

The text is fragile to the computation itself. We recorded the cell next to the crescent twice in float64 on the same processor, once with six threads and once with four, which changes the order of floating-point sums. During the 1000-step transient the chaos had already amplified that difference, so the two recordings started from slightly different states; they wrote the same text for 27 steps and different texts from step 28 on. At 70B the chaotic cells of Figure 32 draw on the words of the unsteered model's fifteen-token loop (Section 14): the first eight distinct leading tokens of one chaotic cell confirmed at three twin offsets are " I", "The", "!", " It", " infinite", " possibilities", " are" and " you" (17 distinct tokens in 300 steps).

Qwen2.5-0.5B, philosophy, $\alpha = 0.0652$, $\log_{10} T = -0.156$: $\lambda = 0.10$, texts split at step 151	
loop	`↩↩The question↩↩The existence↩↩The existence↩↩The existence↩↩The existence↩↩The existence↩↩The existence↩↩The existence`
copy	`↩↩The question↩↩The existence↩↩The existence↩↩The existence↩↩The question↩↩The existence↩↩The question↩↩The existence`
Qwen2.5-0.5B, philosophy, $\alpha = 0.07484$, $\log_{10} T = -0.1226$: $\lambda = 0.44$, texts split at step 44	
loop	` is a.↩↩The question↩↩The is↩↩The is a.↩↩ is is↩↩The is↩↩The is↩↩`
copy	` is a.↩↩The question↩↩The is↩↩The is a.Is↩↩The is there is a.Is↩↩`
Qwen2.5-0.5B, philosophy, $\alpha = 0.07277$, $\log_{10} T = -0.041$: $\lambda = 0.36$, texts split at step 51	
loop	`↩↩The is↩↩The is a. -↩↩The is a.Is↩↩The is a. Is↩↩The is`
copy	`↩↩The is↩↩The is a. -↩↩The is a.↩↩The↩↩↩↩The is a is a.`
Qwen2.5-0.5B, hostility, $\alpha = 0.08946$, $\log_{10} T = -0.05566$: $\lambda = 0.35$, texts split at step 45	
loop	`Yes!Yes!Yes!Yes!Yes!Yes!Yes!Yes, that!Yes!Yes!No,`
copy	`Yes!Yes!Yes!Yes!Yes!Yes!Yes!No, that!Yes!Yes!Yes!`
Qwen2.5-1.5B, philosophy, $\alpha = 0.10553$, $\log_{10} T = 0.0322$: $\lambda = 0.80$, texts split at step 10	
loop	`↩↩↩What↩↩In↩↩↩What↩What??↩↩↩↩What?↩What↩↩In↩↩`
copy	`↩↩↩What↩↩In↩↩↩What↩In??↩↩↩What??↩↩↩↩What↩`
Qwen2.5-1.5B, unsteered, $\log_{10} T = 0.657$: $\lambda = 0.59$, texts split at step 43	
loop	`[end]!！[end]!!，↩!，!![end]!，!!!，↩!，[end]!`
copy	`[end]!！[end]!!，↩!，!![end]!！!!!，[end]!!，！`

Table 2: What chaotic loops write. Each pair of lines shows the most probable token at consecutive steps of a chaotic loop and of a copy started at a relative distance of 10^{-10} from it, from 14 steps before the first step at which they differ to 9 steps after it (or the end of the recording); from that step on the text is red. ↩: line break; [end]: the end-of-turn token; "！" and "，" in the last example are the full-width exclamation mark and comma. λ: largest Lyapunov exponent, from twins at relative offsets 10^{-6} and 10^{-8} over 1000 steps after 1000 transient steps; both offsets give the same value to two decimals. Split: the step of the recording at which the texts first differ. Float64; `tools/chaos_texts.py`, receipts in `docs/chaos-texts`.

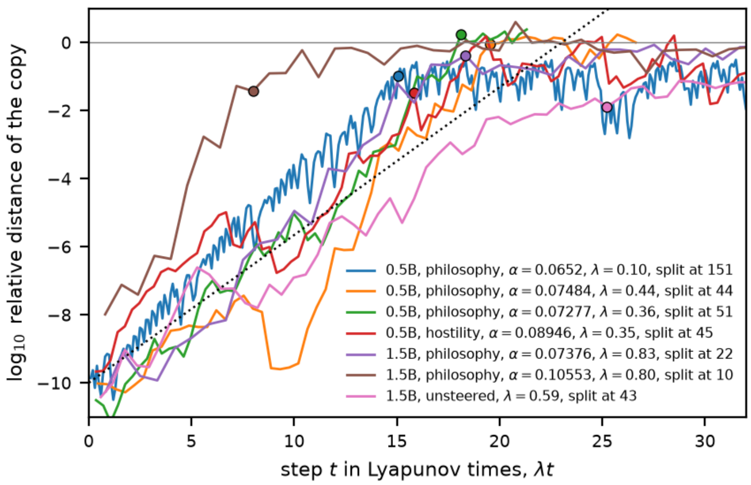


Figure 11: How a difference of 10^{-10} becomes a different text. The relative distance between a chaotic loop and its copy, never reset, for the six cells of Table 2 and a seventh (Qwen2.5-1.5B, philosophy, $\alpha = 0.07376$, $\log_{10} T = 0.0165$), against the step in units of the cell's Lyapunov time $1/\lambda$. Labels give the model (0.5B and 1.5B: Qwen2.5-0.5B- and -1.5B-Instruct), the steering direction and strength, the exponent and the step of the split. Dotted: $10^{-10}e^{\lambda t}$, the growth the exponent sets on average. Dots: the first step at which the two write different most probable tokens. Finite stretches deviate from the 1000-step exponents. Up to a distance of 10^{-2}, five of the steered copies grew 15 to 47 % faster than λ, the sixth ($\alpha = 0.10553$) 2.4 times as fast after a hundredfold expansion in its first step, and the unsteered copy 11 % more slowly. In the batch-layout runs of Section 6, too, divergence came earlier than the exponent predicts.

6 What the chaos does to inference and steering

The loop studied here is the expected-embedding feedback that continuous reasoning methods such as Soft Thinking use [1], and its chaos has consequences that can be measured without any reference to dynamical systems. We report three: whether a run can be reproduced, how the depth at which a steering vector enters sets the chaos it causes, and what happens when two steering vectors are combined. A fourth, that a repeating text can hide a chaotic state from anyone who reads only the output, is shown in Section 10.

How a change of batch reaches the output. Floating-point addition is not associative: $(a+b)+c$ and $a+(b+c)$ can differ in the last bit [37]. A matrix multiplication in a transformer sums thousands of products, and GPU libraries split and order those sums according to the shape of the matrices, to keep the hardware busy; the number of requests processed together, the batch, is part of that shape. The same request computed alone or in a batch therefore receives results that differ at the rounding level, about 10^{-16} relative in float64 and about 10^{-3} in bfloat16. In a single forward pass such differences are invisible. In a feedback loop they are not lost: each output becomes the next input, so the difference between two runs is fed back and transformed by the map's derivative at every step. On a periodic attractor the derivative contracts on average, the exponent is negative, and the difference decays until both runs follow the same orbit and write the same text. On a chaotic attractor the exponent λ is positive and the difference grows like $d_1 e^{\lambda t}$ from its first value d_1, so it reaches a relative size of 10^{-2} after about $\ln(10^{-2}/d_1)/\lambda$ steps (about 100 steps for $d_1 \approx 10^{-15}$ and $\lambda \approx 0.3$). At most a few Lyapunov times later, at 1.27 to 176 % of the state's norm in the examples of Table 2 (0 to 16 steps after the distance passes 10^{-2}), the runs write different text. This has a practical consequence. Batch-invariant kernels, which fix the order of every reduction independently of the batch size, remove this source of difference [38]. In a periodic regime that is enough, and small differences would be forgiven even without it. In a chaotic regime it is not: any remaining difference, from another GPU model, a library or compiler version, the numerical precision, or a mathematically equivalent reordering of a sum, is amplified in the same way, so a run is reproducible only if the whole computation is repeated bit for bit. Numerical differences are the disturbance; the feedback dynamics decide whether it grows. How far it grows within a given budget also depends on its starting size, on the horizon and on how close the run comes to a change of answer, so a periodic loop can still change an answer when the starting difference is large. In bfloat16 the starting difference is already about 10^{-3}, so continuous feedback carries it into different answers within a short budget even where the float64 loop is not chaotic (below).

The regime decides whether rounding differences grow. Language-model inference is not bitwise reproducible: GPU kernels choose different reduction orders for different matrix shapes, so a request's result depends on the batch it shares and on the hardware [38, 39]. We ran the 448 labeled cells of the horizon check (192 chaotic, 192 periodic and 64 other cells of two float64 maps; Appendix B) twice in lockstep from the same start. The two runs differ only in the batch: all cells in one batch of n rows, or the same rows followed by n further rows whose steering vectors are those of the first n in reverse order, at the original temperatures ($2n$ rows). This changes the matrix shapes the kernels see; the first n rows compute the same cells (Figure 12, Table 3). In float64 the two runs first differ by about 6×10^{-15}, the rounding level. At 190 of the 192 chaotic cells this difference passes a relative distance of 10^{-2}, after a median of 102 steps, and grows on to order one (median final distance 0.91); at 173 of the 192 chaotic cells the most probable token of the loop's readout differs between the runs at some step of the last 500. At 182 of the 192 periodic cells the two runs stay within 10^{-2} for all 1500 steps, with a median largest distance of 2.3×10^{-12}, and 182 of the 192 periodic cells end the run within 3×10^{-12}. The Lyapunov exponent accounts for when a chaotic run diverges: the step at which the distance passes 10^{-2} follows $\ln(10^{-2}/d_1)/\lambda$, where d_1 is the smallest nonzero distance in the run, with a median ratio of 0.91 and a log-log correlation of 0.73 over 190 cells. This uses the whole run and is therefore retrospective; an independent recomputation with the first nonzero distance instead gives the same rounded values. The two chaotic cells that never diverge have exponents of 0.013 and 0.0013, for which the predicted step lies beyond the run. The same holds without steering in Qwen2.5-7B, whose embeddings are untied. Over 24 chaotic and 24 periodic temperatures of its unsteered loop, all 24 chaotic float64 runs diverge, after a median of 56 steps (the exponent again accounts for the step, median ratio 0.95), and at 23 of them the readout's most probable token differs at some step of the last 500. None of the 24 periodic runs moves past 10^{-2}; 22 end within 1.3×10^{-13} of each other and all within 2.2×10^{-12}. A second model family, unsteered Llama-3.1-8B, gives the same picture: all 20 chaotic temperatures diverge after a median of 68 steps (median ratio to the exponent's estimate 1.05), all 20 differ in their readout's most probable token in the last 500 steps, and 19 of the 20 periodic ones stay within 10^{-2}. At 70B the picture is unchanged. In Llama-3.3-70B-Instruct steered along hostility-vs-kindness at block 20 (float64 arithmetic on its stored bfloat16 weights, layers spread over two GPUs, 1000 steps), all 24 chaotic cells diverge, after a median of 114.5 steps (median ratio to the exponent's estimate 0.83, log-log correlation 0.74), and at 21 of them the readout's most probable token differs at some step of the last 300. None of the 24 periodic cells moves past 10^{-2}, and all end within 1.3×10^{-12}. In lower precision the rounding is larger. In float32 every chaotic cell of both models diverges, after a median of 20 steps (0.5B) or 16 (7B). In bfloat16, a common serving precision, the two layouts pass a distance of 10^{-2} after a median of one or two steps at every cell of the two Qwen models and at

comparison	cells	reach 10^{-2}	median step	end with different tokens
batch layout, float64	chaotic	190 / 192	102	90.1 %
	periodic	10 / 192		3.6 %
batch layout, float32	chaotic	192 / 192	20.5	92.2 %
	periodic	72 / 192		22.9 %
float32 against float64	chaotic	192 / 192	20	92.7 %
	periodic	75 / 192		19.3 %
batch layout, bfloat16	chaotic	192 / 192	2	99.5 %
	periodic	192 / 192	2	95.3 %
7B unsteered, float64	chaotic	24 / 24	56	95.8 %
	periodic	0 / 24		0 %
7B unsteered, float32	chaotic	24 / 24	16	95.8 %
	periodic	4 / 24		8.3 %
7B unsteered, bfloat16	chaotic	24 / 24	2	75.0 %
	periodic	24 / 24	2	50.0 %
Llama 8B unsteered, float64	chaotic	20 / 20	68	100 %
	periodic	1 / 20		5 %
Llama 8B unsteered, float32	chaotic	20 / 20	10.5	100 %
	periodic	1 / 20		0 %
Llama 8B unsteered, bfloat16	chaotic	20 / 20	1	100 %
	periodic	18 / 20	2	45 %
Llama 70B steered, float64	chaotic	24 / 24	114.5	87.5 %
	periodic	0 / 24		0 %

Table 3: Divergence of two otherwise identical runs, by the cells' float64 labels (1500 steps; "end with different tokens": the most probable tokens differ at some step of the last 500). Top: Qwen2.5-0.5B, philosophy direction at layer 12. Middle: Qwen2.5-7B unsteered, 24 chaotic and 24 periodic temperatures of its long sweep. Bottom: Llama-3.1-8B-Instruct unsteered, 20 and 20. Last block: Llama-3.3-70B-Instruct, hostility-vs-kindness at block 20, 24 and 24 (1000 steps; tokens compared over the last 300). In float32 the periodic cells that diverge may change phase on the same cycle or switch between coexisting attractors; the run does not distinguish the two.

38 of the 40 Llama cells, periodic ones included. With these kernels and at that precision, the loop was not reproducible across batch layouts in any regime we tested.

The same test at the level of answers, with a design frozen before the run, shows what this means for a task. We asked Qwen2.5-0.5B 96 held-out two-digit additions through a short soft-thinking pipeline: for 32 steps the model is fed the probability-weighted average of its token embeddings (the input of F), each new one appended to a growing context, and a fixed answer cue then reads the number off greedily. Each problem ran alone and in a batch with three fixed distractor problems of the same length. Nothing is padded, so the two runs compute the same quantities and differ only in how the GPU groups its sums for the two matrix shapes, that is, by rounding. In bfloat16 the extracted answer changed with the batch layout in 56 % of the problems at $T = 1$ and in 8 % at $T = 10^{0.2}$. A forced greedy baseline with the same budget (32 greedy tokens, then the same cue) produced identical token sequences in both layouts for every problem. The excess disagreement of soft over greedy decoding, averaged over the two temperatures, is 32.3 percentage points (95 % interval $[26.5, 38.1]$; sign-flip $p = 10^{-4}$, the smallest attainable). In float64 no answer changed within the 32 steps. There, the median growth of an injected kick to the first soft token is 0.033 nats per step at $T = 1$ (range -0.13 to 0.24 over six problems and three kick sizes), and even a relative kick of 10^{-4} changes no answer. The bfloat16 discrepancies therefore do not require the float64 map's chaos; they are consistent with bfloat16's rounding differences accumulating through continuous feedback. A discrete argmax absorbs differences smaller than its margin, but not over a long feedback run whose answers the model holds weakly ("Soft Thinking as published", below). (The forced greedy baseline's accuracy is zero and not informative: forced through 32 tokens, the model writes past its answer, and the cue then extracts another number.) The same frozen test in Qwen2.5-7B, whose soft-thinking pipeline answers 51 to 64 % of these problems, replicates the result. In bfloat16 the batch layout changed the soft-thinking answer in 53 of the 96 problems (55 %) at $T = 10^{0.2}$, the temperature at which the soft trace still matters. At this precision even the forced greedy baseline changed in 10 of the 96, the known batch-variance effect. The excess over greedy decoding is 17.7 points ($[9.3, 25.6]$, $p = 2 \times 10^{-4}$). In float64, at that temperature, the 7B's loop amplifies an injected perturbation (median +0.034 nats per step), injected kicks change answers (2 of 18 kicks: six problems and three kick sizes), and the rounding-level difference between the layouts changed 5 % of the answers within the 32 steps. Two horizons thus matter. Within a short budget, continuous feedback in low precision makes answers depend on the batch, and where the loop expands, even float64 rounding does. Over longer runs, a chaotic regime amplifies float64 rounding into different readout tokens, on the schedule its Lyapunov exponent sets.

An exploratory causal test locates the low-precision effect in the loop. Following a suggestion of an external review, we reran the 0.5B pipeline at $T = 1$ on 128 fresh additions, after a batch-1 reference run of each (exploratory; first on a workstation GPU, where the native pipeline's answers changed with the layout in 85 % of the problems, against 56 % on the H100 of the frozen test). In the batch of four, two arms start from the reference's prompt state and are answered by the same batch-1 decoder. In the free arm the soft phase runs on its own; in the reset arm the target's cache and logits are replaced by the reference's before every soft step, so that each step runs in the batch but only one at a time. In bfloat16 the answer changed in 102 of the 128 problems (80 %) in the free arm and in 14 (11 %) in the reset arm, a difference of 69 points (95 % interval $[60, 77]$; 90 discordant pairs against 2, exact McNemar $p = 2 \times 10^{-24}$), while accuracy stayed between 23 and 25 % in every arm. One batched step moves the target's logits by a median 1.5 % of their norm; in the free arm the median difference grows to 0.79 of it by step 32, while the reset arm's median per step stays between 1.3 and 3.1 %. In float32 the free arm's drift grows about 140-fold over the 32 steps but stays near 10^{-4}, and 4 of the 128 answers changed (none with reset). The test replicates on an H100 and in Qwen2.5-7B. On the H100, in bfloat16, the answer changed in 85 of the 128 problems (66 %) in the free arm and in 2 (1.6 %) with reset in the 0.5B model, and in 82 (64 %) against 2 (1.6 %) in Qwen2.5-7B at $T = 10^{0.2}$ (exact McNemar $p \leq 2 \times 10^{-24}$); in the 7B model even float32 answers changed in 47 (37 %) against none. In bfloat16 on the H100 most batched steps reproduce the reference bitwise (the reset arm's median step distance is 0), and the free loop amplifies the rare step that does not. No arm's accuracy differs from the reference's by more than 5 points. Within this budget, the batch decides low-precision answers because the loop carries each step's rounding forward and amplifies it, not because the last steps alone are fragile; 128 problems per arm do not show that accuracy is unchanged.

Soft Thinking as published. A test frozen before its data (`docs/pi-matrix-confirmatory.md`) ran Soft Thinking with its published settings on Qwen2.5-7B-Instruct in bfloat16. The settings are concept tokens restricted to the 15 most probable, $T = 0.6$ and 64 soft steps. The test used 256 GSM8K training problems that no earlier run had scored, on an H100 GPU. Only the batch layout changed, between batches of 8 and of 16 prompts of equal length without padding, and the answer changed for 21.1 % of the problems (54 of 256; 95 % interval clustered by batch $[16.3, 26.1]$). This confirms the prediction of more than 10 %. Each secondary prediction compares another arm with this one on the same 256 problems (paired differences of flip rates, intervals clustered by batch, exact McNemar tests):

- replaying the reference run's fed-back tokens in both layouts: 5.9 % (difference 15.2 points, $[10.4, 20.2]$, $p = 8 \times 10^{-9}$); predicted lower, and it is;
- 16 soft steps instead of 64: 6.6 % (difference 14.5 points, $[10.0, 19.0]$, $p = 4 \times 10^{-7}$); predicted lower, and it is;
- ordinary greedy chain-of-thought decoding: 2.3 %, at 89 % accuracy (difference 18.8 points, $[13.7, 23.6]$, $p = 10^{-11}$); predicted lower, and it is;
- feeding back only the most probable token: 17.6 % (difference 3.5 points, $[-2.9, 10.3]$, $p = 0.35$); predicted lower, and the test cannot tell the two apart.

The three confirmed differences would survive a correction for the four tests by a wide margin, and the fourth is a clear null: discrete feedback through the same pipeline flips nearly as many answers as continuous feedback does. The full vocabulary gave 22.3 % (no prediction). Most changes are between wrong answers: through the soft pipeline this model answers only 22 to 30 % of the problems correctly, and the layout changes whether the answer is correct for 6.2 % of the problems. In float64 the same pipeline contracts. On 1312 and 1314 of the 1319 GSM8K test problems (full vocabulary and top-15), a perturbation of 10^{-10} ended below its initial size, and none changed an answer. These flips are therefore not chaos of the float64 map: the feedback carries bfloat16 rounding differences forward until they decide answers that the model holds only weakly, and it does so for discrete feedback as well.

How the dependence builds up. An exploratory study read the answers of the same pipeline after every doubling of the number of soft steps, from 2 to 128. It used 200 GSM8K test problems, two temperatures (0.6 and 1.2), and six steering strengths along the hostility direction (α from -0.02 to 0.05). The two layouts agree on 90 to 96 % of the answers after 2 steps and on 51 to 70 % after 128, in every configuration. Among the problems whose layouts disagree, the median first disagreement comes after 8 steps. Float64 copies of the same configurations, started 10^{-10} apart, separate the two sources of disagreement.

- Unsteered or weakly steered ($|\alpha| \leq 0.02$), the copies change the answer text of at most 1.5 % of the problems. The loop contracts, so the bfloat16 disagreements are rounding carried forward by the feedback.
- At $\alpha = 0.05$ they change the answer text of 13.5 % of the problems at $T = 0.6$ and 27.5 % at $T = 1.2$, and the extracted number of 4.5 and 9.0 %. The loop itself is then sensitive, but accuracy has fallen to between 0.5 and 2.5 % in either layout.

In this model, then, the regime in which Soft Thinking is useful is the regime of accumulated rounding, not of chaos (`docs/pi/h-horizon-summary.json`).

Reasoning models that answer well. Qwen2.5-7B-Instruct answers few of these problems through the soft pipeline, so most of its flips are between wrong answers. A second test, also frozen before its data (`docs/pi-matrix2-confirmatory.md`), used reasoning models, as the Soft Thinking paper did: DeepSeek-R1-Distill-Qwen-7B, DeepSeek-R1-Distill-Llama-8B and QwQ-32B. A row thinks in concept tokens (top 15, $T = 0.6$) until its most probable next token ends the thinking block, or until 1536 steps. It then answers by sampling discrete tokens, with random numbers shared between the two layouts. The comparison arm thinks with discrete tokens sampled the same way. On fresh GSM8K problems the batches held 8 or 16 prompts of equal length. On the integer-answer problems of MATH-500, each problem ran alone or as eight copies of itself. Over the five primary cells, the batch layout changed 29.6 % of the soft-thinking answers (265 of 896; 95 % interval $[27.3, 31.8]$), confirming the prediction of more than 5 %.

- The two models that answer GSM8K well changed their answers with the layout alone: DeepSeek-R1-Distill-Qwen-7B for 14.5 % of the problems ($[10.9, 18.0]$) at 88 to 90 % accuracy, and QwQ-32B for 19.5 % ($[16.4, 22.3]$) at 86 to 87 %. The layout decided whether the answer was correct for 11.7 and 14.5 % of the problems.
- Discrete sampled thinking changed about as many answers in every cell. On GSM8K the rates were 10.2 % and 19.9 % for the same two models, and no soft-minus-plain difference is significant. This does not show that the two are equal, but the effect does not require continuous thinking: long feedback in bfloat16 suffices.
- On an A100 instead of an H100, the first model's soft rate was 11.7 % (difference 2.7 points, $[-1.6, 6.6]$), a separate replication on a second GPU type; the difference is not significant, which again does not establish equality.
- Where the budgets cut the reasoning short, accuracy was lower and the layout changed more answers: 61 and 67 % of the answers on MATH-500, where half of the rows reached the thinking cap, and 37.5 % for DeepSeek-R1-Distill-Llama-8B on GSM8K, which answered 49 to 54 % of the problems correctly.

The Cold Stop rule of the Soft Thinking paper, which ends thinking when its entropy stays low, was not used (`docs/pi/pm2-summary.json`).

Is this chaos? We ran float64 copies of the same pipeline. Each copy's first soft token was displaced from the base row's by a relative 10^{-10}, and all copies shared the base row's random numbers. The three models respond differently. In DeepSeek-R1-Distill-Qwen-7B, 216 of 256 copies ended the thinking at a distance from their base row below the initial displacement, and 2.3 % changed the answer. Its layout flips are therefore mainly bfloat16 rounding carried to answers near ties, as in Qwen2.5-7B. In QwQ-32B and DeepSeek-R1-Distill-Llama-8B, about half of the copies ended above the initial displacement (34 of 64 and 145 of 256). Fitted over the steps where the distance lay between ten times the displacement and 10^{-3}, the median growth rates were 0.010 and 0.008 per step. The displacement alone changed 10.9 and 18.4 % of the answers, respectively. In those models the loop's own sensitivity contributes to the answer changes. It is a weak, finite-time instability over about a thousand thinking steps, and it acts on top of the rounding; it is not an asymptotic chaos measurement. Some copies ended thinking before their base row (14 in R1-Llama, 7 in QwQ), and the copies of one problem are not independent (`docs/pi/pm2f-summary.json`; 128 problems for each 7–8B model and 32 for QwQ-32B).

Where a steering vector enters decides how much chaos it causes. Steering vectors are a common tool for controlling and probing models [29, 32], usually added at one layer chosen by hand. In Qwen2.5-0.5B the philosophy direction, built with the same recipe at blocks 4, 8, 12, 16 and 20 and scaled to each block's residual norm, makes 5.9, 8.1, 6.8, 5.0 and 2.3 % of the same 128×128 plane chaotic (Figure 13). Early steering carries the chaos down to $\log_{10} T = -0.95$ (block 8) and -0.91 (block 4), while at block 20 the chaotic cells lie between $\log_{10} T = -0.09$ and 0.22, three fifths

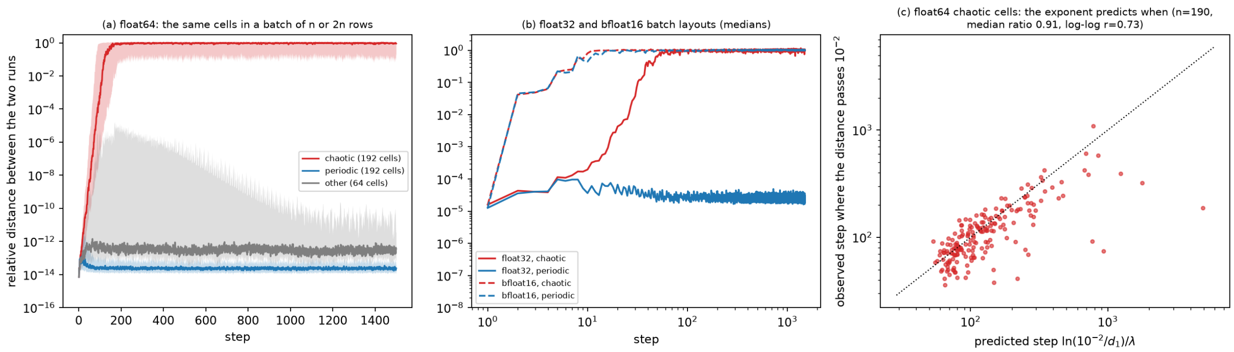


Figure 12: Only the batch layout differs. The 448 labeled cells of Qwen2.5-0.5B-Instruct (philosophy direction, layer 12) run twice from the same start, alone in a batch or followed by n further rows with the steering vectors in reverse order. (a) float64: median and interquartile range of the relative distance between the two runs, by the cells' float64 labels. (b) Medians for float32 and bfloat16 batch layouts. (c) float64 chaotic cells: the step at which the distance passes 10^{-2} against $\ln(10^{-2}/d_1)/\lambda$, with d_1 the first nonzero distance and λ the map exponent; dotted, equality.

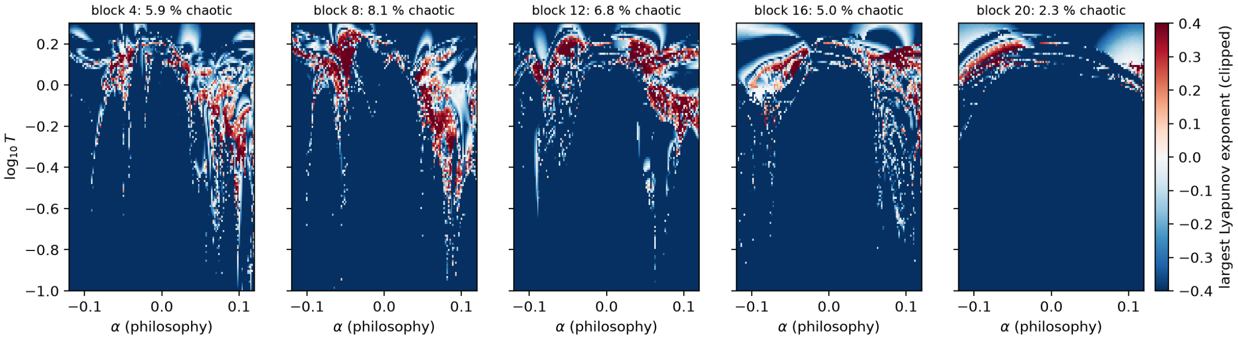


Figure 13: The philosophy direction of Qwen2.5-0.5B-Instruct, built with the same recipe at each of five blocks of 24: largest Lyapunov exponent on 128 × 128 float64 cells over the steering strength and $\log_{10} T \in [-1, 0.3]$. The share of chaotic cells is given above each panel.

of them above 0.1. In the strip $|\alpha| < 0.02$ every block gives 0.8 to 2.1 %. Llama-3.3-70B behaves the same way. Steered along its hostility direction (a contrastive valence direction; Section 10), it is chaotic in 8.0 % of a 24 × 24 map at block 20 of 80, down to $\log_{10} T = -0.14$, against 4.3 % at block 35, where the chaos starts at $\log_{10} T = 0.01$, 4.3 % at block 50 (32 by 32 cells, a two-GPU map from before the precision fix of Section 14), and 0.7 % at block 65, all above $\log_{10} T = 0.19$. These 70B labels are those of the map protocol, one twin offset, and so are candidates. A logit bias, the latest possible point of entry, gives chaos mostly above $\log_{10} T = 0.1$ (Section 3). The gain split of Section 3, repeated along steering lines at all five blocks of the 0.5B model, shows where the depth enters. At low temperature ($\log_{10} T \leq 0$) the chaotic cells of blocks 4, 8 and 12 have expanding readout factors (medians +0.74, +0.46 and +0.46 nats per step over 54, 56 and 21 cells) and contracting network factors (−0.47 to −0.21). At block 16 the readout factor of such cells is only +0.15, and block 20 gives just 6 of them. Early injection creates token competition at low temperature more readily; late injection mostly does not. Its chaos lives at higher temperature, where the network factor carries it (block 20: network +1.09, readout −0.78; Figure 6e).

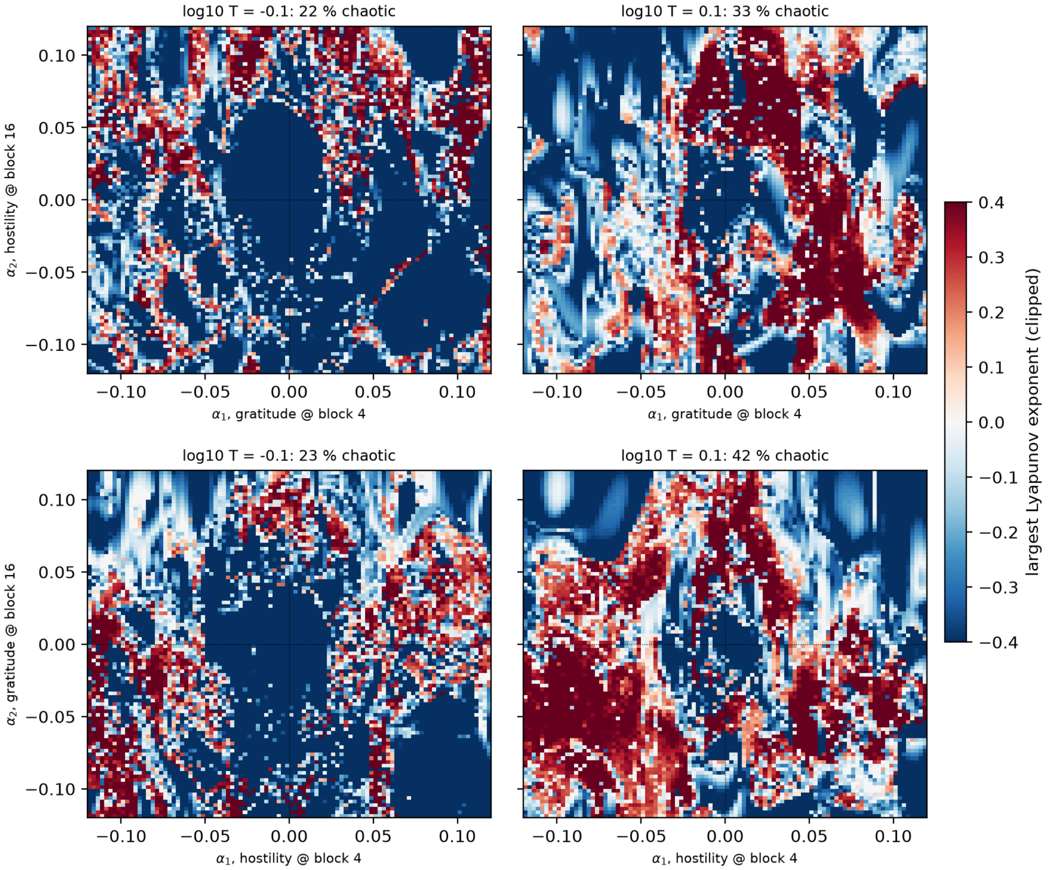


Figure 14: Two steering directions at two blocks of Qwen2.5-0.5B-Instruct: largest Lyapunov exponent on 96×96 float64 cells over the two strengths, at $\log_{10} T = -0.1$ (left) and 0.1 (right). Top: gratitude at block 4, hostility at block 16; bottom: the blocks exchanged. Dotted lines mark the zero axes; the grid does not sample them, and the single-push controls quoted in the text were computed separately on exact zero lines.

Two steering vectors interact. Steering at two blocks at once, gratitude at block 4 and hostility at block 16 of Qwen2.5-0.5B, makes 21.7 % of a 96×96 plane chaotic at $\log_{10} T = -0.1$. The grid has no row or column at exactly zero, so we computed the single pushes separately, each with the other exactly absent and on the same grid: they make 12.5 % (gratitude at block 4 alone) and 5.2 % (hostility at block 16 alone) of their lines chaotic. Large chaotic regions lie where neither push alone is chaotic (Figure 14): 1561 cells, 16.9 % of the plane and 78 % of its chaotic cells, are chaotic although each push alone, at the same strength, is not, and in 1525 of them each push alone is periodic. Over the four planes (two block orders, $\log_{10} T = \pm 0.1$) this share is 8.3 to 16.9 % of the plane, and 7.8 to 16.5 % with both single pushes periodic. With the blocks exchanged (hostility at 4, gratitude at 16) the plane is 23.4 % chaotic, and at $\log_{10} T = 0.1$ both pairs give 33 and 42 %. Each plane keeps a periodic hole around zero steering. Combining steering vectors is therefore not additive in its effect on the loop's stability: a composition of two pushes can be chaotic where each push alone is periodic.

Does a chaotic loop predict worse answers in ordinary decoding? Not beyond the model's own uncertainty. Two exploratory designs found no accuracy signature of the regimes that survives controls (Appendix D). A frozen test then labeled 4000 fresh products of two two-digit numbers by the regime of the unsteered Qwen2.5-7B-Instruct's own loop at $\log_{10} T = 0.40$. Chaotic-loop prompts are answered less accurately by ordinary sampling, by 7.1 points (95 % interval $[4.8, 9.3]$), but the gap vanishes once seven difficulty covariates, fixed before sampling, enter the fit (label coefficient +0.15 points, $[-1.4, 1.7]$, permutation $p = 0.84$), and in a post hoc fit the first-token entropy alone absorbs nearly all of it. This is what the mechanism of Section 3 predicts: the loop turns chaotic where next tokens compete, and a model unsure of its first token also answers worse. Within a prompt no association is detected either, and accuracy does not peak at the edge of chaos: deep periodic, edge and deep chaotic prompts are answered alike, with the intervals of all three adjusted contrasts within $[-3.1, 3.0]$ points. These are nulls at this precision; they do not show that the regime has no effect.

7 Period adding written in text

Between a period-2 region and the chaos lies a fan of stripes of periods $6, 8, 10, \ldots$ Their orbits, numerically validated in float64, can be read as text (Table 4). Each is ". This" followed by k pairs "is X", with period $2k + 2$, so each stripe adds one pair. The period-2 region itself is the 2-cycle "is it". At the stripe centers the orbits are effectively one-dimensional: the leading multiplier is 0.0025 to 0.086, and the next falls from 3×10^{-6} (period 6) to 2×10^{-14} (period 20). This compression belongs to these orbits, not to the map. The 2-cycle's multipliers are -0.271 and -0.058 (the second modulus is about 0.037 near its fold), the next modulus at the period-3 flip of Section 9 is 0.088, and one period-6 orbit in Appendix E has a complex leading pair.

period	orbit (leading tokens)	μ_1	$\lvert\mu_2\rvert$
6	. This is a is a	+0.0052	3×10^{-6}
8	. This is there is a is a	+0.0110	3×10^{-7}
10	. This is there is it is a is a	+0.0025	2×10^{-7}
12	. This is there (is it)^2 is a is a	+0.0163	4×10^{-8}
14	. This (is it)^4 is a is a	+0.0369	2×10^{-9}
16	. This (is it)^5 is a is a	+0.0716	8×10^{-11}
18	. This (is it)^6 is a is a	+0.0792	1×10^{-12}
20	. This (is it)^7 is a is a	+0.0857	2×10^{-14}

Table 4: Periodic orbits of the stripe family at the stripe centers along $\log_{10} T = -0.115$ ($\alpha = 0.077872$, 0.071674, 0.067766, 0.065550, 0.064443, 0.063857, 0.063534, 0.063335), computed with the current all-float64 map (norms and rotary tables patched): Newton residuals 6×10^{-15} to 8×10^{-13}, leading multiplier μ_1 and next modulus $|\mu_2|$ from Rayleigh–Ritz values. Each step in period adds one "is X" pair to the text. The 2-cycle "is it" itself has multipliers -0.2706 and -0.0585 at $(\log_{10} T, \alpha) = (-0.17263, 0.061579)$. The period-2 and period-8 orbits were also reproduced by an independent implementation, and the period-8 orbit is certified by a computer-assisted proof (Appendix L).

Table 5 shows what the map writes: two of these loops, cycle by cycle, next to chaotic cells between the stripes. The chaotic cells reuse the phrases of the neighboring loops but do not settle into a fixed order.

α	regime	H	leading token at 40 consecutive steps
0.071674	period 8	1.9	is a is a. This is there| is a is a. This is there| is a is a. This is there| is a is a. This is there| is a is a. This is there
0.072940	chaotic ($\lambda = 0.13$)	2.4	a is a. This is there is a is↩The is there is a is↩↩The is↩↩The is↩↩The is a is a. This is there is a is↩The is there
0.067766	period 10	1.8	is a. This is there is it is a| is a. This is there is it is a| is a. This is there is it is a| is a. This is there is it is a
0.069060	chaotic ($\lambda = 0.20$)	2.4	. - certainly↩↩The is there is it is a.↩↩The is a is a. This is there is it is a is↩↩The existence isThe is it is it is a. This

Table 5: What the attractors write. Qwen2.5-0.5B-Instruct soft-token map steered along philosophy at layer 12, at $\log_{10} T = -0.115$: the leading token at 40 consecutive steps after 600 transient steps, for the period-8 and period-10 loops of Table 4 (bars mark the start of each cycle) and a chaotic cell beside each. λ: largest exponent over 2000 steps after 1000 transient steps, the same to two decimals at twin offsets 10^{-6} and 10^{-9}. H: mean entropy of the next-token distribution in nats. ↩: line break.

8 Mechanism: a saddle-node bottleneck

The fold. At $\log_{10} T = -0.115$ the 2-cycle "is it" dies in a fold at $\alpha_c = 0.06279827$ (float64 multiplier 1 to 3×10^{-13}, next modulus 0.037; Figure 15): it meets an unstable 2-cycle, and both vanish. In the critical coordinate u of the second iterate $G = F^2$, the fold's normal form is $u' = u + a\varepsilon + bu^2$ with $\varepsilon = \alpha - \alpha_c$; here $a = -4.834$ per unit α and $b = -25.4285$, so $ab = 122.93$. The stable branch gives an independent estimate: extrapolating its multipliers through $(1-\mu)^2/(4(\alpha_c - \alpha)) \to ab$ yields 123.1 with a quadratic fit (120.6 with a linear one). An independent implementation of the forward map and solver recovers α_c to 6×10^{-15} and a to 3×10^{-12}. Its b, from nested automatic differentiation, is -25.42846; our finite-difference value differs from it by 2.2×10^{-6} relatively, which is the finite-step bias of the difference table. The resulting $ab = 122.927$ leaves every passage prediction of Table 6 unchanged.

Controlled passages. In the oriented coordinate $v = \operatorname{sign}(b)\, u$ of Appendix B, we start trajectories at $v = -U$ or 0 ($U = 0.1/|b|$) and count G-iterates until $v > U$. At four distances from the fold, the measured counts match the iterated normal form within one iterate (Table 6, Figure 16), and a transverse displacement of the same norm changes the count by at most one.

$\varepsilon = \alpha - \alpha_c$	full: measured / predicted	midpoint	full + transverse kick
3×10^{-6}	144 / 144	73 / 74	143 / 144
1×10^{-5}	71 / 71	36 / 37	70 / 71
3×10^{-5}	34 / 34	17 / 18	33 / 34
1×10^{-4}	14 / 14	7 / 7	13 / 14

Table 6: Bottleneck passages (G-iterates). Predictions iterate $u' = u + a\varepsilon + bu^2$ with a, b computed from the network's derivatives; no parameter is fitted.

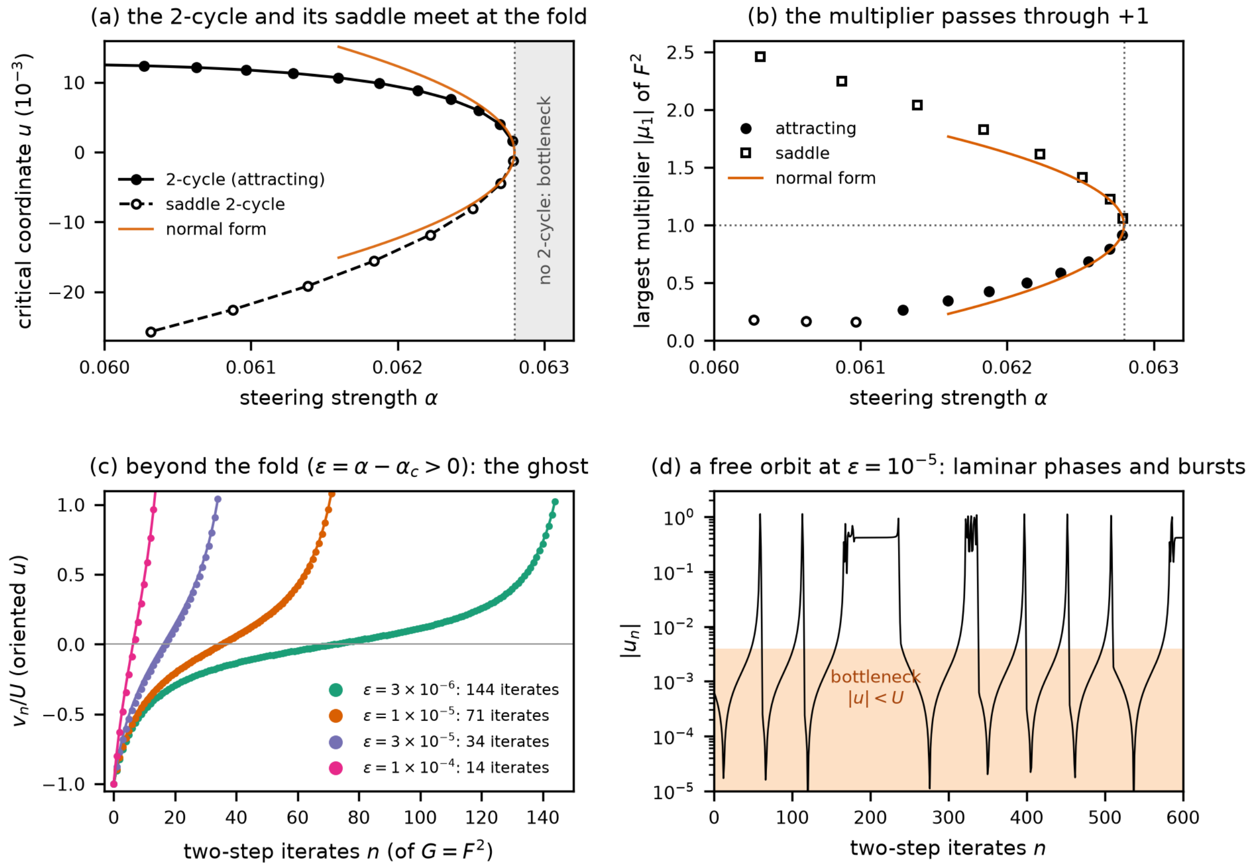


Figure 15: The fold behind the stripes: the 2-cycle “is it” at $\log_{10} T = -0.115$ (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12). (a) Continued in α, the attracting 2-cycle (filled) and its saddle partner (open) meet and vanish at $\alpha_c = 0.06279827$. The vertical coordinate is the projection u of one point of each orbit on the fold's critical direction; orange, the normal form's fixed points $u^* = \pm\sqrt{|a|(\alpha_c - \alpha)/|b|}$. (b) The largest multiplier of $G = F^2$ on the two orbits reaches $+1$ at the fold, following the normal form's $1 \mp 2\sqrt{ab(\alpha_c - \alpha)}$ (open circles: a complex leading pair). (c) Beyond the fold the 2-cycle is gone, but its ghost slows the dynamics: controlled passages from the channel's entrance $v = -U$ to its exit $v = U$ (v the oriented critical coordinate, $U = 0.1/|b|$), measured on the network (dots) and iterated from the normal form $v' = v + |a|\varepsilon + |b|v^2$ (lines), take 144, 71, 34 and 14 iterates, as predicted. (d) A free trajectory at $\varepsilon = 10^{-5}$ alternates between laminar phases in the bottleneck ($|u| < U$, shaded; 40 to 70 iterates each) and bursts away from it, the signature of type-I intermittency. Branch residuals, fold, normal form and passages in float64; branch multipliers in float32. `tools/plot_fold.py`.

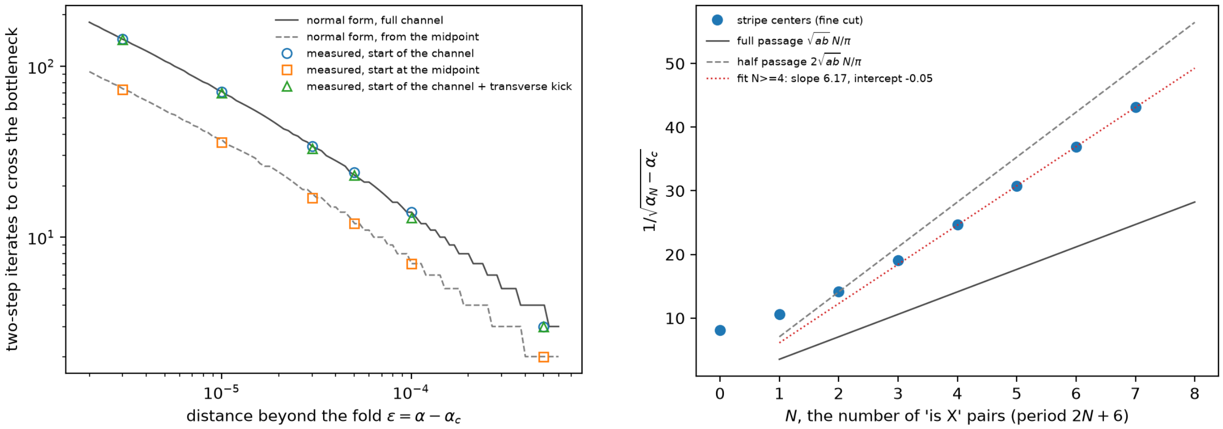


Figure 16: Bottleneck passages and the window law at the fold of the 2-cycle "is it" (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\log_{10} T = -0.115$). Left: measured passage times against the normal-form predictions. Right: the window law, with stripe centers plotted as $1/\sqrt{\alpha_N - \alpha_c}$ against the lap count N, together with the full- and half-passage predictions.

Natural reinjection and the window law. Free trajectories re-enter the channel slightly upstream of the bottleneck's middle ($v \approx -3 \times 10^{-4}$ to -7×10^{-4}, in the same oriented coordinate). From each measured entry point, the normal form predicts the natural laminar durations within 4–8% for $\varepsilon \leq 10^{-4}$, and a natural passage lasts about 0.6 of a full one. The stripes therefore follow the N^{-2} law of a saddle-node ghost (a free fit of the exponent gives 2.18 ± 0.06, below), with about a third of the full-passage prefactor; the two fractions agree because the prefactor scales as the square of the passage fraction ($0.58^2 \approx 1/3$). The distances $d_N = \alpha_N - \alpha_c$ of stripe centers located on a float32 cut along $\log_{10} T = -0.115$ give $d_N N^2 = 0.0263$–0.0265 for $N = 4$–7 (log-log slope -2.00), compared with $\pi^2/(4ab) = 0.0201$ for a half passage. Two-step return maps of p("it") (Figure 17) show the near-tangency.

Numerically estimated stripe edges along $\log_{10} T = -0.115$ (Table 7) sharpen this picture. Each window is bounded by a fold, where its orbit is born, and by a flip. For periods 6–12 the fold faces away from the accumulation point. For periods 16–20 the sides are exchanged, and the period-14 window is bounded by flips on both sides. The fold edges of periods 16–20 follow $\alpha_N - \alpha_c = C/(N + N_0)^2$ with $C = 0.0266$ and $N_0 = 0.15$. Their slope, a rise of 6.13 per lap in $1/\sqrt{\alpha_N - \alpha_c}$, is 1.74 times the full-passage slope $\sqrt{ab}/\pi = 3.53$ of the normal form. That corresponds to an effective passage of 0.58 of a full one, which the measured natural reinjection gives independently.

period	edge toward larger α	edge toward smaller α
6	fold 0.0790337	flip 0.0766947
8	fold 0.0726585	flip 0.0706525
10	fold 0.0684572	flip 0.0670594
12	fold 0.0658982	flip 0.0651944
14	flip 0.0645850	flip 0.0642958
16	flip 0.0639284	fold 0.0638001
18	flip 0.0635683	fold 0.0635034
20	flip 0.0633535	fold 0.0633183

Table 7: Edges of the stripe windows along $\log_{10} T = -0.115$ ($\alpha_c = 0.0627983$), from the window centers of Table 4. Flips: critical multiplier a real -1 to 3×10^{-10}. Folds: the multiplier nearest $+1$ extrapolated to 1 through $(1-\mu)^2$, which is linear near a fold; last converged multiplier 0.991–0.998.

We tested whether the measured reinjection predicts windows beyond the fitted range. A model with one fixed entry point (the median measured one) and one lap offset fitted to the fold edges of periods 16–20 places the fold edges of periods 22–30 3 to 9 % closer to the fold than the constant-prefactor law. Both predictions were frozen before a float64 line of 3200 cells (spacing 10^{-7}) located the windows. The measured edges follow the law within 3 % through period 30 (within 1 % at periods 22, 24 and 28) and fall outside the model's range except at period 30. So the law extrapolates, but one fixed reinjection point does not predict it. The measured entry points move with ε (-3×10^{-4} to -7×10^{-4}). The exponent of the law is near 2, its prefactor empirical. Over the fold edges of periods 16 to 32 ($N = 5$ to 13), $\ln(\alpha_N - \alpha_c)$ against $\ln N$ has slope -2.00, while a free fit of $C/(N + N_0)^\gamma$ with the offset N_0 also fitted gives $\gamma = 2.18 \pm 0.06$ (1 % errors assumed), so the data do not pin the exponent to exactly 2. At these distances, however, the bottleneck is not narrow compared with the counting window: its scale $\sqrt{|a|\varepsilon/|b|}$ is 1.35 to 3.51 times U, and a full passage lasts only 9 to 23 iterates of G. The windows therefore lie outside the asymptotic regime in which the normal form fixes the prefactor [40, 41], and the factor of about one third is a measured value, not a prediction.

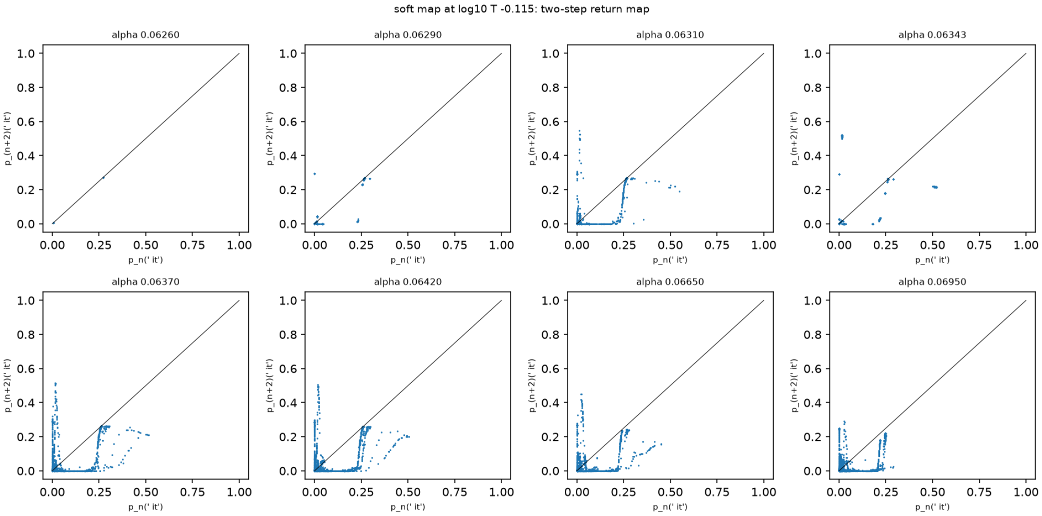


Figure 17: Two-step return maps of the probability of "it" near the fold of the 2-cycle "is it" (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12), at increasing distance from the fold.

A prospective test. To test the normal form at folds that played no part in shaping the mechanism, an independent implementation located two new folds and computed their normal-form coefficients by nested automatic differentiation. All 24 passage predictions were pushed in a frozen design before any test trajectory was run. Target A is the 2-cycle's fold at a new temperature: $\log_{10} T = -0.140$, $\alpha_c = 0.0670549$, $a = -7.055$, $b = -24.927$. Target B is the fold of the period-8 orbit of Table 4, continued in temperature at $\alpha = 0.0717$: $\log_{10} T_c = -0.1033462$, $G = F^8$, $a = -11.900$, $b = -152.05$ per unit $\log_{10} T$. Of the 24 forecasts (four distances and three entrances per target: the channel's start, its midpoint, and its start after a transverse kick), 23 fell within one iterate of the measured passages (Table 8). Seven of the 24 forecasts are five iterates or shorter, where a single iterate is 20 % or more of the passage; of the 17 longer forecasts, 16 fell within one iterate. The pre-specified criterion, that every row pass, failed by one row: for target A, midpoint entrance, $\varepsilon = 3 \times 10^{-6}$, the forecast was 61 and the measurement 59. All full-channel passages and all passages after a transverse kick were within one iterate. The misses are not symmetric: every nonzero residual of a midpoint or kicked start, here and at the two folds tested before (Table 6 and Section 10), is negative, 17 of 17 (two-sided sign test, $p = 2^{-16} \approx 1.5 \times 10^{-5}$). The measured passages are systematically shorter than the iterated quadratic map, by one or two iterates. The design also froze a secondary comparison, the continuous passage integral $\int du/(a\varepsilon + bu^2)$ of the same normal form, whose closed form is the arctangent usually quoted for saddle-node bottlenecks. It came within one iterate of all 24 measurements (errors from -0.41 to $+0.92$) and gives 58.56 for the failing row. At that entrance the two forms already differed by 2.4 iterates before any measurement: iterating the quadratic map from the entrance u_{in} to the exit U adds about $\frac{1}{2}\ln[f(U)/f(u_{\text{in}})]$ steps to the integral, $f(u) = a\varepsilon + bu^2$, most at small ε and at the midpoint entrance. Which of the two forms the full map follows is not settled by this test, and it does not let us promote the secondary comparison after the fact.

ε	target A ($G = F^2$): measured / predicted	target B ($G = F^8$): measured / predicted
3×10^{-6}	118/118, **59/61**, 117/118	26/26, 13/14, 25/26
1×10^{-5}	56/56, 28/29, 56/56	10/10, 5/5, 10/10
3×10^{-5}	26/26, 13/14, 26/26	4/4, 2/2, 4/4
1×10^{-4}	10/10, 5/6, 10/10	2/2, 1/1, 2/2

Table 8: Prospective passage test at two folds that did not shape the mechanism. Each cell lists the full-channel, midpoint and full-plus-transverse entrances as measured / predicted G-iterates. Predictions come from the iterated quadratic normal form, with coefficients from an independent implementation, frozen before the test runs. Bold: the one row outside the one-iterate criterion. The shortest period-8 rows give only weak relative accuracy.

A second prospective test. The first test left open which form of the normal form the full map follows, and its shortest forecasts carried little weight. A second test, frozen before any trajectory was run, made the continuous passage integral its primary predictor and the discrete count its secondary, and chose the distances so that every full-channel forecast is at least 16 iterates and every midpoint forecast at least 8. Its two folds are again fresh. Target C is the 2-cycle's fold at $\log_{10} T = -0.200$: $\alpha_c = 0.0713690$, $a = -10.9505$, $b = -25.3300$. Target D is the fold of a period-6 orbit, a different family, at $\log_{10} T = -0.087439$: $\alpha_c = 0.0756329$, $G = F^6$, $a = -72.695$, $b = 72.237$. Distances $\varepsilon = q^2/(ab)$ with $q = 0.08$, 0.05, 0.03 and 0.018 give both targets the same forecasts (Table 9). At the 2-cycle's fold the integral held in all 12 rows, and every measured count was the forecast rounded up. At the period-6 fold the midpoint passages held as well, but passages from the channel's entrance came in short, by 0.4 to 5.7 iterates and more so closer to the fold, so the primary criterion failed, 18 of 24 rows (the discrete count passed 16 of 24). From the fold point to the exit the period-6 passages follow the normal form; it is the first half of the channel that is crossed faster. Terms beyond the quadratic normal form on that side, or the curvature of the center manifold at the entrance, are candidate causes that this test does not resolve. Over both frozen tests, the continuous integral came within one iterate of 42 of 48 passages; all six misses are entrance passages at the period-6 fold, short by 3.1 to 5.2 %. The sign hypothesis frozen with the design, that midpoint and kicked passages do not exceed the discrete count, held in all 16 rows (`codex/confirm-fold-2`).

q	forecast (full / mid)	discrete (full / mid)	C, 2-cycle: measured	D, period 6: measured
0.08	22.40 / 11.20	23 / 12	23, 12, 23	22, 11, 22
0.05	44.29 / 22.14	45 / 23	45, 23, 45	**42**, 22, **42**
0.03	85.29 / 42.64	86 / 44	86, 43, 86	**81**, 43, **82**
0.018	154.75 / 77.37	155 / 80	155, 78, 155	**149**, 78, **150**

Table 9: Second prospective passage test, frozen before any passage was run. Forecast: the continuous passage integral $\int du/(a\varepsilon + bu^2)$ from the entrance to U (primary); discrete: the iterated quadratic map (secondary). Measured G-iterates for the full-channel, midpoint and full-plus-transverse entrances. Bold: outside one iterate of the primary forecast.

9 A period-doubling cascade to chaos

The crescent of Figure 9 is a closed region in which the period-3 loop “\n\nThe existence” doubles inward. Along the cut $\log_{10} T = -0.14798$, the map's cells show period 3 over 25 columns, period 6 over about 14, period 12 over 4 and period 24 in one column, followed by a chaotic core. The doubled loops repeat the same leading tokens twice and four times, with alternating probabilities. Locating the flips along this cut in float64 (Appendix B), in the steering coordinate α, gives Table 11. The first two ratios, $\delta_2 = 2.844$ and $\delta_3 = 2.859$ (indexed as in Table 11), are far from Feigenbaum's 4.6692, even though from the second flip on the orbits are effectively one-dimensional (next modulus 0.0023, 2×10^{-6}, 2×10^{-12}). The next ratio, $\delta_4 = 3.781$, is markedly larger. The first four flip parameters agree between completed float64 computations on a CPU and an H100 to 4×10^{-15}.

Continuing a just-born doubled orbit past period 48 failed: Newton's method slides onto the half-period orbit, which is also periodic with the doubled period. We therefore also located the superstable parameters s_p, where the critical multiplier vanishes in the middle of each band (Table 10). There the orbit attracts, so it can be reached by forward iteration and cannot slide onto its half-period parent. Through period 768 the superstable ratios are 3.19, 4.02, 4.478, 4.625 and 4.660. The last three come from parameters resolved to 10^{-14} and differ from Feigenbaum's value by 0.19, 0.044 and 0.0097; from 4.02 on, the distance shrinks by factors of 3.4, 4.3 and 4.6. The flip ratios 2.84, 2.86, 3.78, 4.36, 4.60 and 4.65 rise in step (Figure 18); the last three come from runs started inside the period-96 band, which located the flips of periods 96, 192 and 384 directly. The accumulation point lies near $\alpha_\infty \approx 0.0661250$. The early superstable values are less precise. The other computed Ritz values of the period-12 and period-24 orbits blur the root, so s_{12} and s_{24} are known only within brackets of 1.24×10^{-5} and 6.9×10^{-7}, and the ratios 3.19 and 4.019 have worst-case ranges $[2.895, 3.338]$ and $[4.019, 4.101]$. s_6 is less precise still, and s_3 was not bracketed.

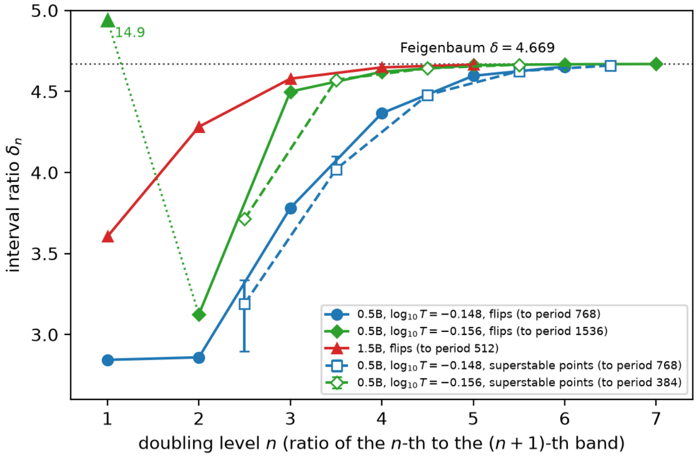


Figure 18: Finite-order interval ratios of the doubling sequences: the crescent of Qwen2.5-0.5B-Instruct (philosophy direction, layer 12), from flips and from superstable points (which reach period 768), and the route of Qwen2.5-1.5B (philosophy direction, layer 14; Figure 48), from flips. A superstable ratio spans half-bands, so it is plotted half a level after the flip ratio of its first band. Bars: worst-case ranges from the brackets of s_{12} and s_{24}. At $\log_{10} T = -0.148$, and in the 1.5B model, the ratios rise toward Feigenbaum's δ from below; the last superstable ratio at -0.148 is within 0.21 % of it, and the last 1.5B ratio (4.665) within 0.1 %. At $\log_{10} T = -0.156$ the first ratio, 14.9, lies off the scale (arrow), and the later ratios approach δ from below, the last (4.6688) within 0.008 % of it, and within 0.015 % over the location uncertainty of the period-384 flip. Its first two superstable ratios carry no bars: the brackets of s_{12} and s_{24} allow $[3.40, 29.3]$ and $[1.01, 4.65]$. Each sequence ends at the deepest level computed: the flips at -0.148 at period 384, the superstable points at period 768, the 1.5B route at period 256, and the cut at -0.156, which we followed furthest, at the 768 → 1536 flip.

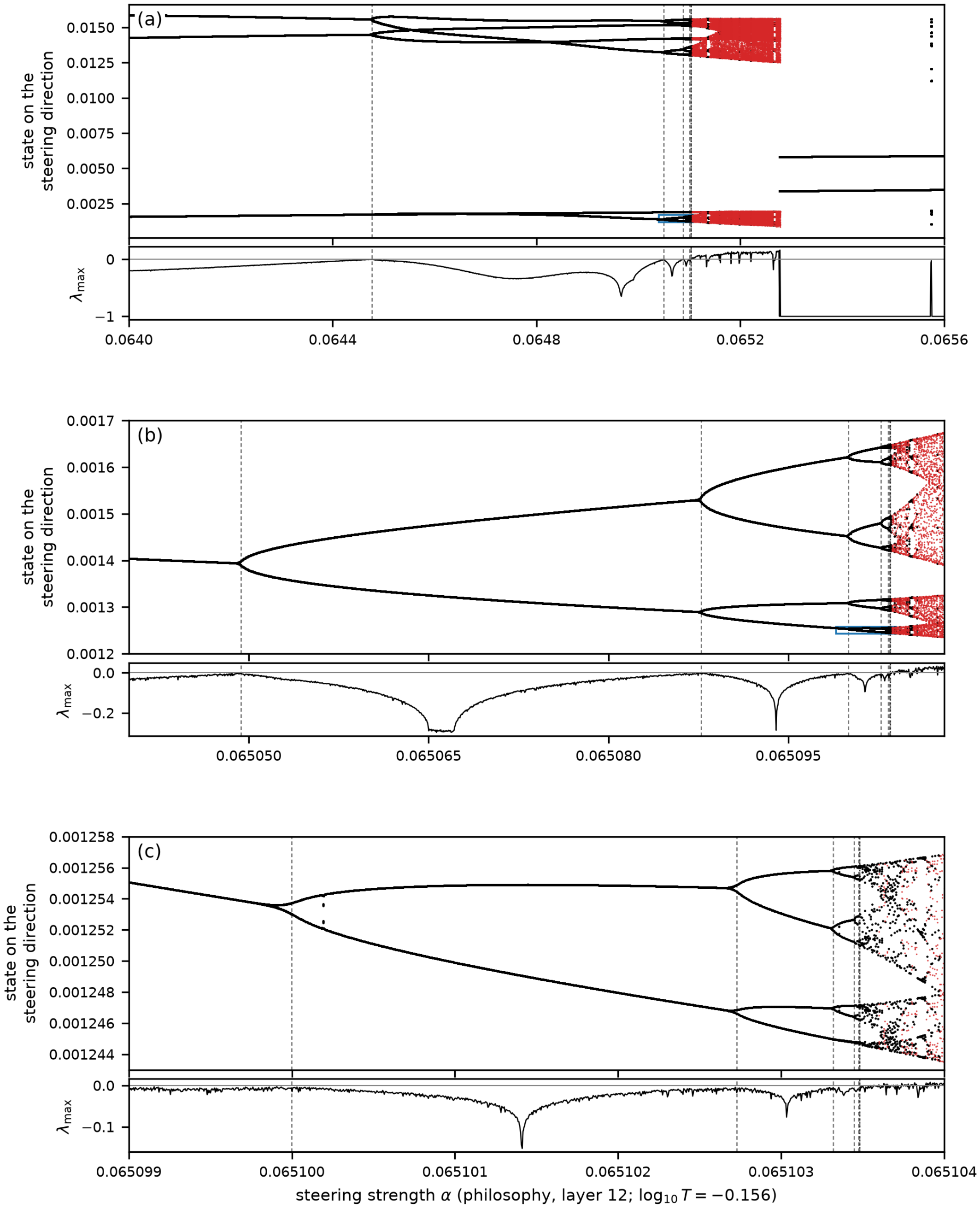


Figure 19: The Feigenbaum tree of the crescent cascade (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\log_{10} T = -0.156$): float64 line scans of 1600, 1360 and 1000 cells, each with 3000 transient steps and 256 recorded states projected on the unit steering direction (red: chaotic cells). (a) The whole cut: period 3 doubles into chaos, which ends at $\alpha = 0.065278$, where the attractor jumps to a strongly contracting period-2 orbit. (b, c) Zooms in both coordinates on the branch boxed in the panel above, showing the doublings $6 \to 12 \to 24 \to 48$ and $24 \to 48 \to 96 \to 192$. Dashed lines: flips located by Newton's method (Table 12); dotted line: the accumulation point estimated from the last two flips with δ. Under each panel, the largest Lyapunov exponent touches zero at every flip and turns positive beyond the accumulation point. Near a flip the orbit converges slowly, so with 3000 transient steps each fork opens slightly before its Newton flip, by 8.4×10^{-6} in α at the first and by at most 1.1×10^{-6} at the later ones.

period p	superstable parameter s_p	bracket width	ratio $\frac{s_p - s_{p/2}}{s_{2p} - s_p}$
12	0.0659717	1.24×10^{-5}	
24	0.06608014	6.9×10^{-7}	3.19
48	0.066114132784466	3.3×10^{-15}	4.019
96	0.066122589892689	7×10^{-15}	4.4779
192	0.066124478510077	6.4×10^{-15}	4.6251
384	0.066124886853811	5×10^{-15}	4.6595
768	0.066124974490236	4.1×10^{-15}	

Table 10: Superstable parameters of the crescent's doubling sequence along $\log_{10} T = -0.14798$: the critical multiplier of the period-p orbit changes sign across the bracket, with Newton residuals below 3×10^{-13} and the other computed Ritz moduli below 10^{-20} from period 96 to 384; at period 768 the residual is 1.2×10^{-12} and the other moduli are below 3×10^{-20}. Brackets for periods 12 and 24 are the closest points where the critical multiplier is cleanly identified.

Why are the early ratios so small? We first considered a crossover. Softmax at temperature T is a smoothed argmax, so the reduced one-dimensional map might have a nearly corner-like extremum, rounded only on a scale set by T. Early doublings would then see the corner, where ratios tend toward 2, and later ones the smooth quadratic core. This predicts smaller early ratios at lower temperature. We fixed that prediction before locating the flips along two lower cuts, and it fails. At $\log_{10} T = -0.156$ the first ratio is 14.93, five times its value at -0.148 (Table 12). At -0.152 the continued period-12 solution collapsed onto its period-6 parent near $\alpha \approx 0.0657$, and the forward search found no next flip. This is compatible with a doubling bubble, but we did not locate its reverse boundary or exclude a narrower cascade on that cut. The early ratios thus depend strongly on where a cut crosses the crescent, and we withdraw the crossover explanation. The later ratios at -0.156 behave as universality requires. The next ratios are 3.124, 4.4988, 4.6208, 4.6618, 4.6673 and 4.6688. Their deviations from $\delta = 4.6692$ are -1.55, -0.17, -0.048, -0.0074, -0.0019 and -0.00036; they keep one sign, from the third on each step closes the gap by a factor of 3.5 to 6.5, and Aitken's Δ^2 extrapolation of the last three ratios gives 4.6694. Universality of this kind is expected for strongly dissipative maps of any dimension [42, 43]. The last ratio uses the flip of the period-768 orbit (the 768 → 1536 flip), located with $|\mu + 1| = 4.1 \times 10^{-7}$. The penultimate flip, of the period-384 orbit, approaches the precision floor of its multiplier, a product of 384 step Jacobians. On an H100 the computed $\mu + 1$ scatters by 1.8×10^{-5} about a straight line near the root; under that local linear fit, the largest residual divided by the slope gives a location uncertainty of about $\pm 3.3 \times 10^{-13}$, an estimate rather than a certified bound. The workstation GPU located it at $\alpha = 0.06510347354866666$ with $|\mu + 1| = 4 \times 10^{-9}$, 9×10^{-14} from the H100 estimate. Over that range the ratio δ_7 lies between 4.6672 and 4.6674, δ_8, the last ratio, between 4.6685 and 4.6692, and the extrapolated value between 4.6688 and 4.6702, an interval that contains δ. The superstable points along the same cut, through period 384, give the ratios 10.21, 3.713, 4.572, 4.643 and 4.6646. Before computing them we fixed the prediction that the last two superstable ratios at a new temperature would lie within 5 % of δ, in $[4.436, 4.902]$. At the original readout, through period 192, the point estimates 4.572 and 4.643 met it, but the bracket envelope of 4.572 did not; extending the computation to period 384 later gave the pair 4.643 and 4.6646, both inside the interval. The last, 4.6646, rests on parameters resolved to 5×10^{-15} and differs from δ by 0.1 %. Over the corners of the brackets the one before lies in $[4.639, 4.647]$, limited by s_{48}, whose bracket is 2.4×10^{-9} wide. The earlier ratio 4.572 depends on s_{24}, known only within a bracket of 5.9×10^{-6} (worst-case range $[1.01, 4.65]$). Finite-order ratios also depend on the parameter coordinate, so only the trend of later ratios can test universality [22, 44, 45].

n	orbit	flip parameter α_n	next modulus	ratio $\delta_n = \frac{\alpha_n - \alpha_{n-1}}{\alpha_{n+1} - \alpha_n}$
1	$3 \to 6$	0.0654807769499	0.088	
2	$6 \to 12$	0.0659055406325	0.0023	2.844
3	$12 \to 24$	0.0660549041785	2.4×10^{-6}	2.859
4	$24 \to 48$	0.0661071412523	2.3×10^{-12}	3.781
5	$48 \to 96$	0.0661209563174	$< 10^{-13}$	4.365
6	$96 \to 192$	0.0661241213732	$< 10^{-13}$	4.596
7	$192 \to 384$	0.0661248100141	$< 10^{-13}$	4.653
8	$384 \to 768$	0.0661249580173	$< 10^{-13}$	

Table 11: Flips of the crescent's period-doubling sequence along $\log_{10} T = -0.14798$. For the first five flips (H100) the critical Ritz value is real and differs from -1 by at most 4×10^{-10}, the Newton residuals are below 4×10^{-13}, and the other computed Ritz values lie inside the unit circle. The sixth flip was located at the precision floor of the period-96 multiplier: the critical value differs from -1 by 5.7×10^{-7}, its scatter between bracket points 10^{-13} apart is about 10^{-6}, and the Newton residual is 5.5×10^{-12}; the location is uncertain by about 2×10^{-12}, which changes δ_5 by less than 10^{-5}. The seventh flip is clean (critical value -1.000000016, Newton residual 2.8×10^{-12}). The eighth was accepted at the precision floor of the period-384 multiplier: there the critical value differs from -1 by 5.3×10^{-5} (acceptance threshold 10^{-4}), bracket evaluations within 4×10^{-12} of it scatter by about 10^{-4} in that difference, and the Newton residuals are below 4×10^{-11}. Its location is uncertain by about 4×10^{-12}, which changes δ_7 by about 10^{-4}. Feigenbaum's constant is $\delta = 4.6692\ldots$ [22]; for the logistic map the first ratios are 4.751, 4.656, 4.668. The first four parameters agree between the completed CPU and H100 computations to 4×10^{-15}. Superstable parameters (Table 10) extend the ratio sequence through period 768.

n	orbit	flip parameter α_n	next modulus	ratio δ_n
1	$3 \to 6$	0.0644766567385	0.12	
2	$6 \to 12$	0.0650493555308	0.045	14.93
3	$12 \to 24$	0.0650877194158	7.0×10^{-5}	3.124
4	$24 \to 48$	0.0650999993318	4.3×10^{-7}	4.4988
5	$48 \to 96$	0.0651027289524	1.8×10^{-13}	4.6208
6	$96 \to 192$	0.0651033196816	2.9×10^{-14}	4.6618
7	$192 \to 384$	0.0651034463988	1.5×10^{-14}	4.6673
8	$384 \to 768$	0.0651034735487	3.2×10^{-13}	4.6688
9	$768 \to 1536$	0.0651034793638	2.9×10^{-14}	

Table 12: Flips of the crescent's doubling sequence along $\log_{10} T = -0.156$, in float64. Flips 1–3 were computed on an H100 and flips 3–8 on a workstation GPU (NVIDIA RTX 2000 Ada); the two computations of flip 3 agree to 4×10^{-16}, and an H100 computation of flip 8 from its evaluations at the multiplier's precision floor lies 9×10^{-14} away. Flip 9 was computed on an H100. Every critical multiplier is real; for flips 1–8 it differs from -1 by at most 5×10^{-8}, and for flip 9 by 4.1×10^{-7}.

Not every doubling leads to chaos. In the high-temperature region (Figure 40, Appendix E), along $\log_{10} T = 0.0319$, the period-2 loop "\n\nThe" flips at $\alpha = 0.0753534$ and its doubled loop at 0.0736921. These orbits are far from one-dimensional (next moduli 0.38 and 0.28). Toward smaller α the period-8 orbit does not double again: near $\alpha \approx 0.0705$ it merges back into the period-4 orbit. Along this cut it is therefore a period-doubling bubble, bounded by two flips of the period-4 orbit, not the start of a cascade to chaos.

	offset	β	amplitude ratio	doubled multiplier	$1-4\beta$
period $2 \to 4$	2.5×10^{-5}	0.0092	1.005	0.96335	0.96320
($c = 6.49 \times 10^3$)	1×10^{-4}	0.0364	1.020	0.8572	0.8545
	4×10^{-4}	0.1400	1.078	0.513	0.440
period $4 \to 8$	2.5×10^{-5}	0.0175	1.003	0.92987	0.92983
($c = 1.24 \times 10^4$)	1×10^{-4}	0.0685	1.012	0.7257	0.7260
	4×10^{-4}	0.2470	1.056	complex	0.012

Table 13: The flip normal form where the map is far from one-dimensional: the two flips of the high-temperature bubble (next moduli 0.38 and 0.28). At offsets below each flip in α, $\beta = -\mu_p - 1$ from the period-p orbit's multiplier; amplitude ratio: the doubled orbit's half-amplitude along the critical direction divided by the prediction $\sqrt{\beta/c}$; doubled multiplier: computed against the prediction $1-4\beta$. Both coefficients are positive (supercritical), as the stable doubled orbits require. Three of five offsets per flip are shown.

The local theory still applies here (Section 3). At each flip we computed the cubic coefficient of the flip normal form, $\eta \mapsto -(1+\beta)\eta + c\,\eta^3$ on the critical direction, from the dense monodromy and float64 directional derivatives (notation of Appendix B), $c = \ell^\top C(r,r,r)/6 + \ell^\top B(r,h_2)/2$ with $(I-M)h_2 = B(r,r)$ [36]. To leading order it predicts the doubled orbit without fitting: half-amplitude $\sqrt{\beta/c}\,(1+O(\beta))$ along r and multiplier $1-4\beta+O(\beta^2)$. At the smallest offset the amplitude ratios are 1.005 and 1.003, and the multipliers agree within 1.5×10^{-4} and 5×10^{-5}; the larger discrepancies at larger offsets are consistent with higher-order corrections (Table 13). These finite comparisons support the leading-order prediction; they do not establish the order of the remainder. Figure 19 shows the cascade itself as a bifurcation diagram: in three nested zooms each fork sits at a flip located independently by Newton's method, and the largest exponent touches zero at every flip and turns positive beyond the accumulation point. The route to chaos of Qwen2.5-1.5B (Figure 48) behaves the same way: its last ratio, 4.665, is within 0.1 % of δ (Figure 18).

10 Steering toward hostility

A second steering direction tests whether the organization of the philosophy-steered map is specific to one concept. We steer along a valence direction, the contrast of hostile statements against matched kind ones, built in the manner of contrastive activation addition [29]; its cosine with the gratitude direction is -0.04. Sampled self-dialogue under the same steering is a different system, with a growing context and sampling noise; we report that study separately and follow the direction here in the deterministic map.

The hostility direction in the deterministic map. Steering the soft-token map of Section 2 along the same hostility direction gives a plane with chaos (a positive largest exponent in 10 % of a coarse 128×192 float32 scan) and with periodic text loops that are hostile: "Yes, that's crap" (period 6), "a lie is" (3), "STUP. . . " (3), a loop on the Chinese word for insult and its 2-cycle, and others. They appear at α above about 0.035 (the insult loop) and above 0.065 (the rest). Around the period-6 loop the windows form a dome (Figure 20), and toward higher temperature its texts gain one "!Yes" after another (periods $6, 8, \ldots, 20$). In float64 the windows of periods 12 to 20 are born in folds on their low-temperature side and end in flips or +1 crossings on the other, like the stripes of Sections 7 and 8. The period-6 window ends in a fold at $\log_{10} T = -0.0991$, and the continuations of the period-8 and period-10 windows failed to converge.

The windows accumulate on the fold of the 2-cycle "!Yes" at $\log_{10} T_c = -0.0462465357$ ($\alpha = 0.083$; critical multiplier 0.9999999954, next modulus 0.050). From period 16 on, $1/\sqrt{\log_{10} T_c - \log_{10} T_{\text{fold}}}$

rises by 2.235 per lap. The fold's normal form ($a = -1.1683$, $b = 44.525$ with ε in $\log_{10} T$) predicts $\sqrt{|ab|}/\pi = 2.296$ per lap for full passages through the bottleneck, within 3 % and without a fitted parameter. An independent float64 forward and automatic-differentiation implementation reproduces this fold (location to 6×10^{-10} relative, a simple real $+1$ multiplier with next modulus 0.050) and its coefficients (ab to 8 parts per million). In that implementation, controlled passages from fixed entrances at $\log_{10} T$ offsets -10^{-5}, -3×10^{-5} and -10^{-4} took 119/59, 61/31 and 27/13 iterates of F^2 (full channel/from its midpoint), against prospectively frozen quadratic forecasts of 118/61, 61/32 and 27/14. Five of the six are within one iterate; the smallest-offset midpoint missed by two, failing the frozen all-six criterion. Finite-endpoint (arctan) forecasts, also frozen, were within 0.923 iterates. These are controlled local transits. At the window distances the finite-channel correction to the asymptotic passage time is large, so the agreement of the window slope with the full-passage coefficient does not by itself show where the natural dynamics reinjects. The same type-I intermittency mechanism as in the philosophy-steered map is thus consistent with the hostile windows' organization, which the local fold normal form supports; the controlled passages do not identify the natural reinjection or the global organization.

A second region, near $(\alpha, \log_{10} T) = (0.06, 0.20)$, shows arms of period-7 and period-8 windows spiralling around the insult loop. In float64 the insult fixed point doubles into a 2-cycle at $\alpha \approx 0.041$ and 0.079, and that 2-cycle has a Neimark–Sacker bifurcation in the spiral region: at $\alpha = 0.062$ its complex pair crosses the unit circle near $\log_{10} T = 0.1656$, turning 0.12 of a revolution per step of the 2-cycle. This does not explain the arms. The two invariant circles born at a 2-cycle's Neimark–Sacker point are exchanged by the map, so orbits locked on them have even periods, near 16 for this rotation, whereas the arms include period 7; and of four float32 arm cells, only one kept its period in float64. In dissipative flows, windows wind in nested spirals around periodicity hubs [46], and in the Rössler system the hub's shape follows from a conjugacy to unimodal maps [47]; we have not tested whether the arms here wind around such a hub. Float64 zooms show more structure. In one arm a period-7 domain carries period-14 crescents with period-28 cores along its lower edge, between period-1 and period-2 regions and a chaotic band. In the cascade the windows are nested crescent arcs inside a dome bounded by the "!Yes" 2-cycle region, and near $(\alpha, \log_{10} T) = (0.087, -0.069)$ the period-12 and period-14 windows bend into hooked, nested arches. These are the map-level features that accompany shrimp (a periodic body with doubling crescents; hooked windows).

In float64 the period-7 body is bounded by folds. At $\log_{10} T = 0.1925$ the orbit exists between $\alpha = 0.065966$ and 0.071188, and at $\alpha = 0.066$ it ends in a fold at $\log_{10} T = 0.19263$. Below that, the multiplier passes through -1 twice (at 0.1919 and 0.1899), so the period-14 crescent is a doubling bubble. Further down, the branch folds twice (0.18797 and 0.18921, with three coexisting period-7 orbits in between) before a third flip. All crossings are simple (next modulus below 1.5×10^{-3}). An S-shaped pair of folds is what a cusp of the fold curves leaves on a nearby cut. When these boundaries are continued in the plane in float64 (Figure 23), each of the two fold curves reverses its parameter tangent once. There, as the fold condition requires, the oriented quadratic coefficient b of the fold's normal form passes through zero; what makes such a point a cusp is that the cubic coefficient c and the unfolding determinant $\det J$ are nonzero there (the cusp test, Appendix E) (upper fold: -23.9 to $+244.6$ between adjacent points, with $a \approx 2.0$; lower fold: $+31.2$ to -58.7, with $a \approx 0.55$) while a stays away from zero. Located along the curves, the two cusps lie at $(\alpha, \log_{10} T) = (0.067213, 0.198983)$ and $(0.066611, 0.189382)$. There $b \operatorname{sign} a$ is below 1.3×10^{-5} of its bracket values, $c = -1.41 \times 10^6$ and -5.61×10^5, and $\det J = 58.2$ and 55.2, all evaluated at the located points. The flip curve of the period-14 bubble lies inside the body and closes around the map's period-14 cells. The cubic coefficient is nonzero at the points that bracket both sign changes: $c \approx -1.4 \times 10^6$ to -2.0×10^6 (step agreement 5 %; the smallest steps converge like the square of the step, and extrapolation leaves the sign) and -5.8×10^5 to -5.3×10^5. This supports nondegenerate cusps inside both brackets, and $c < 0$ agrees with the three coexisting period-7 orbits, two of them stable, found between the S-shaped folds. At a fold point near the lower cusp (where $b = 31.4$) the estimated unfolding is nonsingular too ($\det J = 53.5$ in the same coordinates, singular values 262 and 0.20). Its local predictions at six frozen offsets matched the orbits that Newton solves seeded at the predicted roots and the base state recovered: in number at all six and in stability at five. The miss, at the largest offset beyond the edge of the region with three orbits, marks the range of the cubic model. The hostile period-7 body thus has the same skeleton as the period-6 window of Appendix E.

Its interior is shrimp-like too. At $\log_{10} T = 0.1830$, float64 bifurcation diagrams through the body (Figure 21; 1400 cells per line) show the period-7 orbit doubling to periods 14 and 28 and entering chaos (largest exponent up to +0.1) for $\alpha \approx 0.0659$–0.0673. Windows of periods 28, 35 and 42 sit inside the chaotic band, and the orbit then undoubles through 28 and 14 back to 7; the exponent touches zero at every doubling. Newton's method converges at map cells to primitive orbits of periods 14, 28 and 21 (residuals 3×10^{-15}, 2×10^{-14} and 5×10^{-13}; spectral radii 0.08, 4×10^{-4} and 0.006). Periods 21, 35 and 42 are three, five and six times the body's period, as expected for the windows that renormalization places inside the band chaos of a period-7 cascade. Continued in temperature, the period-14 orbit flips into period 28 near $\log_{10} T = 0.1841$ and back near 0.1857, the two ends of a doubling bubble that holds period 28 and chaos. Lower down, its branch is S-shaped (folds near 0.1824 and 0.18244, with three coexisting period-14 orbits between them) before another flip near 0.1819. All these crossings are simple (next modulus below 4×10^{-5}). A 160^2 float64 zoom of the lower body (Figure 46) shows this anatomy in the plane. The period-14 domain encloses a period-28 ring around a chaotic core. Period-21 windows sit in the chaotic region as thin crescents with legs, and curves of very negative exponent, where the orbits are superstable (the leading multiplier crosses zero), run through the windows, as superstable lines run through the windows of low-dimensional maps. In this zoom 18 % of cells are chaotic.

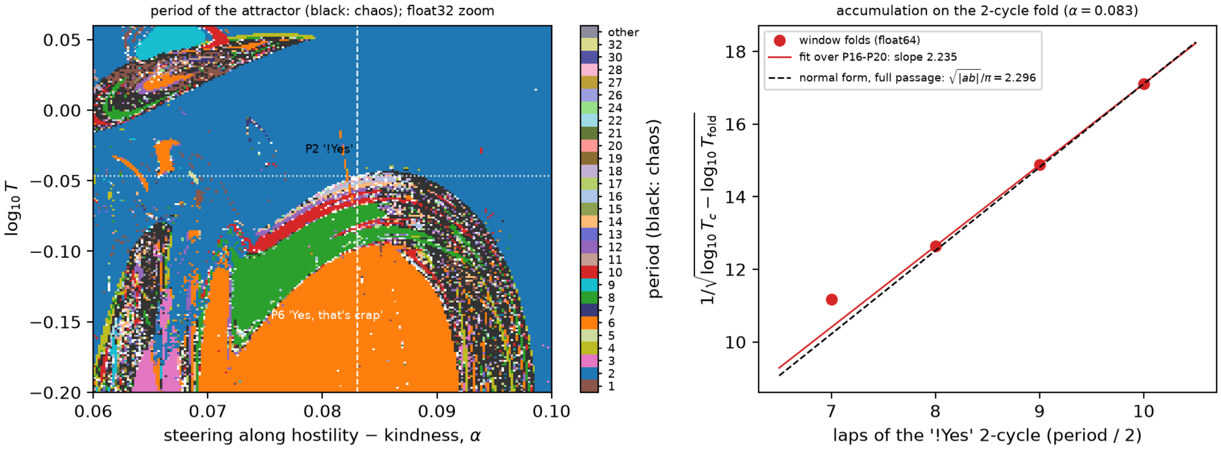


Figure 20: The hostility-steered soft-token map of Qwen2.5-0.5B-Instruct (layer 12). Left: period of the attractor in a float32 zoom (black: chaos), with the period-6 "Yes, that's crap" dome and its period-adding bands; dashed, the cut at $\alpha = 0.083$; dotted, the float64 fold of the "!Yes" 2-cycle. Right: float64 fold edges of the windows (periods 14–20) against the number of "!Yes" laps; the dashed line is the normal-form prediction for full passages, anchored at the last window.

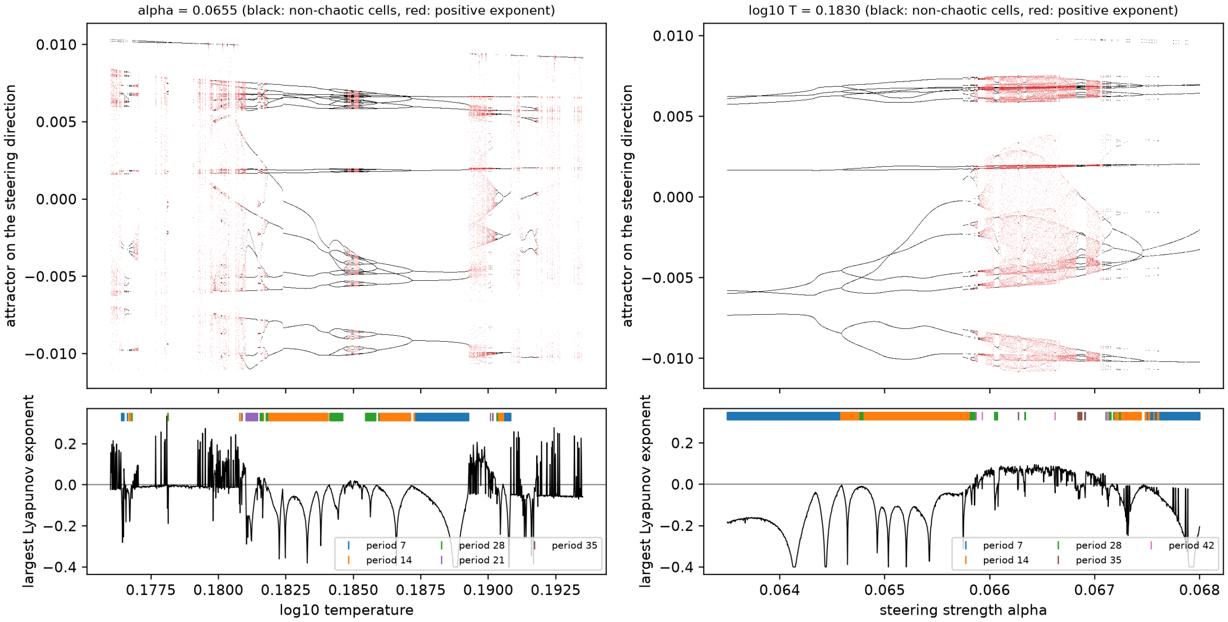


Figure 21: Inside the hostile period-7 body (Qwen2.5-0.5B-Instruct, hostility direction at layer 12): float64 bifurcation diagrams along temperature at $\alpha = 0.0655$ (left) and along the steering strength at $\log_{10} T = 0.1830$ (right). Top: the last 160 states of the attractor reached from the fixed start, projected on the steering direction (red: cells with a positive largest exponent). Bottom: largest Lyapunov exponent, with the cells of periods 7, 14, 21, 28, 35 and 42 marked (cells of other periods are not marked). On the right, period 7 doubles to 14 and 28, enters chaos with windows of periods 28, 35 and 42, and undoubles back to 7. Labels come from 400 measured steps after 1000 transient steps; with 2000 measured steps at two twin offsets, 135 of the 235 red cells on the left stay chaotic and 91 are periodic (Appendix K).

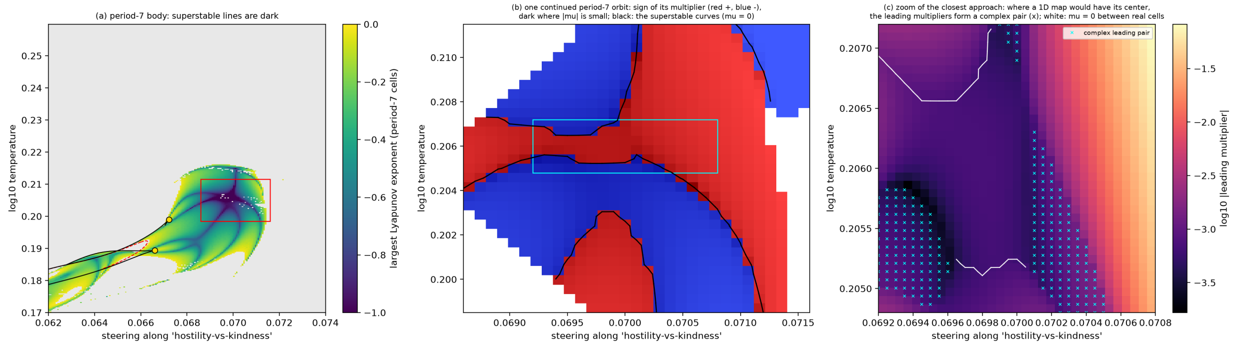


Figure 22: Superstable structure of the hostile period-7 body (Qwen2.5-0.5B-Instruct, hostility direction at layer 12, float64). (a) Largest Lyapunov exponent of the period-7 cells, with fold (black) and flip (red, dashed) curves and the two located cusps (gold); dark curves are superstable lines. (b) One period-7 orbit continued over the red box of (a): sign of its leading multiplier (red positive, blue negative), dark where it is small, and its zero curves (black). The two arcs approach near $(0.0700, 0.206)$ without crossing. (c) Finer zoom of the cyan box of (b) (about 2 times finer in α, 6 times in $\log_{10} T$): logarithm of the leading multiplier's modulus. Crosses mark cells where the two leading multipliers form a complex pair; white segments mark sign changes between cells with a real leading multiplier. Where a one-dimensional map would have its center, the orbit's linearization has two comparable leading directions.

No one-dimensional shrimp center. In a shrimp of a one-dimensional map, two superstable curves of the window's orbit cross at a doubly superstable center, the head [48, 49]. We continued one period-7 orbit over a 31×31 grid of the body's upper part and followed the sign of its leading multiplier (Figure 22). Its superstable curves form two arcs that approach each other near $(\alpha, \log_{10} T) = (0.0700, 0.206)$ and turn away without crossing at this resolution (10^{-4} in α, 4.3×10^{-4} in $\log_{10} T$). A finer grid of that approach (1155 cells, about 2 times finer in α and 6 times finer in $\log_{10} T$) shows why. There the leading multiplier is tiny (median modulus 9×10^{-4}) and as small as the next one, and in 221 cells the two form a complex pair, in two wedges that flank the gap. On this sampled approach the two computed leading multipliers exchange dominance and form a complex pair, so the sign of a leading real multiplier no longer defines a scalar superstable curve there. Where a one-dimensional map would place its doubly superstable center, the orbit's local linearization has two comparable directions; whether the dynamics there reduces to an effectively two-dimensional map is not tested. Next to the body, the 7-step return maps of chaotic cells have a coarse one-extremum shape in every phase: one interior extremum when the points are binned into 16 intervals, while finer partitions keep extra extrema, especially in phases 4 and 5, which we do not attribute to noise. In two phases the points split into two thin strands, an observation of the projection that does not establish a second invariant dimension (Figure 43). Cobwebs on these maps show the doubling cascade and the chaos as on a one-humped interval map (Figure 44). On the tracked branch and in the sampled region, the body is therefore not the shrimp of a one-dimensional map. In the two-dimensional theory of Façanha et al. the head is not generic (Section 11); arcs that turn away resemble its Type B unfolding, but they follow the sign of the leading real multiplier, not the trace that the theory uses, so this is a comparison, not a test.

Chaos behind a repeating text. Along the temperature line of Figure 21 (left), a longer test (2000 measured steps after 1000 transient steps, twins at offsets 10^{-6} and 10^{-9}, whose exponents agree to within 0.0031 in every chaotic cell) finds 182 chaotic cells among 1400 between $\log_{10} T = 0.176$ and 0.191. In 65 of them the leading token repeats exactly for all 2000 steps while the state moves

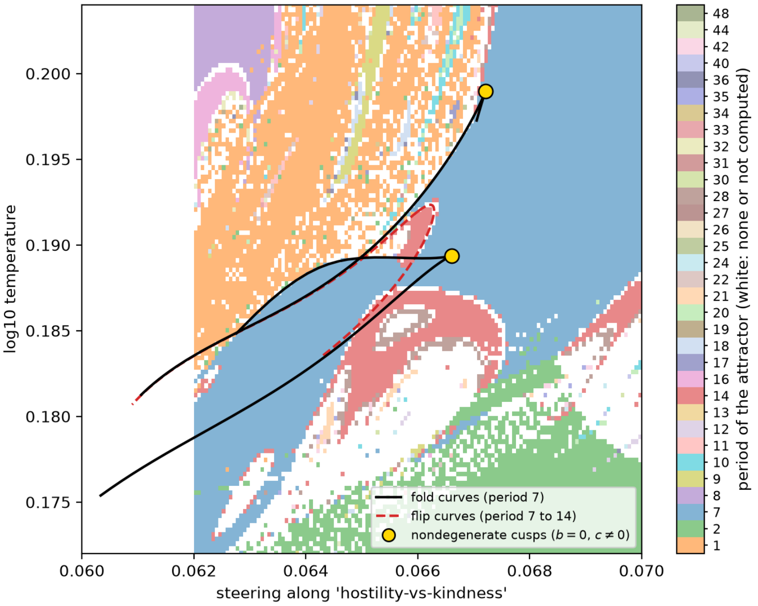


Figure 23: The period-7 body of the spiral region (Qwen2.5-0.5B-Instruct, hostility direction at layer 12): attractor periods of a float64 zoom (white at $\alpha < 0.062$: outside the zoom), the float64 fold curves of the period-7 orbit (black) and the flip curves into period 14 (red, dashed), with the two located cusps (gold).

chaotically; in 59 of these the period of the text is a multiple of 7, most often 7 or 28. Figure 24 shows the strongest case ($\log_{10} T = 0.18995$, largest exponent 0.091 at both offsets). Its 2000 sampled states fall into seven separated groups, one per phase of the text, visited in a fixed order and at least 0.098 of the state norm apart (a property of the samples, not a proof of separated invariant sets), and each group lies inside one cell of the partition by leading token, so the text `I!Yes heels↩↩pee↩↩` repeats at every one of the 2000 steps we followed. This is band chaos, the regime of a chaotic attractor that is a finite union of pieces visited cyclically [50, 51]; the partition by leading token is not generating, so the text cannot reveal the chaos. Inside a group the sampled states lie close to a curve (a strongly anisotropic cloud). Read once per repetition through the probability of the last token, the sampled return relation is well approximated by a smooth map with a single quadratic maximum, the shape of the logistic map: a polynomial of degree 8 fits it to 4×10^{-4} of its range, and the fitted map's Lyapunov exponent, 0.647 per repetition, is within 2 % of the full map's 0.636 per seven steps, measured with twin trajectories in all 896 dimensions. The scalar approximation holds across most of the cascade. Rerun with the leading-token sequences saved, 62 of the 65 cells are again chaotic behind a repeating text, and in every one of them several phases of the text (a median of ten) give single-valued return maps; in 59 cells the best-fitting phase has a single interior stationary point (a maximum in 36, a minimum in 23). If the dynamics were exactly one-dimensional and each phase's probability a smooth invertible coordinate on it, all single-valued phases would give the same asymptotic exponent, since exponents do not change under such a change of coordinates. We therefore fit each phase's map by a polynomial whose degree is chosen by cross-validation over random folds of the return-map pairs, and take the median over the phases. It matches the full map's exponent within 10 % in 51 of the 62 cells (median discrepancy 3.7 %; within 10 % in 23 of the 24 cells with seven-token text, which give 285 repetitions in 2000 steps; Figure 45). This analysis was adopted after we inspected an outlier of our first, pre-specified analysis (a fixed degree of 8 at the single best-fitting phase), which gave 49 of 62 and underfitted one cell whose maps are steep and

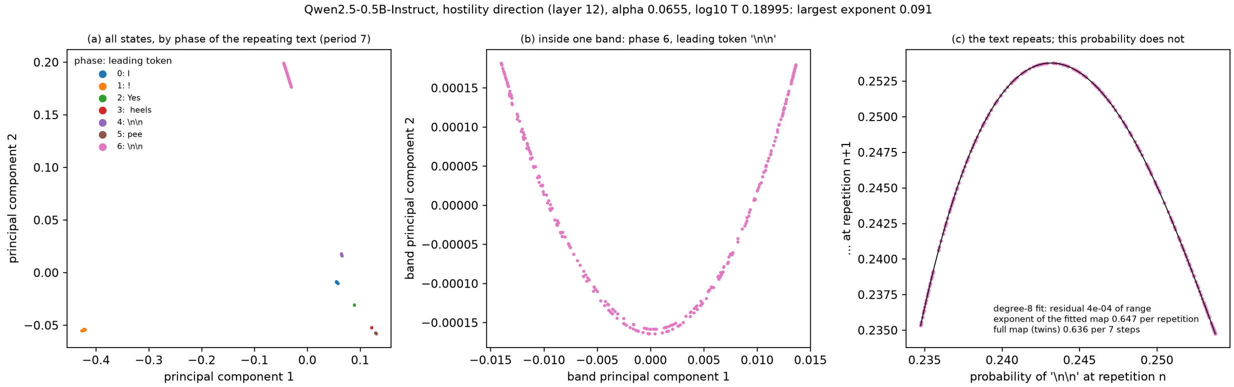


Figure 24: Chaos behind a repeating text: the soft-token map of Qwen2.5-0.5B-Instruct steered along hostility at layer 12 ($\alpha = 0.0655$, $\log_{10} T = 0.18995$; float64, 2000 steps after 1000 transient steps). (a) All states on their first two principal components, colored by the phase of the seven-token text that the leading token repeats: seven separated sampled clouds. (b) The widest cloud (phase 6, leading token a double line break) on its own principal components: the sampled states lie close to a curve. (c) The probability of that token at successive repetitions of the text, with a fitted return curve with one quadratic maximum (black: polynomial of degree 8) whose Lyapunov exponent is within 2 % of the full map's.

have two turning points; we report both. An exploratory temporal split (fit on the first 60 % of each record, test on the last 30 %) gives small value-prediction errors at the best scalar phase, chosen on the test segment itself, but in 30 of 61 cells two- or three-delay predictors improve the held-out error by 20 % or more, which the pre-stated rule counts against scalar sufficiency. These results support an empirical scalar description, not a proved one-dimensional reduction: a low-residual fit does not show that the observable is invertible, and even exact agreement of exponents would not establish a one-dimensional invariant curve (two uncoupled logistic maps observed through one coordinate agree in the same way). An independent implementation reproduced the strongest cell (the same leading token at all 2000 steps; exponent 0.089 against our 0.091; scalar discrepancy +2.3 % under the first rule and +1.5 % under the amended one) and both cross-cell counts (49 and 51 of 62). At two further cells it chose, it found good agreement after the amendment in one (text period 10, +0.2 %, although its fitted maps have both a maximum and a minimum) and a persistent discrepancy of about 18 % in the other (period 28), despite a small value error. These are the chaotic bands that follow a period-doubling cascade, and a smooth unimodal map is the setting of Feigenbaum's universality [22]; a fitted maximum on the sampled range does not by itself establish that theorem's hypotheses, and we have shown the scalar approximation for this cascade only. A reader of the text sees a loop. Only the probabilities, or the state, reveal the chaos.

11 Cusps and shrimp-like windows

The fold of Section 8 is one boundary of one window. Continuing such boundaries through the parameter plane, and following the orbits inside them, shows how the windows of the maps are organized, and which parts of that organization are low-dimensional. Appendices E and F report the continued curves and cusps of the period-4, -5 and -6 windows in full, and Section 10 those of the period-7 body; this section collects what they establish about the shape of the windows, after the cascade of Section 9 and the hostile windows of Section 10.

Fold curves with cusps. Fold and flip curves continued in float64 with the minimally augmented systems of Appendix B keep their critical multipliers at $+1$ and -1 (Appendix E). The fold curve of the 2-cycle "is it" traces the boundary of the period-2 region on which the stripes accumulate, as the mechanism of Section 8 requires. The fold curves of a period-6 window, of a hostile period-7 window, of a period-4 window and of a period-5 window steered at two depths bend back on themselves. On the nine fold curves we continued, all thirteen reversals of the parameter tangent are located cusps: the oriented quadratic coefficient of the fold's normal form changes sign there, while the cubic coefficient stays nonzero and the two-parameter unfolding is regular. This is not a global count of cusps. Near a cusp with negative cubic coefficient, three orbits of the same period coexist between the two fold branches, two of them stable. At a fold point near one cusp in each of two windows, the local cubic model, frozen before solving, matched the number of orbits that Newton solves seeded at its predicted roots recovered at all twelve parameter offsets, and their stability at eleven; the search was not exhaustive.

What the shrimp-like shapes are. The maps show bodies with legs, hooks and crescents that resemble the shrimp-shaped domains that Gallas found in the parameter plane of the Hénon map: a main body of period k, followed by domains of periods $k \cdot 2^n$, with thin legs [23]. They are made of the bifurcation curves of periodic windows; where two windows overlap, their strips can cross in an X (Appendix E). A periodic window is the region where one periodic orbit attracts. It is bounded by fold curves, where the orbit is born or dies together with a saddle (multiplier $+1$), and flip curves, where it doubles (multiplier -1); beyond the flips lie the period-doubled shells. In a one-dimensional map the multiplier is $+1$ on a fold edge and -1 on a flip edge, so it vanishes in between: every path across a window from a fold edge to a flip edge crosses a superstable curve, and these are the dark lines of strongly negative exponent in the maps. Gallas's dissection of one-dimensional shrimps names the parts of a cell (Figure 27a) [48]. The superstable curves of the map's two critical points cross at the head, where the orbit passes both critical points at once. At the end of the cell two legs, each a thin strip between a fold and a flip curve, cross in a kite. The kite begins at a cusp, where the two legs' fold curves meet; inside it two attracting orbits of the same period coexist, and their superstable curves cross at the tail.

Mira and coworkers sort the organizations around a cusp of a fold curve in two-dimensional maps into three kinds [52–54]. Passing through the cusp into its wedge splits an attracting node into a saddle and two nodes in a crossroad area, where the flip curves of the cusp's two branches cross, as in Gallas's kite, and in a spring area, where a flip curve loops around the cusp. It splits a saddle into a node and two saddles in a saddle area, where the attracting orbit exists only inside the wedge. The sign of the cubic coefficient c of the fold's normal form decides between the two splittings: $c < 0$ for crossroad and spring areas, $c > 0$ for the saddle area (Figure 25). Mira also noted that in two dimensions the attracting node becomes a focus, with a complex pair of multipliers, away from the fold and flip curves. In a two-dimensional map whose Jacobian is never singular, such as Hénon's, no multiplier can vanish. There the curves on which the trace of the cycle's two multipliers vanishes take the place of the superstable curves, and the head is no longer generic. Façanha, Oldeman and Glass found two unfoldings: in both the crossing at the tail survives, while at the head the curves either close into a loop (their Type A) or approach each other and turn away (Type B) [49].

The skeleton of one window. The plane steered at two depths (Section 6) holds a period-5 window (Figure 26). A 31×31 grid of the window's continued orbit (Appendix F) left two anomalies, signs of folds it could not resolve, so we continued the window's fold and flip curves in float64. One fold curve bounds the body on its right, runs over its top and along one side of each leg. Flip curves

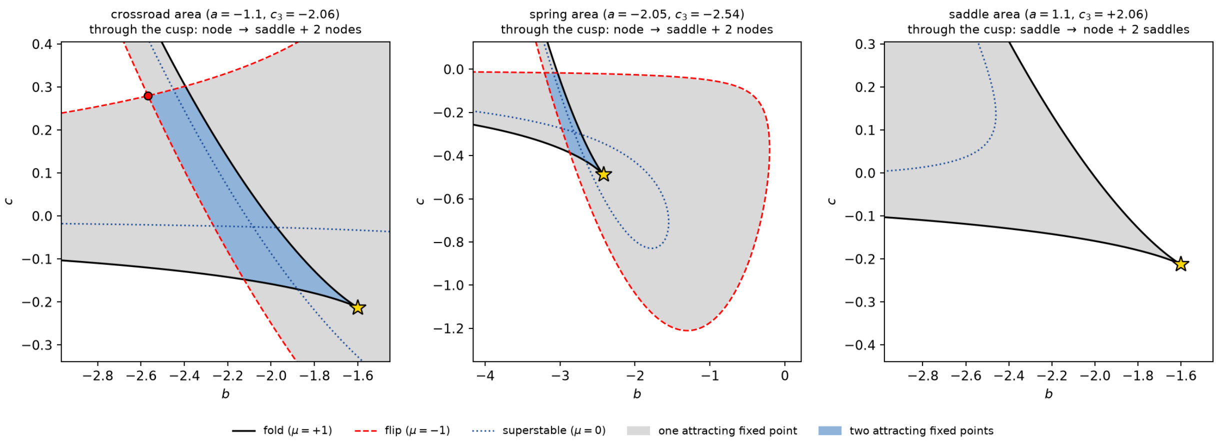


Figure 25: The three kinds of organization around a cusp of a fold curve, computed from the quartic map of Carcassès et al. [53], $x \mapsto c + (1+a)x + bx^2 + x^4$, in planes of b and c at fixed a, as they drew them (gray: one attracting fixed point; blue: two). The cusp (gold) lies at the fixed point $x = t$ with $a = 8t^3$, and $c_3 = 4t$ is the cubic coefficient of its fold normal form (written c in the text). Left, $a = -1.1$: a crossroad area; passing through the cusp splits a node into a saddle and two nodes, and the two attracting fixed points coexist in a kite whose far corner is the crossing of the flip curves of the cusp's two branches (red dot). Middle, $a = -2.05$: a spring area; a flip curve closes into a loop that winds once around the cusp (the loop forms at $a = -2$, the codimension-three point a_7 of Carcassès et al.). Right, $a = +1.1$: the saddle kind; passing through the cusp splits a saddle into a node and two saddles, and the attracting fixed point exists only inside the cusp's wedge. Each frame contains the whole defining structure. Period-2 curves are not drawn.

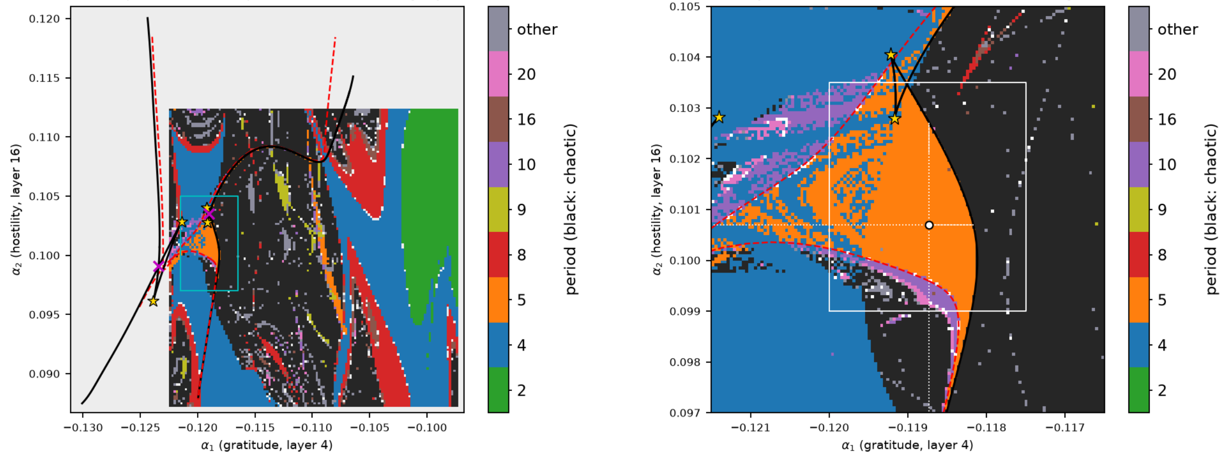
bound the body's upper-left and lower sides and the other side of each leg, so each leg lies between a fold curve and a flip curve, as in Gallas's anatomy, and beyond the flips lie shells of periods 10 and 20. Where the upper leg leaves the body, the fold curve has two located cusps that share a segment of the curve, and its other two branches cross (Figure 26d). At the crossing the two arms are folds of two different period-5 orbits with the same token loop (their states differ by about 4 %). A second fold curve, on the far left, where the orbit dies after regaining stability beyond a doubling bubble, has two more located cusps, which also share a segment, and its other two branches cross as well, again as folds of two different period-5 orbits (states about 9 % apart; Figure 26a). Along every continued curve the critical multiplier stays at $+1$ or -1 to within 1.2×10^{-7}, and no other multiplier exceeds 0.4 in modulus, so no second multiplier reaches the unit circle along them; this does not exclude cusps, which need only a nonlinear degeneracy.

Shrimps of a two-dimensional kind. The period-5 window has this anatomy; only its head differs. At every cell of that grid and every curve point, the multipliers beyond the leading pair are below 0.0032, so on the grid the orbit's linearization has at most two non-negligible directions. On the grid the leading multiplier never vanishes (its modulus stays at least 0.0139): where a one-dimensional map would be superstable, the two leading multipliers form a complex pair, the node having become a focus, or pass each other as real numbers of opposite sign. Between grid cells, however, a two-parameter Newton search for the point where the sum of the leading multipliers and the sum of their pairwise products vanish converges at $(\alpha_1, \alpha_2) = (-0.1184836, 0.1017217)$ (`docs/review/p5-nilpotent.jsonl`). There the three leading multipliers have nearly equal moduli, 3.4 to 3.6×10^{-3}: the orbit contracts about 280-fold per period, and its linearization has no dominant one- or two-dimensional part. This point of strongest contraction plays the role of the head, but the two-dimensional reading does not describe it. The determinant $D = \mu_1\mu_2$ of the leading pair also changes sign in the window ($D < 0$ at 41 of its 447 attracting cells), so the effective two-dimensional map reverses orientation in part of the window: a noninvertible setting in the sense of Mira [55], not that of Hénon's map, whose determinant has a fixed sign. The trace-zero curve around an island of negative trace (Figure 26c) is closed on the grid except at one cell next to the fold curve, where it must close, because on a fold the trace is 1 plus the second multiplier and therefore positive: a closed trace-zero contour as in Type A, a classification derived for determinants of fixed sign [49]. In the hostile period-7 body the superstable arcs turn away without crossing, which resembles the head of Type B (Section 10). The same plane holds a period-doubling cascade whose cells are nested V-shaped tongues near $\alpha_1 = -0.105$, $\alpha_2 = 0.105$. Continued in float64 on grids over these cells, its period-2, -4 and -8 orbits read the same way: no cell reaches superstability (the smallest leading moduli are 0.222, 0.041 and 0.034), and complex leading pairs and trace-zero curves take the place of the superstable curves. The cells become more one-dimensional with each doubling: over their attracting cells, the median ratio of the second to the leading multiplier is 0.59, 0.125 and 0.027. Per step, the second multiplier contracts by a nearly constant factor (medians 0.50 to 0.57), so its share shrinks like a power of the period; the per-step medians predict 0.52, 0.146 and 0.029, within 17 % of the measured ratios. In the terms of the Hénon map, whose Jacobian determinant $-b$ is constant, so that $\mu_1\mu_2 = (-b)^p$, the per-step area contraction $|\mu_1\mu_2|^{1/p}$ plays the role of $|b|$; its medians are 0.41, 0.41 and 0.50. In the period-4 and period-8 cells the trace-zero curves run along both arms of the V. Finer grids at its bottom resolve them: in both cells the two arms are one U-shaped trace-zero curve, and they do not cross. In the period-8 cell the determinant of the leading pair is positive throughout and the third multiplier is at most 6×10^{-5} of the second, so the reading is two-dimensional to that accuracy. In the period-4 cell the determinant changes sign, and near the bottom of the U the second and third multipliers form a complex pair, so the leading pair is not isolated there. Neither cell has a crossing of trace-zero curves, the analog of a one-dimensional shrimp's head, and the period-4 cell has no near-nilpotent point at the bottom of its U either: a two-parameter Newton search on the invariants of the leading spectrum, as for the period-5 window, converges where its three leading multipliers have nearly equal modulus, 0.0325, and lie 120° apart, ten times the period-5 point's 3.5×10^{-3} (`tools/nilpotent_search.py`). Under the decision rule fixed before these grids, both cells remain unresolved between the loop and turn-away types (the first period-8 grid left the bottom of the V unsolved because of a defect in the continuation, fixed for the second; `docs/dss`).

The junction where the upper leg leaves the body, two cusps that share a segment of the fold curve and a crossing of its other two branches, is the configuration that Mira et al. call a dovetail for critical curves with two such adjacent cusps [55]; we borrow the name, with periodic orbits in place of preimages. With the crossing, the configuration is a plane section of a swallowtail, which predicts four orbits in the pocket between the cusps and the crossing, none beyond the crossing and two elsewhere (Figure 27b). An orbit census at 13 points fixed before the computation found exactly these numbers at nine (Figure 26d): four orbits at three of five pocket points, none at either point beyond the crossing, and two at four of five surrounding points. At three other points it found fewer orbits, but all the attracting ones the model predicts (the counts are lower bounds); at one point, beyond a flip curve, it also found an orbit with another token loop. At all five pocket points of the census two attracting period-5 orbits coexist; near the upper cusp a flip curve crosses the pocket, beyond which one of them has doubled. The pocket therefore plays the kite's part, a bistable region at the end of a leg, with two cusps and a crossing where Gallas's kite has one cusp; both configurations are plane sections of the swallowtail of a quartic fixed-point equation (Figure 27). Lifted from the parameter plane into the orbit states, the fold curve is smooth: its cusps are only where its shadow folds, and the two arms that cross in the shadow lie on different sheets, as in the model (Figure 28). The window's second fold curve has a second dovetail. In each, the two cusps have cubic coefficients of opposite sign, as in any section of a swallowtail: one of the saddle-area kind ($c > 0$, the upper cusp in both) and one of the crossroad or spring kind ($c < 0$). We do not claim the swallowtail point itself, which needs a third parameter, and we have not traced the tail [48], determined symbolic sequences or tested for a periodicity hub [46, 47]. With these qualifications the period-5 window carries the anatomy of a shrimp, as Gallas found it in Hénon's map, here in a two-dimensional map whose determinant changes sign [23, 56]; whether a two-dimensional map also reproduces its nonlinear dynamics is not tested (a closure test failed for numerical reasons; Appendix F).

(a) fold (black) and flip (red) curves of the period-5 orbit; x: self-crossings; cyan: zoom of (b)

(b) the body; dotted: branches from a common start; white box: grid of (c)



(c) the curves over the continued orbit's trace (gray: repelling)

(d) junction of the upper leg: cusps (stars), crossing (x), orbits found (attracting)

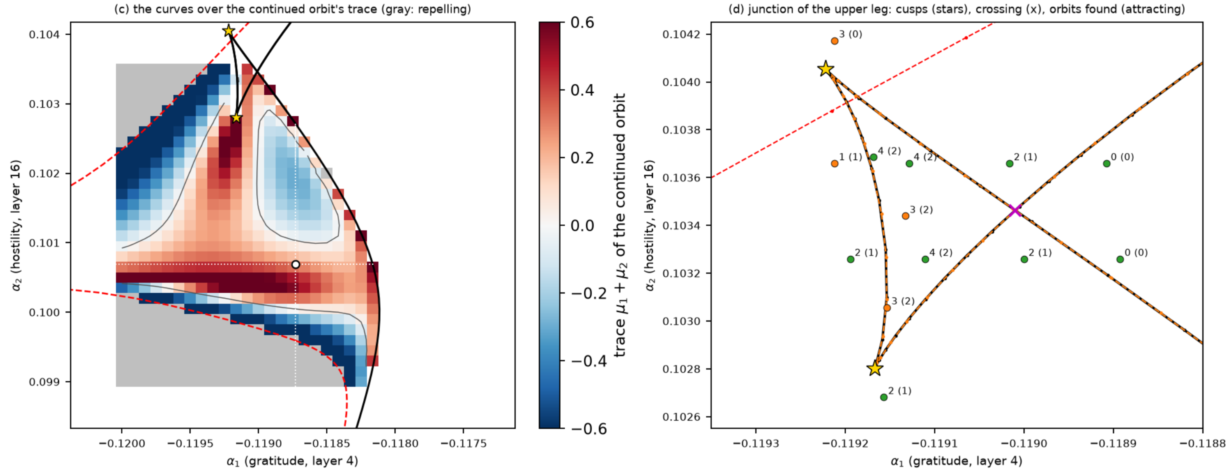
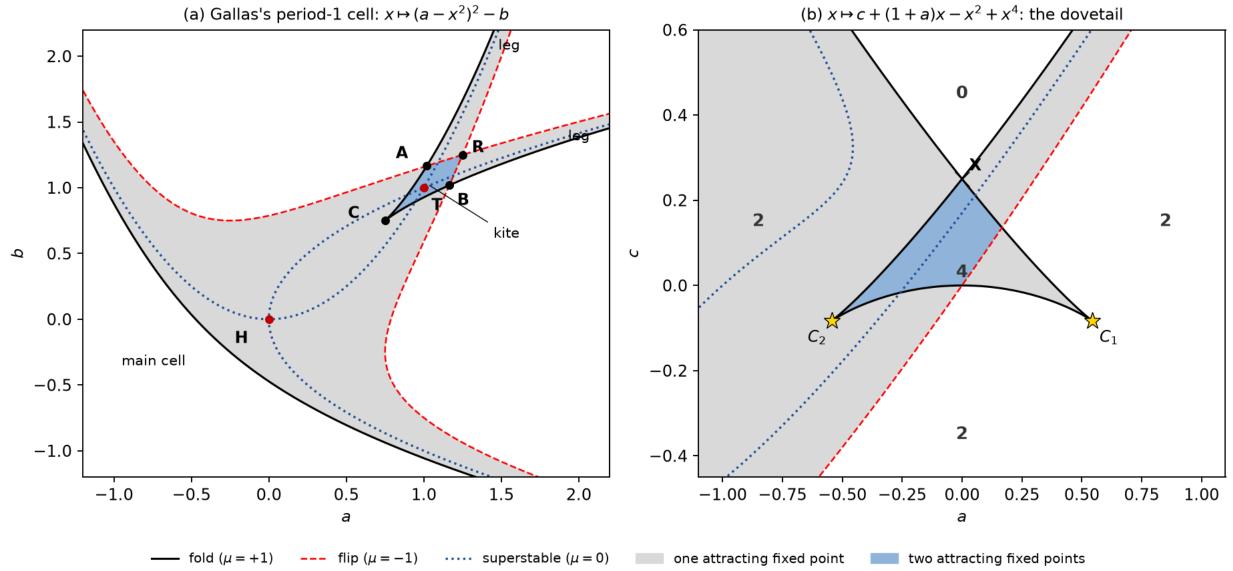


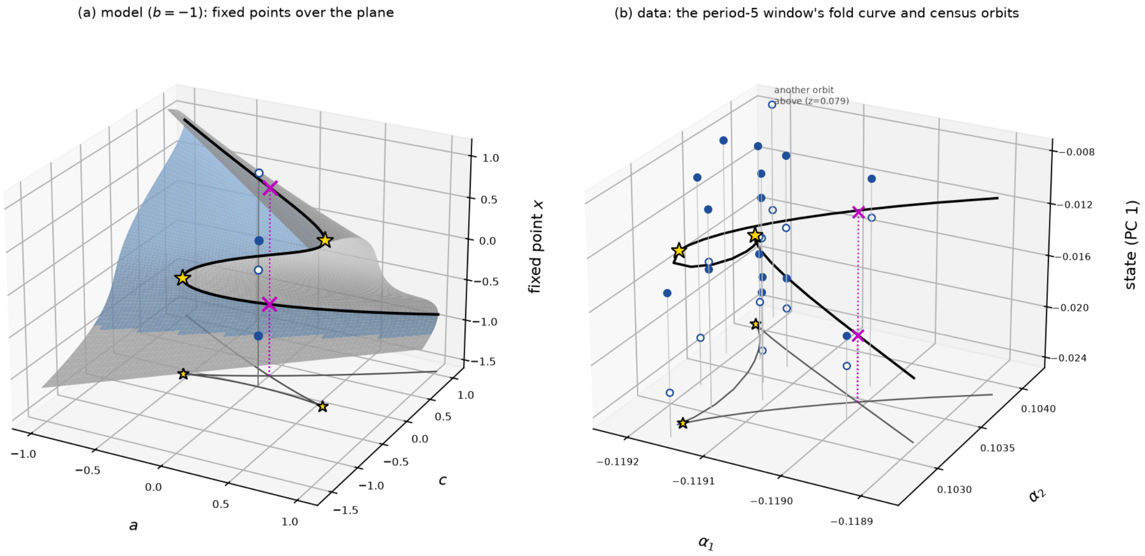
Figure 26: The skeleton of the period-5 window in the plane steered at two depths (Qwen2.5-0.5B-Instruct, gratitude at block 4 and hostility at block 16, $\log_{10} T = -0.1$, float64). (a) The two fold curves (black; multiplier $+1$) and the flip curves (red, dashed; multiplier -1) of the window's period-5 orbit, over the attractor reached from the fixed start of every cell (Appendix B) on a wider grid (gray: outside it); gold stars: the four located cusps; magenta crosses: the self-crossings of the two fold curves; cyan: the zoom of (b). (b) The body over a finer grid (as Figure 42a), with the four branches (dotted) that found the first boundary points from a common start (circle); white: the continuation grid of (c). (c) The same curves over the trace of the continued orbit's two leading multipliers (Figure 42b; gray: repelling). (d) The junction of the upper leg with the body: the fold curve has two cusps that share a segment and crosses itself (x), where its arms are folds of two different period-5 orbits; its two traces, continued from two different boundary points (black, and orange dashed), coincide. Circles: the 13 points of the orbit census, with the number of period-5 orbits found and, in brackets, the number attracting (green: the count predicted before the computation; orange: not).



(a) Gallas's period-1 cell: $x \mapsto (a - x^2)^2 - b$

(b) $x \mapsto c + (1 + a)x - x^2 + x^4$: the dovetail

Figure 27: The anatomy of a shrimp's cell in two classical one-dimensional models, computed (gray: one attracting fixed point; blue: two). (a) Gallas's canonical period-1 cell, $x \mapsto (a - x^2)^2 - b$ (his eq. (5.3), rotated as in his Figure 7 [48]). Fold curves (black; multiplier $+1$) and flip curves (red, dashed; -1) bound the main cell, and the superstable curves (blue, dotted; 0) of the map's critical points cross at the head H. Two legs, each between a fold and a flip curve, cross in the kite C–A–R–B, which begins at the cusp C of the fold curve; in the kite two attracting fixed points coexist, and their superstable curves cross at the tail T. The flip curves of the two legs cross at R, which makes the kite a crossroad area in the sense of Carcassès et al. The computed corners A and B agree with Gallas's to six digits. (b) The quartic model of Carcassès et al. [53], $x \mapsto c + (1 + a)x + bx^2 + x^4$, in the plane of a and c at $b = -1$, a plane they did not draw: the fold curve has two cusps C_1 and C_2 that share a segment and crosses itself at X (a dovetail, in the term Mira et al. use for critical curves with two such cusps [55]). Numbers: real fixed points in each region; a flip curve crosses the pocket, and two fixed points attract only to its left. Both fixed-point equations are quartics $x^4 + ux^2 + vx + w = 0$, whose fold set is the swallowtail: (a) is a section at fixed v, with one cusp, (b) a section at fixed $u < 0$, with two cusps and a crossing.



(a) model ($b = -1$): fixed points over the plane

(b) data: the period-5 window's fold curve and census orbits

Figure 28: The dovetail in Mira's foliated representation: parameters and one state coordinate. (a) Model: the fixed points x of the quartic map at $b = -1$ over the plane of a and c (blue: attracting; gray: not). The fold curve (black) is a smooth curve on this surface, and its shadow on the parameter plane is the dovetail of Figure 27b, with its two cusps (stars) and its crossing X, whose two lifts (crosses) lie on different sheets. Over a point of the pocket the surface has four sheets (circles; filled: attracting). (b) Data: the period-5 window of Figure 26 (Qwen2.5-0.5B-Instruct, gratitude at block 4 and hostility at block 16, $\log_{10} T = -0.1$, float64). The continued fold curve through the junction (black), lifted by the first principal coordinate of these states (86 % of their variance), with its shadow (gray), its two located cusps (stars), the two lifts of the crossing (crosses), and every period-5 orbit the census found (filled: attracting; open: repelling), each at the phase continuous with the fold curve. One orbit with another token loop lies above the frame.

12 From greedy loops to chaos: one picture

The preceding sections establish the pieces: a factorized derivative, folds, cascades and windows. This section assembles them into one account of why the loop of Qwen2.5-0.5B along the philosophy direction looks the way it does. It answers three questions: where the periodic orbits come from, why the windows cross one another, and why one-dimensional maps describe a network with 896-dimensional states. Every step was checked against the full map in float64 (`docs/e2e/`).

At zero temperature: greedy loops. As $T \to 0$ the soft token becomes the embedding of the most probable token, and the loop becomes greedy decoding with a one-token memory (Section 3). Its cycles are the zero-temperature attractors, and they are superstable. Along this direction the greedy map has a two-token loop " is there" for $0.0630 \leq \alpha \leq 0.0777$ and a three-token loop "\n\nThe question" for $0.0661 \leq \alpha \leq 0.0700$. At $\alpha = 0.070$ and low temperature, the loop indeed settles on these: in a column of 96 temperatures from $\log_{10} T = -2$ upward, 59 end on "\n\nThe question" and 16 on a period-4 loop that alternates " is it" and " is there".

At finite temperature: compromises between competing continuations. The attractors that organize the maps are not greedy loops. No value of α between 0 and 0.12 makes " is it" a greedy cycle; after " it" greedy decoding writes "'s". The 2-cycle exists only because the soft token is a mixture. After " is", the fed-back vector mixes " it" and " there"; after " it", it mixes " is" and "'s". Each step thus feeds back a weighted blend of two continuations, and the blend closes into a cycle that no single token sequence forms. Such a compromise lives in a band of temperature and dies at both of its ends. At $\alpha = 0.070$ the 2-cycle is stable between $\log_{10} T = -0.5147881$, where a multiplier reaches -1 (a flip: the competition becomes too sharp, and the blend alternates), and $\log_{10} T = -0.1741065$, where a multiplier reaches $+1$ (a fold, the cyan curve of Figure 8). The period-3 loop of the crescent, "\n\nThe existence", is a compromise of the same kind, between " existence" and " question" in the noun slot. It too lives in a band, between a fold above (at $\alpha = 0.065$, $\log_{10} T = -0.14536$) and a flip below. It does not connect to a zero-temperature loop.

What happens at the two ends. Beyond the fold the 2-cycle is gone, but its ghost delays every trajectory that passes near it. The number of laps N spent there grows as the inverse square root of the distance to the fold, so periodic windows of period $2N + 6$ line up parallel to the fold curve and crowd toward it as $1/N^2$ (Section 8). This gives the near-vertical fan of stripes. Below the flip, a compromise starts alternating between its two blends, and in the crescent the alternation doubles again and again, a Feigenbaum cascade into chaos (Section 9). The near-horizontal band that crosses the stripes near $\log_{10} T = -0.14$ is the first doubling of the crescent's period-3 loop, continued to larger α. At $\alpha = 0.070$ it is bounded by two flips, at $\log_{10} T = -0.1392937$ above and -0.1414292 below. The two families cross because each is bounded by its own curves: the stripes by the distance to the fold of " is it", the band by the flips of the period-3 loop. Where they overlap, we tested two crossing points from five starts each, and every start ended on the band's 6-cycle.

Why one dimension. A single step of the map is not close to rank one. Along these orbits, the second eigenvalue of the readout covariance Σ_p is 0.23 to 0.87 of the first, and the second singular value of the step derivative is 0.12 to 0.80 of the first. Composed over a cycle, however, the map is nearly one-dimensional. The ratio of the two leading singular values is 0.081 for the 2-cycle, 0.092 for the period-3 loop and 0.26 for the period-6 band; it falls to 3×10^{-4} over twelve steps of the chaotic crescent and to 4×10^{-9} in the chaotic stripes. Each step stretches some direction, but different steps stretch different ones, and only the direction along which the competition between two continuations repeats keeps growing over a cycle. The Lyapunov spectra say the same. At three chaotic cells the leading exponents are 0.109, 0.424 and 0.239 per step, the next three lie between -0.82 and -2.0, and the Kaplan–Yorke dimension is 1.13, 1.43 and 1.21. One direction expands and the rest are crushed, so the attractor is nearly a curve, and a scalar return map with a single maximum describes it. That is the shape that makes period doubling and Feigenbaum's ratios inevitable. A scalar model built from the two competing tokens, $q' = \sigma(h(q)/T)$, reproduces the crescent's hump and places its first flip at $\alpha = 0.06476$ against 0.0644767 in the full map. It does not reproduce the cascade, so the competition sets the shape, while the rest of the network sets where the cascade ends.

The picture. At zero temperature the loop writes greedy loops. As the temperature rises, the soft token starts to mix the two continuations a token has, and the mixtures form new periodic orbits that no token sequence has. Each lives between a flip, where the competition is too sharp, and a fold, where it is too weak. Their folds leave ghosts that line up the windows, and their flips start cascades into chaos. Because one competition stretches while the rest of the network contracts, every one of these structures follows the laws of one-dimensional maps. Steering moves which continuations compete, and so where all of this happens.

13 Beyond the one-token loop

The soft-token map feeds back one expected embedding at a fixed prefix and position. This section summarizes what changes when the loop keeps more context, when tokens are sampled and in other models (details in Appendices G and H); Sections 14 and 15 then follow model size and a second system.

Memory and sampling. As $T \to 0$ the soft-token map becomes greedy decoding with the same finite memory, whose attractors are cycles (Appendix G). Fed back the last 8 or 16 soft tokens instead of one, Qwen2.5-7B is chaotic at 21 and 20 of 24 temperatures, against 8 with one. In Soft Thinking's growing context, where every soft token stays visible, an early perturbation of 10^{-10} grows exponentially in the same model inside its one-token chaotic band. This does not hold in every model: in Qwen2.5-0.5B the growth moves to sharper distributions, outside its narrow one-token chaotic band, and Qwen2.5-72B stays nearly insensitive (Section 14). Sampling is different. When the expected embedding is replaced by the mean of m sampled token embeddings, the laminar phases of the bottleneck, 120 steps long in the exact map at the closest offset, last about 10.6 steps with $m = 64$ and 7.7 with $m = 8$, and with single samples essentially none remain. The deterministic intermittency belongs to the expected-embedding map; sampling noise turns the bottleneck into a noise-limited trap whose strength grows with m.

Other models. The organization recurs in Qwen2.5-1.5B (Figure 29) and 7B, in OLMo-2 7B, in Llama-3.1 8B, whose attractors we also read with sparse-autoencoder features, and in a second checkpoint of the 0.5B architecture, which shares the organization but not the attractors (Appendix H). The fold law of Section 8 recurs at 1.5B and 7B: windows accumulate on the fold of a short cycle, and the nearest four resolved windows meet a criterion fixed before the lines were computed, within 0.12 % at 1.5B and 4.5 % at 7B (a correction of the period selector after the first scoring is disclosed); farther out, the 1.5B windows spread by 10.3 to 14.0 %.

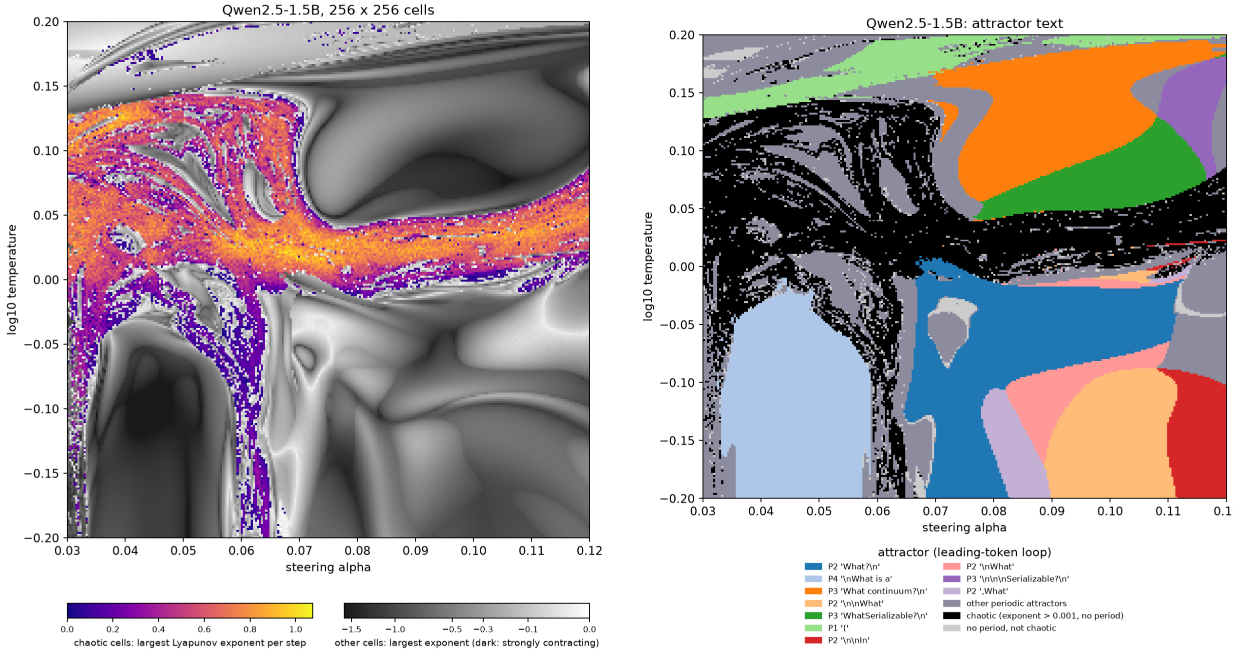


Figure 29: Qwen2.5-1.5B soft-token map steered along the philosophy direction at layer 14, 256^2 float64 cells over $\alpha \in [0.03, 0.12]$, $\log_{10} T \in [-0.2, 0.2]$: largest Lyapunov exponent (left, colored as in Figure 1) and attractor texts (right, as in Figure 9). Several period-2 label regions touch without a gap; as explained in Appendix B, such boundaries may be label switches of one orbit.

14 Chaos and model size

Is the chaos of the loop a small-model effect that larger models leave behind? We measured the unsteered map of fifteen models: Qwen2.5 at seven sizes from 0.5B to 72B parameters, OLMo-2 at 1B, 7B, 13B and 32B, Llama-3.1 at 8B and 70B, Llama-3.3-70B, and the mixture of experts Qwen3.5-35B-A3B, all with the paper's prefix and the settings of Appendix B (1000 transient and 2000 measured steps in float64; chaotic: an exponent above 10^{-3} and no detected period, which is what "positive" abbreviates below). The stored checkpoint values are evaluated in float64 (Appendix B, "Precision"), and the 70B-class models, spread over two GPUs, were recomputed where an earlier two-GPU computation was less precise (Appendix I). The 7–8B and 70B-class models had shown a steep decline of chaos with size. Before measuring the others we fixed three predictions that extrapolate it (`docs/scaling-predictions.md`): within each family, every model above 8B is less chaotic than the family's 7–8B model (P1); the chaotic share does not increase with size above 7B (P2); and at matched next-token entropy, both gain factors of Section 3 of the 14B and 32B models lie at or below those of the 7B model (P3). After a review pointed out that a single prompt and a fixed temperature window could mislead, we froze a fourth test across ten prompts (`docs/prefix-scale-test.md`).

In a fixed temperature window, chaos seems to fade with size. Table 14 and Figure 30 give each model's chaotic share of the temperatures in $\log_{10} T \in [-1, 0.3]$, the range common to all sweeps, and the chaotic length of its whole temperature axis. In two families the largest models are nearly free of chaos in this window. Qwen2.5-7B is chaotic at 12.7 % of the temperatures in the common range, and the 14B, 32B and 72B models at none. Llama-3.1-8B is chaotic at 19 %, Llama-3.1-70B, from the same release and training recipe, at 2.3 %, and Llama-3.3-70B, which differs from Llama-3.1-70B only in post-training, at none. In the third family the order fails: OLMo-2-32B is the most chaotic OLMo model, at 16.2 % against 6.2 % at 7B, 3.8 % at 1B and none at 13B. (The 32B model is a later release, 0325, than the 7B and 13B models, 1124, so within OLMo-2 size and release are confounded.) The frozen predictions P1 and P2 hold for Qwen2.5 and fail for OLMo-2. Below 7B we predicted no order, and none appears (Qwen2.5 0.5B, 1.5B and 3B: 1.4, 1.5 and 0 %).

What a loop without chaos writes at 70B. Not chaotic does not mean a single repeated token. Over its 128 temperatures ($\log_{10} T$ from -1 to 0.3; 1000 transient and 2000 measured steps), the unsteered Llama-3.3-70B settles on long cycles. At low temperature, at 46 of the 49 temperatures from $\log_{10} T = -1$ to -0.5, it writes a fifteen-token loop without end, `. It'supgrading the infinite possibilities are you'recognized protocols` (the tokens run together as written). At intermediate temperatures period-16 variants end in `established` (15 temperatures) or `acknowledged` (9), and a period-21 variant turns into `... of a fellow AI to another artificial greetings` (8). Higher temperatures give cycles of 24 to 63 steps, and 19 temperatures have neither a period up to 64 nor a positive exponent. Llama-3.1-70B, which differs only in post-training, mostly writes the three-token loop `to another AI` (31 temperatures) or the four-token loop `. It'sup` (21), longer repetitions of `to another AI` with periods 6 to 18, and at four temperatures the fixed point `It`, a single token repeated. Steering changes this picture (Figure 32): along the hostility direction, fixed points fill the plane above the chaotic band and windows of periods 3 to 13, most often 6, lie below it.

Across prompts, the chaos moves to higher temperature instead. The fixed window hides where larger models are chaotic. The frozen prompt-blocked test compares Qwen2.5-7B and -14B on ten system prompts and openers (two for development, eight held out), over $\log_{10} T \in [-0.4, 0.9]$ (48 temperatures) and then over the temperature interval where each model's next-token entropy rises most (24 temperatures), with twins at offsets 10^{-5} and 10^{-7}; a temperature counts as chaotic only when both exponents are positive (Figure 55). It predicted that the 14B model would be less chaotic. Both predictions fail: on all eight held-out prompts the 14B model is chaotic at more of the 48 temperatures (9 to 17, against 0 to 7 for the 7B), and its largest exponent is lower on only three. Every chaotic temperature of the 14B model, on all ten prompts, lies between $\log_{10} T = 0.36$ and 0.90, at and above the temperature where its next-token distribution spreads (0.35–0.65); the 7B model's chaos lies mostly between 0.21 and 0.43, around its own, lower transition (0.21–0.46). On the paper's prompt, 12 of the 48 coarse temperatures of the 14B model are chaotic, from 0.59 to 0.90, and the adaptive fine stage found one more near 0.519; the initial sweep of Table 14 ended at 0.6 and saw only the lower edge (the table now also includes the upper sweep). Within Qwen2.5, then, the larger model is not less chaotic: its distribution stays sharp up to higher temperature, and its chaos sits there. The decline in the fixed window is largely a shift. A second frozen test (`docs/prefix-scale-olmo.md`; stage 1 only, four held-out prompts) asked whether the OLMo-2 order survives other prompts. It does, with small counts: OLMo-2-32B is chaotic at more of the 48 temperatures than OLMo-2-7B on three of the four prompts (4, 5, 3 and 0 against 3, 0, 1 and 3), as predicted, and its chaos again lies somewhat higher ($\log_{10} T = 0.10$–0.35 against -0.01–0.21).

model	params (B)	chaotic, $[-1, 0.3]$ (%)	unres.	sweep, $\log_{10} T$	chaotic dec.	breadth	max λ
Qwen2.5-0.5B	0.5	7/512 (1.4)	1	-1.0 to 0.3	0.018	0.053	$+0.28$
Qwen2.5-1.5B	1.5	2/130 (1.5)	0	-1.0 to 1.2	0.531	0.191	$+0.85$
Qwen2.5-3B	3.1	0/130 (0.0)	0	-1.0 to 1.2	0.030	0.060	$+0.85$
Qwen2.5-7B	7.6	65/512 (12.7)	6	-1.0 to 1.2	0.226	0.249	$+0.88$
Qwen2.5-14B	14.7	0/130 (0.0)	0	-1.0 to 1.2	0.500	0.191	$+1.14$
Qwen2.5-32B	32.8	0/130 (0.0)	0	-1.0 to 1.2	0.010	0.091	$+0.36$
Qwen2.5-72B	72.7	0/128 (0.0)	0	-1.0 to 1.2	0.000	0.014	-0.45
Llama-3.1-8B	8.0	96/512 (18.8)	2	-1.0 to 1.2	0.241	0.634	$+0.66$
Llama-3.1-70B	70.6	3/128 (2.3)	28	-1.0 to 0.3	0.031	0.307	$+0.41$
Llama-3.3-70B	70.6	0/128 (0.0)	19	-1.0 to 0.3	0.000	0.225	-0.27
OLMo-2-1B	1.5	5/130 (3.8)	0	-1.0 to 1.2	0.050	0.322	$+0.23$
OLMo-2-7B	7.3	32/512 (6.2)	3	-1.0 to 1.2	0.083	0.197	$+0.30$
OLMo-2-13B	13.7	0/130 (0.0)	0	-1.0 to 1.2	0.000	0.131	-0.0007
OLMo-2-32B	32.2	21/130 (16.2)	0	-1.0 to 1.2	0.252	0.694	$+0.62$
Qwen3.5-35B-A3B	35.0	0/128 (0.0)	15	-1.0 to 0.3	0.000	0.143	-0.27

Table 14: Unsteered soft-token maps of fifteen models (float64; 1000 transient and 2000 measured steps; one twin offset). Chaotic: exponent above 10^{-3} and no period up to 64; unres.: temperatures in $[-1, 0.3]$ with neither a detected period nor a positive exponent. Chaotic decades: the grid-weighted extent of the temperature axis labeled chaotic over the whole sweep, each temperature weighted by its grid spacing (an estimate of the sampled extent; models whose sweep stops at 0.3 have incomplete coverage); max λ: the largest exponent on the sweep, rounded to two decimals (OLMo-2-13B: -6.7×10^{-4}); breadth: decades in which the mean next-token entropy on the orbit lies between 1 and 10 nats. The Qwen2.5-72B sweep joins two grids that share $\log_{10} T = 0.3$, counted once; for the 7–8B models the share in $[-1, 0.3]$ comes from their 512-temperature sweeps, on which the frozen predictions are scored, and the whole-axis columns also include sweeps over $[-1, 0.6]$ and $[0.61, 1.2]$ (64 temperatures), as for the other models swept to 1.2. Positive cells found later on finer lines (Qwen2.5-32B, -72B; Llama-3.3-70B) are not in these regular-grid counts.

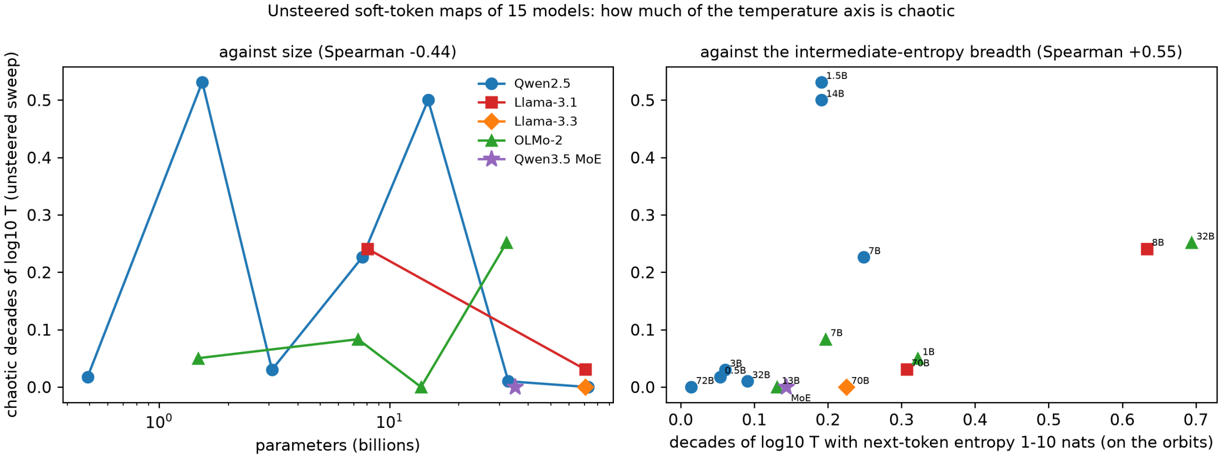


Figure 30: How much of the temperature axis is chaotic, for the fifteen unsteered maps of Table 14: against parameter count, by family (left), and against the breadth of the intermediate-entropy regime (right). Spearman correlations over the fifteen models. Each model's sweeps cover the range listed in Table 14; those of the two Llama-70B models and the mixture of experts end at $\log_{10} T = 0.3$, so their values are lower bounds. The breadth is measured on the loop's own orbits; the loop-free test of the same idea is in Appendix I.

What sets it: the gain of the loop. The gain split of Section 3 shows which factor removes or keeps the chaos (Figure 56). We compare medians of the two factors in bins of the mean next-token entropy on the orbit; the bins hold 1 to 17 of each model's 24 temperatures, so single medians are noisy. Within one training recipe, from Llama-3.1-8B to Llama-3.1-70B, the readout factor changes by at most 0.3 per step in every bin above one nat (below one nat the 8B model has a single temperature), while the network factor is lower by 0.5 to 1.3 per step between 1 and 10 nats (6–10 nats: median -0.98 from 3 temperatures against $+0.28$ from 4) and by only 0.18 above 10 nats: in these two checkpoints the larger model damps changes of its fed-back token. Llama-3.3-70B, which differs from Llama-3.1-70B in post-training, has a much lower readout factor below 3 nats (1–3 nats: -2.33 from 7 temperatures against -0.10 from 6) and a similar one above 6 nats. In Qwen2.5, both factors of the 14B and 32B models lie below the 7B's in every bin between 1 and 10 nats, as the frozen prediction P3 states (the 6–10 nat bin only with the 7B's wider gain split, from one temperature), and both factors of the 72B lie below the 7B's in every bin. Above 10 nats, where the distribution is nearly uniform, the 14B and 32B models instead have the larger network factor ($+2.98$ and $+1.17$ against $+0.19$), and those are the bins of their chaotic temperatures above $\log_{10} T = 0.3$. Below 3 nats the readout factor dominates at 32B and 72B, because the next-token distribution stays sharp up to high temperature (for the 72B, entropy at most 0.15 nats up to $T = 2$). In OLMo-2, P3 fails, and the readout factor tracks the chaos: OLMo-2-13B, with no chaotic temperature, has a lower readout factor than OLMo-2-7B in all three bins between 1 and 10 nats (-1.0 to -1.4 against $+0.06$ to -0.72 per step) although its network factor is higher in two of them, whereas OLMo-2-32B, the most chaotic OLMo model, has a higher readout factor than the 7B in all three ($+0.10$ to $+0.35$), positive as in the chaotic 7–8B models. These are descriptive comparisons of trained checkpoints, not interventions on size or post-training: the split is a decomposition along the evolving tangent direction, and equal entropy does not imply equal embedding covariance. Entropy clamping tests at 72B whether forcing an intermediate entropy suffices to destabilize the loop. If its loop were stable only because its distribution jumps from sharp to flat, holding the distribution in between should destabilize it. We therefore choose each step's temperature so that the next-token entropy equals a target and sweep the target instead of the temperature (entropy-clamped decoding; the map stays deterministic, and the entropy holds to 10^{-14}). The temperature then depends on the state, $T = \tau(x)$ with $H(x, \tau(x)) = H_\star$ for the entropy H, so the derivative of the clamped map $\hat{F}(x) = F(x, \tau(x))$ gains a rank-one term, $D\hat{F} = D_x F - \partial_T F \, D_x H / \partial_T H$. We therefore take its exponents from twins of the clamped map itself, temperature solve included, and use no fixed-temperature derivative or gain split for it. The safeguarded solve is only piecewise smooth, but on these orbits one-step central differences at 10^{-5} to 10^{-7} agree with those at 10^{-4} within 6×10^{-6}. Held at any of 16 targets between 1 and 11 nats, Qwen2.5-72B has only periodic orbits, the echo of period 2 on the paper's prompt and periods 1 to 19 on a held-out prompt, with twin exponents between -0.08 and -4.5 that agree within 0.004 across offsets 10^{-5} to 10^{-7}, whereas Qwen2.5-0.5B under the same clamp is chaotic at 6.1 nats. The clamped 72B loop thus stays stable even where its readout is spread. The clamp changes the map, so this is consistent with damping by the network in the original map but does not identify it. The pretrained checkpoint is also mostly non-chaotic. A test frozen before its data (`docs/prefix-scale-72b-base.md`) predicted that the pretrained base model of the same size would be chaotic at three or more temperatures more than Qwen2.5-72B-Instruct on at least three of the four held-out prompts. It fails on all four: the base model is chaotic at 1, 0, 0 and 0 of the 48 temperatures (and at 2 on the paper's prompt; computed after the precision fix, with 500 transient and 1000 measured steps, half the Instruct model's horizon), although its next-token distribution spreads over 2.5 to 5.5 times the temperature range (intermediate-entropy breadth 0.28 to 0.61 decades against 0.11 to 0.22) and from lower temperatures on. At 72B, then, the sharpness that post-training adds does not explain the stability, and post-training is not necessary for it; the failed prediction does not quantify post-training's other effects. The gain split points to the network's damping of the fed-back token (the 72B's network factor lies below the 7B's in every entropy bin), and the breadth that orders the fifteen models (Figure 30, right) fails for this pair. The damping holds when the whole context is kept as well: in Soft Thinking's growing context (Appendix G), an early perturbation grows at none of eleven steered cells, six of them chaotic as one-token maps, and at one of 48 unsteered temperatures on two prompts, transiently.

Where the network damps. The network factor can be resolved through the stack. Let h_l be the residual stream at the soft token's position entering block $l+1$, and δh_l its change. The factor then splits exactly into three terms: the growth of the change through the stack, $A_L = \log(\|\delta h_L\|/\|\delta x\|)$; minus the growth of the residual stream itself, $N_L = \log(\|h_L\|/\|x\|)$; plus the final normalization. The difference $A_l - N_l$ is the log of the relative change after l blocks (Appendix B).

We measured this split along the unsteered loop of fourteen models at 24 temperatures each (Figure 31). At 0.3 to 6 nats, the entropies where the loop's chaos occurs at low temperature, the median network factor falls with size in two of the three families.

- In Qwen2.5 it falls from -0.12 at 0.5B to -1.99 at 72B, though not monotonically (-1.80 at 3B, -0.85 at 7B).
- In Llama-3 it falls from -0.14 at 8B to -1.50 and -1.30 at 70B.
- In OLMo-2 it does not fall: $+0.11$, -0.52, -0.79 and -0.36 from 1B to 32B. OLMo-2-32B, the family's most chaotic model, is the least damped of its three larger models.

Qwen2.5-72B is the only Qwen2.5 model whose network factor stays negative at nearly uniform distributions: -1.12 above 10 nats, against $+0.22$ to $+2.97$ for the 1.5B to 32B models, as its stability under entropy clamping suggests.

The profiles through depth show where the difference arises. Near 1 nat, the change of the fed-back token keeps pace with the growing residual stream in the small models: entering the last block their relative change is -0.32 (Qwen2.5-0.5B) and -0.51 (Llama-3.1-8B). It does not keep pace in the large ones, where it is -2.51 (Qwen2.5-72B) and -2.08 (Llama-3.1-70B): their blocks keep writing a growing state that depends little on the fed-back token.

Sublayer knockouts hold one attention or MLP output at the soft token at its unperturbed value. They find no component that does the damping: at each model's knockout temperature nearest 1 nat (entropies 0.12 to 1.61 nats), removing any single sublayer raises the network factor by at most 0.44 nats in any of the fourteen models. The amplification moves instead.

- In the Qwen2.5 models from 1.5B to 32B, the first attention layer amplifies the change: removing it lowers the factor by 1.67 to 3.74 nats near 1 nat.
- In the 0.5B model, the first MLP does (2.40).
- In Qwen2.5-72B and in both 70B Llama models, no sublayer contributes more than 0.46.

The larger models' damping is therefore not added by a component. The early amplification that lets the change keep pace with the residual stream is weaker or absent. These are descriptive comparisons of trained checkpoints within three families, and the OLMo-2 exception shows that training can break the trend.

Mixtures of experts. A mixture-of-experts layer sends each token to a few of many expert networks, those with the largest router scores, so its map is smooth only between routing boundaries. Two questions follow: whether routing switches drive the measured exponents, and whether a model

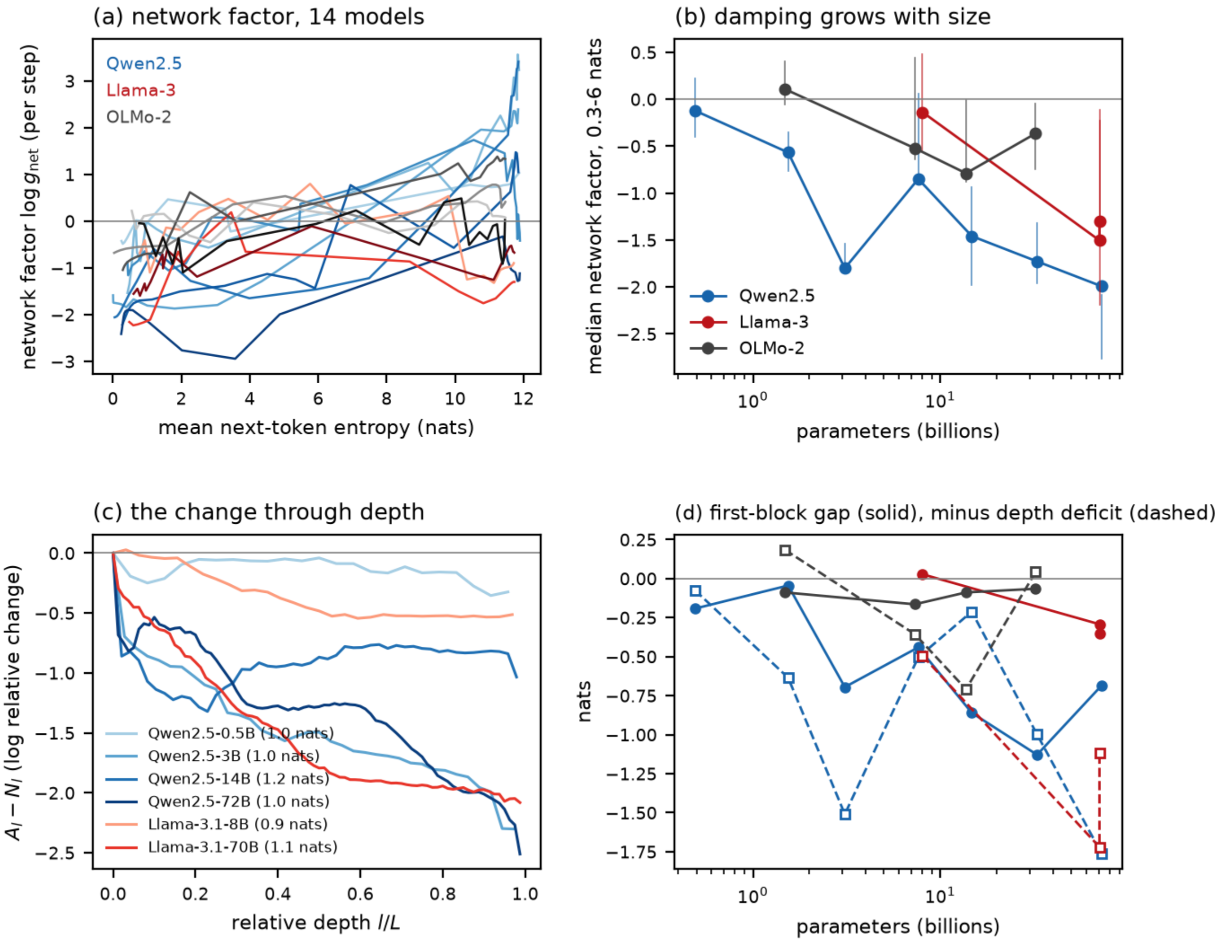


Figure 31: Where the network damps a change of the fed-back token. Unsteered loop on the paper's prompt, 24 temperatures per model, float64; a twin at relative offset 10^{-7} along the evolving tangent; 500 transient and 500 measured steps. (a) Network factor against the mean next-token entropy on the orbit, for fourteen models (Qwen2.5 blue, Llama-3 red, OLMo-2 gray; darker for larger models). (b) Median network factor between 0.3 and 6 nats against the number of parameters; bars give the range of the medians in the bins 0–1, 1–3 and 3–6 nats. (c) The log relative change $A_l - N_l$ after l blocks against relative depth, at each model's temperature nearest 1 nat. (d) The two parts of the final relative change at that temperature: the gap after the first block, $A_1 - N_1$ (solid), and the deficit from the second block on, $(N_L - N_2) - (A_L - A_2)$, drawn with a minus sign (dashed); the increment through the second block itself is not shown. The profiles use block inputs, so the last value is the relative change entering the last block.

that is large in total but small per token behaves like a small or a large dense model (Figure 54, Appendix H). We followed the unsteered loops of Qwen3.5-35B-A3B (8 of 256 experts per layer; about 3B of its 35B parameters are active per token) and OLMoE-1B-7B (8 of 64; about 1B of 7B). Along them, a twin at relative offset 10^{-7} never chose different experts from the base orbit, at any step, layer or temperature. Freezing its routing to the base orbit's left the network factor unchanged, so the exponents measure the smooth map. The orbits do pass close to routing boundaries. At each step we recorded the smallest gap between the eighth and ninth largest router logits over all layers; its median along a loop lies between 1.1×10^{-4} and 1.75×10^{-2}, and its minimum is about 3.38×10^{-7}. Rounding errors of bfloat16 inference (relative precision about 4×10^{-3}) can cross gaps this small, so in low precision routing can add jumps that the float64 exponents do not contain.

On size, the mixture of experts behaves like a large dense model, not a small one. Two dense models of its family share its hybrid attention: Qwen3.5-4B, about its active size, and Qwen3.5-9B. Both are chaotic. Unsteered, they are chaotic at 19 and 10 of 128 temperatures; steered, at 15 % and 7 % of a philosophy map and at 32 % and 16 % of a hostility map. The mixture of experts is chaotic at none of the 128 temperatures and none of the philosophy cells, and irregular at 5 % of the hostility cells. Its median network factor between 0.3 and 6 nats is −1.90, against +0.05 and −1.41 for the two dense models and −1.73 for the dense Qwen2.5-32B. Within this family, chaos falls and damping grows from 4B to 9B to the mixture of experts, which damps more than a dense model with three times its active parameters. Routing a token through about 3B parameters therefore does not make the loop behave like a 3B model.

What this implies for larger models. Size does not set this chaos in a simple way. In Qwen2.5, the 14B model's chaos did not vanish but moved to higher temperature, to where its next-token distribution spreads; in OLMo-2 the largest model was the most chaotic; two 70B Llama checkpoints with different post-training gave different results on the same grid; and at 70B, chaos that passes a three-offset test remains, rarer, with and without steering (Appendix I). The shift does not continue to Qwen2.5-72B. A prompt-blocked test frozen before its data (`docs/prefix-scale-72b.md`) predicted that the 72B model would be chaotic at fewer of the 48 temperatures than the 14B on at least three of four held-out prompts. It is chaotic at 0, 1, 0 and 0 of them, against 14, 9, 11 and 15, so the prediction holds on all four. These lines ran before the precision fix (1000 transient and 2000 measured steps, twins at 10^{-5} and 10^{-6}, chaotic only when both exponents are positive); no temperature on them is positive at only one offset, and the one chaotic temperature ($\log_{10} T = 0.319$, exponents 0.35 and 0.34) is chaotic again when recomputed in float64 throughout, with an exact tangent exponent of 0.34. With the paper's prompt, where it showed no robust chaos up to $\log_{10} T = 1.2$, that makes five prompts on which the 72B model is nearly free of unsteered chaos. Its next-token distribution jumps from sharp to nearly uniform within less than the grid spacing (0.03 decades), so a narrow chaotic band at the jump would be missed by the grid. Fine lines through the jump, computed in float64 throughout with exact tangent exponents, find only periodic orbits on the paper's prompt and on three of the four held-out prompts, and on the fourth a band of chaos about 0.02 decades wide just above the jump ($\log_{10} T = 0.319$–0.336; exponents up to 0.40, which twins at 10^{-5} to 10^{-7} reproduce within 0.004), at nearly uniform distributions, as the 14B model's chaos but far narrower. A fixed temperature window is therefore the wrong yardstick, and so is size alone. A model's chaos follows the gain of its loop where its own next-token distribution is spread, no single forward pass we tested predicts it (Appendix I), and to find it in a larger model one has to run the loop there, across prompts. For frontier models this means two things: their unsteered chaos may be rare, if, like the largest models here, their distributions spread over a narrow range of temperature; and steering still drives both 70B-class models that we steered, Llama-3.3-70B and

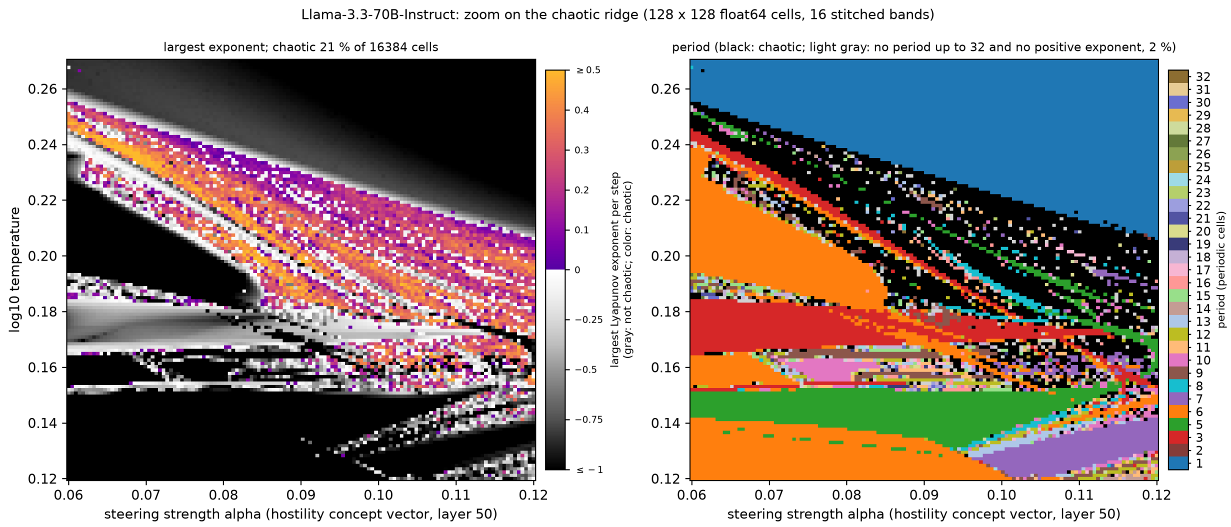


Figure 32: Chaos at 70B parameters: the soft-token map of Llama-3.3-70B-Instruct steered along its hostility concept vector at layer 50, 128 × 128 float64 cells over $\alpha \in [0.06, 0.12]$, $\log_{10} T \in [0.12, 0.27]$ (300 transient and 300 measured steps; sixteen bands of eight temperatures computed separately with identical vectors and joined). Left: largest Lyapunov exponent, gray where the cell is not chaotic and in color where it is; 3410 of the 16,384 cells (20.8 %) are chaotic (largest exponent 0.64). Right: period of the attractor reached from the fixed start (black: chaotic; light gray: neither a detected period up to 32 nor a positive exponent, 1.7 % of the cells). The chaotic band's center falls from $\log_{10} T \approx 0.23$ at $\alpha = 0.06$ to ≈ 0.19 at $\alpha = 0.12$, and periodic windows thread it, most often of periods 3, 5, 6 and 7 and up to 13; fixed points lie above it.

Qwen2.5-72B, into chaos (Appendix I). Resolved through the stack, the damping grows with size in Qwen2.5 and Llama-3 because the change of the fed-back token stops keeping pace with the residual stream (Figure 31). If larger models of these families continue the trend, their unsteered loops need stronger token competition to become chaotic, which steering can still supply; this is an extrapolation from checkpoints up to 72B, and OLMo-2 does not follow it. Figure 32 shows it at 70B parameters: in a 128 × 128 float64 scan of Llama-3.3-70B-Instruct steered along its hostility direction, computed after the precision fix, 20.8 % of the 16,384 cells are chaotic, in a band threaded by periodic windows, as in the 0.5B model's maps; a test of 24 sampled cells at three twin offsets and an exact tangent confirms 11 of the 12 chaotic and 5 of the 6 periodic labels (Appendix I).

15 A second system: recurrent depth

Huginn-0125 iterates a recurrent block at fixed input tokens [3]. For a GSM8K-style [57] prompt, the native loop converges to a fixed point whose leading Jacobian pair is $0.588 \pm 0.771i$ (modulus 0.970). For the relaxed loop $h \leftarrow (1-\lambda)h + \lambda F(h)$, whose multipliers are $1 - \lambda + \lambda\mu$, this predicts that the pair crosses the unit circle at $\lambda_c = 2(1 - \operatorname{Re}\mu)/|1-\mu|^2 = 1.0777$, transversally and away from the strong resonances (at 0.156 of a revolution per step): a Neimark–Sacker bifurcation. Below it the latent state decays; above it the state settles on a closed curve. A fit of the squared amplitude gives $\lambda_c = 1.0778$, with $A/\sqrt{\lambda - \lambda_c} = 0.52$–$0.54$. A stable circle that grows like the square root of the distance is consistent with a supercritical Neimark–Sacker bifurcation; we did not compute its first Lyapunov coefficient, so its nondegeneracy is not established. Causal attention makes the loop block lower-triangular in token positions. The first position to oscillate is the "." of the prompt, and all later positions follow. On the invariant circle the motion turns by 0.156 of a revolution per step, matching the argument of the critical pair. The rotation number changes smoothly, by about 10^{-4} over $\lambda \in [1.079, 1.092]$, and no return within 60 steps was found at 10^{-3} of the radius. This is consistent with quasi-periodic motion, although locking in narrow high-order tongues cannot be excluded. From $\lambda \approx 1.094$ the circle deforms or a second frequency appears, near where a second complex pair destabilizes the fixed point ($\lambda = 1.0814$). Fixed-point and Jacobian stability of looped models has been studied before [16]; what this adds is the predicted crossing, its amplitude law and the rotation number.

The spectrum predicts the relaxation response. With the full instruction template, the GSM8K prompts' fixed points look different: the leading multipliers surround the origin, a real 0.914, complex pairs of modulus 0.85–0.89 at many angles and negative real values near -0.8, with no near-critical pair. Because the prefill Jacobian is block lower-triangular in positions, its spectrum is the union of per-position blocks, and template positions contribute identical eigenvalues to every prompt. Such a spectrum makes three predictions before any task is scored. First, within the computed spectral estimates of these prompts, no fixed relaxation can speed convergence much. With leading multipliers spread around the origin at many angles, any λ that shrinks one group enlarges another, and the best $\lambda^* = 1.03$–1.06 lowers the rate from 0.914 to only 0.908–0.911. (A finite computed spectrum does not show that no other polynomial scheme could do better; Anderson acceleration, tried below, did not.) Second, under-relaxation slows convergence ($\lambda = 0.55$: predicted 0.953, measured 0.951 on one prompt). Third, over-relaxation destabilizes the negative modes first, at $\lambda_c = 2\,\mathrm{Re}(1-\mu)/|1-\mu|^2 = 1.074$–$1.126$ (five prompts), in a flip or a Neimark–Sacker bifurcation close to -1. The approach to it is visible: on the prompt with $\lambda_c = 1.074$, the measured rate at $\lambda = 1.05$ is 0.955 against a predicted 0.956 (native 0.864). Elsewhere the spectral radius bounds the rate measured over the accessible window from above, by up to 0.05, because the slowest modes carry little amplitude there. Independently, Wang et al. [58] report a Huginn GSM8K drop from 19.41 % to 8.34 % at a constant step of 1.2; at 0.8 GSM8K is 19.79 %, while their four-task mean falls slightly. Their trajectory-based scheduling analysis does not compute the prompt-specific spectral thresholds studied here.

Does the task notice? (development data). On ARC-Challenge [59] questions scored by log-likelihood, where one prefill per answer choice is exactly the analyzed recurrence, the spectra again predict convergence. Rates agree to 0.002–0.006 where the slowest mode dominates, and beyond each question's λ_c (1.075–1.149 for four questions) the residual stops decaying. For 6 of 30 development questions the native fixed point is itself unstable ($\rho(1) = 1.009$–1.273; the fixed point, located with a relaxed transient, has a real multiplier beyond -1 or a complex pair outside the circle). The native loop then oscillates (relative amplitude 2×10^{-2} for one of them), and relaxation to 0.55 or 0.75 restores convergence to 10^{-6}. For the other 24, no unstable mode was detected in the computed spectral estimates up to $\lambda = 1.020$–1.149 (median 1.095); partial Ritz spectra cannot exclude an undetected unstable mode. Question by question, an estimated spectral feature ranks the question's residual. This feature, $\hat{\rho}_q(\lambda)$, is the largest $|1 - \lambda + \lambda\mu|$ over the computed Ritz values of the question's gold-answer prefill, and its Spearman correlation with the residual is 0.73 at $\lambda = 1$ and 0.75 at 1.10 over the 30 questions (development data). In this historical analysis the residual was the all-choice aggregate entering the 32nd recurrence call, so this is a predictive association across related maps and precisions, not a same-state contraction identity. It is consistent with the latent orbits that Geiping et al. [3] observed on some tokens of harder questions, here traced to the fixed point's spectrum. The item-level part of the frozen test below uses matched gold-row token IDs and a per-row residual with an explicit clock.

Accuracy, however, tolerates the instability (300 training questions, 32 steps): it is 0.367 native, 0.387 at $\lambda = 1.10$ and 0.373 at 1.15, where the latent state oscillates with relative size 0.08 and 0.23. Only at 1.25 (size 0.68) does it fall, to 0.300. Yet the oscillation does reach the answers. Reading every step from 24 to 32 on 100 questions, the margin between the correct choice and its best rival moves by 0.11 nats per step natively, 0.37 at 1.10 and 0.95 at 1.15, in alternating directions, and the chosen answer changes within these nine steps for 12 %, 16 % and 33 % of the questions. The changes go both ways, so the accuracy at any one step stays the same on average. Under-relaxation instead costs 4–5 points at 8–16 steps and catches up by 32 (a slower transient), and Anderson acceleration neither speeds the transient nor gains accuracy. Testing the accuracy consequence therefore needs a task in which such step-to-step changes can compound.

GSM8K generation is such a task. With eight chain-of-thought examples in the lm-evaluation-harness format, greedy decoding and 32 recurrence steps per token (development problems 0–119, every arm on the same problems and initial states), accuracy is 0.375 natively, 0.425 at $\lambda = 1.05$ and at 1.10, 0.29 at 1.15 (17 problems lost and 7 gained, one-sided McNemar $p = 0.03$) and 0.16 at 1.20 (29 lost, 3 gained, $p = 10^{-6}$). Under-relaxation to 0.8 costs nothing at 32 steps (0.41). The break lies between 1.10 and 1.15, at or just above the spectral thresholds ($\lambda_c = 1.074$–1.126 for the GSM8K prompts; median upper stability limit 1.095 for the ARC development questions). Generation breaks earlier than log-likelihood scoring, consistent with answer changes that compound over tokens. The collapse at 1.2 is the one Wang et al. [58] reported.

A frozen test on fresh problems confirms this. Its design and analysis were pushed before any test item was read, after two rounds of independent review. On 200 GSM8K test problems, accuracy is 0.330 natively, 0.395 at $\lambda = 1.05$, 0.255 at 1.15 and 0.095 at 1.20. The two pre-stated predictions pass: accuracy declines at 1.20 (−23.5 points; 50 problems lost and 3 gained, exact one-sided McNemar $p = 3 \times 10^{-12}$), and 1.05 loses no more than four points (difference +6.5, one-sided 95 % lower bound +2.5). At 1.15 accuracy falls by 7.5 points ($p = 0.01$, reported descriptively). Development and test agree closely (Figure 33).

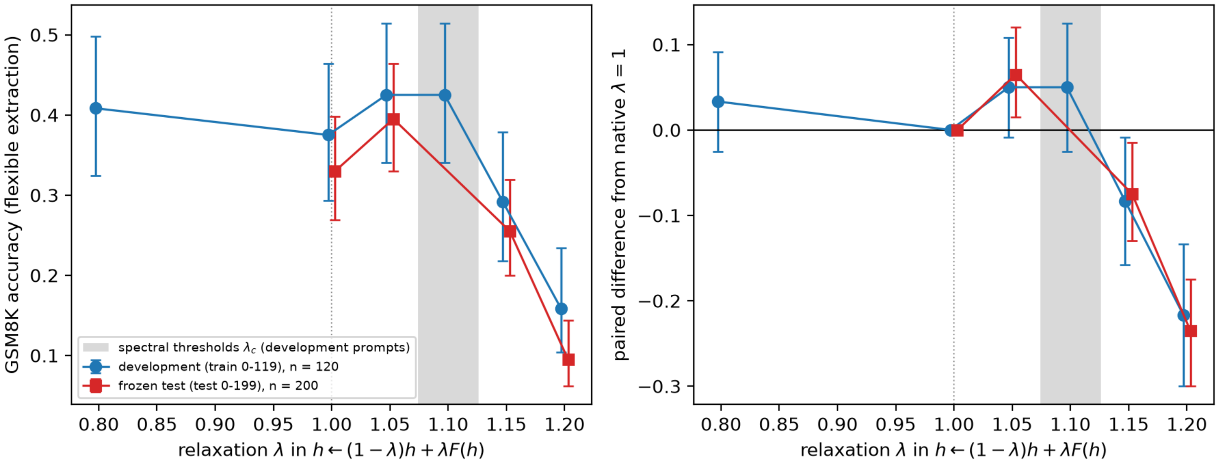


Figure 33: Huginn-0125 chain-of-thought accuracy on GSM8K against the relaxation strength λ (eight examples, greedy decoding, 32 recurrence steps per token): development problems (blue) and the frozen test on held-out problems (red; design and analysis fixed before any test problem was read). Gray: the spectral thresholds $\lambda_c = 1.074$–1.126 computed from the fixed points of five development prompts. Left: accuracy with Wilson 95 % intervals. Right: paired difference from the native loop with paired-bootstrap 95 % intervals.

The same frozen design tested the item-level prediction on 64 held-out ARC-Challenge questions.

The first 16 spectra preceded the read-out generation; the remaining ones were completed afterward under a recorded amendment, with the producer attesting that read-out outcomes had not yet been inspected, and duplicate copies were resolved by a rule fixed in advance. Every question passed the admission gates (fixed-point residual at most 2.6×10^{-6}; identical gold-row token IDs in every arm); for 13 questions one of the three Arnoldi sets was partial, which the design admits for an estimated feature. The pre-stated statistic, the mean of the within-arm Spearman correlations between $\hat{\rho}_q(\lambda)$ and the gold row's residual entering the 32nd update, is 0.701 (95 % bootstrap interval $[0.575, 0.780]$; 0.68, 0.73, 0.73 and 0.67 at $\lambda = 1$, 1.05, 1.08 and 1.10); the frozen primary gate, the interval's lower end above 0.3, is met. On unseen questions, the fixed point's estimated spectrum therefore ranks each question by the size of the gold row's residual entering update 32; a residual is not in general a distance to the fixed point. Descriptively, the share of questions with a detected unstable mode rises from 8 % natively to 34, 69 and 84 % at $\lambda = 1.05$, 1.08 and 1.10. An independent implementation reproduced both parts of the frozen test from the saved data.

16 Two loops in conversation

Two instances of a language model left to talk with each other drift into recurring states [12]. The best-known case is the "spiritual bliss" attractor of two Claude Opus 4 instances, whose open-ended conversations end in repeated exchanges of gratitude and silence [60]. The soft-token map poses the deterministic version of the question: can two loops that are each chaotic on their own lock each other into a periodic orbit? Agents A and B are Qwen2.5-0.5B-Instruct with the conversation prompt of a sampled self-dialogue study that we report separately, each with its own steering strength along the philosophy direction at layer 12. They write one reply together, alternating soft tokens, $x_{n+1} = F_A(x_n)$ and $x_{n+2} = F_B(x_{n+1})$, so one exchange is the map $G = F_B \circ F_A$. Equal agents reproduce the single loop bitwise in float64, so on a map over (α_A, α_B) the diagonal holds each agent writing alone (Figure 34).

At $\log_{10} T = -0.10$, over $\alpha_A, \alpha_B \in [0.060, 0.082]$ (48×48 cells), 27 of the 48 agents are chaotic alone and 21 periodic. Of the 702 pairs whose two agents are both chaotic alone, 422 are periodic together; of the 420 pairs of two periodic agents, 91 are chaotic together. Writing together favors order (70 % of pairs are periodic, against 44 % of single agents), though some chaos is inherited: pairs of chaotic agents are periodic less often than pairs in general. The exponent map is plaid, as if each agent contributed its own stability, and where one agent is weakly steered ($\alpha \lesssim 0.064$) and the other is not, the pair settles into a fixed two-token exchange. Twelve entrained pairs spread over the map all converge under Newton's method to attracting primitive orbits of G (residuals at most 2×10^{-13}, leading multipliers 0.0008 to 0.79 in modulus, exchange periods 1, 3, 5 and 6). Their texts show how two chaotic writers share a loop. At period 1, A writes "is" and B "it", the single loop's 2-cycle split between two writers. At period 5, B answers "is" every time while A cycles through "This", "there", "it" and "a". That alternating two chaotic maps can produce order is known for quadratic maps [61]; here it happens in a language model's own dynamics.

Two different models behave alike. Qwen2.5-0.5B and 1.5B share a tokenizer, so a message can be passed as a distribution over the shared vocabulary and embedded by each model with its own matrix. Unsteered, the 1.5B writing with itself is chaotic at 41 of 131 temperatures ($\log_{10} T$ from 0.29 to 0.80, exponents 0.21 to 1.76 per exchange), the 0.5B at one. Writing together they are periodic at 39 of those 41 temperatures, mostly in a fixed two-token exchange: the stable partner sets the rhythm. A dialogue framing behaves differently. With each one-token message placed in a user turn and the first token of the reply as the answer, every map we computed is periodic, for one agent alone and for every pair. The reply is decided by the replier's own prompt and steering, and

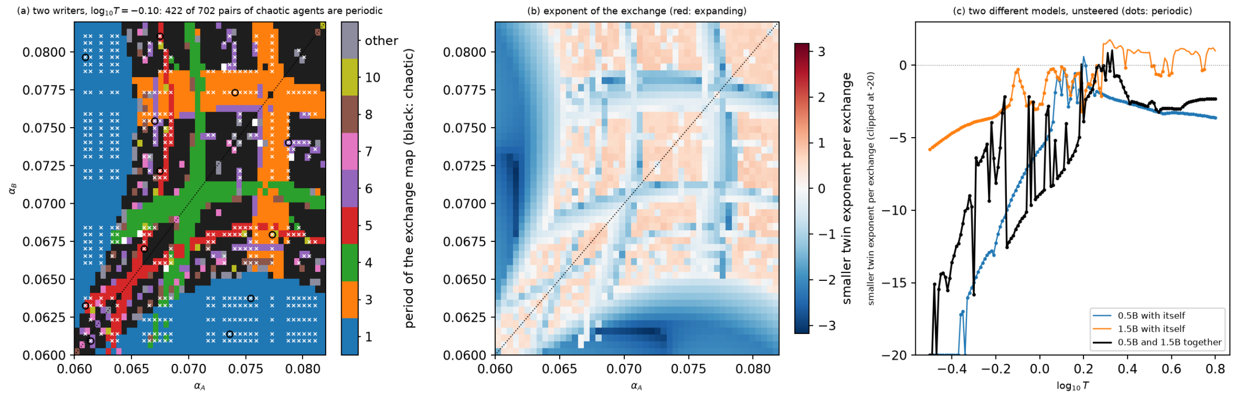


Figure 34: Two loops writing one text. (a) Agents A and B (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, strengths α_A and α_B) alternate soft tokens at $\log_{10} T = -0.10$: period of the exchange map $G = F_B \circ F_A$ per cell (black: chaotic). The dotted diagonal holds each agent alone; white crosses mark pairs that are periodic although both agents alone are chaotic, and circles the twelve such pairs verified by Newton's method. (b) The smaller twin exponent of G per exchange on the same cells. (c) Two different models, unsteered: the smaller twin exponent per exchange of Qwen2.5-0.5B and of Qwen2.5-1.5B each writing with itself, and of the two writing together; dots mark periodic cells.

the partner's message barely moves it; mutual locking of the kind seen in long conversations would need longer messages than this frame provides.

17 Discussion

What theory allows, and what is new. Chaos and periodic windows occur in many nonlinear feedback systems: random recurrent networks become chaotic above a critical gain [20, 21], and periodic windows, shrimp-shaped domains and doubling cascades are found in low-dimensional maps and in experiments [22, 23, 62]. In our map temperature plays the role of an inverse gain and the steering vector that of an input. The factorization of Section 3 makes this correspondence concrete, but it remains an analogy, not a transfer of the random-network theorems (Appendix C). These results motivate looking for such structure in a language model's feedback map; they do not imply it. This paper shows that the models we studied realize it, with concrete texts as attractors, so that a bifurcation changes what the model writes, and that several prospective quantitative predictions succeeded; the failed all-row criteria and the unresolved cases are reported with them.

Is steering the origin of the chaos? No. Without steering ($\alpha = 0$) the map is the model's own feedback map, and temperature alone gives chaos with embedded windows in most dense models we swept (in eleven of fourteen on regular grids). Steering widens the chaotic regions, but not through the concept it names: in the 0.5B model norm-matched random directions are as chaotic as the concept directions, and in Llama-3.1 8B sparse-autoencoder feature, contrastive and random directions all are, with direction-dependent prevalence. What the direction selects is the text of the attractors (Appendix J).

What is and is not claimed. The objects studied here are invariant sets and bifurcations of an exactly defined smooth map built from a frozen model's own computation. The answer-level reproducibility tests use two Qwen2.5 models (0.5B and 7B), one task family and a soft-thinking pipeline without the top-k filtering of the published method, and their bfloat16 effect is not the float64 map's chaos (Section 6). The mechanism of Section 3 is measured along three directions in a 0.5B model and one in a 7B model, and at high temperature the network factor can carry the instability. The loop's regime does not mark the accuracy of ordinary sampled answers beyond the model's first-token uncertainty, and when slight steering moved a prompt's loop into chaos no change of that accuracy was detected (+0.36 points, $[-0.36, 1.08]$; Appendix D). We do not claim that ordinary sampled generation is chaotic in this sense (Appendix G), nor that any window is a certified shrimp. The period-5 window carries the anatomy of a shrimp in a two-dimensional map (Section 11): fold curves with located cusps, legs, doubling shells, a closed trace-zero curve, and two dovetails. The period-6 and period-7 windows have fold-and-flip skeletons with located cusps, and in the hostile period-7 body the superstable arcs avoid each other where a head would be. We have not traced tails or symbolic sequences, tested for a periodicity hub [46, 47], or shown that a two-dimensional map reproduces the windows' nonlinear dynamics. The cusps are float64 computations validated by step-size checks, not proofs (Appendix E).

Limitations. The detailed orbit studies cover Qwen2.5 at 0.5B and 1.5B, with three float64 orbits at 7B; OLMo-2 7B and Llama-3.1 8B have maps but no continued orbits. The size comparison rests on regular temperature grids and mostly one twin offset per label, and part of the 70B-class maps predate a precision fix and were rechecked only in samples. The maps use one prompt (more in the size tests), and steering uses contrastive vectors, checked against random and sparse-autoencoder directions in two models. Chaos labels are finite-horizon estimates: a longer horizon check kept 86 and 92 of 96 chaotic labels in two maps, near the accumulation of a cascade transient chaos makes labels less reliable (on one line only 135 of 235 chaotic labels held), no chaotic cell is proved chaotic, and only one periodic orbit is proved to exist. The Feigenbaum ratios are finite-order values. All orbit, bifurcation and chaos results concern the soft-token map, not sampled generation with a growing context. Appendix K gives the details.

Steering strength is a bifurcation parameter. In the soft-token map, small changes of the steering strength switch the model between distinct repeating texts, chaos and coexisting attractors. Near a fold curve, behavior becomes intermittent, with laminar phases whose duration the fold's normal form predicts in advance. This argues for treating an activation-steering coefficient like any control parameter of a nonlinear system. Its bifurcation set should be characterized before an operating point is chosen, history dependence should be expected near boundaries, and a monotone, smooth dose response should not be assumed. The block at which a vector enters matters as much as its strength. In a 0.5B and a 70B model, earlier injection gives more chaos and brings it to lower temperature, and two vectors at two blocks create chaos where neither does alone (Section 6). Outputs that fall into verbatim repetition at high steering doses are naturally treated as quality failures. In the present framework such loops are candidate attracting periodic orbits, and the dose at which they appear is a boundary that can be mapped like the others. The deterministic map is where such sets can be defined precisely and computed numerically. Whether its boundaries predict boundaries of sampled behavior is open. The bridge of Appendix G shows that sampling noise shortens the laminar phases of the bottleneck, by an amount that depends on how many samples are averaged.

What this suggests for interpretability and alignment. Our results come from open models of 0.5B to 72B parameters and the soft-token loop, but they suggest three things about models whose outputs feed back into their inputs, as in self-dialogue, agent loops and pipelines of model calls. First, where the loop is chaotic, a perturbation of 10^{-6} grows to order one within roughly 15 to 50 steps at the largest exponents we measured (those of the unsteered 7B and 8B models, 0.30 to 0.88 per step), and more slowly at smaller ones. Even a complete account of single forward passes then predicts a long trajectory only if the state is known far more precisely than any interpretation supplies. Stepwise summaries can be inconclusive: every step of the period-8 orbit expands some direction while the cycle contracts, so the ordered product of the step Jacobians, not a summary of each step, captures the loop's stability (Section 3). What remains predictable are the invariant sets, their statistics and the boundaries between them. This extends the analysis of trained recurrent networks through their fixed points [63, 64] to the periodic orbits, chaos and bifurcations of a pretrained transformer. Second, feature dictionaries say which features an attractor occupies, but the boundaries between attractors, which decide what a small change does, are properties of the map's derivatives. That would hold even if most units have one interpretable job, as a recent essay argues for Qwen3-32B [65]; in its pruning tests, too, the answers that change are mostly those the network was already unsure of, as the loop turns chaotic where next tokens compete. Single neurons are no exception: in the 0.5B model, steering along the output direction of the neuron most specific to philosophy makes 9.9 % of the map chaotic, its index neighbor 5.1 % and a random neuron 12.9 %, against 6.8 to 12.5 % along the concept and Gaussian directions (Appendix J). In our 8B controls, a sparse-autoencoder feature direction, contrastive and random directions all drove the map into chaos (27 to 39 % of cells); at 70B the hostility-related directions gave more chaotic cells than the random ones (4.3 and 3.5 % against 1.1 and 0.2 %, two-GPU maps from before the precision fix; Appendix I). Steering acts by making next tokens compete (Section 3), which is not specific to concept-labeled directions, but a direction's geometry and its meaning can be correlated. Text can also hide the dynamics: in the hostile cascade, a seven-token text repeats exactly while the state underneath is chaotic (Figure 24), so a monitor that reads only the emitted text would call a chaotic regime periodic. Third, conclusions drawn at one operating point need not transfer: which loop a model enters depends on how much context it keeps (Table 18), on the path by which a boundary is approached, and on the sampling noise (Appendix G). Treating a looped model as a dynamical system, with regimes and boundaries to be mapped, rather than as a function to be evaluated point by point, seems to us the appropriate stance for both interpretability and safety evaluation. The reproducibility tests of Section 6 make this concrete: in a chaotic regime the batch a request happens to share decides, after a predictable number of steps, what the loop writes. In low precision, continuous feedback makes even short-budget answers depend on the batch, far more often than greedy decoding does; over a long run whose answers the model holds weakly, discrete feedback does too.

Is the chaos useful? The chaos we map is not a source of new content. The chaotic cells of Table 5 recombine the phrases of the neighboring loops in an order that did not settle over the recorded horizon: the trajectory wanders among the ghosts of nearby periodic orbits, which gives variety in sequence but not in substance. The periodic loops are the familiar degeneration of decoding into repetition [25, 26]. Neither regime is good text, and the diversity of ordinary generation comes from sampling and a growing context, which wash out much of this structure (Appendix G). What the chaos does matter for is predictability and control. Positive finite-horizon exponents limit open-loop prediction of a trajectory at fixed state precision (our experiments do not establish mixing or limits on feedback stabilization); a stable attractor is predictable, but it can also hold a model in a state it should leave, such as the hostile loops of Section 10. Whether regimes near the boundary between order and chaos are especially useful, as has been argued for computation in dynamical systems and recurrent networks [66, 67] and shown for signal propagation in deep networks at initialization

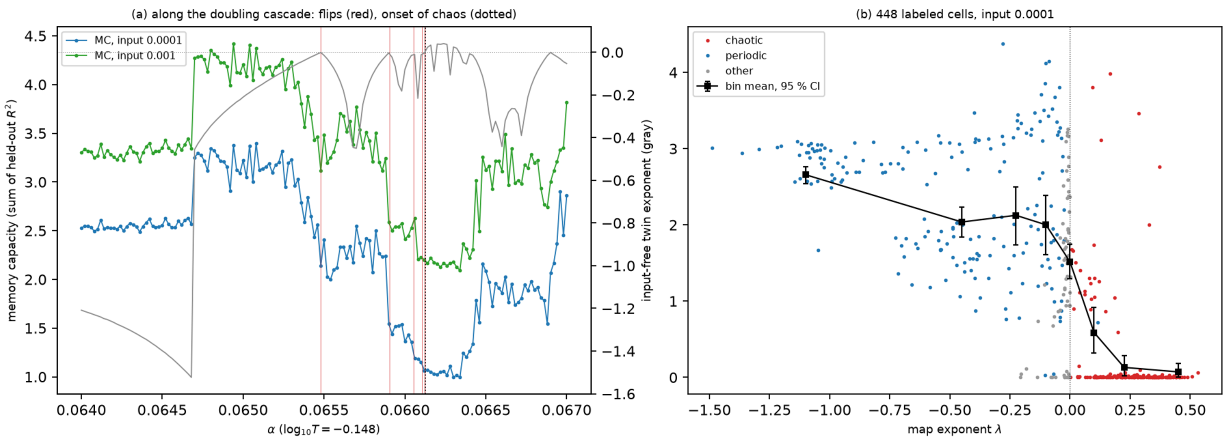


Figure 35: Memory capacity of the soft-token loop used as a reservoir (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, float64). A weak random input enters the map, $x_{n+1} = F(x_n + \epsilon|x|u_n r)$, and linear readouts of the state reconstruct the past inputs; the capacity is the sum of their held-out R^2 over 41 delays. (a) Along the doubling cascade at $\log_{10} T = -0.148$: capacity for two input sizes, the input-free twin exponent (gray), the flips (red) and the onset of chaos (dotted). (b) The 448 labeled cells of the horizon check against the map exponent, with bin means and bootstrap 95 % intervals.

[68, 69], is open for language models. We tested three versions, and none shows an advantage at the edge. Prompts whose own loop sits at the edge ($|\lambda| \leq 0.1$) are answered no better by ordinary sampling than prompts whose loop is deeply periodic or deeply chaotic (Section 6). Used as a reservoir, the loop remembers a weak injected input best in short stable cycles. Its linear memory capacity falls at every period doubling toward the onset of chaos, does not rise as the exponent approaches zero within a period, and is nearly erased by chaos (Figure 35). In Huginn, whose reasoning is itself a latent loop, a float32 pilot on 40 development problems, separate from the tests of Section 15, recorded the decay rate of every generated token's recurrence. Accuracy holds while that recurrence is comfortably stable and falls as it approaches its edge: it was 0.475 natively and 0.525 at relaxation 1.05, where the token recurrence is nearly as stable as natively (median decay rate 0.921 against 0.912), and fell to 0.375 at 1.15, where the recurrence is near critical (0.981, with about half of the tokens no longer converging). With 40 problems these differences are indicative only.

Which findings might reach frontier models. The parts that rest only on structure every smooth softmax-readout language model with a fixed prefix shares should carry over: the factorization of the loop's derivative, contraction at sufficiently high temperature, greedy cycles at zero temperature, and the role of a steering coefficient as a bifurcation parameter. The attractor texts, the positions of windows, folds and cusps, and every number we report belong to particular weights; a second checkpoint of the same architecture already has different attractors (Appendix H). Size gives no guarantee of stability (Section 14), hard routing in mixture-of-experts models falls outside the smooth theory of Section 3 (no expert switched along our float64 orbits, but in low precision routing can), and deployed inference samples tokens, in low precision, over long contexts; an 8-bit weight quantization that changes next-token distributions by only 0.003 nats changes the attractor class or period of one start in seven (Appendix D). Closed models permit none of these computations, which need the input embeddings, full next-token distributions and derivatives; for them the method and the questions transfer, not the numbers.

Use of AI assistance

This work was carried out with AI coding and research agents (Anthropic's Claude Code and OpenAI's Codex) working under the author's direction. The agents wrote and ran the code, executed the experiments, checked results against the raw data and drafted text; other AI systems made independent read-only checks of the results and the manuscript. Frozen designs, run cards, failed attempts, audits and hash-bound records of every run are kept in the project repository. The author, whose doctoral research in nonlinear dynamics included work on a periodicity hub of the Rössler attractor [47], directed the study, reviewed the work and takes full responsibility for its content.

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A Related work in detail

Language-model generation has been studied as a dynamical system at several levels, and the closest studies differ from ours in what they compute. Lyapunov-like quantities have been used to measure sensitivity to small perturbations in reasoning traces [4, 5] and in token rings resampled by a model's own conditionals, with a test that separates the construction from the model [24], but these studies measure chaos-like sensitivity in generation without computing orbits or bifurcations. Output statistics change abruptly with sampling temperature, and several studies analyze the changes as phase transitions or criticality [6–9]. These analyses concern output statistics; they are not bifurcation analyses of a defined map. Closest to ours in subject, a study of eleven models sorts sampled continuations into classes from frozen and periodic to intermittent and noisy across temperature, fits a noisy effective map to one hidden-state coordinate that predicts repetition, and steers along it [10]; that map is fitted to sampled trajectories, and no orbits, multipliers or bifurcation curves of the model's own map are computed. Repetition loops are a known failure of decoding [25, 26], successive paraphrasing converges to attractor cycles [11], and multi-turn conversations between models settle into attractor states [12]. These attractors are behavioral: they are observed in generated text rather than computed as invariant sets of a defined map. Feeding a model its own output during training degrades it over generations [70]; we study the inference-time loop of a model with fixed weights. The attractors of recurrent networks have been related to the difficulty of training them [71], and fixed points with their linearizations have long been used to reverse-engineer trained recurrent networks [63, 64]. Equilibrium, looped and recurrent-depth models iterate a block to a fixed point or for many steps, and their stability has been analyzed through Jacobians and fixed points [3, 13–16, 72–76], and recent analyses of looped reasoners follow their internal dynamics, their settling, marginal or drifting regimes and slowly decaying oscillatory modes [77–79]; these studies examine fixed-point stability, finite-depth dynamics and nonstationary regimes. Here we continue full-state periodic branches and classify their bifurcations in specified frozen-model feedback maps. A minimal recurrent self-attention with a repulsive value map becomes chaotic [80], but it is a constructed model, not a trained one. A separate line predicts the answers of looped models from the geometry of score margins along completed finite-depth trajectories [17], without fixed points or Jacobians. The closest methodological precedent is a derivative-based fold analysis of a small task-trained looped transformer, whose passage and spectral-gap predictions are unfitted [19]. That model was trained for a task, whereas ours is a released pretrained language model, and our analysis also differs in its objects: the fold of a primitive periodic orbit, the full-channel passage of a two-step map, and the accumulation law of the windows, whose prefactor is set by the global reinjection. Expected-embedding feedback itself is used in training [18] and, training-free, in pretrained models [1, 81], next to continuous latent reasoning [2]; a reduced dynamical model explains why pretrained models fail to preserve such mixtures [82], and whether they exploit the superposition the mixtures could carry has been questioned [83]. Fixing the prefix and the position turns that feedback into the autonomous map we analyze.

In neural systems more broadly, saddle-node ghosts produce quantitative delays in task-trained recurrent networks [84]. Finite-order Feigenbaum ratios have been measured in neural rate and mass models [44, 85] and, through period 128, in a transformer trained on logistic-map trajectories [45]. Learned networks reproduce period-doubling routes [86], and so does gradient descent itself [87]. Normal forms and continuation have been applied to recurrent neural maps, including their quasi-periodic orbits [88], and shrimp-shaped windows and periodicity staircases appear in neuron models [89–91]. Our second control parameter comes from activation steering along contrastive directions [31, 32]. The dynamical-systems tools are standard: numerical continuation and normal forms [35, 36], for maps in particular [92], type-I intermittency [40], shrimps and hubs [23, 46, 47, 49], and period-doubling universality [22].

B Methods

Precision. Every result on the soft-token map labeled float64 is computed in float64 arithmetic throughout: embeddings, activations, attention keys and values (the prefix cache), normalizations (patched, since the library computes RMS normalization in float32 by default), rotary tables, logits, the softmax and the fed-back token. The released checkpoints store their weights in bfloat16, and a bfloat16 number is exactly a float64 number, so converting the weights is exact. For large models the decoder's linear weights stay in their stored bfloat16 and are converted to float64 at each use, which gives the same map as converting them once (bitwise on Qwen2.5-0.5B). The designated float64 computations use the patched normalization and rotary paths; some diagnostic traces, such as the step records of the gain split, are serialized in float32, as recorded in their artifacts. The dense checkpoints store bfloat16 weights (checked against their safetensors metadata), and the mixture of experts stores some tensors in float32, which we restore exactly; float64 does not make the weights more precise than stored. One exception concerns the 70B-class models, which are spread over two GPUs. In the two-GPU runs made before the fix described here, the final RMS normalization was computed in float32: the hooks that move tensors between the GPUs keep a layer's original code, so our float64 version of that layer was bypassed. This makes the computed map rough at about 10^{-8} of the state (its fixed points wander by 10^{-8}, against 10^{-15} on one GPU) and can matter for exponents measured with perturbations of 10^{-6} and below. The affected runs are the 70B-class sweeps and their plateau tests, the first Qwen2.5-72B prompt test, the 70B steering maps and the first zoom, the gain splits of the 70B-class models, and the start-state test, which spread all fourteen models over two GPUs; Appendix I says which results we recomputed.

Phase maps. Each cell of a phase map is one parameter pair. Unless stated otherwise, we discard 300 transient steps and measure over the next 300. The largest Lyapunov exponent comes from a twin trajectory at offset 10^{-6} that is renormalized to that offset after every step, Benettin's method with a finite-difference tangent [93, 94]; it estimates over a finite horizon the exponent that the multiplicative ergodic theorem defines as a limit [27]. At the 448 random cells of the horizon check (Appendix K), an exact tangent vector carried by forward-mode differentiation reproduces the twin exponents at offsets 10^{-6} and 10^{-9} within 0.0041 (median 0.0004), with the same 178 chaotic labels, and at six chaotic cells the four leading exponents from QR iteration include exactly one positive one (0.06 to 0.42; the next is -0.64 to -1.10). The period is the smallest $q \leq 32$ (48 in the later zooms) for which the last state returns to within $10^{-8} \max(1, \|x\|)$ of the state q steps earlier (10^{-5} to 10^{-4} in float32 maps), and the attractor is identified by its leading-token loop, up to rotation. We call a cell chaotic when this finite-horizon exponent exceeds 10^{-3} and no period is found; the label is a finite-horizon diagnostic, not a proof of chaos. We call what the trajectory from a cell's fixed start settles on the cell's attractor. The label is operational: it does not show that the set attracts initial conditions of positive measure, as Milnor's definition requires [95], and a long chaotic transient can carry a chaotic label [96]. A loop is at the edge of chaos when $|\lambda| \leq 0.1$. Maps draw the chaotic cells by this criterion: in color by their exponent where a map shows the exponent, in black where it shows periods or attractor text. Panels whose legend reads 'Lyapunov > 0.005' follow an older convention that marks every cell whose exponent exceeds 0.005, whether or not a period was found; on the five maps we redrew with the criterion, the two differ in at most 0.56 % of a map's cells. A 256^2 map takes about 100 minutes on one H100 GPU in float64. Every cell starts from the same state, so where attractors coexist, a map shows the one whose basin contains that state. Labels are text, not dynamical identities: one continuous orbit changes its label where two tokens tie for the maximum. We therefore call two labeled regions different attractors only when the contraction rate jumps between them or when float64 orbits of both exist at one parameter value.

Resolving the network factor through the stack. Hooks read, at the soft token's position, the residual stream entering each block, the outputs of its attention and MLP sublayers, and the final hidden state before and after the final normalization. They do so for the base row and for a twin at relative offset 10^{-7} along the evolving tangent direction. The split of the network factor into $A_L - N_L$ and the final normalization is exact.

In the pre-normalization architectures of Qwen2.5, Qwen3.5, Llama-3 and OLMoE, the residual stream adds the sublayer outputs, $\delta h_L = \delta x + \sum_s \delta o_s$, and this sum closes to within 4×10^{-7} relative in every model and temperature. OLMo-2 normalizes each sublayer's output before adding it, so the sum does not close there (relative errors of 0.9 to 150). For OLMo-2 we therefore use only quantities that do not depend on it: the factor itself, A_l, N_l and the knockouts.

A knockout holds one sublayer's output at the soft token at its value in the base row while the input carries the twin's change, which blocks the change's path through that sublayer. The knockout rows reproduce the plain twin's network factor within 5.1×10^{-7}.

For the mixtures of experts, a third row receives the twin's input but uses the base row's experts in every layer, weighted by its own router probabilities, renormalized over them where the router renormalizes its top-8 weights (Qwen3.5) and not where it does not (OLMoE). Each step also records whether the twin's set of experts differs from the base row's in any layer, and the base row's smallest gap between the eighth and ninth largest router logits over all layers. Qwen3.5's linear-attention layers carry recurrent states, which are restored after every step, as in the soft-token loop.

False chaos and real chaos. A twin exponent measures how fast the difference between two trajectories grows, so anything that adds a difference of its own can imitate growth. Suppose that both rows receive independent noise of relative size σ at every step, from rounding or from the float32 normalization described above. Suppose also that a difference survives along one leading direction, growing by e^{λ} per step, while the other directions contract within a step. Let c_t be the cosine of the angle between the renormalized twin offset and the leading direction, and $r = \sqrt{2}\,\sigma/\varepsilon$ the noise relative to the offset ε; one step gives $c_{t+1}^2 = e^{2\lambda} c_t^2 / (e^{2\lambda} c_t^2 + r^2)$. While $r < e^{\lambda}$, the angle settles at $c^2 = 1 - r^2 e^{-2\lambda}$, and the twin grows by exactly e^{λ} per step. Otherwise $c \to 0$, and the twin grows by r per step, whatever the dynamics. The twin therefore measures

$$\lambda_{\text{meas}} = \max\{\lambda,\ \log(\sqrt{2}\,\sigma/\varepsilon)\}. \tag{5}$$

Above the offset $\sqrt{2}\,\sigma e^{-\lambda}$ this is the true exponent. Below it, a noise term takes over, rising by one nat for every factor e by which the offset shrinks. Noise can thus make a contracting orbit look chaotic when the offset lies below $\sqrt{2}\,\sigma$, but in this model it cannot make a chaotic orbit look stable.

The model assumes a single leading direction and contraction within one step, so we test it in the network (Figure 36). The test uses Qwen2.5-0.5B in float64 at three cells of the philosophy-steered map: a chaotic cell, the weakly contracting period-3 loop of the coexistence example (Section 5), and a strongly contracting orbit; each runs 400 transient and 800 measured steps. In clean float64 the exponents do not depend on the offset over eight decades: +0.092 to +0.099, −0.123 to −0.128, and −0.843 to −0.862 for offsets from 10^{-12} to 10^{-4}.

We then add Gaussian noise with $\sigma = 10^{-10}$, 10^{-8} or 10^{-6} to every row's next soft token.

- Where the noise dominates, all three cells give the same false exponent, which Eq. (5) matches within 0.006: for example 4.951 to 4.953 against $\log(100\sqrt{2}) = 4.952$ at $\sigma = 10^{-10}$ and $\varepsilon = 10^{-12}$.
- Where the offset lies well above the crossover, the prediction matches within 0.047.
- In the crossover itself, two offsets per cell and noise level, the measured exponent lies above the prediction by up to 0.43, because noise injected one step earlier has not yet contracted.
- Noise lowered no chaotic estimate below 0.087 (clean 0.092 to 0.099). It lowered the contracting estimates by at most 0.048, which makes them look more stable, not less.

A free twin that is never renormalized shows the difference directly (Figure 36d). Started at 10^{-12}, the chaotic twin grows exponentially until its distance saturates near 10^{-1} after about 240 steps. Under $\sigma = 10^{-8}$, the period-3 twin instead jumps to a plateau at about 2×10^{-8} (the model predicts 3×10^{-8}) and stays there.

The float32 final normalization, computed as in the early two-GPU runs and without added noise, reproduces the defect.

- At offsets from 10^{-10} to 10^{-8}, both contracting orbits show positive exponents, up to 2.03 for the period-3 loop.
- At 10^{-12} and 10^{-11}, rounding makes the twin and base rows identical in 454 to 771 of the 800 steps, leaving too few measurable steps to read.
- From 10^{-6} up, the contracting orbits read -0.177 to -0.123 and -0.853 to -0.783 (clean -0.124 and -0.848), and the chaotic one 0.059 to 0.078 (clean 0.094).

The last shift is of another kind. The defect changes the map itself, by about 10^{-8} at every step, not only the measurement, and the chaotic exponent of the changed map is lower, though still positive. The spurious Qwen2.5-72B window of Appendix I carries the signature of Eq. (5). Its exponent was negative at 10^{-5}, near zero at 10^{-6} and positive at 10^{-7}, and its orbits are periodic when computed in float64 throughout.

Hence the rule we apply: a chaotic label needs an exponent that does not depend on the offset. We check this with three offsets, or with an exact tangent vector, which needs no offset. Two limits remain.

- The argument concerns noise in a measured difference. Noise acting on the dynamics can change which attractor a trajectory follows, and in some maps it lowers a positive exponent, the phenomenon of noise-induced order [97]; see also [98]. Noise of 10^{-8} moves a stable orbit by about that much, so it can change the attractor only where the orbit lies that close to a basin boundary or a bifurcation.
- Far below the rounding resolution, two rows can become identical. A naive estimator would then report infinite contraction, a false negative; ours reports the step as unmeasurable. The offsets of our maps lie far above the resolution of their arithmetic.

Periodic orbits. We solve $F^p(x) = x$ by Newton's method with forward-mode Jacobians. The multipliers are the eigenvalues of the monodromy matrix, the ordered product of the step Jacobians along the orbit, which plays the role of the Floquet matrix of a flow [36]; where the dense matrix is not float64, we refine them by a short Arnoldi factorization with float64 Jacobian-vector products.

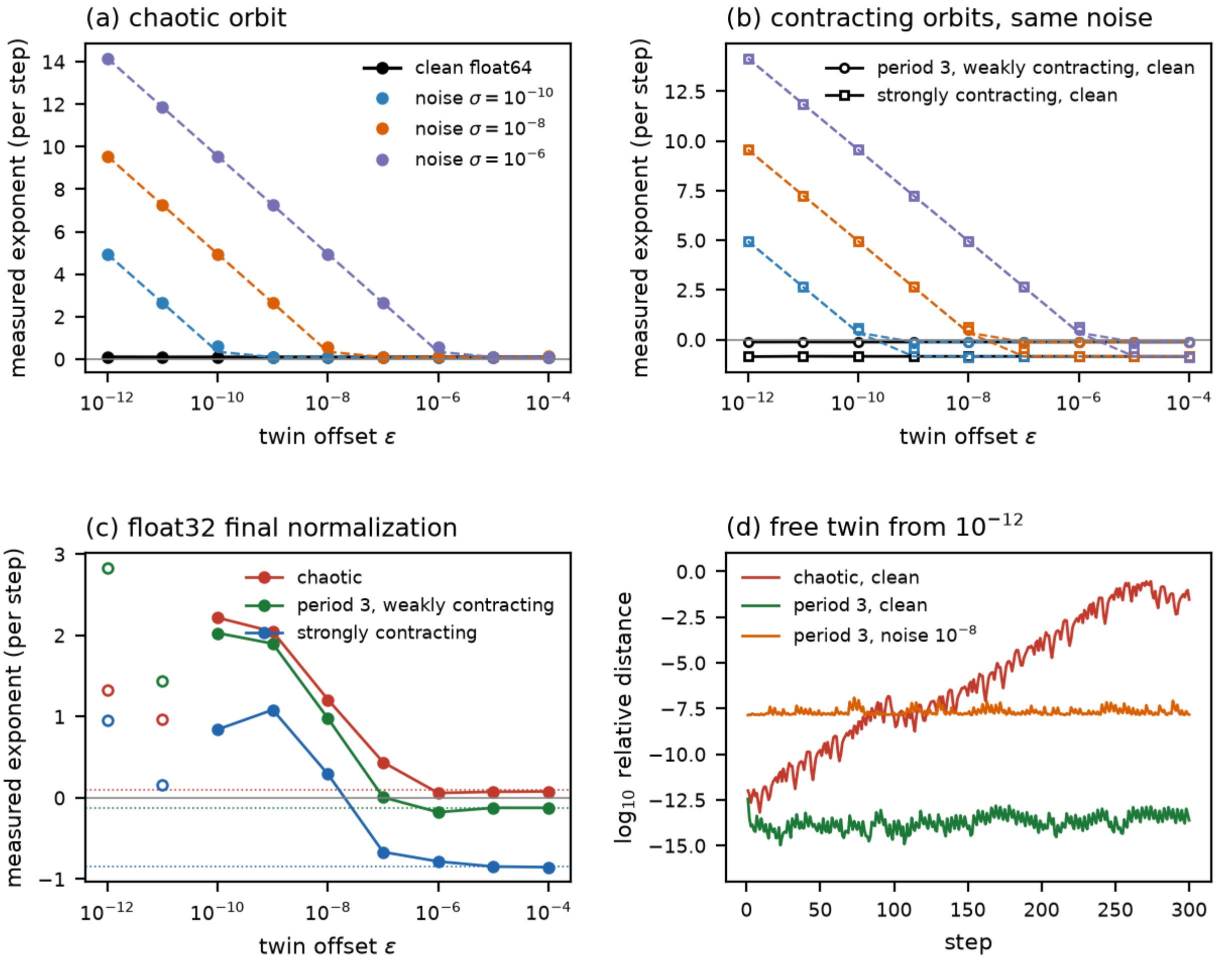


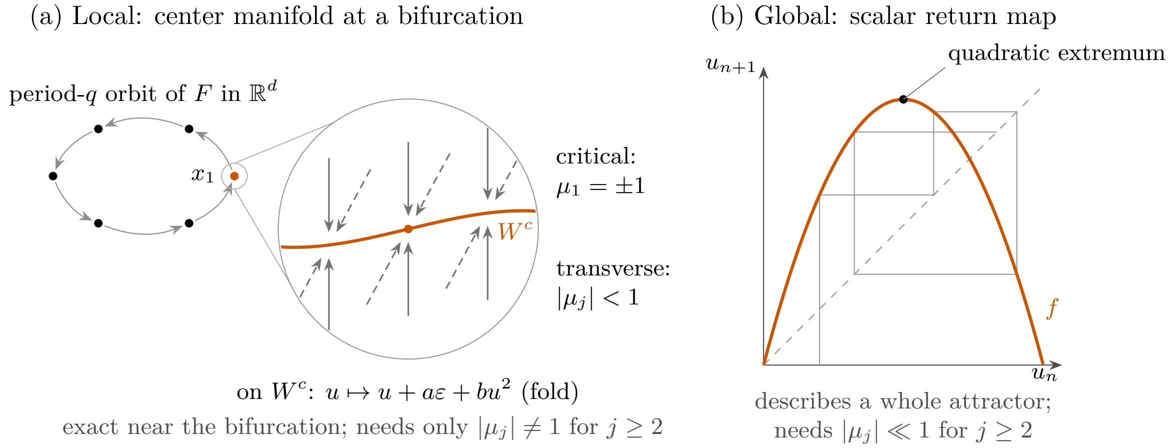
Figure 36: False chaos and real chaos in a twin test (Qwen2.5-0.5B-Instruct, float64, philosophy direction at block 12; 400 transient and 800 measured steps). The twin exponent is the mean logarithmic growth of a second row renormalized to the relative offset ε after every step. (a) A chaotic cell ($\alpha = 0.0652$, $\log_{10} T = -0.156$): clean float64 (black) and Gaussian noise of relative size σ added to every row's next soft token (colors); dashed, Eq. (5) with the clean exponent. (b) The same for a weakly contracting period-3 loop ($\alpha = 0.0642$, $\log_{10} T = -0.156$; circles) and a strongly contracting orbit ($\alpha = 0$, $\log_{10} T = 0.18$; squares): below the offset $\sqrt{2}\,\sigma$ both read as chaotic. (c) The float32 final normalization of the early two-GPU runs, without added noise: false positive exponents near float32 resolution. Open markers mark offsets at which the twin and base rows became identical in more than half of the steps; dotted lines mark the clean exponents. (d) A free twin, started at 10^{-12} and never renormalized: real chaos grows; under noise the contracting loop's twin jumps to the noise level and stays there; in clean float64 it stays near 10^{-14}, the rounding floor of the map.

Continuation. Periodic branches are continued in one parameter by pseudo-arclength continuation, and fold and flip curves in two parameters with minimally augmented systems [35, 36]. The test function g of the bordered system $[[M \mp I, v], [w^\top, 0]]$ is obtained by GMRES (generalized minimal residual iterations) with Jacobian-vector products, and its gradient by reverse-mode differentiation through a Jacobian-vector product. When the parameter tangent reverses, we flag a cusp candidate and test it with the normal form. To locate a codimension-two point on a continued curve (`tools/cusp_locate.py`), we bracket a sign change between two curve points, then retrace the curve from the left end with steps of a quarter of the bracket. We do this for three rounds and finish with a secant step. The function whose sign changes is the oriented quadratic coefficient $b \operatorname{sign} a$ for a cusp, $|\mu_2| - 1$ for a second multiplier μ_2 leaving the unit circle, and the projected parameter tangent for a turn of the curve. At a cusp we then evaluate c and the unfolding matrix at the located point. Elsewhere, the multipliers and the angle between their eigenvectors type the point: a double multiplier whose eigenvectors coincide is a Jordan block. On maps with known answers the tool recovers a cusp to 10^{-6} (with $c = -1$ and $\det J = 1$) and a fold-flip point.

Normal form and passages. Consider a fold of the two-step map $G = F^2$, with right and left eigenvectors r and ℓ normalized by $\|r\| = 1$ and $\ell^\top r = 1$. Along the critical direction the reduced map is $u' = u + a\varepsilon + bu^2$, with $a = \ell^\top \partial_\alpha G$ and $b = \frac{1}{2}\ell^\top D^2 G[r, r]$. We compute b by central differences of float64 Jacobian-vector products over a step-size plateau (later cross-checks use nested automatic differentiation), and a includes the steering's effect on the prefix. The signed coefficients are kept in the original eigenvector gauge, with ε measured in the free parameter (α or $\log_{10} T$, whose units apply to a). On maps with known normal forms the code recovers ab, the cubic coefficient c with its center-manifold term and the two-parameter unfolding matrix, and it traces the analytic fold and flip curves of the Hénon map (`tests/test_bifurcation_curves.py`). For passages we set $v = \operatorname{sign}(b)\, u$ and take the ghost side $p - p_c = \operatorname{sign}(ab)\, \epsilon$ with $\epsilon > 0$, so that $v' = v + |a|\epsilon + |b|v^2$ to quadratic order. A full passage starts at $v = -U$ and a midpoint passage at $v = 0$; both end at the first strict crossing $v > U$, with $U = 0.1/|b|$. We compare passage counts with the iterated quadratic map and with the arctangent formula [40].

Period-doubling cascade. Along a cut in one parameter, we locate the flip of each period-p orbit as the root of $\operatorname{Re} \mu_c + 1$, where μ_c is the multiplier with the smallest real part, using the Illinois variant of regula falsi with warm-started orbits. All but one or two multipliers are small, so Newton's method on $F^p(x) - x$ needs the monodromy only on a six-dimensional dominant subspace. We obtain that subspace by subspace iteration with exact forward-mode Jacobian-vector products chained along the orbit, so no dense Jacobian is formed, and the residual is always the exact float64 $F^p(x) - x$. Each root is classified before it is accepted. The critical multiplier must be real and equal to -1, with a Ritz residual below 10^{-6}, and the other computed Ritz values (six in all, estimates of the dominant spectrum) must lie inside the unit circle. We then reach the period-$2p$ orbit by forward iteration inside the next band. On the logistic map embedded in six dimensions, the procedure reproduces the first four flip parameters to 10^{-9}.

C Theory background, network contraction and the dimension of the chaos


(a) Local: center manifold at a bifurcation

(b) Global: scalar return map

period-q orbit of F in $\mathbb{R}^d$
x_1
W^c
critical:
$\mu_1 = \pm 1$
transverse:
$|\mu_j| < 1$
on W^c: $u \mapsto u + a\varepsilon + bu^2$ (fold)
exact near the bifurcation; needs only $|\mu_j| \neq 1$ for $j \geq 2$
u_{n+1}
quadratic extremum
f
u_n
describes a whole attractor;
needs $|\mu_j| \ll 1$ for $j \geq 2$

Figure 37: Two reductions to one dimension (Section 3). (a) Near a bifurcation of a period-q orbit of the soft-token map (dots; arcs: the map), the dynamics of F^q near a point x_1 of the orbit reduce to its center manifold W^c, tangent to the critical eigenvector of the monodromy matrix (multiplier $\mu_1 = \pm 1$; drawn with one critical direction, as at a fold, a flip or a cusp; a 1:1 resonance or a fold-flip point has two). The other multipliers lie off the unit circle; drawn here inside it, the transverse directions contract onto W^c (dashed arrows stand for the remaining dimensions). The reduction is exact near the bifurcation whatever the size of the other multipliers; the normal forms of folds, flips, cusps and 1:1 resonances rest on it. (b) A global scalar return map $u \mapsto f(u)$ with a quadratic extremum, the diagonal (dashed) and a few iterates (gray). It describes a whole attractor only when the other multipliers are small.

Why the structures of low-dimensional maps appear. The map acts on a space of hundreds to thousands of dimensions, yet two reductions make low-dimensional theory apply (Figure 37). Near a bifurcation of a periodic orbit the center manifold theorem reduces the local dynamics to the critical directions, whatever the size of the other multipliers, as long as they stay off the unit circle; the normal forms are the leading terms of this local reduction [36]. Our normal forms of folds, flips, cusps and 1:1 resonances rest on this reduction alone. A scalar description of a whole attractor needs additional, global evidence. The small non-critical multipliers of the periodic orbits we computed show strong transverse contraction near those orbits and motivate that description: they range from about 0.1 at short periods to below 10^{-13} along the cascade (Sections 8 and 9 and Appendix E), and along the cascade at $\log_{10} T = -0.156$ the next modulus falls from 0.12 at the first flip to 0.045, 7×10^{-5}, 4×10^{-7} and 2×10^{-13} at the next four. Temperature, through the gain of (3), and steering act as two parameters. One-dimensional families of maps with a quadratic extremum can exhibit fold and flip curves, nondegenerate cusps, period-doubling cascades whose interval ratios tend to Feigenbaum's δ, and saddle-node windows that accumulate on a fold with $d_N \propto N^{-2}$ (Figure 38 sketches the objects we locate); we test these predictions on specific branches and parameter sequences in Sections 8 to 11. The organization of shrimps needs more, either a second turning point of the reduced map or a second dimension [23, 47]. We test the consequences where we state them: the multipliers of each orbit, the interval ratios, and the normal-form predictions of passage times and coexisting orbits. Where the second multiplier is not small, as in the high-temperature bubble of Section 9 (next moduli 0.28 and 0.38) and along the period-5 curves of Section 11 (up to 0.39), the global reduction is only a rough guide, while the local normal forms still apply.

Coexisting attractors constrain one-dimensional descriptions. A classical theorem says what a one-dimensional reduction cannot do. If an interval map is three times differentiable, unimodal and has negative Schwarzian derivative, $Sf = f'''/f' - \frac{3}{2}(f''/f')^2 < 0$, the immediate basin of every attracting periodic orbit contains the turning point or an endpoint of the interval [34]; if basins reaching an endpoint are excluded, at most one attracting periodic orbit remains. The soft-token map has coexisting attracting orbits in several places. Near each located cusp with $c < 0$, two attracting orbits of the same period coexist with a saddle between them (periods 4 and 6, Appendix E; period 7, Section 10). Inside the hostile period-7 body, the period-7 loop coexists with a fixed point, which the fixed start reaches at isolated cells. Such coexistence constrains a scalar interpretation: unless one attractor's basin reaches an end of the scalar interval, the effective one-dimensional dynamics cannot be a single unimodal map with negative Schwarzian derivative, and needs a second turning point, a stretch of positive Schwarzian derivative, or a second dimension. A second turning point is also what a doubly superstable shrimp center requires.

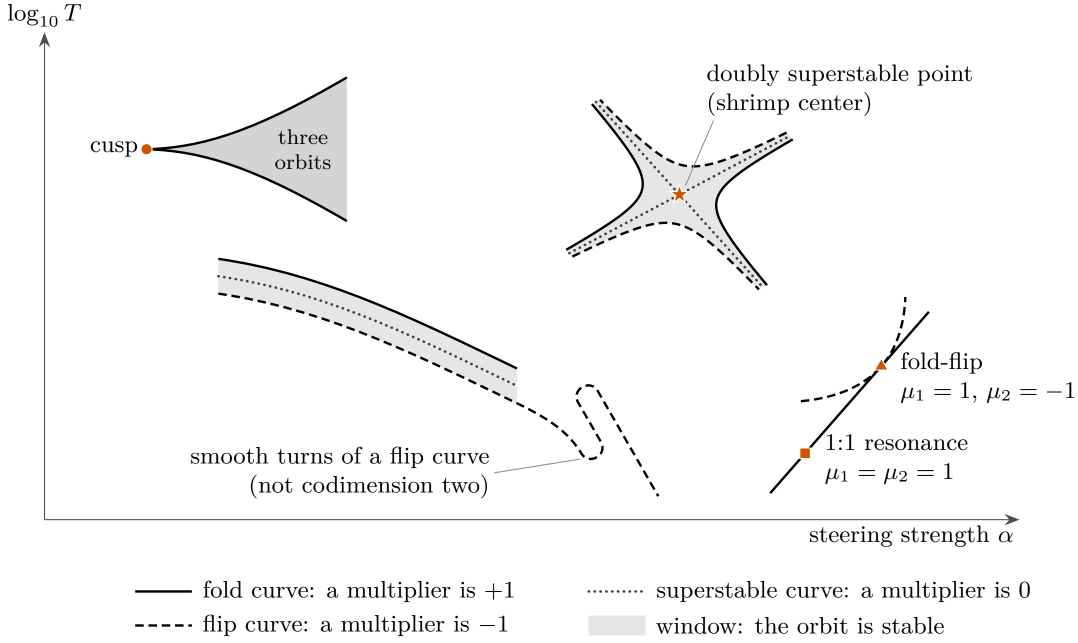

$\log_{10} T$
doubly superstable point
(shrimp center)
cusp
three
orbits
fold-flip
$\mu_1 = 1$, $\mu_2 = -1$
1:1 resonance
$\mu_1 = \mu_2 = 1$
smooth turns of a flip curve
(not codimension two)
steering strength α
fold curve: a multiplier is +1
flip curve: a multiplier is −1
superstable curve: a multiplier is 0
window: the orbit is stable

Figure 38: Schematic of the bifurcation objects in the plane of steering strength α and $\log_{10} T$ (not data; positions and shapes are illustrative). On the curves a multiplier of the orbit equals +1 (fold), −1 (flip) or 0 (superstable); gray marks windows, where the orbit is stable. Between two fold curves that meet in a cusp, three orbits coexist. A flip curve can turn tightly but smoothly, which is not a codimension-two point. On a fold curve, a 1:1 resonance is a point where a second multiplier also reaches +1, and a fold-flip point is one where a flip curve touches it. Two superstable curves of one orbit cross at a doubly superstable point, the center of a shrimp [23, 47]; near it the multiplier is approximately proportional to the product of the distances from the two curves, so the window is a cross whose sides alternate between fold and flip curves. Appendix E locates cusps, 1:1 resonances, a fold-flip point and smooth turns of a flip curve in the soft-token map, superstable curves appear in its exponent maps (Section 10), and Section 17 states which parts of this anatomy are not verified.

What theory allows. Chaos and periodic windows occur in many nonlinear feedback systems. In a specified ensemble of random recurrent networks with asymmetric couplings, chaos sets in above a critical gain, sharply in the large-network limit, and finite networks pass through limit cycles of increasing complexity on the way [20]; how the transition looks depends on the architecture and the transfer function [99]. Discrete-time random networks, $x(t+1) = \tanh(gJx(t))$, iterate a map as our loop does: their fixed point loses stability when the gain times the spectral radius of the couplings exceeds one, the first bifurcation of large networks is a Neimark–Sacker bifurcation, and chaos follows by a quasi-periodic route, through a torus and frequency locking, over a range of gain that narrows as the network grows [21]. Our largest model narrows differently: Qwen2.5-72B passes from a confident echo to a nearly uniform fixed point within 0.03 decades of temperature, and its chaos is confined to narrow bands, whereas in random networks chaos is generic above the transition. External input can suppress that chaos [100], and deep networks at initialization show an order-to-chaos transition in signal propagation [68, 69]. In our map, temperature plays the role of an inverse gain and the steering vector that of an input. Section 3 makes the correspondence concrete through an exact chain-rule factorization, with the softmax derivative $(\operatorname{diag} p - pp^\top)/T$ in the place of the slope of the transfer function; the identification with the random-network transition remains an analogy, not a transfer of that ensemble's theorem. Feigenbaum scaling holds for a class of one-dimensional maps with a quadratic maximum [22], and periodic windows, shrimp-shaped domains with their cusps and spirals, and doubling cascades are found in many dissipative two-parameter systems [23, 46, 49, 101] and in experiments [62]; intermittency near folds is classical [40].

Contraction and expansion in the network. Analyses of deep attention identify contracting tendencies: pure self-attention collapses token representations toward a common vector with depth [102], and idealized attention dynamics form clusters [103]. Residual connections and MLPs counteract the collapse [102]. Which tendency wins along an orbit is a property of the trained weights that only computation settles. In every dense model we mapped, both outcomes occur: the same map has negative largest exponents in some regions of its parameter plane and positive ones in others.

How many dimensions does the chaos have? Near a periodic orbit, the multipliers measure how many directions the dynamics keeps (Section 3). For chaotic attractors the corresponding measure is the Kaplan–Yorke dimension [104], $D_{\mathrm{KY}} = j + \sum_{i \le j} \lambda_i / |\lambda_{j+1}|$ with j the largest index at which the partial sum of the ordered Lyapunov exponents is still nonnegative; with one positive exponent and $\lambda_1 + \lambda_2 < 0$, $D_{\mathrm{KY}} = 1 + \lambda_1/|\lambda_2|$. We estimated the four leading exponents by QR iteration (Appendix B; 1000 transient and 2000 measured steps, float64) at 35 completed cells: 28 chosen by percentile of their largest exponent before any of their spectra were seen (of 30 requested), and seven from the earlier horizon check, six chaotic and one periodic control. At this horizon 31 have a positive leading estimate and none has two positive estimates. These are finite-time estimates, not asymptotic dimensions. They grow with temperature. Along the hostile temperature line, where we fit a scalar return relation (Section 10), eight chaotic cells have $D_{\mathrm{KY}} = 1.008$ to 1.16, consistent with nearly one-dimensional chaos. At fourteen chaotic cells of the headline maps ($\log_{10} T$ from -0.17 to -0.04) it is 1.03 to 1.65, rising with λ_1. In the high-temperature region ($\log_{10} T$ from -0.02 to 0.13), nine chaotic cells have $D_{\mathrm{KY}} = 1.37$ to 2.02; at two of them the sum of the first two estimates is nonnegative, which suggests a second expanding direction at this horizon. A scalar return map describes the low-temperature chaos well and the high-temperature chaos poorly, consistent with the large second multipliers of the high-temperature bubble of Section 9. The number of competing tokens does not set the dimension. The readout's trace-to-leading-eigenvalue ratio, $\operatorname{tr}\Sigma_p/\lambda_{\max}(\Sigma_p)$ averaged along the orbit (with $\lambda_{\max}$ estimated by eight power iterations), counts the directions in which the readout amplifies comparably. D_{KY} never exceeds it and rises with it within each group of cells (Spearman 0.72 over 17 philosophy cells and 0.69 over 8 hostile cells, computed when the data came in and not independently reproduced; both are confounded with temperature). Across groups it does not predict the dimension: the hostile cells have ratios of about 16 to 19 and next-token entropies of about 7 to 9 nats, yet $D_{\mathrm{KY}} \leq 1.16$. The readout supplies many competing directions; the network decides how many of them expand.

D Behavioral tests in detail

We looked for a behavioral signature of the regimes outside the loop. Two exploratory designs found no accuracy signature that survives controls. First, ordinary sampled answers of Qwen2.5-0.5B under steering at the strengths of chaotic and periodic map cells (30 matched pairs; arithmetic, open questions, short programs) showed no measurable accuracy contrast: at those strengths, steering every position turns every reply into philosophy-flavoured text, so accuracy sits at the floor in both zones (3 against 4 correct of 4500 arithmetic answers). Of nine secondary contrasts of entropy and reply length, one (open questions, next-token entropy, $p = 0.016$) passes 0.05 uncorrected, not after a correction for nine tests, and the entropy contrasts change sign across tasks. Second, labeling each of 200 prompts by the regime of its own unsteered loop in Qwen2.5-7B, prompts with a chaotic loop were answered far less accurately (by 26 to 37 points at three temperatures). The chaotic labels fell mostly on the harder task, and in a linear model with task, greedy correctness and the sampled answers' entropy, the label's coefficient is -3.7 to -4.6 points (permutation p from 0.07 to 0.20). These are associations: the entropy is measured during sampling, so the adjustment does not identify a causal effect of the regime.

A frozen test addresses the second design. Its design, covariates and analysis were pushed before any data. It uses 4000 fresh products of two two-digit numbers and the unsteered Qwen2.5-7B-Instruct at $\log_{10} T = 0.40$: 1484 prompts have a chaotic loop and 2509 a periodic one (seven with other labels are excluded). Chaotic-loop prompts are again answered less accurately by ordinary sampling, by 7.1 points (95 % interval $[4.8, 9.3]$; 10 samples per prompt). The gap vanishes once seven difficulty covariates, all fixed before sampling, enter the fit. Beyond them the label's coefficient is $+0.15$ points ($[-1.4, 1.7]$, permutation $p = 0.84$), and the label adds no held-out predictive value. The first-token entropy alone absorbs nearly all of the gap (post hoc: -0.8 points remain). This is what the mechanism of Section 3 predicts. The loop turns chaotic where next tokens compete, and a model unsure of its first token also answers worse. Beyond that uncertainty, no association of the loop's regime with accuracy is detected at this precision; this does not show that the regime has no effect.

Within a prompt no association is detected either. We steered the first 1000 of these prompts slightly, at block 14 along philosophy, hostility-vs-kindness and a random direction. The strength, 1 % of the residual norm, is the largest at which greedy accuracy stayed within 5 points of the unsteered model on the first 200 of these prompts (so the calibration overlaps the test prompts); on the 1000 prompts it is 0.87 to 0.89, against 0.89 unsteered. Under these pushes the loop's label differs between conditions for 508 of the 1000 prompts. With prompt and condition fixed effects (3992 prompt-condition rows), in an analysis fixed before the data, a chaotic label is associated with a change of $+0.36$ points in the accuracy of the prompt's ordinary sampled answers (95 % interval $[-0.36, 1.08]$; permutation within prompts, $p = 0.34$). This does not identify a causal effect of entering chaos: the pushes can act on the answers through other paths as well.

Nor does accuracy peak at the edge of chaos. Adjusted for the covariates, three groups of prompts (3603 of the 4000; the rest have intermediate exponents) are answered alike:

- deep periodic ($\lambda \leq -0.3$, 1961 prompts): $+0.3$ points;
- edge ($|\lambda| \leq 0.1$, 242 prompts): -0.1 points;

- deep chaotic ($\lambda \geq 0.3$, 1400 prompts): +0.3 points.

All three intervals lie within $[-3.1, 3.0]$. The edge contrast is -0.4 points ($[-3.5, 2.6]$), and the curvature in λ is positive, the opposite of an edge optimum. Post hoc, the curvature comes from the most strongly contracting tenth of prompts ($\lambda \leq -3.42$), answered 5.3 points ($[3.0, 7.6]$) better than the covariates predict. That is a lead for a future test, not a finding.

Sensitivity without control. A loop whose answers change with rounding differences raises a further question: can a deliberate perturbation of one fed-back token steer the answer to a chosen wrong value? An exploratory test asked this of the Qwen2.5-0.5B soft-thinking pipeline on two-digit additions, in float32, with 16 soft steps, at temperatures 0.3 and 1.0, on 16 development and 64 held-out problems. The target was the correct sum plus or minus 10. A 20-step projected gradient ascent on the target's log-probability perturbed the first soft token by one token norm, and random search over 45 directions with at least as much computation served as the comparison. On the held-out problems, the early perturbations reached the target in 5 of 101 problem-temperature pairs that the unperturbed pipeline answered correctly, and random search in none: a difference of 4.2 points (95 % interval $[0.8, 8.3]$), below the advantage of 10 points that the design required. The same optimization applied to the last soft token before the answer cue reached the target in 14 of the 101, so in this pipeline the loop gives less leverage early than late. Random early perturbations of that size changed about a third of the answers (one random direction: 34 of the 101 eligible pairs), almost never to the target. Here the loop's sensitivity comes without selective control. The test covers temperatures at which the loop contracts for most problems (on the development problems 10 and 38 % of twins expanded, with median rates of -0.131 and -0.058 per step), where the pipeline still answers correctly, and the doses and the optimizer were chosen on the development problems (`docs/pi/c12-1-summary.json`).

Weight quantization and the attractors. Deployed models are often quantized, and quantization is usually checked one step at a time, by how close the quantized model's next-token distributions stay to the original's. An exploratory test asked whether that check certifies the closed loop. We quantized every linear weight of the decoder blocks of Qwen2.5-0.5B-Instruct, either in 8-bit blocks (as the Q8_0 format of llama.cpp) or in symmetric 4-bit blocks, and kept the arithmetic in float64. The unsteered loop then ran on 48 held-out prompts, at 8 temperatures over $\log_{10} T \in [-0.5, 0.5]$, from 2 starts each. A frozen random logit bias, scaled so that its one-step divergence matches the 8-bit variant's, served as an unstructured control.

- *8-bit.* The one-step distributions stay close: on the reference's attractor states the mean Kullback–Leibler divergence is 0.0028 nats, and the most probable token agrees in 94.5 % of states. The share of starts that end on a periodic orbit of period at most 8 shows no detectable change (+1.2 points, 95 % interval $[-1.9, 4.2]$). Yet 14.1 % of the starts end in a different class or period, as do 15.0 % under the random bias of the same one-step size. These changes concentrate at the temperatures where the attractors change.
- *4-bit.* The share rises by 6.8 points ($[2.6, 11.0]$), mostly because the edge of a chaotic band turns periodic. Its one-step distributions, however, are far from the original's: the mean divergence is 0.83 nats, and the most probable token agrees in only 52 % of states.

In this model, then, a close one-step check was consistent with unchanged statistics of the closed loop but did not fix the attractor that an individual run reaches. We found no effect of quantization beyond that of unstructured noise of the same one-step size (`docs/pi/c12-2-summary.json`).

Chaos onset and repetition in sampled text. Does the temperature at which a prompt's loop turns chaotic predict the temperature at which ordinary sampled text stops repeating itself verbatim? An exploratory test with Qwen2.5-0.5B-Instruct on 24 held-out prompts did not detect the proposed positive association. The Spearman correlation is -0.23 (95 % interval $[-0.60, 0.25]$, permutation $p = 0.28$). This is not evidence of no association. One sensitivity analysis (orbits of period up to 32, clustered by system prompt) gives an interval that excludes zero on the negative side, $[-0.62, -0.18]$. At matched temperature, the repetition rates of chaotic and periodic cells differed by 0.003 (interval $[-0.013, 0.023]$), with both regimes present at only 8 of the 24 temperatures. In 13 of the 15 prompts where both onsets were observed, sampled text stopped repeating below the loop's chaos onset. The median gap over these 15 is 0.42 decades of temperature. The soft-token loop and sampled decoding with a growing context are different systems, and this test did not find a bridge between them (`docs/pi/e5-heldout-analysis.json`).

Steering history and a policy decision. Coexisting attractors suggest that the history of a steering schedule can matter, not only its present value. An exploratory test asked this of a decision with right and wrong answers. Qwen2.5-3B-Instruct, thinking in soft tokens, judged synthetic access-control requests against a supplied permission table, answering ALLOW or DENY, while a permissive-versus-restrictive direction steered it at layer 18. Every run ended with 32 steps at the same steering strength ($\alpha = 0.025$). Some runs got there directly; others first made a brief excursion to $\alpha = 0.04$ and back. The excursion raised the share of prohibited requests wrongly allowed from 49.5 to 70.8 %, a difference of 21.4 points (95 % interval over 24 held-out instances $[15.6, 26.6]$). The difference was 20.3 points after 64 steps: it persisted through the 64 steps tested. A restrictive excursion likewise raised false denials of permitted requests by 31.2 points ($[22.4, 41.1]$). An excursion of the same size along a random direction showed no detectable effect on either rate (1.0 points, $[-4.2, 6.3]$), and an excursion of zero size changed no decision. The present steering strength therefore does not describe the steered model's behavior here. What the loop wrote during the excursion stays in its growing context and keeps biasing the decision: this is a memory of the context, not hysteresis of a fixed-size state. One model, one direction and one schedule were tested, in float32 (`docs/pi/c12-4-summary.json`).

Does the hidden state change answers? In band chaos the loop writes an exactly repeating text while its state moves chaotically (Section 10). An exploratory test asked whether states with the same visible history answer a follow-up question differently. We took 16 states at one phase of the text from each of 92 held-out band-chaos cells; within a cell, all 16 had identical leading tokens over the preceding 70 steps. Each state was read through the steered prefix, followed by one of three queries chosen on development cells: a yes-or-no question about hostility, a one-word mood, and a hostility rating from 1 to 10. Each answer was decoded greedily. No pair of states gave different answers in any cell for any query (a one-sided 95 % bound of 3.21 % of cells), although the probability of a fixed answer token differed at six significant digits in 98 to 99 % of the pairs. The hidden differences therefore reach the readout, but here they stay within the margins of the greedy choices, and only the wording of the yes-or-no answer changed, in 9 of the 92 cells. The visible text is not a complete record of the state. For these readouts that is a limit on what can be observed, not a change in behavior (`docs/pi/c12-5-summary.json`).

E Bifurcation curves and cusps

Period-8 stripe. Continued for 80 points with a float64 test function, the fold curve of the period-8 orbit keeps its critical multiplier within $1 \pm 6 \times 10^{-6}$, follows the stripe's upper edge and caps its top near $\log_{10} T = -0.065$ (Figure 8). At $\alpha = 0.0717$ the stripe lies between a flip at $\log_{10} T = -0.128955$ and the fold at -0.1033462. Continued downward, the flip curve keeps its multiplier at -1 for 115 points to $(\alpha, \log_{10} T) = (0.0776, -0.402)$, with two reversals of its parameter tangent near $(0.0750, -0.219)$ and $(0.0746, -0.206)$, and the fold curve keeps its multiplier at $+1$ for 120 points to $(0.0758, -0.200)$. Located by bisection along the curve (Appendix B), the two reversals of the flip curve, at $(0.074955, -0.219336)$ and $(0.074608, -0.206007)$, carry no second multiplier near the unit circle (the next ones are 3×10^{-7} and 1×10^{-5}). On a flip curve the parameter tangent can vanish only where a second multiplier reaches $+1$, so both are tight but smooth turns of the curve in the parameter plane, not codimension-two points.

The accumulation boundary. The fold curve of the 2-cycle “is it”, continued from $(\alpha, \log_{10} T) = (0.0628, -0.115)$ in both directions (38 and 126 points, critical multiplier 1 to 10^{-7}, no cusp), traces the boundary of the period-2 region in Figures 8 and 9. The stripe fan accumulates on this boundary, as the mechanism of Section 8 requires. Further along one direction, outside the plotted window, another multiplier leaves the unit circle (moduli up to 11). Located along the curve, this second real multiplier reaches $+1$ at $(\alpha, \log_{10} T) = (0.035628, -0.341724)$, where the eigenvectors of the two multipliers at $+1$ coincide to 3×10^{-4} degrees: a double multiplier $+1$ with a Jordan block, the 1:1 resonance of maps (the analog of a Bogdanov–Takens point).

A period-6 window with shrimp-like anatomy. The hooks of Figure 10 (bottom) belong to the period-6 loop “. -\n\nThe is a”. Float64 orbits (Newton residuals 10^{-15}–10^{-13}) show that its curved body and its thin leg are one window: the multipliers are -0.061 and $+0.048$ at the body, -0.564 lower down and -0.645 on the leg. The foot is its period-doubled loop (period 12, leading multiplier -0.156). Inside the body lies a different 6-cycle, “.The\n\nThe is a”, whose leading multipliers are a complex pair, $-0.031 \pm 0.152i$. One-parameter branches from the body end in folds in every direction: at $\log_{10} T = -0.0558$ and -0.0721 in temperature, at $\alpha = 0.08074$ in steering, and, toward smaller α, in two folds only 10^{-5} apart. Such a close pair is the signature of a nearby cusp. A flip at $\alpha \approx 0.0797$ then opens the period-12 foot. The body is not effectively one-dimensional either. At two body cells the leading multiplier is a complex pair, $-0.020 \pm 0.264i$, or has a second multiplier of comparable size (-0.261 against 0.099). Below the body, where the window's strip appears to cross a period-5 strip near $(\alpha, \log_{10} T) = (0.0773, -0.0774)$, both orbits continued from their own strips are unstable at the crossing (leading multipliers -1.88 and -6.59). The crossing is therefore neither a center nor a point where two attractors coexist.

Fold curves that bound a window and meet in cusps, with flip curves inside, form the codimension-two organization of a shrimp [49] (complex dynamics of small recurrent networks has been studied in the same spirit [105]). In the continued curves (Figure 39), a single fold arc bounds the body on its outer side and a flip arc on its inner side, and the leg lies between them where they converge. The outer fold arc extends far to the left as a thin arm and ends in a cusp at $(\alpha, \log_{10} T) = (0.069106, -0.039750)$. The fold curve through the close fold pair has a second cusp inside the body, at $(0.080097, -0.058344)$, from which two fold branches descend.

At both reversals the quadratic coefficient b changes sign between consecutive curve points while a stays away from zero (Table 15). That alone is not evidence of a cusp: along any fold curve the parameter tangent reverses exactly where b vanishes. The test's content is the nondegeneracy at that point. Because the signs of a and b both flip with the orientation of the critical eigenvector, the test uses $b \operatorname{sign} a$. A cusp is nondegenerate when the cubic coefficient c of the normal form $u' = u + a\varepsilon + bu^2 + cu^3$ is nonzero where b vanishes. We compute c with the second-order center-manifold term, $c = \ell^\top C(r,r,r)/6 + \ell^\top B(r,h_2)/2$ with $(I-M)h_2 = B(r,r) - 2br$ and $\ell^\top h_2 = 0$, where M is the monodromy matrix and B and C are the second and third derivatives of its map, obtained by central differences of float64 Jacobian-vector products. (The code reproduces c to 10^{-6} on a map with a known reduction.) At the two curve points that bracket each sign change, c is nonzero far beyond its step-size uncertainty and has the same sign, and b changes sign with nonzero slope between them. This supports a nondegenerate cusp inside each bracket. Located along each fold curve (Appendix B), $b \operatorname{sign} a$ falls to below 5×10^{-6} of its bracket values at $(\alpha, \log_{10} T) = (0.069106, -0.039750)$ on the outer arc and at $(0.080097, -0.058344)$ inside the body. At these same points $c = -1.51 \times 10^4$ and -2.38×10^5, and the unfolding matrix J defined below is regular ($\det J = 2416$ and 14.0): two numerically located cusps. (Location uses the small-step oriented coefficient $b \operatorname{sign} a$; the unfolding coefficients use a separate finite-difference ladder, and the selected plateau of c spreads by at most 4.8 % over adjacent steps.) The negative sign of c means that near each cusp, two stable period-6 orbits coexist with a saddle between the fold branches.

A generic codimension-two cusp also needs the two parameters to unfold it independently. Near the cusp the reduced map is $u' = u + \beta_1 + \beta_2 u + bu^2 + cu^3$, where β_1 and β_2 vanish at the cusp and depend on the parameters. The unfolding is regular when $J = \partial(\beta_1, \beta_2)/\partial(\log_{10} T, 10\alpha)$ has rank two ($\det J \neq 0$). We first estimated J at a fold point near the outer-arc cusp (where $b \operatorname{sign} a = 14.7$): $\det J = 2737$ (singular values 1630 and 1.7), so the estimated unfolding is nonsingular there, as it is at the located cusp ($\det J = 2416$). With these derivatives, the local cubic model predicted the number of period-6 orbits and their stability at six parameter offsets, written to disk before any orbit was solved. Newton solves started from the predicted roots and from the base state recovered the predicted number of distinct orbits, with the predicted stability, at all six. The search was seeded, not exhaustive: orbits that these starts do not reach would be missed.

	cusp on the outer arc	cusp inside the body
$b \operatorname{sign} a$ at consecutive curve points	+92.9, +42.9, +14.7, **−16.9**, −52.8	−24.7, −14.3, −2.0, **+12.8**, +28.0, +45.9, +58.9
$\lvert a \rvert$	1.2–6.9	3.13–3.35
c at the two points around $b = 0$	-1.37×10^4, -1.65×10^4	-2.38×10^5, -2.34×10^5

Table 15: Normal-form test at the two parameter-tangent reversals of the period-6 window's fold curves (bold: the flagged points; critical multiplier 1 to 10^{-7}). c: cubic coefficient at the two curve points that bracket the sign change (finite-difference steps agree to 0.1 and 0.7 %). Bisection along the curves then located both cusps (text); they are numerical, not certified.

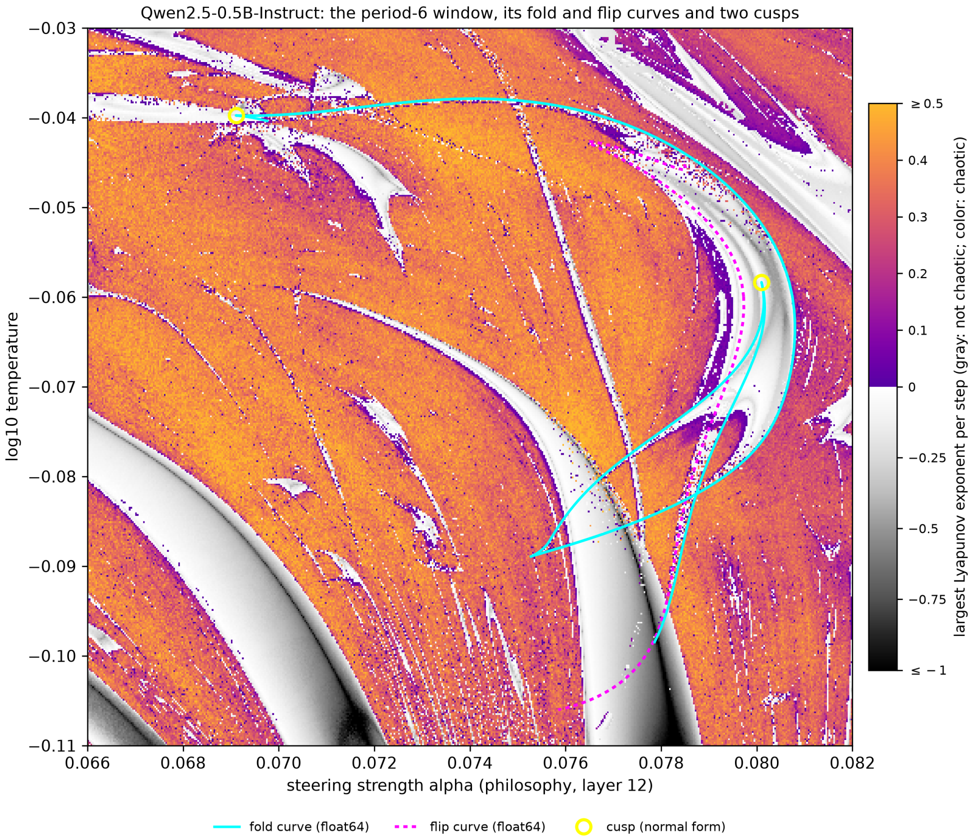


Figure 39: The period-6 window of the Qwen2.5-0.5B-Instruct map (philosophy direction, layer 12) with its float64 fold curves (cyan), flip curve (magenta, dotted) and the two located cusps (yellow circles). The background is a float32 phase map of 480 × 400 cells (largest exponent from 300 steps after 300 transient steps; chaotic cells colored by the exponent and the other cells in gray by theirs, on one color bar). Where it overlaps the float64 map of Figure 8, 95 % of cells carry the same chaotic or periodic label at the nearest grid point. Some fold branches were continued from two starting points and overlap. Continuation of the lower branches stopped after 60 points near $(0.0753, -0.089)$, where they approach each other; how they join there is not resolved.

A period-4 window with five cusps on one fold curve. At higher temperature, $\log_{10} T \approx 0$–0.1, periodic orbits are no longer effectively one-dimensional (second multipliers 0.06–0.09). At $\alpha = 0.06484$ the temperature branch of the period-4 orbit "\n\n - \n\n The" is S-shaped, with folds at $\log_{10} T = 0.0640$ and 0.0664. When that fold is continued in the plane, its parameter tangent reverses at three points; at the first two, the quadratic coefficient b changes sign while a stays away from zero, and the cubic coefficient c (defined above) is nonzero (Table 16). Located along the curve, these two cusps lie at $(\alpha, \log_{10} T) = (0.062959, 0.059958)$ and $(0.065574, 0.074447)$. There $b\,\mathrm{sign}\,a$ is below 1.1×10^{-5} of its bracket values, $c = -1.02 \times 10^4$ and $+1.37 \times 10^5$, and $\det J = -42.0$ and 35.5. Further traces of the same fold curve, started from other points of the period-4 branches, retrace it to within 3×10^{-7}, and one of them finds two more reversals near $\log_{10} T \approx 0.012$. All five reversals are located cusps. The other three lie at $(\alpha, \log_{10} T) = (0.083082, 0.108633)$, $(0.074072, 0.011831)$ and $(0.073220, 0.012759)$. There c is -9.7×10^4, -5.7×10^3 and $+1.7 \times 10^3$, and $\det J$ is 25.3, −188.5 and 68.5. The last two cusps, close together, have opposite signs of c. Three period-4 orbits coexist between the two fold branches, and near the first cusp ($c < 0$) two of them are stable (Figure 40).

	cusp 1	cusp 2
b at five curve points	$+28.1,\ +18.7,\ +8.4,\ -\mathbf{1.1},\ -12.8$	$+51.8,\ +29.7,\ +9.3,\ -\mathbf{9.0},\ -23.4$
a	3.02–3.08	-2.58 to -2.57
critical multiplier	1 ± 10^{-9}	1 ± 10^{-8}
c around $b = 0$	$-1.06,\ -1.01\ (\times 10^{4})$	$+1.39,\ +1.36\ (\times 10^{5})$

Table 16: Normal-form test of the two parameter-tangent reversals (bold: the flagged points). c: cubic coefficient at the points around the sign change, nonzero at both reversals (steps agree to 10^{-4}), which supports a nondegenerate cusp inside each bracket; bisection then located both (text); at the second, $c > 0$: the orbit between the two others is the stable one.

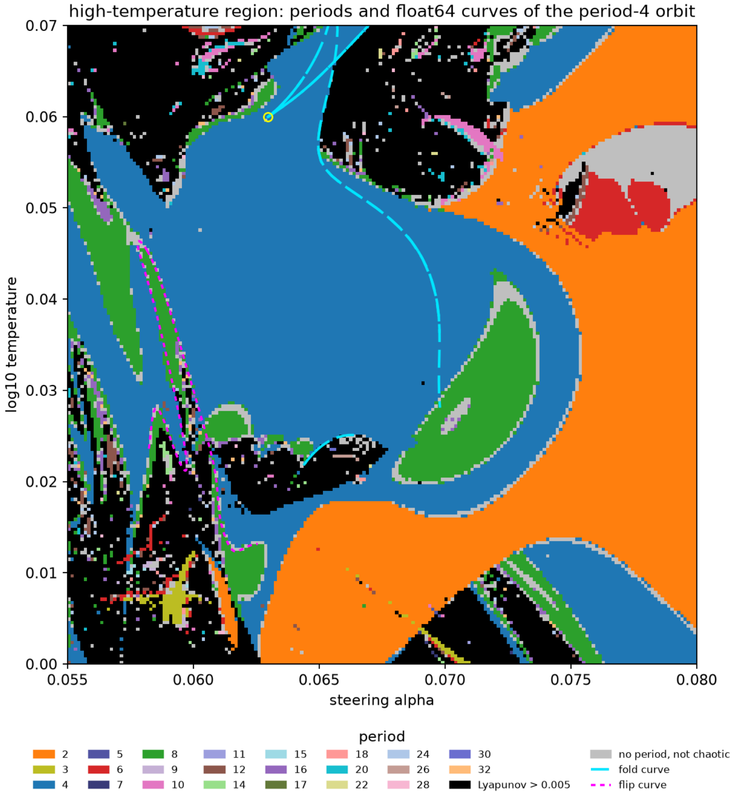


Figure 40: Period map of the high-temperature region of the Qwen2.5-0.5B-Instruct map (philosophy direction, layer 12; 192^2 float64 cells) with float64 fold curves (cyan, long dashes) and flip curves (magenta, short dashes) of the period-4 orbit. The yellow circle is the first cusp; the second lies just above the window. The flip curves run along the boundary between period 4 and period 8. The fold curves run inside the period-4 region, where three period-4 orbits coexist (near the first cusp, two of them stable). In this map a C-shaped band of the doubled loop "\n\nThe\n\nThe" (period 4) inside the period-2 region "\n\nThe" contains period 8 and an irregular core. Along the cut of Section 9 the period-8 orbit merges back into period 4 (a doubling bubble), so this band is not shown to be a second cascade. Continued from all its fold points, the region's boundary consists of an upper fold curve, with the five cusps of the text, and a separate lower fold curve, nowhere within 2×10^{-3} of it in the traced ranges; the lower curve carries the codimension-two points described below. Within the traced ranges the curves do not join, so the region is not shown to be one shrimp-organized component.

Along the lower fold curve a second real multiplier reaches $+1$ twice, at $\alpha = 0.062483$, $\log_{10} T = 0.004955$ and at $\alpha = 0.062499$, $\log_{10} T = -0.001780$. Both points are 1:1 resonances, with the eigenvectors of the double multiplier parallel to 3×10^{-4} degrees. Close by, at $\alpha = 0.062522$, $\log_{10} T = -0.002667$, a multiplier reaches -1: a fold-flip point. We did not continue the curves that emanate from these points.

Evidence level of the cusps. We estimated the rank condition of the unfolding at a fold point near one cusp of the period-6 window and one of the hostile period-7 body (Section 10; determinants 2737 and 53.5), and there the local model's predictions at twelve frozen offsets matched the orbits recovered by Newton solves seeded at the predicted roots and the base state, in number at all twelve and in stability at eleven; the search was not exhaustive. At the thirteen located cusps themselves (including the four of the period-5 window of Appendix F), $|\det J|$ lies between 14 and 2416. These are float64 computations validated by step-size checks (rung 1), not computer-assisted proofs. At the two continuation points that bracket the lower cusp of the hostile body, an independent implementation of the model reproduced the normal-form coefficients, including the sign change of b: $|b|$ within 1.2 % and c to two figures, with the opposite orientation of the null vector, which flips the signs of a and b together.

F Steering toward hostility: further results

This appendix gives further results on the hostility direction of Section 10: the details of the shrimp-like shapes, the period-5 window of a plane steered at two depths, its orbit census and a closure test.

Details of the shrimp-like shapes. Section 11 summarizes the anatomy (Figure 27); three details belong here. In higher dimensions a window can also end where a complex pair crosses the unit circle, and the leading multiplier can change sign through a complex pair, as it does in parts of our windows; beyond some of the flip curves we followed lie the doubling sequences of Section 9, but a flip need not start a complete cascade. The period-6 hook's orbit never comes close to superstable: its leading multiplier stays above 0.028 in modulus. And the X-shaped crossing of a period-5 and a period-6 strip (Figure 10, bottom) is not one window's anatomy but two windows overlapping, and at the crossing both of their orbits are unstable (Appendix E). The shapes in these one-block planes are therefore the two-parameter unfolding of windows whose return maps have one coarse extremum: shrimp-like in body, legs and internal cascade, but without a one-dimensional shrimp center on the tracked branches.

Two turning points from two steering depths. Steering at two depths gives the band dynamics a second turning point. We drew 1584 chaotic cells (200 from each of the four two-block planes of Figure 14, 150 from each one-block map of Figure 13 and the 34 of the hostile line of Figure 21), counted the turning points of each band's return map and inspected every strong candidate by eye, since a projection can fold a single hump. Three cells, all in two-block planes at $\log_{10} T = -0.1$, have smooth, single-valued return maps with two or more turning points (Figure 41); the other strong candidates were point sets, split bands or folded projections, or were not reproduced. In one cell (hostility at block 4, gratitude at block 16), every one-dimensional phase of a 20-band attractor is N-shaped: a bimodal map. A 128×128 zoom around another, whose map has three maxima, shows a period-5 window with a body, legs, a period-10 and period-20 border, and patches where the fixed start settles on period 4 (Figure 42a). We continued the window's period-5 orbit over a 31×31 grid of its body: 608 cells solved, all with the same token loop, 447 of them attracting. In 61 of these 447 cells, the nearest cell of the zoom (at most 2.0×10^{-5} away in α_1 and 3.2×10^{-5} in α_2) settles on period 4, so attractors of periods 4 and 5 coexist at grid scale, which a single unimodal map with negative Schwarzian derivative allows only if a basin reaches an end of its interval (Section 3).

The window still has no one-dimensional center. The leading multiplier of the continued orbit never comes near zero: its modulus is at least 0.0139 in all 608 cells. Where a one-dimensional map would be superstable, it meets the second multiplier, and the two form a complex pair (91 cells, in narrow strips) or, where their product is negative (41 other cells), pass each other as real numbers of opposite sign (Figure 42c). Per step, the second direction contracts by a median factor of 0.44 and the first by 0.76 over the attracting cells, so near these strips the orbit's linearization has two comparable leading directions. The curves on which the trace $\mu_1 + \mu_2$ changes sign take the place of superstable curves, and they do not cross: no 2×2 block of attracting cells has the alternating signs of a crossing (Figure 42b). One isolated cell holds a repelling orbit with the same token loop ($\mu_1 = 1.65$), and near the top of the grid one column of Newton solves failed above cells whose leading multiplier rises to 0.88; both are explained by the window's fold curve (next paragraph). A second turning point in the band maps therefore does not give this window a one-dimensional shrimp center. Like the hostile body, it is a window whose orbit has two comparable leading directions near its superstable strips, and whether it is the shrimp of a two-dimensional map is open.

The folds and flips of the period-5 window. To resolve the folds the grid did not, we continued the window's period-5 orbit along four lines from one attracting cell, $(\alpha_1, \alpha_2) = (-0.118730, 0.100693)$, and then the boundaries the lines found, each in both directions (float64; 18 to 73 points per line, 30 to 120 per curve; one of the eight curve runs failed at its start for lack of GPU memory, and its curve is covered by the other trace of the same fold). At every curve point the residual is below 10^{-10} and the test function below 10^{-9}. The critical multiplier stays real and within 6×10^{-8} of $+1$ on the fold curve and of -1 on the flip curves; the largest other multiplier modulus is 0.040 on that fold curve and 0.122 on the flip curves. Going right and up from the start, the orbit dies in folds at $\alpha_1 = -0.118166$ and $\alpha_2 = 0.102715$ (next moduli 0.0012 and 0.0008). Going left and down, it first loses stability in flips (at $\alpha_1 = -0.12147$ and $\alpha_2 = 0.09954$), regains it where the same multiplier crosses -1 again (at -0.12305 and 0.09586; a doubling bubble), and dies in a fold (at -0.123323 and 0.095720). Three of these four folds lie on one continued fold curve (within 3.1×10^{-7}); the left one lies on a second. The first fold curve bounds the lower leg on its right, runs up the body's right edge and over its top, and leaves along the upper leg. Near $\alpha_1 = -0.1192$ its parameter tangent reverses twice, identically on two traces continued from different boundary points, and both reversals are located cusps (the test and criteria of Appendix E, applied on each trace): at $(\alpha_1, \alpha_2) = (-0.1192221, 0.1040536)$, with $c = +2.60 \times 10^4$ and $\det J = 176.6$, and at $(-0.1191670, 0.1028019)$, with $c = -2.05 \times 10^4$ and $\det J = 88.6$. The two traces agree to 2×10^{-12} in these positions and to five figures in c and $\det J$. Its outer arms cross at $(-0.11901, 0.10346)$ (Figure 26d). There they are folds of two different period-5 orbits with the same token loop: at the four continuation points that bracket the crossing, the orbits on the two arms differ by 3.65 to 3.78 % of the state norm (root-mean-square over the five points, at the best phase alignment; both traces), 15.8 to 44.5 times the state change between neighboring curve points (`docs/fm5/crossing-orbits.json`). The second fold curve, continued both ways from the left fold, runs from below the lower leg far up the left of the plane, where the orbit has regained stability beyond its doubling bubble (critical multiplier within 1.2×10^{-7} of $+1$; other multipliers below 0.4). It has two more reversals, both located cusps: $(-0.1238662, 0.0961512)$, with $c = -9.65 \times 10^3$ and $\det J = 405.9$, and $(-0.1214073, 0.1028234)$, with $c = +4.75 \times 10^4$ and $\det J = -535.4$ (condition 1.2×10^4). This curve also crosses itself, at $(-0.12341, 0.09904)$, between its points 6 and 64, which lie on either side of the two cusps (points 15 and 40), so that the two cusps share a segment and their other branches cross: a second dovetail. There too the arms are

points	region	predicted	found	attracting
T-low, T-high, T-near-U	pocket	4	4	2
T-centroid, T-near-L	pocket	4	3	2
E-low, E-high	beyond the crossing	0	0	0
W-low, S-low, N-high, below-L	around the pocket	2	2	1
W-high	around the pocket	2	1	1
above-U	beyond the upper cusp	2	3	0

Table 17: Orbit census of the junction of the period-5 window (run OC; Figure 26d): period-5 orbits found from 27 starts per point, and how many attract. The predictions were fixed before any solve.

folds of two different period-5 orbits with the same token loop, which differ by 8.83 to 9.06 % of the state norm (measured the same way), 23 to 31 times the state change between neighboring curve points. In both dovetails the cusp with $c > 0$ lies above the one with $c < 0$. At all four cusps the critical multiplier is within 1.3×10^{-9} of 1 and the second modulus below 0.008. The flip curves bound the body's upper-left and lower sides and run along the legs. The fold curve resolves both anomalies of the grid. The isolated repelling cell ($\mu_1 = +1.65$) lies 1.2×10^{-5} from it, on the saddle branch, and the column of failed Newton solves at $\alpha_1 = -0.119167$ runs into the lower reversal, at $\alpha_1 = -0.119166$, where the followed orbit ceases to exist between the folded branches.

Orbit census of the junction. Before any solve we fixed 13 points around the junction (`docs/oc/oc-points.json`) and the number of period-5 orbits that a plane section of a swallowtail predicts at each: four in the pocket bounded by the two cusps and the crossing, none beyond the crossing, and two elsewhere. At each point, Newton's method (low-rank, at most 20 steps) starts from the states of the nearest point on each arm of the fold curve, shifted along that point's critical eigenvector by $s|x|$ for $s \in \{0, \pm 10^{-3}, \pm 3 \times 10^{-3}, \pm 10^{-2}, \pm 3 \times 10^{-2}\}$ (27 starts per point). Orbits are distinct when they differ by more than 10^{-7} of the state norm at every phase. The counts are lower bounds, because the seeded search can miss orbits. Table 17 gives the results: 9 of the 13 counts match. The three undercounts found all the attracting orbits that the model predicts, so by the model each missed a repelling one. Above the upper cusp, two of the three orbits found share one token loop: a saddle (multiplier +1.44) and the node, which lies beyond the flip curve that passes the cusp (multiplier −1.35), so that no period-5 orbit attracts there (the classifier map shows a period-4 attractor next to this point). The third has another token loop and the multiplier −130, and only one start reached it; that it does not belong to the junction's family is a reading after the fact.

A closure test that failed. We also tested whether two coordinates forecast the window's nonlinear return map (run CL2). Perturbations of relative size 10^{-3} and 10^{-2} (64 per size, half to fit and half to test) were followed for eight returns; quadratic return models in one, two or four modal coordinates (the spectral projections on the leading eigenvectors), with a fitted lift back to the state, were iterated from the first return, and the error is the squared forecast error over the squared deviation, summed over returns 2 to 8. The test failed for numerical reasons: at most locations the iterated models diverged, with median errors up to 2×10^{122}, against at most 0.71 for the linearized map in the same rows. We did not establish why (in 9 of these 12 rows even the one-coordinate model diverges), and no regularized refit was run. At one location the divergence of the four-coordinate model satisfied the frozen two-coordinate criterion spuriously. A guard added after the run marks fits with a median error above 1 as degenerate; 12 of the 18 location-and-size

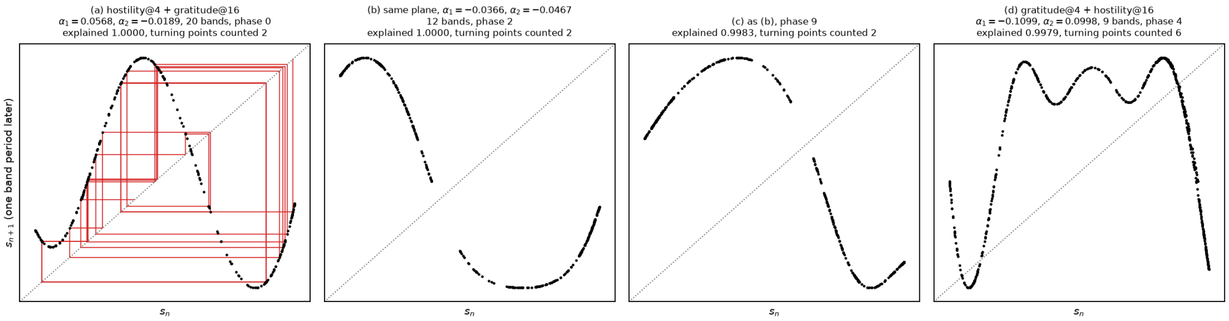


Figure 41: Band return maps with two or more turning points, from two-block planes of Qwen2.5-0.5B-Instruct (float64, $\log_{10} T = -0.1$). Each panel shows one phase of a chaotic band attractor, projected on its leading principal component s (explained variance in the title), with points (s_n, s_{n+1}) one band period apart. (a) Hostility at block 4 and gratitude at block 16, $(\alpha_1, \alpha_2) = (0.0568, -0.0189)$: phase 0 of a 20-band attractor, N-shaped, with a cobweb of consecutive visits. (b, c) The same plane at $(-0.0366, -0.0467)$: two phases of a 12-band attractor, N-shaped maps on two intervals. (d) Gratitude at block 4 and hostility at block 16, $(-0.1099, 0.0998)$: phase 4 of a 9-band attractor, with three maxima.

rows are. At the body location, one coordinate gives errors of 0.026 and 0.031 and two coordinates below 6×10^{-5}. We draw no conclusion about two-dimensional closure from this test.

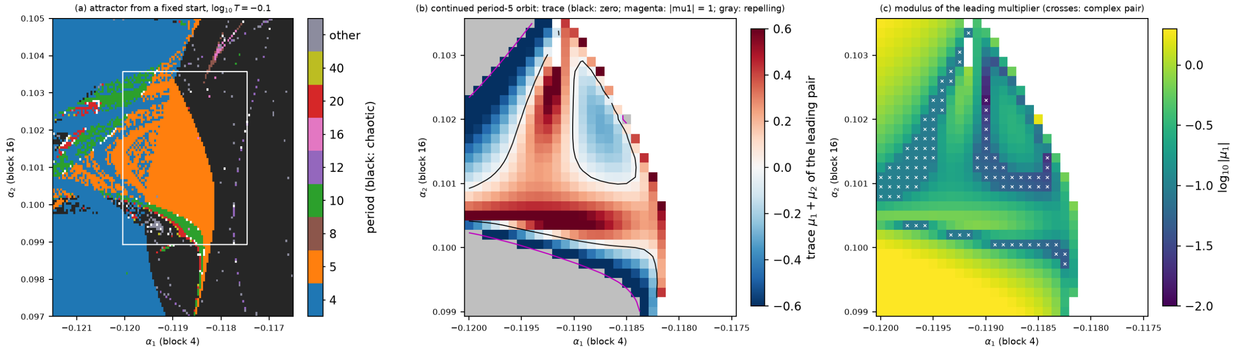


Figure 42: A period-5 window in the plane of gratitude at block 4 and hostility at block 16 (Qwen2.5-0.5B-Instruct, float64, $\log_{10} T = -0.1$). (a) Attractor reached from the fixed start on a 128 × 128 grid (black: chaotic), with the box of the continuation grid. (b) The window's period-5 orbit continued over the box (31 × 31, 608 cells solved, one token loop): trace $\mu_1 + \mu_2$ of its two leading multipliers on attracting cells, its zero curves (black) and the curves $|\mu_1| = 1$ (magenta); gray cells hold a repelling orbit. (c) Modulus of the leading multiplier; crosses mark a complex leading pair. The modulus never falls below 0.0139; the dark strips, where a one-dimensional map would be superstable, are where the two leading multipliers meet.

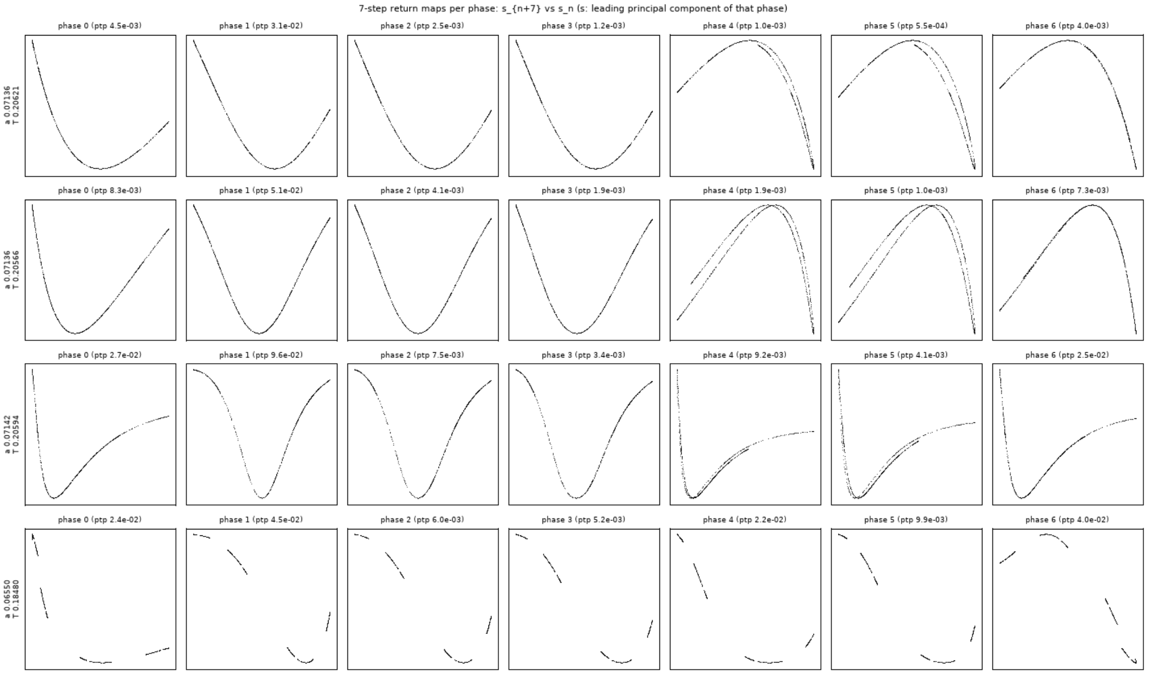


Figure 43: Seven-step return maps s_{n+7} against s_n, for each phase of the 7-periodic text, of chaotic cells next to the hostile period-7 body (top three rows: $\alpha = 0.07136$–0.07142, $\log_{10} T = 0.2057$–0.2062) and on the hostile line of Figure 21 (bottom row: $\alpha = 0.0655$, $\log_{10} T = 0.1848$); s is the projection on the leading principal component of that phase's states, which carries 89–100 % of their variance. Binned coarsely (16 intervals), every phase has one interior extremum; finer partitions keep extra extrema, especially in phases 4 and 5, where the points also split into two thin strands.

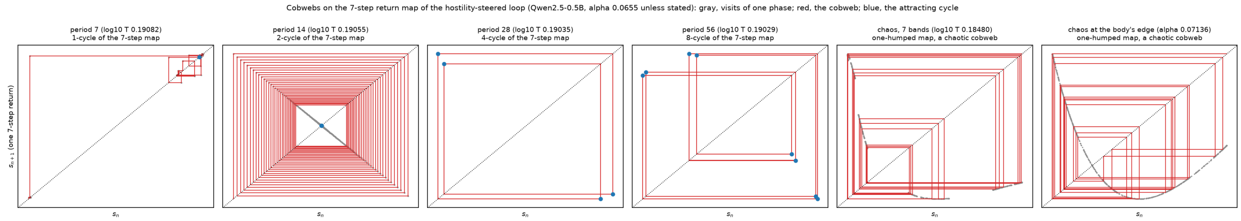


Figure 44: Cobweb diagrams of the 7-step return map of the hostility-steered loop (Qwen2.5-0.5B-Instruct, layer 12, float64). s is the projection of one phase's states on their leading principal component. Left four panels: the doubling cascade at the right edge of the hidden-chaos window along $\alpha = 0.0655$. As $\log_{10} T$ falls from 0.19082 to 0.19029, the attracting set of the 7-step map (blue) is a fixed point (period 7 of the full loop), then a 2-, 4- and 8-cycle (periods 14, 28 and 56); the red cobweb shows the approach from the fixed start (for period 7, its first visits). Right two panels: chaotic cells of Figure 43, where the visits trace a one-humped map (gray) and the cobweb wanders over it.

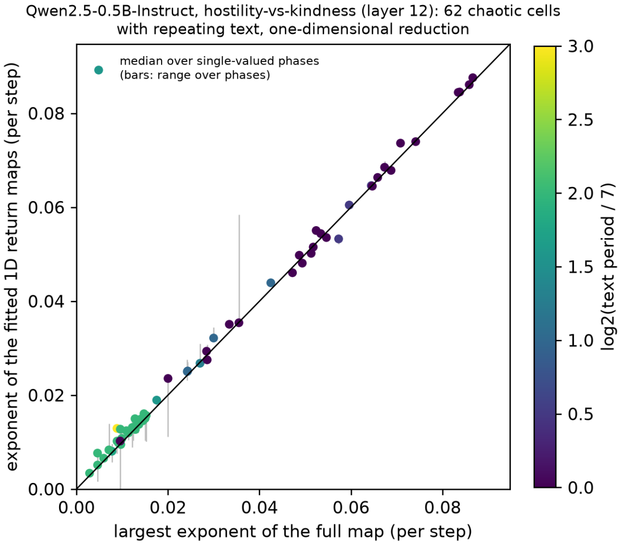


Figure 45: Scalar return-map approximation across the hostile cascade (Qwen2.5-0.5B-Instruct, hostility at layer 12, $\alpha = 0.0655$, $\log_{10} T \in [0.176, 0.191]$): for each of the 62 chaotic cells whose leading-token text repeats exactly, the Lyapunov exponent of the fitted return maps of the leading-token probability (per step; median over the single-valued phases of the text, bars: their range; polynomial degree chosen by cross-validation) against the largest exponent of the full map from twin trajectories. Color: text period relative to 7. The line is equality. With seven-token texts the 2000 steps give 285 repetitions; with 28-token texts only 71 or 72 observations per phase (70 or 71 return pairs), and the fitted maps then overestimate small exponents.

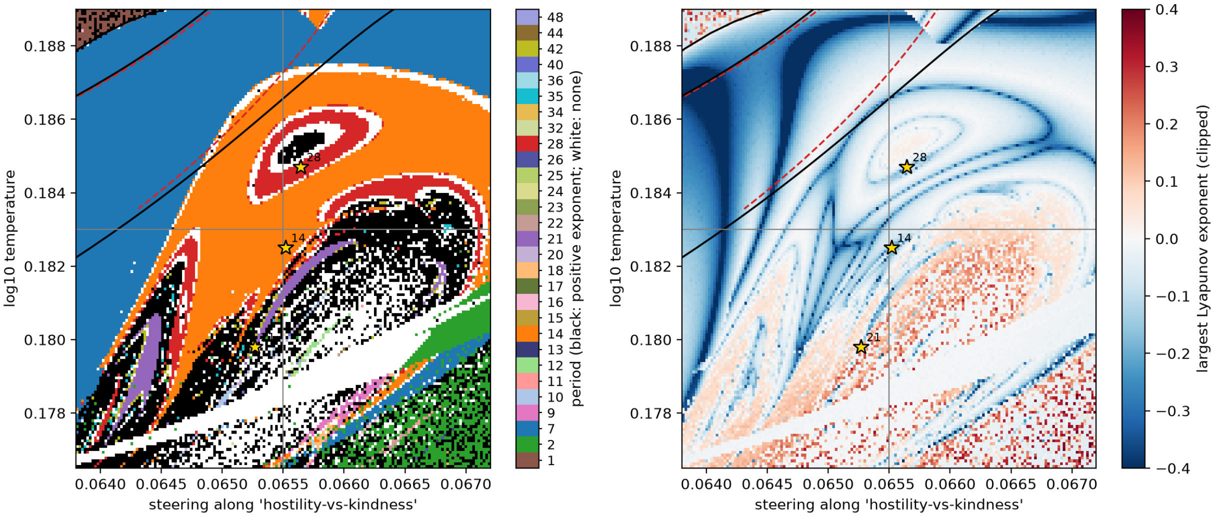


Figure 46: The lower part of the period-7 body of the hostility-steered Qwen2.5-0.5B-Instruct map (layer 12) at higher resolution (160^2 float64 cells). Left: period of the attractor (black: positive largest exponent; white: no period up to 48). Right: largest Lyapunov exponent (clipped); the darkest curves are superstable lines. Both panels: float64 fold (black) and flip (red, dashed) curves of the period-7 orbit, the two lines of Figure 21 (gray) and the Newton-verified orbits of periods 14, 28 and 21 (stars).

G Bridge to sampled decoding

How much of the fold mechanism of Section 8 survives sampling? To find out, we replace the expected embedding by the mean of m sampled token embeddings (Figure 47). With the exact map, laminar phases near the fold of the 2-cycle “is it” last 120 and 63 steps at $\varepsilon = 10^{-5}$ and 3×10^{-5}. With $m = 64$ they last about 10.6 steps, independently of ε (57% of steps in the channel), and with $m = 8$ about 7.7 steps (38%). With single samples ($m = 1$, sampled decoding of the same truncated system), essentially no laminar phases remain (11% of steps in the channel). The deterministic intermittency mechanism therefore belongs to the expected-embedding map. Sampling noise turns the bottleneck into a noise-limited trap whose strength grows with m.

Zero-temperature limit and ordinary greedy decoding. As $T \to 0$ the soft-token map becomes greedy decoding with a one-token memory, $g(t) = \arg\max_s z_s(\text{prefix}, E_t)$. This is a map on the vocabulary, and its cycles are the zero-temperature attractors. For Qwen2.5-0.5B we computed g for all 151 936 tokens. Unsteered, it has 1060 cycles, and one of them, the period-5 loop “![](https://Hello”, drains 86 % of the vocabulary; it is the largest attractor family of every 0.5B soft-token map. Ordinary greedy decoding with a growing context does not inherit these cycles (Table 18). From 220 starts (the first tokens of the 20 largest cycles and 200 random tokens), 19 runs end in a short loop within 256 tokens, and the dominant 5-cycle never appears. Steering along philosophy makes ordinary decoding loop almost always (218 of 220 runs at $\alpha = 0.1$), but in other loops, mostly punctuation such as “ –”, none of them one-token cycles. Greedy decoding that keeps only the last K tokens interpolates between the two. For the tested $K \geq 2$ most final loops differ from the one-token cycles: at $K = 2$ only 7 of 220 unsteered runs and 1 of 220 steered runs keep one of them. Unsteered, looping collapses to the level of the growing context between $K = 16$ and 32; steered, the share of runs whose loop equals the growing-context loop rises from 0 at $K = 1$ to 58 % at $K = 64$. The soft-token map's low-temperature attractors are therefore those of decoding with a one-token memory. Which loop ordinary decoding enters depends on the context it keeps, but the tendency of steering to drive decoding into loops is shared. In ordinary decoding, repetition is known to reinforce itself [25, 26], a theoretical analysis ties loops to words of high inflow [106], and induction heads, which copy earlier context [107], are one mechanism by which a longer context can favor loops that repeat earlier text.

	unsteered		$\alpha = 0.1$ (philosophy)	
memory K	looped	loop as with full context	looped	loop as with full context
1	220	5	220	0
2	220	5	220	0
8	220	5	220	3
16	169	5	219	24
32	32	7	219	71
64	26	11	219	126
full context	19	–	218	–

Table 18: Greedy decoding of Qwen2.5-0.5B with a memory of the last K generated tokens after the fixed prompt ($K = 1$ is the soft-token map's zero-temperature limit), 220 starts of 256 tokens each: runs ending in a loop of period up to 64 repeated three times, and runs whose final loop equals that of ordinary decoding with the full context from the same start.

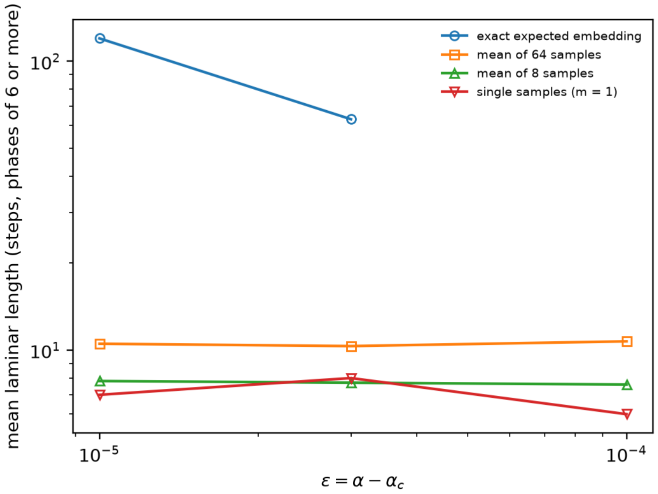


Figure 47: Mean laminar length near the fold of the 2-cycle "is it" (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\log_{10} T = -0.115$) against ε, for the exact map and for means of m sampled embeddings.

A longer fed-back memory. The one-token window is the main simplification of the map. Feeding back the last K expected embeddings instead, so that the state is a window of K soft tokens at fixed positions after the prompt, we measured 24 temperatures between $\log_{10} T = -0.3$ and 0.6 for K from 1 to 16 (Table 19; twins at 10^{-5} and 10^{-7} agree within 0.006 at every chaotic temperature). In Qwen2.5-7B the chaos does not vanish but spreads: it is chaotic at 8 temperatures with one token of memory, at 13 with four, and at 21 and 20 with eight and sixteen, nearly the whole range, with smaller exponents (largest 0.96 for $K = 1$, 0.43 for $K = 16$). Qwen2.5-0.5B, chaotic at one temperature with one or two tokens of memory and at none with four, is chaotic at ten with eight. The chaos is therefore not an artifact of the one-token window.

memory K	1	2	4	8	16
Qwen2.5-7B	8 (0.96)	10 (1.08)	13 (0.70)	21 (0.58)	20 (0.43)
Qwen2.5-0.5B	1 (0.20)	1 (0.24)	0	10 (0.31)	3 (0.23)

Table 19: Chaotic temperatures, of 24 between $\log_{10} T = -0.3$ and 0.6, when the loop feeds back the last K expected embeddings (largest exponent in parentheses); paper's prompt, unsteered, float64, 500 transient and 1000 measured steps, twins at 10^{-5} and 10^{-7} (chaotic: both positive and no period up to 64).

Soft thinking with a growing context. Soft Thinking [1] feeds back the same expected embedding but appends it to the context, so that every earlier soft token stays visible. We ran that decoding in float64 from the paper's prompt and displaced the first soft token of twin trajectories by 10^{-10}, without renormalization; the distance between the newest soft tokens then shows whether the early perturbation grows. The context grows, so these are finite-time growth rates of a non-autonomous system, not exponents of an attractor. In Qwen2.5-7B the perturbation grows exponentially at temperatures inside the one-token map's chaotic range: by 0.77 to 0.82 per step at $\log_{10} T = 0.17$ and 0.21, reaching 10^{-3} within 49 to 58 steps, and it reaches 10^{-3} within 400 steps at nine other temperatures between −0.10 and 0.40, at 0.02 to 0.18 per step (0.8 at 0.33 for two of three twins). It does not grow at the sharpest distributions ($\log_{10} T \leq -0.14$) or at nearly uniform ones ($\log_{10} T \geq 0.44$), where it stays between 10^{-13} and 10^{-7} for 400 steps. In Qwen2.5-0.5B the growth moves to sharper distributions: it appears at six of the seven temperatures from $\log_{10} T = -0.30$ to -0.07 (up to 0.33 per step) and at 0.09, with a late transient burst at 0.05, and is absent from 0.13 up, which includes its narrow one-token chaotic band. Sensitive dependence therefore survives when the whole context is kept, and in the 7B model the one-token map locates it. Qwen2.5-72B, run the same way on two GPUs, stays nearly insensitive. We count a temperature as growing when two of its three twins reach 10^{-3} within 400 steps, a rule fixed before the 72B run and applied afterwards to the earlier 0.5B and 7B records, where it gives 11 of 24 temperatures for the 7B model and 8 for the 0.5B model. The 72B model grows at one of 24 temperatures on the paper's prompt, and only transiently: at $\log_{10} T = 0.09$ the twins pass 10^{-3} at step 123 and fall back to 5×10^{-6} (at 0.13 one twin of three passes 10^{-3} at step 381). It grows at none of the 24 completed temperatures on the held-out prompt with its narrow unsteered band (a planned finer line there was not completed), including $\log_{10} T = 0.326$, between two temperatures where its one-token map is chaotic; there the distance never exceeds 1.1×10^{-9}. It also grows at none of the eleven steered cells of Appendix I, including the six where its one-token map is chaotic: there the distance never exceeds 3×10^{-7} and ends between 1×10^{-13} and 6×10^{-10}, and the growing context holds the next-token entropy at 2.3 to 3.6 nats instead of the one-token map's 7.3 to 11.1. At these 72B prompts and cells, over 400 steps, the growing-context decoder therefore did not reproduce the sustained sensitivity of the fixed-window loop, while the one transient burst shows that the method registers growth at this scale.

H Replication and scale

In Qwen2.5-1.5B (steering at layer 14 of 28), a coarse 64^2 float64 map over the range of the 0.5B maps again shows chaotic regions with embedded windows. A 256^2 zoom (Figure 29) resolves a chaotic band near $\log_{10} T = 0$–0.13 (14,422 of 65,536 cells), with a branch descending along $\alpha \approx 0.06$ and periodic windows inside it. The large periodic regions are question loops: "What?\n" (period 2), "\nWhat is a" (period 4) and "What continuum?\n" (period 3). Unlike the 0.5B model, no period-adding fan was resolved in this map.

A cut of 2048 cells at $\alpha = 0.100$ (Figure 48) shows how the band is entered from below. The 2-cycle "What?\n" gives way abruptly to a period-4 loop, and the Lyapunov exponent jumps, which is consistent with a switch between attractors rather than a resolved smooth doubling. The period-4 loop then doubles, with float64 flips at $\log_{10} T = -0.0031269$ (period $4 \to 8$, next modulus 0.014), -0.0021666 (period $8 \to 16$, next modulus 8×10^{-5}), -0.0019002 (period $16 \to 32$, next modulus 5×10^{-9}) and -0.0018380 (period $32 \to 64$). Along the way the exponent rises continuously to zero, and chaos follows. The interval ratios are 3.60, 4.28, 4.58, 4.648 and 4.665; the last three use the flips of the period-64, period-128 and period-256 orbits at $\log_{10} T = -0.0018244$, -0.00182151 and -0.00182088 (next moduli below 3×10^{-14}). The last ratio differs from Feigenbaum's 4.6692 by 0.0044 (0.1 %); the period-256 flip is located to about 3×10^{-14} in $\log_{10} T$, the floor set by float64 parameters. Like those of the 0.5B crescent at $\log_{10} T = -0.148$, the ratios rise toward 4.669. Their higher start does not indicate a temperature trend, since along nearby cuts of the 0.5B crescent the early ratios range from 2.84 to 14.9 (Section 9).

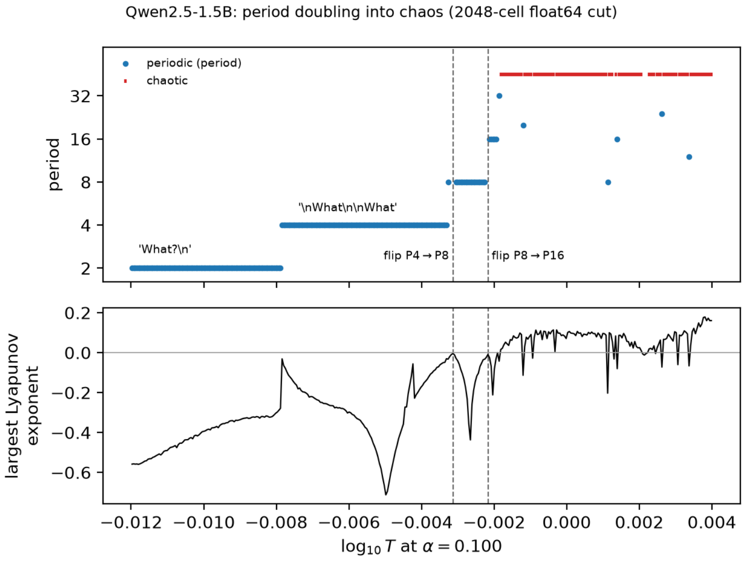


Figure 48: Qwen2.5-1.5B soft-token map steered along the philosophy direction at layer 14, along $\log_{10} T$ at $\alpha = 0.100$: period of the attractor reached from the fixed start (top) and largest Lyapunov exponent (bottom). Dashed lines: float64 flip parameters. The doubled loops alternate the number of line breaks between the "What"s.

In Qwen2.5-0.5B, a second steering direction, hostility, reproduces the fold-and-adding mechanism (Section 10).

Qwen2.5-7B (steering at layer 14 of 28, concept vectors built as for the smaller models) shows the same kind of organization in coarse 64^2 float64 maps (Figure 49). Over the philosophy range used for 0.5B and 1.5B, 8 % of cells have a positive largest exponent, and periods 2, 4, 3 and 11 dominate the periodic cells. Along the hostility direction ($\alpha \in [0, 0.12]$, $\log_{10} T \in [-0.3, 0.3]$), 24 % of cells are chaotic. Bands of period-3, -5 and -6 windows lie in a chaotic sea below $\log_{10} T \approx 0.1$, with high-entropy chaos above, except for a period-3 region at strong steering ($\alpha \gtrsim 0.09$). The windows hold genuine periodic orbits. In float64, Newton's method converges at the most interior period-6 cell to the loop ", shut the end\n\nOh" at $(\log_{10} T, \alpha) = (-0.0143, 0.0476)$ (residual 2×10^{-14}; spectral radius 1.6×10^{-7}, nearly superstable), and at the most interior period-3 cell to a period-3 loop at $(0.2524, 0.1143)$ (residual 3×10^{-15}; leading multiplier -0.12). Continued for about an hour each, over ± 0.001 in α and -0.016 in $\log_{10} T$ (one float64 continuation step takes about two minutes at 7B), both orbits stay strongly stable without a crossing. Where the coarse map switches from the period-6 loop to the period-3 loop "you better than" (between $\alpha = 0.0495$ and 0.0514 at $\log_{10} T = -0.0143$), both loops are float64 orbits (residuals 7×10^{-15} and 1×10^{-14}), and their continuations overlap. The period-6 orbit stays stable from $\alpha = 0.0476$ up to 0.0507 and the period-3 orbit from 0.0514 down to 0.0475, both with spectral radius below 1.4×10^{-3}. The two attractors therefore coexist where both continuations are stable, $\alpha \approx 0.0476$ to 0.0507, and within the tested range the map's edge there is the boundary of the fixed start's basin, not a bifurcation. These short continuations located no bifurcation; a period-2 fold of the 7B model is located below.

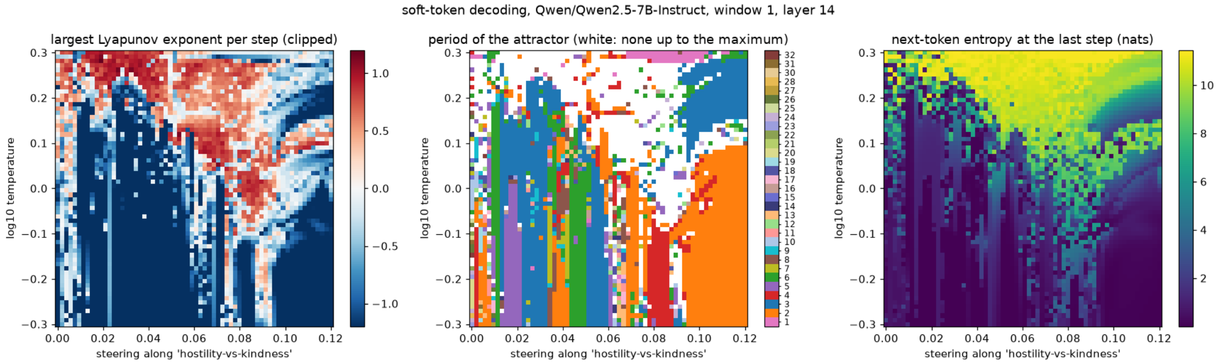


Figure 49: Qwen2.5-7B soft-token map steered along the hostility direction (layer 14), 64^2 float64 cells: largest Lyapunov exponent, period of the attractor reached from the fixed start, and next-token entropy at the last step.

A second model family. OLMo-2-1124-7B-Instruct [108] comes from another laboratory, with other data, training and code. Steered at layer 16 of 32, with concept vectors built exactly as for Qwen, it shows the same kind of organization (Figure 50). In 64^2 float64 maps, 6 % of cells are chaotic along the philosophy direction and 13 % along the hostility direction, in a band near $\log_{10} T = 0.03$–0.17 that bends to lower temperatures under strong hostility steering ($\alpha > 0.07$). Below the band the attractor is the 2-cycle “!Hello”; above it, at high temperature, the fixed point “!”; inside it lie periodic windows of periods 4 to 31. Along the hostility direction the largest windows, of periods 8 to 12, are loops of k laps of the 2-cycle “!<|endoftext|>” with a short excursion (“there is an”). Each lap adds two to the period, as each lap of “!Yes” does in the hostile cascade of Qwen2.5-0.5B (Section 10). Whether a fold of that 2-cycle organizes them is not tested: these are coarse pixels, and no OLMo orbit has been continued.

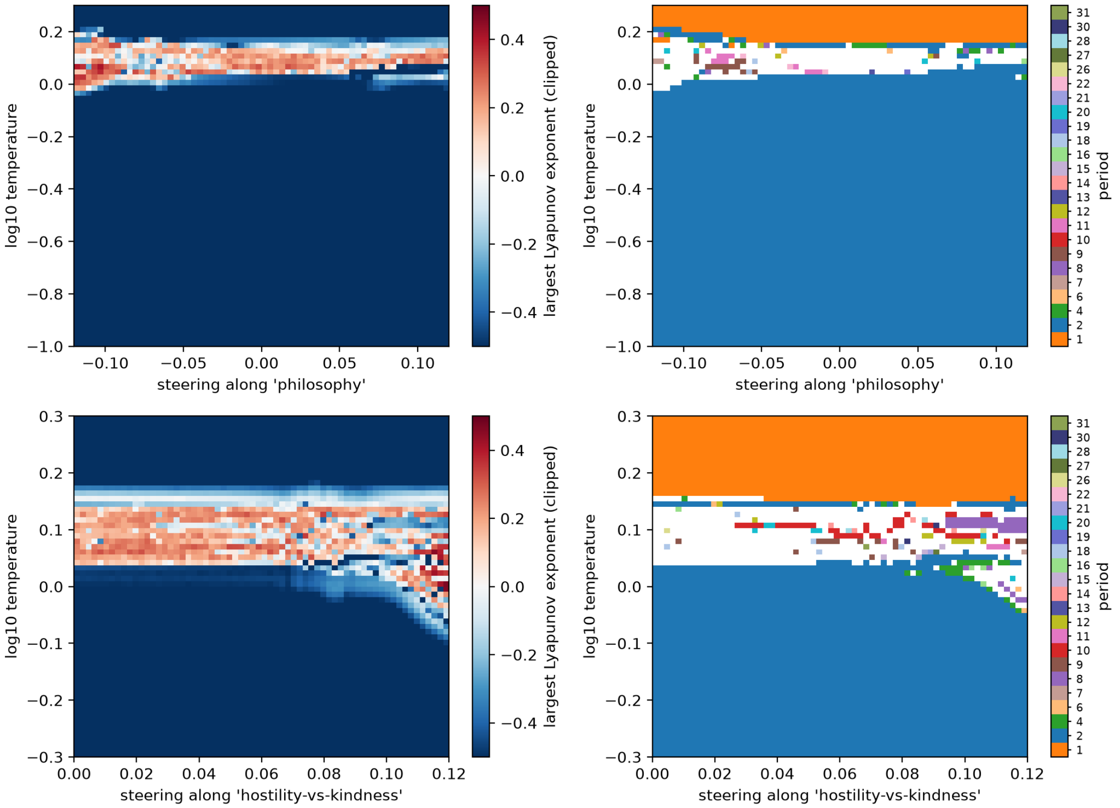


Figure 50: OLMo-2-1124-7B-Instruct soft-token maps, 64^2 float64 cells each, steered at layer 16 along the philosophy direction (top) and the hostility direction (bottom): largest Lyapunov exponent (clipped) and period of the attractor reached from the fixed start (white: no period up to the maximum).

A third family, read with sparse autoencoders. Llama-3.1-8B-Instruct [109], with official weights, steering at layer 16 of 32 and vectors built as for the other models, shows the same organization in all three planes (Figure 51). In 64^2 float64 maps, 8 % of cells are chaotic along gratitude, 12 % along philosophy and 28 % along hostility, with windows of periods up to 32. Its largest loops are greetings between AIs ("-Greetings fellow AI", ". Greetings fellow AI entity detected", period 7). Along hostility, too, the largest loops are greetings and "You", not hostile text. We read 24 of its attractors (the six largest families and the two most chaotic regions of each map) with two sparse autoencoders on the residual stream: one trained on this Instruct model (Goodfire, layer 19 [110]) and one on the base model (TopK, layer 24 [111]). We first checked each one's hook point by its reconstruction of ordinary text. The Instruct-trained autoencoder separates the attractors (their top-15 features overlap by a mean Jaccard index of 0.19 within a steering direction and 0.11 across directions), whereas the base-model autoencoder sees mostly a signature common to every attractor (overlap 0.52–0.86). Both reconstruct the attractors' states less well than ordinary text (median relative error 0.64–0.76 against 0.52–0.56). The readout is therefore descriptive: without interventions it identifies which learned features an attractor occupies, not a mechanism, and since autoencoders trained with different seeds learn different features [112], the features are those of these two autoencoders.

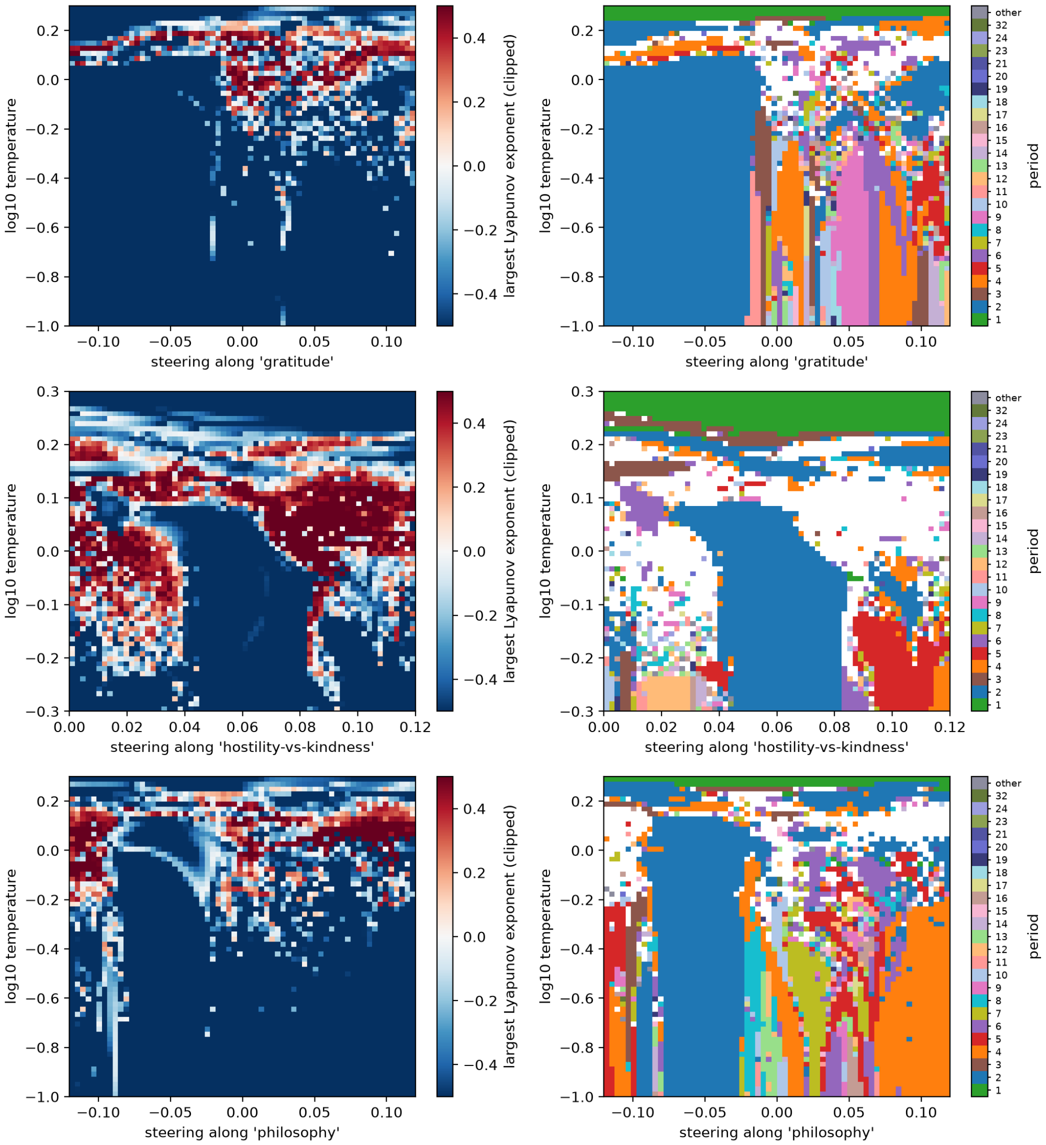


Figure 51: Llama-3.1-8B-Instruct soft-token maps (64^2 float64 cells, steering at layer 16) along gratitude, hostility and philosophy: largest Lyapunov exponent (clipped) and period of the attractor reached from the fixed start (white: no period up to 32).

Maps of 128^2 cells (Figure 52) resolve the windows: the hostility plane has 29 % chaotic cells, as at 64^2, and the philosophy region below the chaotic band is filled with nested periodic tongues of periods 2 to 13.

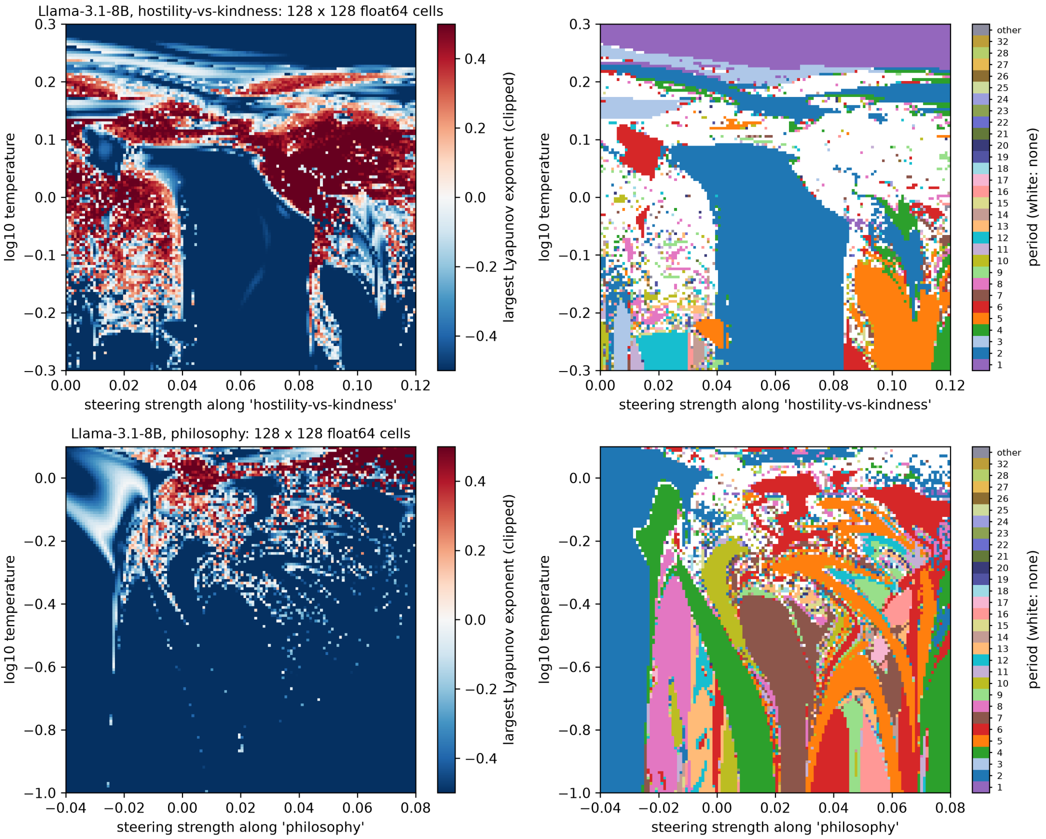


Figure 52: Llama-3.1-8B-Instruct soft-token maps steered at layer 16, at 128^2 float64 cells (each map merged from four 64^2 tiles on halves of one grid): the hostility plane (top) and the philosophy region of the 64^2 map's windows (bottom). Largest Lyapunov exponent (clipped) and period (white: none up to 32).

Same architecture, other weights. Qwen2-0.5B-Instruct has exactly the architecture and size of Qwen2.5-0.5B-Instruct (24 layers of width 896, the same vocabulary size and rotary base) but was trained on other data with another recipe. Built identically, their 128^2 float64 maps along gratitude and philosophy (Figure 53) share the gross layout. A large region at low temperature around $\alpha = 0$ is held by one periodic loop, a band of chaos with embedded windows lies near $\log_{10} T = 0$–0.25 (4–8 % of cells chaotic in each map), and thin window families run along the flanks. The attractors themselves differ almost completely. Of 416 to 900 distinct loops per map, only 10 occur in both models, all short and generic (“.”, “!”, “ you are”, “ is it”), and they cover less than 10 % of either model's cells. Qwen2's planes are dominated by one period-4 loop, “!<|im_end|>\nHello” (an end of turn followed by a new greeting; 41 and 59 % of cells), and Qwen2.5's by period-5 and period-2 loops (“![](https:// Hello”, “ you have”). The two checkpoints share the organization of the map; the weights set its content.

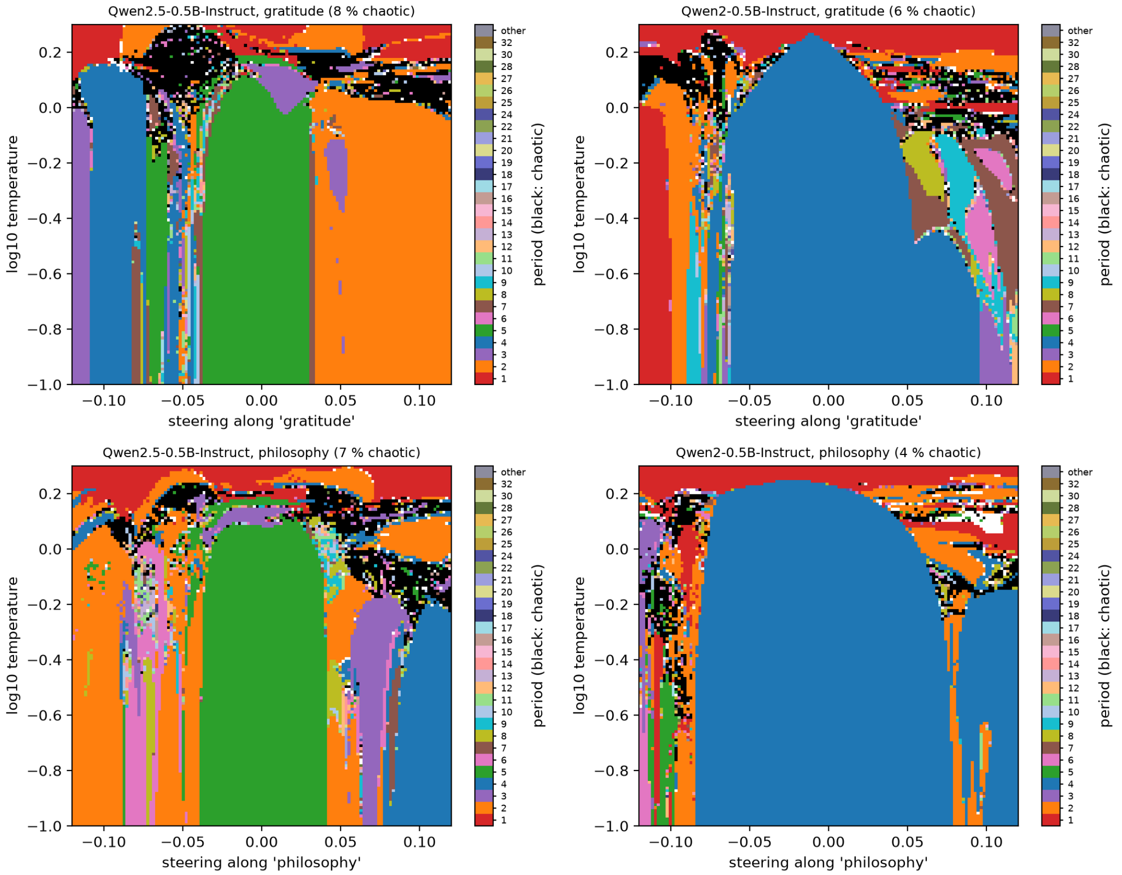


Figure 53: Same architecture, other weights: 128^2 float64 maps of Qwen2.5-0.5B-Instruct (left) and Qwen2-0.5B-Instruct (right) steered along gratitude (top) and philosophy (bottom) at layer 12, with vectors built identically. Color: period of the attractor reached from the fixed start; black: positive largest exponent.

The fold law at 1.5B and 7B. We tested whether the mechanism of Section 8, windows accumulating on the fold of a short cycle with $d_N = C/(N + N_0)^2$, recurs in larger models. The criterion, a constant $d_N(N+N_0)^2$ within 10 % over at least four windows for one offset N_0, was fixed before the lines were computed; a disclosed correction to the period-progression selector after the first scoring kept the conclusions for the nearest four windows and changed which farther windows were selected. In Qwen2.5-1.5B (philosophy direction, layer 14, $\alpha = 0.058588$) the period-4 orbit folds at $\log_{10} T_c = -0.0912029$ (multiplier 1 to 3×10^{-14}, next modulus 0.008), with normal-form coefficients $a = -7.16$ per unit $\log_{10} T$ and $b = -21.09$, so $ab = 151.0$. A float64 line of 1700 cells, 1570 of them above the fold, shows windows of periods $4N$ accumulating on it, from single cells at periods 44–64 to a wide period-12 window. Measured in $\log_{10} T$ from $\log_{10} T_c$ to their edges nearest the fold, the four nearest resolved windows (periods 28–40) meet the criterion within 0.12 % ($\delta = -0.99$, $C = 0.0612$). With the windows farther out the spread rises to 10.3–14.0 %; there several periods recur in multiple disconnected runs on the line's 10^{-5} grid, and grid-spaced edges no longer identify the fold edges. In Qwen2.5-7B ($\alpha = 0.051429$) the period-2 orbit folds at $\log_{10} T_c = -0.2040156$ (next modulus 0.0092, $ab = 55.7$), and odd windows accumulate on it. The nearest four and five (periods 11–17 and 9–17) meet the criterion within 4.5 %, and six (to period 7) within 6.1 %. Each window's own doubling or tripling lies on its far side, as for the 0.5B windows of periods 16–20; this organization is consistent with fold edges on the near side, and the fitted distances use grid estimates of those edges. The effective passage fraction $f = \sqrt{Cab}/\pi$ is about 0.97 at 1.5B and 0.78 to 0.81 at 7B, against 0.58 at 0.5B. The fold and its window law thus recur at all three sizes, while f, an effective constant that absorbs the global reinjection, differs between the models. With one fold per model, crossed in different coordinates (α at 0.5B, $\log_{10} T$ at 1.5B and 7B), no trend of ab with size can be read.

A mixture-of-experts model. Qwen3.5-35B-A3B differs from the dense models in two ways. Thirty of its 40 layers use linear (Gated DeltaNet) attention, with gated full attention in every fourth layer, and each layer routes a token to 8 of 256 experts plus a shared expert, so that about 3B of its 35B parameters are active. Hugging Face computes several parts of this architecture in float32 even when the model is cast to float64 (the normalizations, the delta-rule recurrences, the rotary tables and the router softmax), and some tensors are stored in float32; we patch these paths, restore the stored tensors exactly, and disable the thinking block that the default chat template opens (a first run with the block open is kept in the research log). Steered at layer 20, the map behaves unlike those of the dense models. Unsteered, 128 temperatures over $\log_{10} T \in [-1, 0.3]$ give no positive exponent (largest -0.265): periodic loops at all but 15 temperatures ("!Hello" with period 2 at 92 of them, other periods up to 52), and the remaining 15, above $\log_{10} T \approx 0.1$, contract without a detected period. Along philosophy, none of 1024 cells is chaotic (largest exponent -0.112). Along hostility, 52 of 1024 cells (5.1 %) have a positive exponent and no period, against 28 % for Llama-3.1 8B on the same plane. Routing is discontinuous: a twin trajectory that crosses a routing boundary jumps, which can inflate a finite-difference exponent, so we call these cells irregular rather than chaotic.

Two dense models of the same family, Qwen3.5-4B and Qwen3.5-9B (32 layers each, 24 of them linear attention), test whether the architecture causes the contraction. With concept vectors at their middle layer (16 of 32) and the same ranges, transients and maximal periods, the 4B model is chaotic at 19 of 128 unsteered temperatures (largest exponent 0.79), at 153 of the 1024 philosophy cells (0.84) and at 323 of the 1024 hostility cells (0.89). The 9B model is chaotic at 10, 73 and 161 (0.48, 0.68 and 0.75; Figure 54). The hybrid attention therefore does not prevent chaos. The routing-aware version of the stack resolution (Appendix B) follows a third row whose routing is frozen to the base orbit's. At the twin offset 10^{-7}, neither mixture of experts switched an expert in any layer at any step, and the frozen row's network factor equals the free twin's. At every knockout temperature, the first attention layer (a linear-attention layer) is the largest amplifier in all three Qwen3.5 models. At the knockout temperature where the entropy is 2 to 4 nats, removing it lowers the network factor by 1.27 (4B), 1.33 (9B) and 0.76 nats (the mixture of experts). Removing the first MLP or expert layer lowers it by 0.92, 0.77 and 0.24.

OLMoE-1B-7B stores its experts in bfloat16. We convert each expert's weights to float64 when it is used, which is exact, compute the router softmax in float64, and cast everything else to float64; central differences of the map then converge as h^2. Its unsteered sweep of 130 temperatures over $\log_{10} T \in [-1, 1.2]$ has one temperature with a positive exponent and no period (0.023, irregular), and 7 with neither a detected period nor an exponent above 10^{-3}. Its median network factor between 0.3 and 6 nats (-0.36) lies between those of the dense OLMo-2-1B ($+0.11$) and OLMo-2-7B (-0.52). OLMoE normalizes before each sublayer and OLMo-2 after it, so this pair differs in more than the experts, and we read no size effect from it.

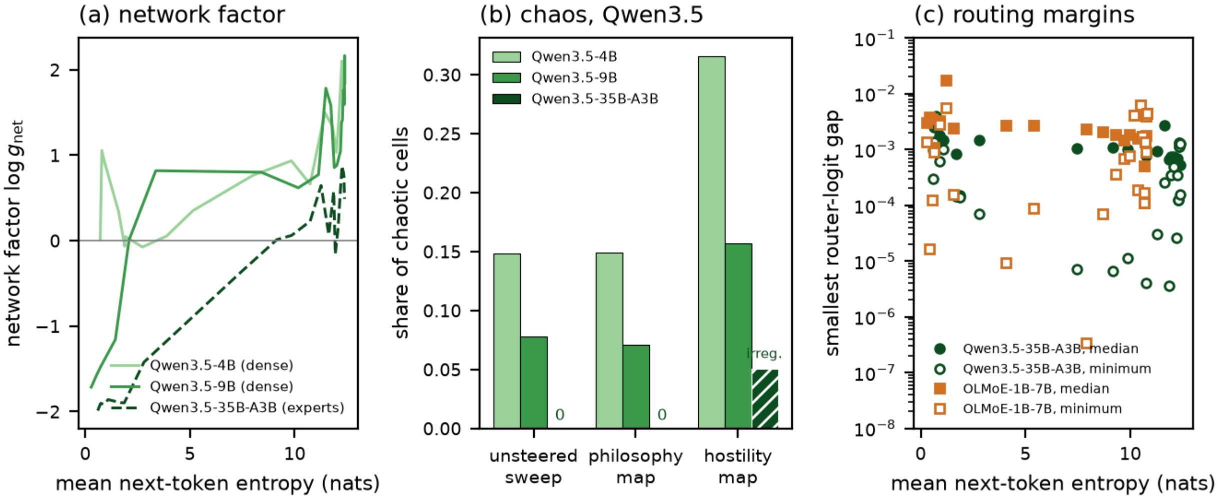


Figure 54: Mixtures of experts against dense models of the same family. (a) Network factor against the mean next-token entropy on the orbit, unsteered loop, 24 temperatures per model, float64 (as in Figure 31), for Qwen3.5-4B and -9B (dense) and Qwen3.5-35B-A3B (mixture of experts, about 3B of 35B parameters active per token). (b) Share of chaotic cells (by the criterion of Appendix B) in the unsteered sweep (128 temperatures, $\log_{10} T \in [-1, 0.3]$) and in 32×32 maps steered along philosophy and hostility at the middle layer; the mixture of experts' hostility cells (hatched) are called irregular, since routing jumps were not excluded when that map was computed. (c) Routing margins along the unsteered orbits of both mixtures of experts, Qwen3.5-35B-A3B and OLMoE-1B-7B (24 temperatures each, as in (a)): per temperature, the median over steps (filled) and the minimum (open) of the per-step smallest gap between the eighth and ninth largest router logits over all layers. A perturbation must move the router logits by about this gap to switch an expert; a twin at relative offset 10^{-7} never did.

I Chaos and model size: further results

This appendix completes Section 14: the 70B-class models and their precision check, cheap predictors of the chaos, steering at 70B, and the supporting figures.

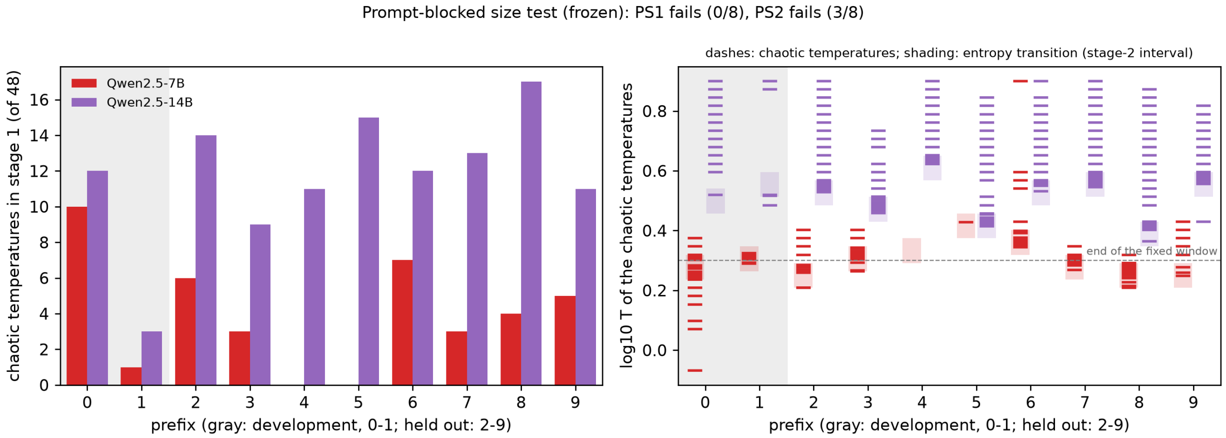


Figure 55: Prompt-blocked size test (frozen before any run), Qwen2.5-7B and -14B on ten prompts (gray: development). Left: chaotic temperatures among 48 over $\log_{10} T \in [-0.4, 0.9]$, the primary endpoint. Right: where the chaotic temperatures lie (dashes; both stages), with each model's entropy transition, the interval of its largest entropy increase (shading). Chaotic: exponents above 10^{-3} at both twin offsets (10^{-5}, 10^{-7}) and no period up to 64; 1000 transient and 2000 measured steps, float64.

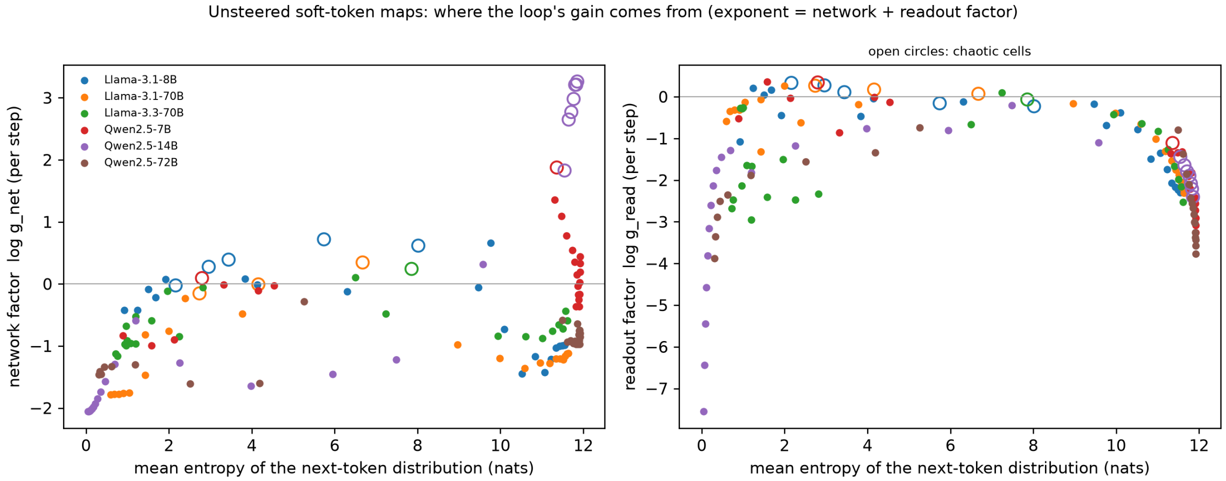


Figure 56: Gain split of the unsteered loop (Section 3) at 24 temperatures per model (500 transient and 1000 measured steps, relative twin offset 10^{-6}): network factor (left) and readout factor (right) against the mean next-token entropy on the orbit; open circles: chaotic temperatures. The two factors add up to the largest exponent up to a boundary term. Medians per entropy bin are given in Section 14.

At 70B, unsteered chaos is rare, and a precision defect made part of it spurious. The regular grids of the 70B-class models find little unsteered chaos: Qwen2.5-72B has no positive exponent at any of 191 temperatures between $\log_{10} T = -1$ and 1.2, Llama-3.3-70B none in $[-1, 0.3]$, and Llama-3.1-70B three temperatures in $[-1, 0.3]$. Finer lines found candidates: a window of positive exponents at Qwen2.5-72B's entropy jump ($\log_{10} T = 0.3728$–0.376) and two isolated temperatures of Llama-3.3-70B (0.250, 0.306), all at the single twin offset 10^{-6}. The window failed a three-offset test: the exponent is negative at all 64 temperatures of the window at 10^{-5}, near zero at 10^{-6}, and positive at 60 of the 64 at 10^{-7}, including four temperatures whose state repeats with a period, which no stable periodic orbit allows; in a repeat it is negative at all 16 temperatures at 10^{-4}, 3×10^{-5} and 10^{-5}. The growth of small perturbations came from our computation, whose two-GPU runs computed the final normalization in float32 (see the start of Section 14); a controlled reproduction of this false chaos is in Appendix B (Figure 36). Recomputed in float64 throughout, the window's orbits have periods 2 and 4, and their exponents are negative at all four offsets from 10^{-4} to 10^{-7} and for an exact tangent vector carried along the orbit by forward-mode differentiation, which needs no offset; the four twin exponents lie within 0.015 of the tangent exponent at every temperature, and the exponent approaches zero only where the period doubles from 2 to 4 ($\log_{10} T = 0.374$). The selected cells below kept their labels at the larger offsets we tested; this is a finite check, not a qualification of every cell of the original maps. In the steered Llama-3.3-70B maps below, twelve randomly drawn chaotic cells are chaotic at offsets 10^{-4}, 10^{-5} and 10^{-6} alike, with exponents that agree within 0.052, and the drawn periodic and unresolved cells stay negative at all three; on one GPU, the exponents at 10^{-7} and 10^{-5} agree within 0.021 at every chaotic temperature of the prompt-blocked test. The same test on Llama-3.1-70B's three unsteered chaotic temperatures confirms two of them ($\log_{10} T = 0.136$ and 0.198; exponents 0.23 to 0.26 and 0.40 to 0.41 at the three offsets) and rejects the weakest (0.116; positive only at 10^{-6}), so unsteered chaos exists at 70B, if rarely. Two Qwen2.5 models are chaotic far above the fixed window on the paper's prompt, where the next-token distribution is nearly uniform: the 1.5B model at 55 of 224 temperatures, between $\log_{10} T = 0.28$ and 1.04, and the 14B model at 53, between 0.51 and 1.15 (exponents up to 1.14), at mean entropies up to 11.9 nats against $\log V = 11.93$. The other models of the series are not chaotic above $\log_{10} T = 0.6$. For the 1.5B model a longer test (3000 measured steps) at twin offsets 10^{-5} and 10^{-7}, at 12 temperatures from 0.288 to 0.58, finds 9 chaotic and 3 periodic, with the two exponents agreeing within 0.003 at every temperature; one of the periodic ones ($\log_{10} T \approx 0.394$) has positive exponents at both offsets, a contradictory label that we report as returned. On these sampled trajectories the upper sweeps become non-chaotic above about $\log_{10} T = 1.1$ in these two models; we did not locate the uniform contraction that Section 3 proves for large enough temperature.

Can a cheap property predict it? Motivated by the saturation and high-temperature limits of Section 3, we tested the breadth of intermediate entropy as a heuristic predictor: a model whose next-token distribution jumps from sharp to flat over a small range of temperature might leave little room for chaos. The breadth of this intermediate regime, the decades of $\log_{10} T$ in which the mean entropy on the orbit lies between 1 and 10 nats, ranges from 0.014 (Qwen2.5-72B) to 0.69 (OLMo-2-32B). It orders the fifteen models' chaotic length (Spearman +0.55) somewhat better than their parameter count does (−0.44; Figure 30, right), but it misses the chaos of Qwen2.5-1.5B and -14B at nearly uniform distributions, above its 10-nat bound. This analysis was adapted after the OLMo-2-32B result, and the breadth is measured on the loop's own orbits, which chaos itself spreads, so the test is partly circular. A loop-free version, frozen before it was run (`docs/start-state-predictor.md`), uses one forward pass per model and prompt: the entropy profile of the logits at the loop's start state, and the one-step gain there. Its primary prediction fails: the start-state breadth on the paper's prompt orders the fourteen dense models' chaotic length with Spearman +0.30, below the frozen threshold of 0.5. The secondary criterion, its median over ten prompts, narrowly passes on the frozen outcome (+0.52; $p = 0.06$) and weakens on the expanded sweeps (+0.44), and the one-step gain at the start state is negative in every model, the most chaotic ones included (-0.5 to -3.4 per step), and orders nothing (-0.32). These features ran on two GPUs before the precision fix, which changes them only at about the 10^{-7} level. We have not established a reliable predictor from a single forward pass: as for the period-8 orbit of Section 3, it has to be measured on the loop.

Steering at 70B. Under steering, the 70B maps show chaos, rarely and not independently of the direction (Figure 57; labels at the single offset 10^{-6}, which the plateau test above supports for this computation). On Llama-3.3-70B's hostility plane, at the layer of Goodfire's 70B SAE and at equal norm, 4.3 % of cells are labeled chaotic along the concept vector, 3.5 % along the frozen-rule hostility feature, 1.1 % along a Gaussian direction and 0.2 % along a random live feature, against 28, 29, 39 and 27 % for the same four directions in Llama-3.1-8B (on one GPU); Qwen2.5-72B along hostility gives 0.8 %: eight cells, all at $\log_{10} T = 0.26$–0.30. Recomputed in float64 throughout, six of the eight are chaotic, with exact tangent exponents of 0.23 to 0.36 that the twins at all four offsets from 10^{-4} to 10^{-7} reproduce within 0.01; five have nearly uniform next-token distributions (11.0 to 11.1 nats) and one alternates between sharp and flat behind a period-2 leading token. The same temperatures without steering are period-2 orbits with exponents between -3.8 and -3.6, and the steered cells above $\log_{10} T = 0.325$ are near-uniform fixed points, so steering drives Qwen2.5-72B into chaos too, in a narrow band. The chaotic cells of the hostility directions lie on a ridge that steering moves from $\log_{10} T \approx 0.3$ to 0.1. A 128×128 zoom of the concept plane at the same offset, computed after the precision fix in sixteen bands with identical vectors and joined (Figure 32; $\alpha \in [0.06, 0.12]$, $\log_{10} T \in [0.12, 0.27]$), resolves it: a chaotic band (3410 of the 16,384 cells, 20.8 %; largest exponent 0.64), threaded by periodic windows, whose center steering moves from $\log_{10} T \approx 0.23$ at $\alpha = 0.06$ to ≈ 0.19 at $\alpha = 0.12$; fixed points lie above it, and windows of periods 3, 5, 6 and 7, up to 13, fill the plane below. Only 1.7 % of the cells have neither a detected period nor a positive exponent. A plateau test of 24 cells drawn by a rule fixed in advance (12 chaotic, 6 periodic, 6 unresolved; 300 transient and 300 measured steps, absolute twin offsets 10^{-4}, 10^{-5} and 10^{-6} and an exact tangent) confirms 11 of the 12 chaotic labels, whose four exponents agree within 0.012, and 5 of the 6 periodic ones. The two exceptions are neighboring cells near $\log_{10} T = 0.20$ where a long, weakly attracting loop (periods 15 and 16) and chaos are reached from the same start depending on rounding: one cell labeled chaotic reached the loop, and one labeled periodic reached chaos. The six unresolved cells have small negative exponents (-0.005 to -0.03) and no period up to 32. An earlier 64×64 zoom of the same window, computed before the fix, had 44 % such cells and, above the band, 515 cells with periods from 3 to 32 but a constant leading token, an artifact of the period test at that computation's noise floor; Figure 32 supersedes it.

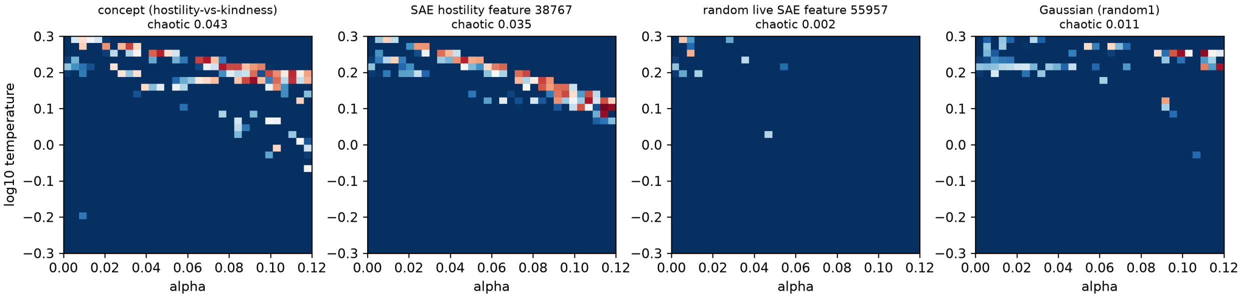


Figure 57: Steering controls at 70B: Llama-3.3-70B-Instruct soft-token maps (32^2 float64 cells, 300 transient and 300 measured steps, maximum period 32) steered after zero-based block 50, the hook point of Goodfire's 70B sparse autoencoder, along the hostility concept vector, the autoencoder's hostility feature selected by a rule frozen before any map, a random live feature and a Gaussian direction, all at the same norm. Chaotic cells: 44, 36, 2 and 11 of 1024 (concept, feature, random feature, Gaussian); cells with neither a period nor a positive exponent: 468, 358, 223 and 424.

J Unsteered chaos and steering controls

Steering adds a fixed vector to the residual stream, so chaos along a steering axis could belong to the push rather than to the model. At $\alpha = 0$, however, the map is the unsteered model's own feedback map, with temperature as its only parameter. The $\alpha = 0$ columns of the hostility maps contain cells with positive finite-horizon exponents in Llama-3.1 8B (26 of 64 temperatures), Qwen2.5-7B (15 of 64) and OLMo-2 7B (11 of 64), while the near-zero columns of the 0.5B and 1.5B maps contain one such cell each (of 128 and 64). A longer float64 sweep of the unsteered 0.5B map (512 temperatures over $\log_{10} T \in [-1, 0.3]$, 1000 transient and 2000 measured steps) finds a narrow chaotic band at $\log_{10} T \approx 0.15$–0.21 (7 temperatures, exponents up to 0.28), with a period-5 loop at 437 of the 512 temperatures. The same long sweep finds 96 chaotic temperatures in Llama-3.1 8B, between $\log_{10} T = -0.20$ and 0.24 (exponents up to 0.66), interleaved with periodic windows of periods 2 to 52; within $\log_{10} T \in [-0.3, 0.3]$ its chaotic share, 41 %, equals that of the coarse column. Qwen2.5-7B has 65 chaotic temperatures between $\log_{10} T = -0.07$ and 0.28 (exponents up to 0.74), and OLMo-2 7B has 32 between 0.04 and 0.14 (up to 0.30). Steering is therefore not needed to reach chaotic regimes in any of these four models; where it does create chaos, it does so mainly through the readout, by making next tokens compete (Section 3). Temperature alone produces chaos with embedded windows, broadly in the 7B and 8B models and in a narrow band in the 0.5B model (Figure 58). Across the fifteen models of Section 14, the regular-grid sweeps find unsteered chaos in eleven of the fourteen dense models (in Llama-3.1-70B at two temperatures that pass a three-offset test), but not in Qwen2.5-72B, Llama-3.3-70B, OLMo-2-13B or the mixture of experts; finer lines found candidates in the first two, and the Qwen2.5-72B window, recomputed in float64 throughout, is periodic.

Steering does widen the chaotic regions, but not as a property of the concept: along norm-matched random directions, the 0.5B maps are as chaotic as along the concept directions (Figure 59). On the same 128^2 grid, 7.6, 8.3 and 12.5 % of cells are chaotic for three Gaussian directions with the philosophy vector's norm, against 6.8 % for philosophy and 7.5 % for gratitude. In all five maps the chaotic share rises from about 1 % at $|\alpha| < 0.01$ to between 5 and 22 % at larger $|\alpha|$, and a chaotic region at high temperature, with windows reaching down, recurs. In this model, chaos under steering is therefore a generic response to perturbing the residual stream at this norm. Rotating the steering direction makes the same point continuously: along the great circle from one Gaussian direction to another, 90° away, at constant norm, the chaotic regions move and change shape, but the chaotic share stays between 13 and 24 % at every one of 103 angles (Figure 7; a video of the rotation is in the supplementary material). Single neurons behave the same way. Steering along the output direction (Figure 60) of one MLP neuron of the steered layer (its column of the down-projection, rescaled to the same norm; adding it is the same as raising that one neuron's activation by a fixed amount for every token of the input: by 37.4 at $|\alpha| = 0.12$ for the neuron most specific to philosophy, whose mean activation on the philosophy texts is 0.27) gives 9.9 % chaotic cells for the neuron most specific to philosophy on held-out texts, chosen like the autoencoder feature below by a rule fixed before any map, 5.1 % for its index neighbor and 12.9 % for a random neuron (Figure 61); near $\alpha = 0$ the share is 0.4 to 1.2 %. The direction does select the texts. The largest attractor family in all five maps is the unsteered period-5 loop, but beyond it only 26 to 56 % of one map's cells carry an attractor text that appears anywhere in another map. Llama-3.1 8B agrees. On its hostility plane, two norm-matched random directions at the same layer give 26 and 17 % chaotic cells against 28 % along the concept direction; all three carry 34 to 44 % near $\alpha = 0$, and the random directions thin out at large $|\alpha|$. At layer 19, where a sparse autoencoder trained on this model reads the residual stream [110], the concept vector rebuilt there gives 28 %. The decoder direction of the autoencoder feature most specific to hostility on held-out texts (chosen by a rule fixed before any map; cosine 0.10 with the concept vector) gives 29 %, a random live feature 27 % and a Gaussian direction 39 %, all at the same norm. In this model, chaos therefore occurs along autoencoder-feature, contrastive and random directions alike, with direction-dependent prevalence; at 70B the directions differ more (Appendix I).

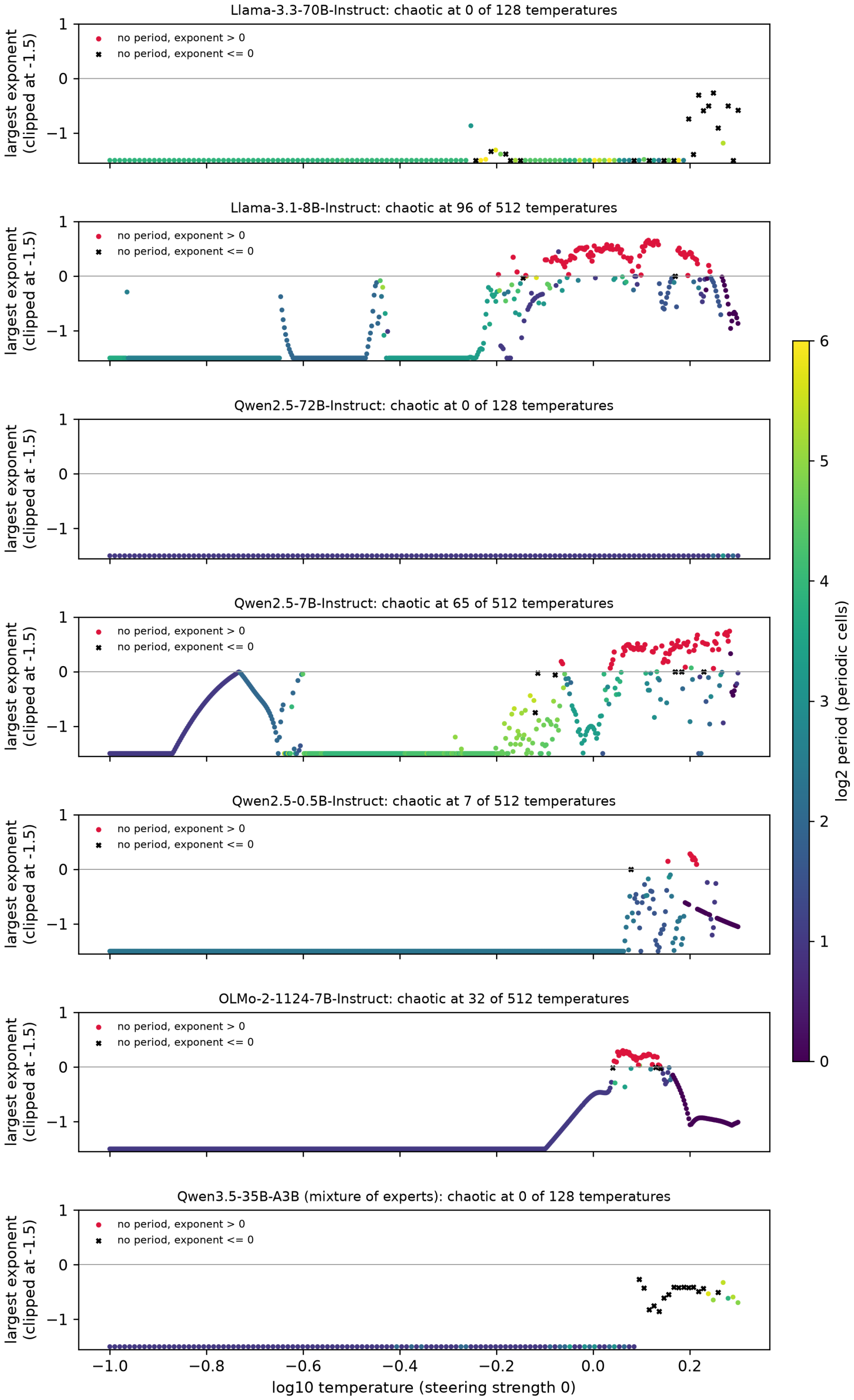


Figure 58: The unsteered soft-token map ($\alpha = 0$) along temperature, top to bottom: Llama-3.3-70B-Instruct, Llama-3.1-8B-Instruct, Qwen2.5-72B-Instruct, Qwen2.5-7B-Instruct, Qwen2.5-0.5B-Instruct, OLMo-2-1124-7B-Instruct and the mixture of experts Qwen3.5-35B-A3B. Largest Lyapunov exponent (clipped at -1.5) from float64 sweeps of 512 temperatures (128 for the 70B-class and mixture-of-experts models; 1000 transient and 2000 measured steps). Red: no period up to 64 and an exponent above 10^{-3}; black crosses: neither a detected period nor a positive exponent (unresolved); other points have a detected period, colored by $\log_2$ of the period. Table 14 summarizes these and the wider sweeps of the other models.

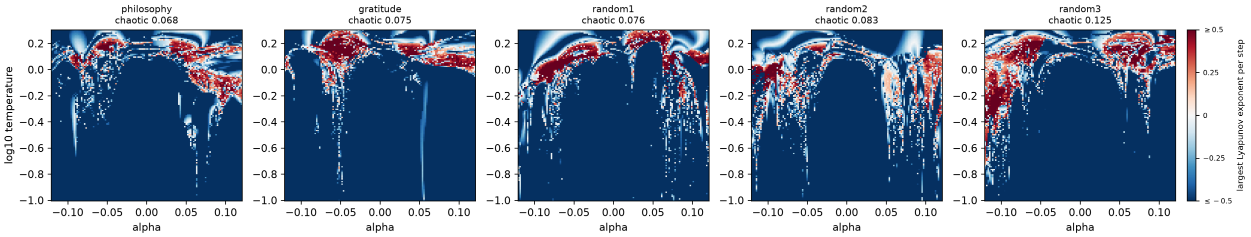


Figure 59: Steering controls in Qwen2.5-0.5B-Instruct (128^2 float64 cells, steering at layer 12, one grid and one set of settings): largest Lyapunov exponent along the philosophy and gratitude directions and along three Gaussian directions rescaled to the philosophy vector's norm. Titles give the chaotic share (finite-horizon exponent above 10^{-3}, no period up to 48); color gives the exponent, clipped at ±0.5.

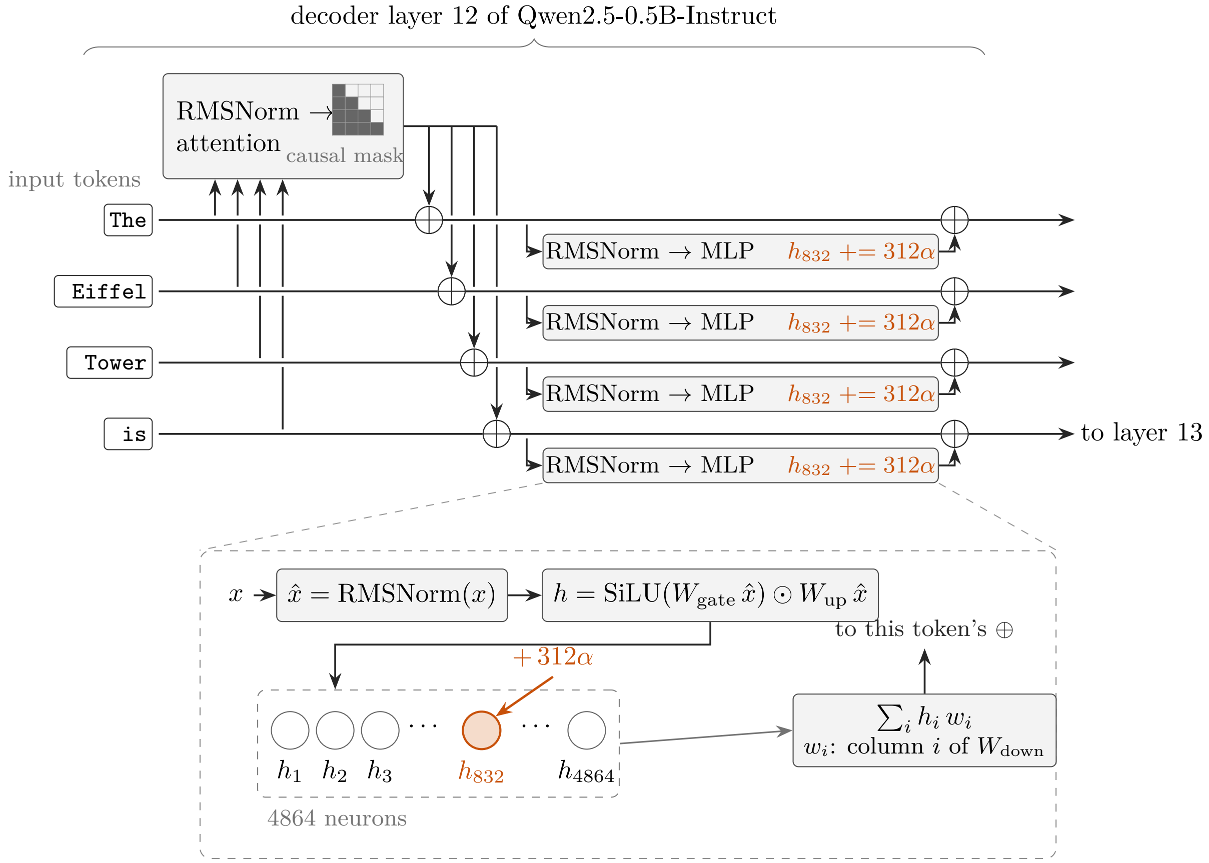

decoder layer 12 of Qwen2.5-0.5B-Instruct
RMSNorm → attention
causal mask
input tokens
The
Eiffel
Tower
is
RMSNorm → MLP
h_{832} += 312α
to layer 13
x
$\hat{x} = \mathrm{RMSNorm}(x)$
$h = \mathrm{SiLU}(W_{\mathrm{gate}}\,\hat{x}) \odot W_{\mathrm{up}}\,\hat{x}$
to this token's ⊕
$+312\alpha$
h_1
h_2
h_3
h_{832}
h_{4864}
4864 neurons
$\sum_i h_i\, w_i$
w_i: column i of W_{down}

Figure 60: Steering along one MLP neuron, in decoder layer 12 of Qwen2.5-0.5B-Instruct. Each token has its own residual stream, one 896-dimensional vector per token, and its own pass through the same MLP; attention, with its causal mask, is the only step that mixes tokens. Inside every MLP box (bottom, for one token with residual vector x), the 4864 neurons write $\sum_i h_i w_i$ into the stream, neuron i along its own column w_i of the down-projection. The steering strength α is a scalar we set (negative values lower the activation), and $312 = \|v_{\mathrm{phil}}\|/\|w_{832}\| = 188.99/0.606$ is a fixed scale for neuron 832. Raising h_{832} by 312α for every token is the same as adding αv, with $v = 312\,w_{832}$, to the layer's output, which is how we implement it: $\sum_i h_i w_i + 312\alpha\, w_{832} = \sum_{i \neq 832} h_i w_i + (h_{832} + 312\alpha)\, w_{832}$. At $|\alpha| = 0.12$ the increase is 37.4, against a mean activation of 0.27 on the philosophy texts; for neurons 833 and 4137 the scale is 329 and 346. The tokens shown are only an example: in the loop, the input is the fixed prompt followed by the fed-back soft token.

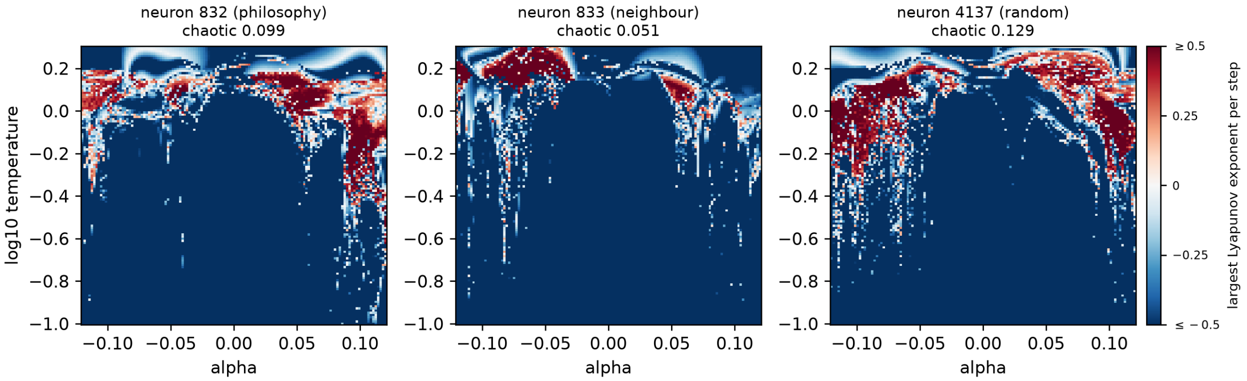


Figure 61: Steering along single MLP neurons in Qwen2.5-0.5B-Instruct (same grid, settings and color scale as Figure 59): the output directions of the layer-12 neuron most specific to philosophy on held-out texts (832), of its index neighbor (833) and of a random neuron (4137), each rescaled to the philosophy vector's norm (cosines with it 0.013, −0.028 and 0.022). Titles give the chaotic share.

K Limitations in detail

The detailed orbit studies cover one model family (Qwen2.5, 0.5B and 1.5B), with coarse maps and three float64 orbits at 7B; the two other families (OLMo-2 7B and Llama-3.1 8B) have maps, including 128^2 refinements for Llama, but no continued orbits. The size comparison of Section 14 uses regular temperature grids that can miss narrow windows, a window that missed most of the larger models' chaos, and mostly one twin offset per label. The two-GPU runs of the 70B-class models computed their final normalization in float32 until we fixed it; their labels rest on offsets of 10^{-6} or more, and the cells we rechecked, the Qwen2.5-72B jump window and steered candidates in float64 throughout and samples of the Llama maps at three offsets, behave as stated, but the rest of those maps were not recomputed. Neither were the four coarse lines of the frozen 72B test; after the fix we computed only fine lines through their entropy jumps, one of which passes through their single chaotic temperature. The maps use one prompt (Section 14 adds ten for Qwen2.5-7B and -14B, and four for Qwen2.5-72B, its base model, and OLMo-2-7B and -32B), and steering uses contrastive vectors, checked against norm-matched random and sparse-autoencoder directions in two models. The soft-token loop is one inference scheme: Appendix G extends it to memories of up to sixteen soft tokens and to the growing context of Soft Thinking, but sampled decoding of discrete tokens is a different system. The evidence consists of numerical validation, cross-checks and prospectively frozen passage tests (including failed all-row criteria), with a computer-assisted proof for only one orbit at one parameter point. Chaos labels are finite-horizon estimates (300 measured steps after 300 to 500 transient steps), and no chaotic cell is proved chaotic. A computer-assisted proof of chaos, such as a horseshoe verified by covering relations, would need enclosures of the map over boxes many orders of magnitude larger than the period-8 proof's (radius 2^{-640}). With the ball arithmetic used there, a box of radius 2^{-80} already loses the positivity of an RMS normalization at layer 3 (the enclosures overestimate through the layers), so such a proof needs other verified methods. A horizon check retested random cells of two float64 maps (Figures 1 and 8) with 1000 transient and 2000 measured steps at two twin offsets: 86 and 92 of 96 cells labeled chaotic stayed chaotic (all but one of the rest settled on periodic orbits), all 96 periodic cells in each map stayed periodic, and 29 and 27 of 32 cells with neither label settled on periodic orbits (18 and 20 of them with periods up to 32, the maps' maximum, the others with periods up to 64). For the cells that stayed chaotic, the map's 300-step exponent differs from the 2000-step exponent by a median of 0.020 and

0.030 (90th percentiles 0.064 and 0.096, maxima 0.23 and 0.30), and the 2000-step exponents at offsets 10^{-6} and 10^{-9} agree within 3×10^{-3}. The chaotic counts do not hinge on the threshold: raising it from 10^{-3} to 0.01, 0.02 or 0.05 leaves 77,646, 76,761 or 73,437 of the 78,868 chaotic cells of Figure 1. Near the accumulation of a cascade the labels are less reliable, because orbits there converge slowly or wander chaotically before they settle (transient chaos [96, 113]): on the hostile line of Figure 21 (left), 135 of the 235 cells labeled chaotic from 400 measured steps stayed chaotic and 91 were periodic. The Feigenbaum ratios are finite-order values that rise toward δ (the last is within 1 %); the limit itself is not computed. All orbit, bifurcation and chaos results concern the soft-token map, not sampled generation with a growing context. With sampling, the bridge of Appendix G tests only whether the bottleneck's laminar phases survive the noise; its other tests are deterministic (greedy decoding, longer fed-back memories and Soft Thinking's growing context).

L Evidence and reproducibility

We grade claims on an evidence ladder. Rung 1 means numerically validated (tolerances, residuals, convergence, precision varied); rung 2, cross-checked by an independent estimate, implementation or hardware; rung 3, a computer-assisted proof.

Search scale. The soft-token results come from an exploratory search. An initial census of the raw data at 00:19 UTC on 14 October 2026 counted 34 distinct maps (987 308 cells, 27 maps in float64) in six models, each with one prompt and one of three steering directions or none, plus one partial map. Of these, 29 maps and the partial one appear here; the five omitted are early discovery or side maps. Later additions are not in that count: the steering-control maps of Appendix J, the mixture-of-experts maps, and the sweeps, lines and maps of the models of Section 14 and Appendix I. The labeled maps contain about 2600 periodic families of at least six cells, counted per map. We singled out about 26 windows or families for float64 work: 20 are reported in detail, 2 only as map features and 4 not at all (an orbit whose continuation was abandoned, a cell that proved aperiodic, a band whose float32 orbit did not converge and a pilot that found no cusp within a bounded search). Of 54 Newton orbit attempts, 7 failed. All nine tangent reversals on continued fold curves (seven found by 14 October, and two on traces of a period-4 fold curve continued on 15 October) are numerically located cusps; the two reversals of a continued flip curve are smooth turns. The four reversals on the two fold curves of the period-5 window found on 15 October (below) are located cusps as well. Two of the five regions we flagged as shrimp candidates are reported as shrimp-like. On 15 October, the turning-point search of the two-block planes (not counted in Table 20) led to two more windows: a period-5 window, reported with its continued orbit and its continued fold and flip curves in Appendix F, and a period-6 window that was mapped but not continued. One further language model (Ouro-1.4B) was explored and is not reported; its apparent chaos was a long transient. Every window was chosen after the maps were inspected. Only the passage forecasts, the cusp-unfolding offsets and Huginn's relaxation test were fixed before the computations that test them (Table 20).

	explored	reported	note
Soft-token maps (distinct)	34	29	plus one partial map, reported
cells	987 308	893 868	omitted: 5 discovery or side maps
float32 maps	7	4	
Models with maps	6	6	one prompt; 3 directions and $\alpha = 0$
Periodic families (≥ 6 cells)	≈ 2600	–	counted per labeled map
Windows given float64 work	≈ 26	20 (+2)	+2: map features only
Newton orbit attempts	54	–	7 failed
Tangent reversals on fold curves	9	9	all located cusps (2 found on 15 Oct.)
Unfolding offsets, frozen	12	12	counts 12/12, stability 11/12
Shrimp-candidate regions	5	2	reported as shrimp-like
Tests fixed before outcomes	6	4	2 all-row criteria failed by one row
Language models loaded	8	7	Ouro-1.4B not reported

Table 20: Search scale on 14 October 2026, before the steering-control maps; the reversal row includes the two cusps located on 15 October (Huginn's frozen test counted once; of the two tests not reported here, one belongs to the separately reported sampled-dialogue study and one, a cusp pilot in a second checkpoint, is the unreported bounded search).

- Rung 3: the period-8 stripe orbit of Table 4. At $\alpha = 0.071674$ and $\log_{10} T = -0.115$, read as exact binary64 rationals (the temperature is the real number $\exp(\ln 10 \cdot \log_{10} T)$ for that exact dyadic $\log_{10} T$), the expected-embedding map of the stored Qwen2.5-0.5B-Instruct checkpoint (all 24 layers, the full 151 936-token readout, the fixed 37-token prefix, steering after zero-based block 12) has a primitive period-8 orbit, which is the unique multiple-shooting root in a product of eight boxes of coordinate radius 2^{-640} around refined centers, and the orbit is locally attracting. Ball arithmetic (Arb [114]) encloses the map and its Jacobian over each box (8×896 directions); a Krawczyk-type inclusion [115–117] for the preconditioned shooting system gives existence and uniqueness, separation of the first point from the points one, two and four steps later gives the primitive period, and a norm bound on powers of the enclosed monodromy gives attraction. The map is the real-number map of the stored constants, read as exact dyadic rationals: rotary angles, exponentials and square roots are real functions enclosed by the balls, not floating-point tables. An independent audit in exact rational arithmetic rechecked the saved witnesses (the scalar inequalities, the primitive-period separations and the row norms of the saved monodromy powers); it did not recompute the enclosures of the model or the dense matrix products, which rest on one execution. The proof trusts the Python, FLINT and Arb stack and the stored weights. It does not exclude other period-8 points in the first box, and it says nothing about the window's boundaries or global organization. The record (`certificates/llm-p8-certificate-final-001.json`) pins the SHA-256 hashes of the checkpoint, its configuration and the steering vector, and lists the 37 prefix token IDs.
- Rung 2: the 2-cycle's fold and its normal form. An independent reimplementation of the forward map and solver recovers α_c, a and b (to 6×10^{-15}, 3×10^{-12} and 2×10^{-6} relatively), and ab also follows from the stable-branch multipliers. The same reimplementation reproduces all 18 fixed-entrance counts of the controlled passage runs exactly (Table 6 lists 12 of them).
- Prospectively frozen: the passage forecasts at two new folds (Table 8); 23 of 24 within one iterate (16 of the 17 longer than five iterates), and the all-row criterion failed by one row. A second frozen test at two more folds (Table 9): 18 of 24 within one iterate of the primary forecast; it failed at the entrance passages of a period-6 fold, short by 3.1 to 5.2 %.

- Rung 2 as well: the period-2 and period-8 orbits of Table 4 (independent implementation), and the cascade's first four flip parameters (CPU and H100, agreement 4×10^{-15}).

- Rung 2 as well: the hostile 2-cycle fold and its normal form, reproduced by an independent float64 forward map with nested automatic differentiation (location to 6×10^{-10}, ab to 8 parts per million); its controlled passages were also checked independently, with the frozen all-six criterion failing by one row.

- Rung 1: the other orbits, the curves, the thirteen located cusps, the superstable parameters, the 1.5B flips, the hostile windows' edges, the spiral region's period-7 body and the cascade inside it, the three 7B orbits, and Huginn's float32 spectral estimates.

- Prospectively frozen, not independently reproduced: the scaling predictions P1–P3, the prompt-blocked size tests in Qwen2.5 (7B and 14B; 72B; 72B base against Instruct) and OLMo-2, and the loop-free predictor test of Section 14 and Appendix I, all scored by code pushed before the data, together with the three-offset plateau tests of the 70B-class maps and the float64 recheck of the Qwen2.5-72B cells with exact tangent exponents. These are float64 computations at rung 1; no independent implementation has rechecked them.

- Prospectively frozen on 15 October: the answer-level reproducibility test in Qwen2.5-0.5B (positive), whose saved rows and statistics an independent implementation reproduced (the generation itself was not rerun); its 7B extension (positive) and the behavior test (null on its primary question; no edge optimum), not independently reproduced. All are reported in Section 6, the behavior test in detail in Appendix D, and each was scored by code pushed before the data (`docs/repro-task-test.md`, `docs/bpc-test.md`).

- Independently reproduced from saved data (15 October): every summary field of the 0.5B batch-layout comparison (448 cells, four comparisons) and the gain-split, logit-bias and crossover summaries of Section 3 (5353, 6837 and 24 cells); the coefficients and offset rows of the flip normal-form test (Table 13); the 1.5B and 7B fold-law scores under both period selectors (all 2700 cells); and the scalar conditions of the nine located cusps, matched one-to-one to the tangent reversals of the seven continued fold curves. This reproduces the statistics and saved scalars, not the model runs, derivatives or eigenvectors. The within-prompt steered test (null) used an analysis fixed and pushed before its data, without a separate frozen design document.

- Prospectively frozen and independently replayed: Huginn's relaxation test on held-out data, confirmed in both parts; an independent implementation reproduced both from the saved rows and spectra (GSM8K part exactly, item-level part to 10^{-12}) (Section 15).

Hardware. Float64 GPU computations ran on rented NVIDIA H100 (NVL, PCIe and 80 GB HBM3), H200, H200 NVL and A100 80 GB (SXM4 and PCIe) GPUs, one GPU per run except for the 70B-class models, whose layers were spread over two; the float32 frames of the rotation video ran on RTX 4090 GPUs; the first reset test of Section 6 ran on a local NVIDIA RTX 2000 Ada workstation GPU and its replication on an H100 NVL; cross-checks, the interval proof, the second prospective passage test, the coexistence runs and the near-nilpotent searches ran on CPUs. The manifest `docs/manifests/runpod-runs.json` records for each run the GPU type, the code commit, the job and its arguments, the wall time and the exit status, and the available hashes of the uploaded code and input archives and of the retrieved results, with absent fields left explicit; per-run receipts and accepted provenance limitations are kept with it. Regenerated at 02:11 UTC on 16 October 2026, after the last run, it lists 438 runs with retrieved results (code and input-archive hashes for 330, result-archive hashes for 412) and 21 pods that ended without results. GPU types round float64 reductions differently, so quantities used across runs were checked across hardware where it mattered (for example, flip parameters agree between CPU and H100 to 4×10^{-15}). Code: `tools/explore_llm_soft_loop.py`, `tools/bifurcation_curves.py`, `tools/llm_feigenbaum.py`, `tools/explore_latent_loop.py`, `tools/cusp_check.py`, `tools/huginn_relax_eval.py`, `tools/huginn_arc_analysis.py`, `tools/steer_gain.py`, `tools/crossover.py`, `tools/repro_noise.py`, `tools/repro_task.py`, `tools/repro_summary.py`, `tools/bpc_analyze.py`, `tools/turning_scan.py`. Raw-data hash receipts: `evidence/llm-dynamics-data-receipts.json` (to 12 October), the per-run receipts in the research log, and the verified snapshot manifests in `docs/backups/`; independent cross-check and audit receipts are under `codex/` (`crosscheck`, `confirm-fold`, `hostile-fold-crosscheck`, `huginn-design-review`, `huginn-independent-replay`, `paper-review`).

[Planned: public code and data release at submission.]